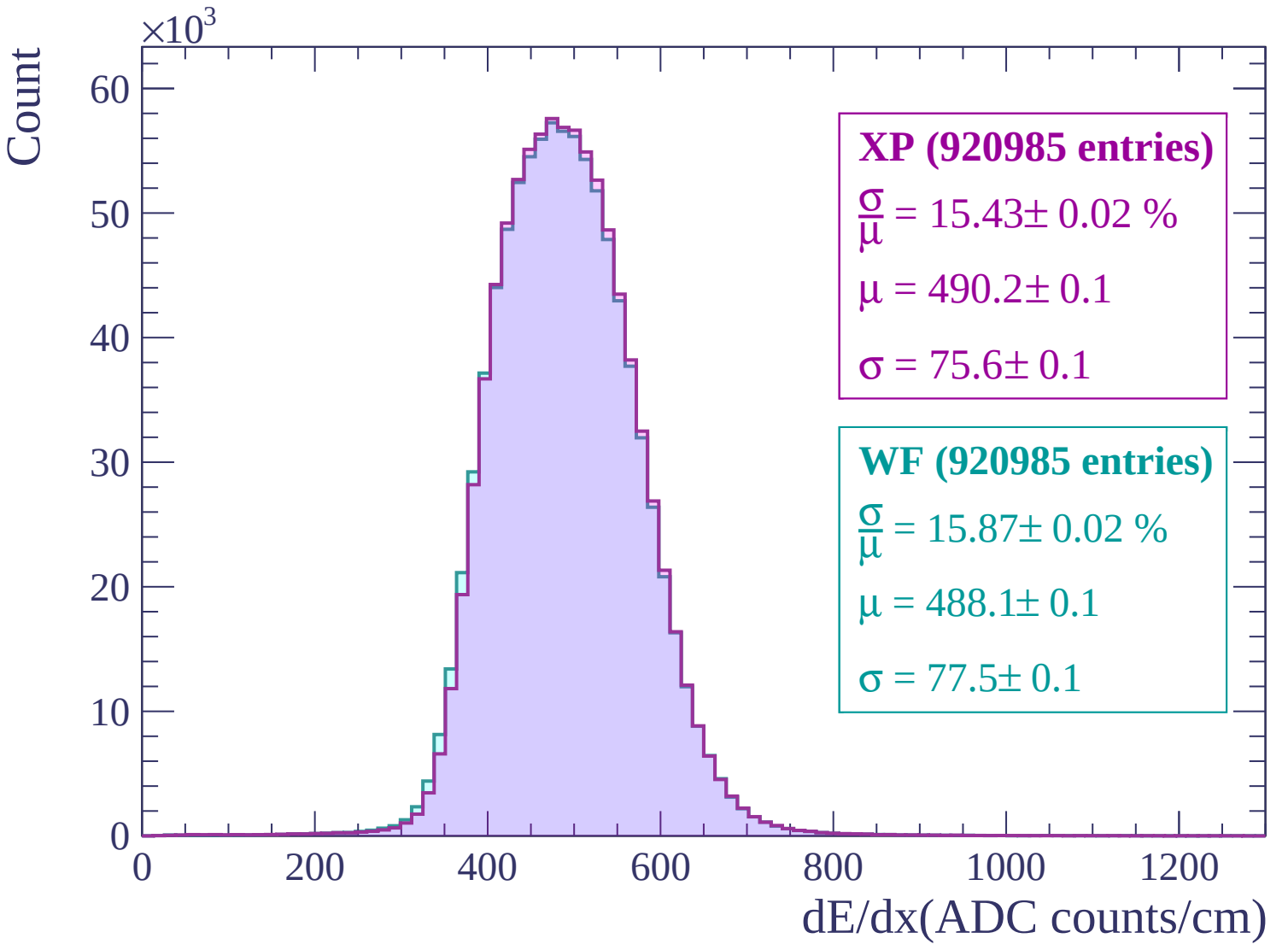
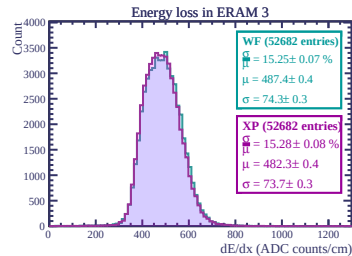
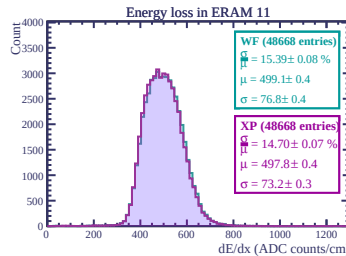
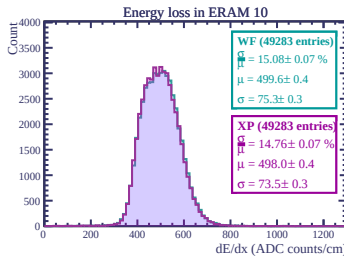
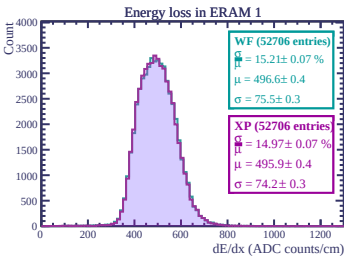
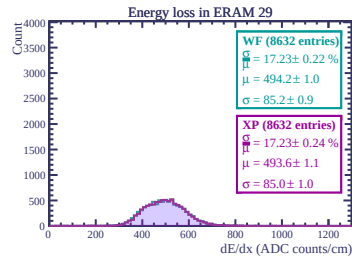
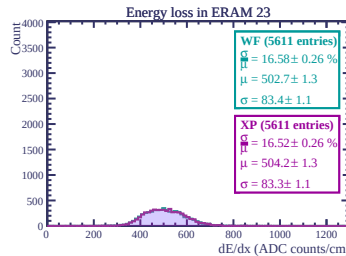
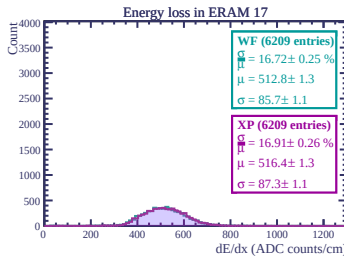
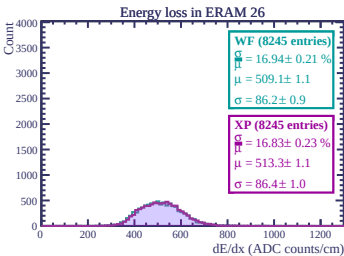
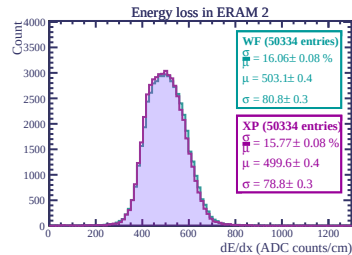
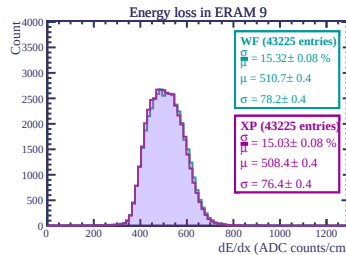
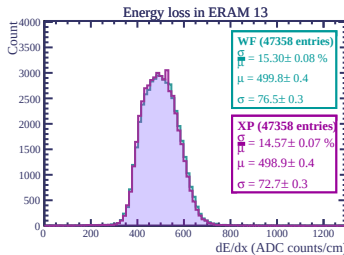
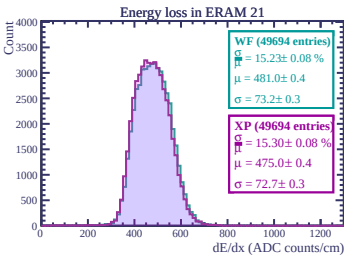
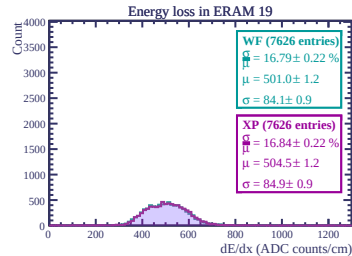
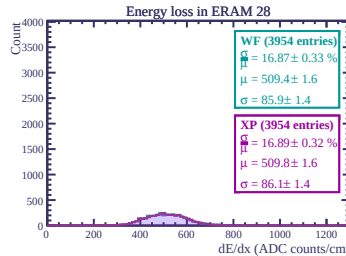
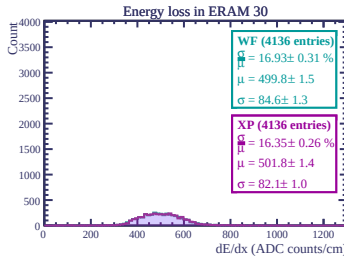
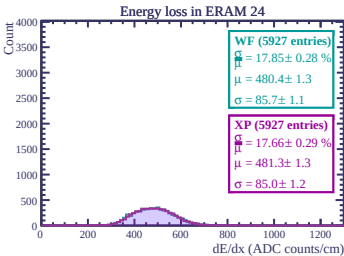
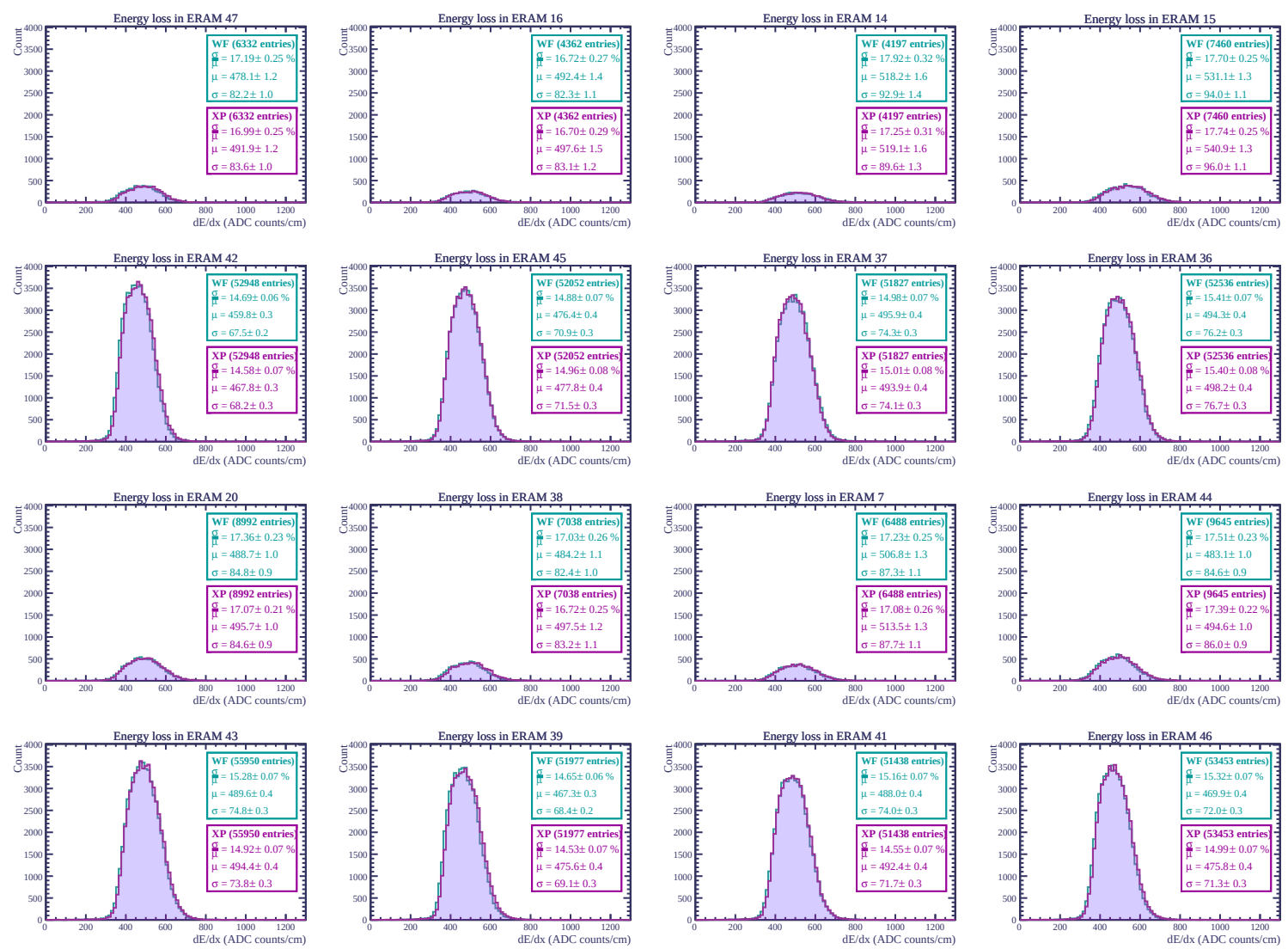
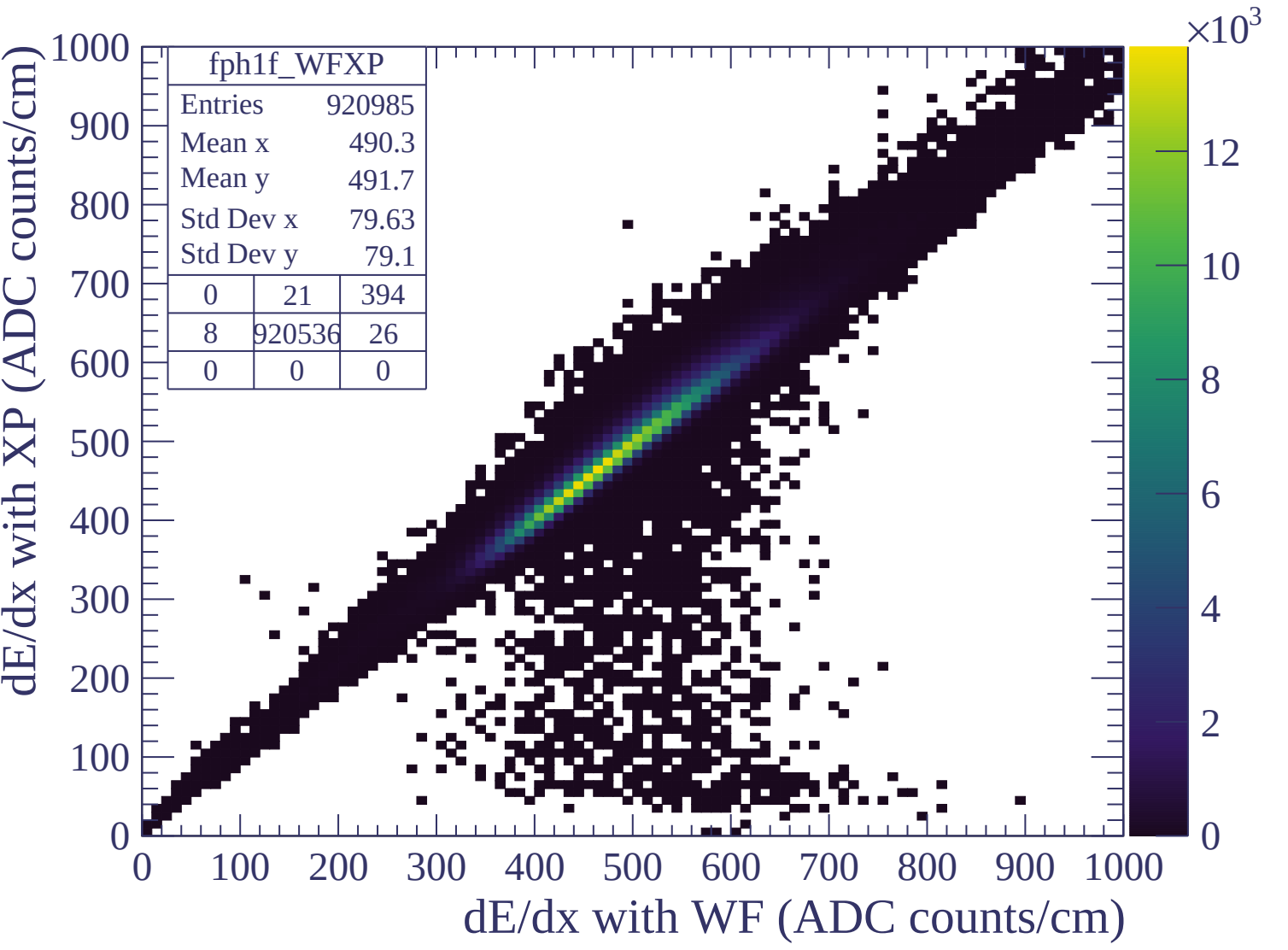


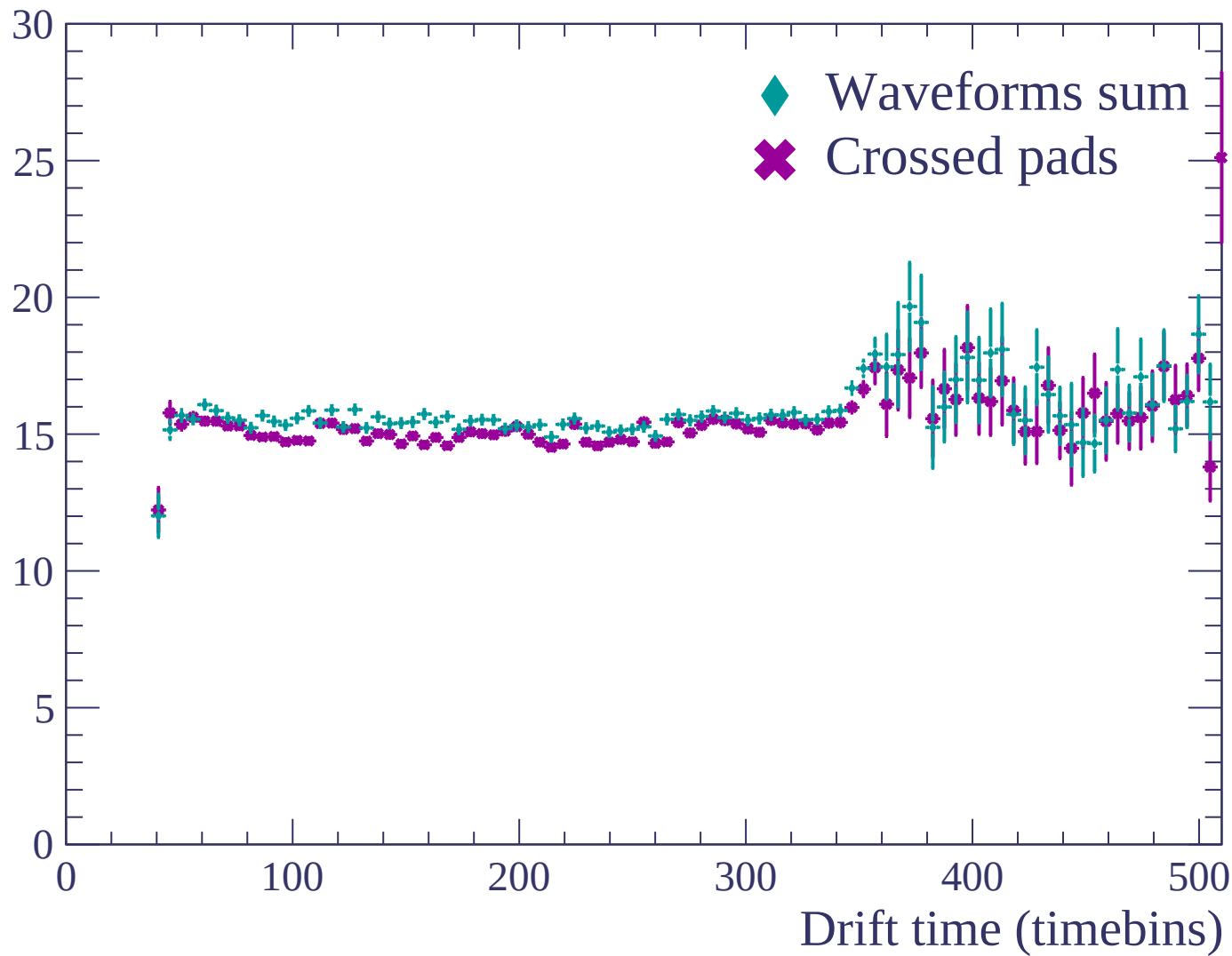


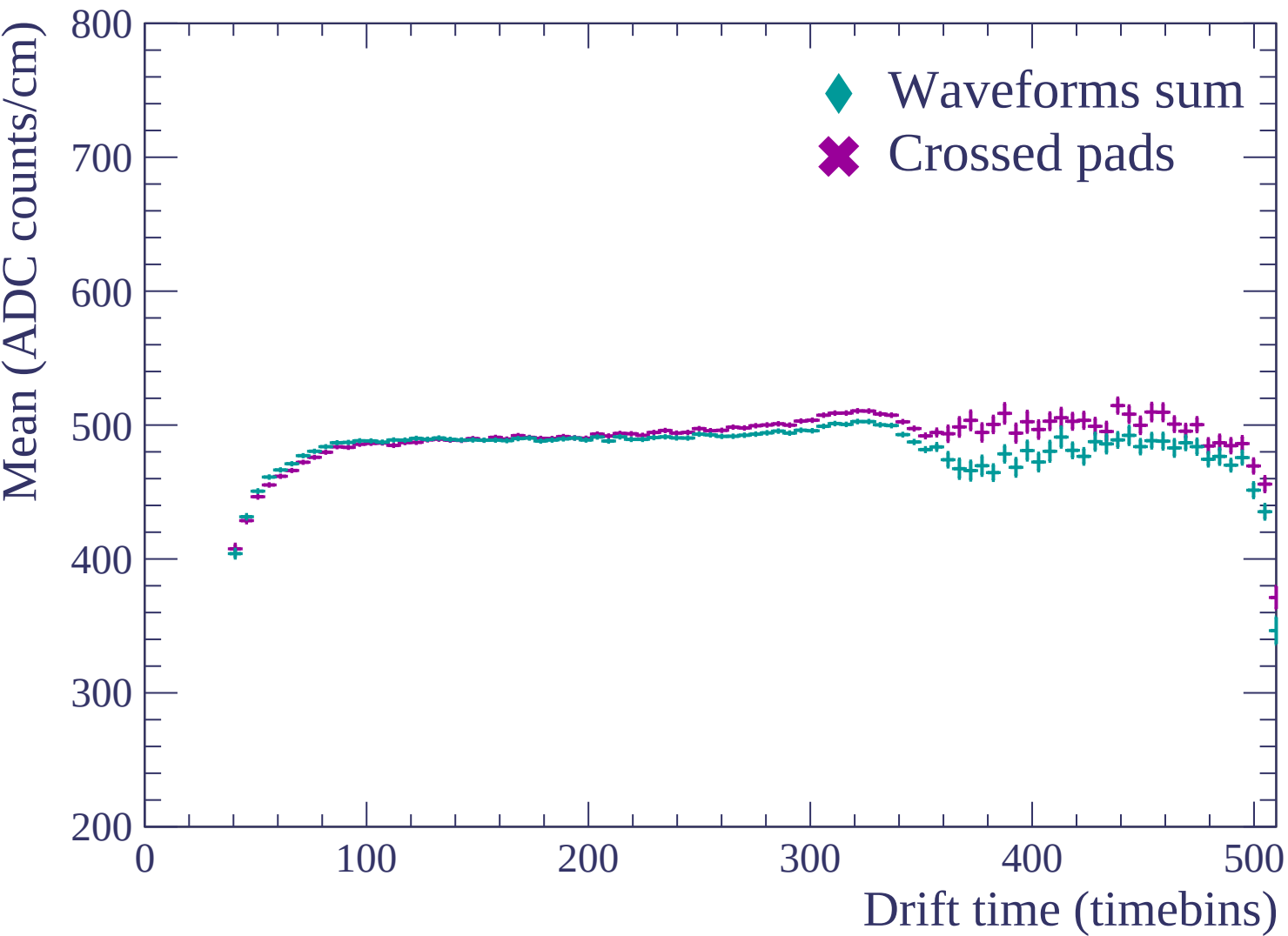


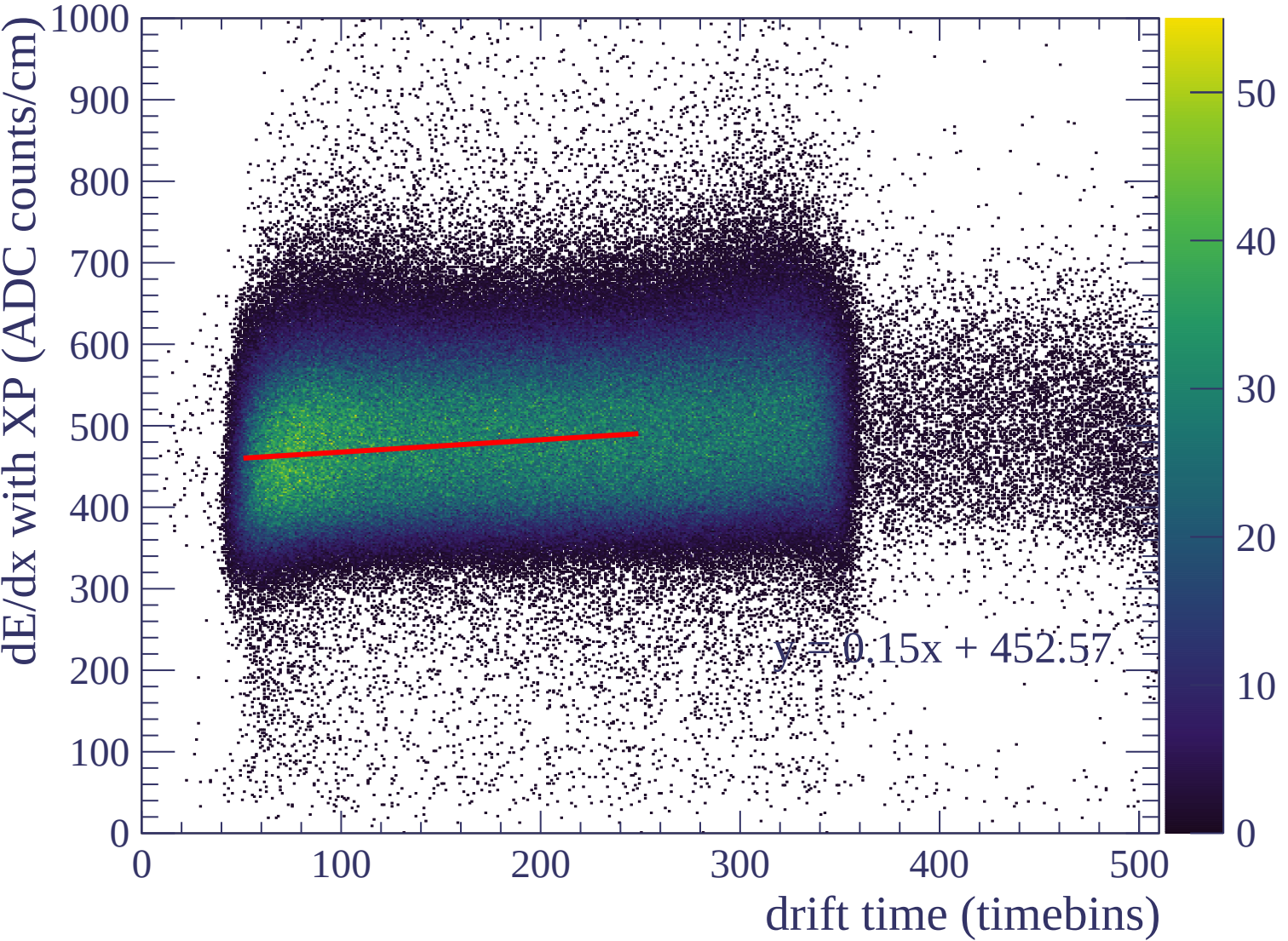


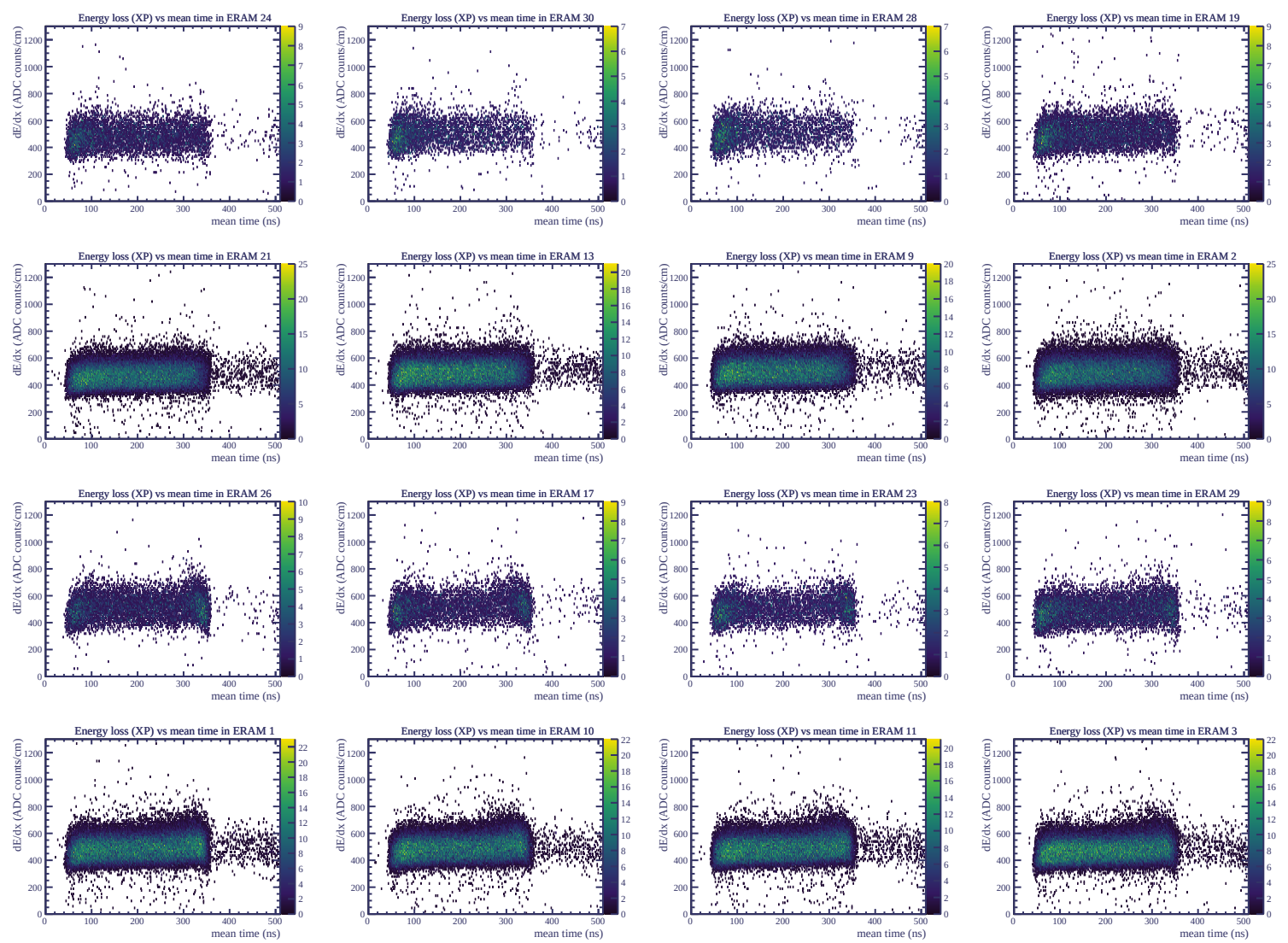


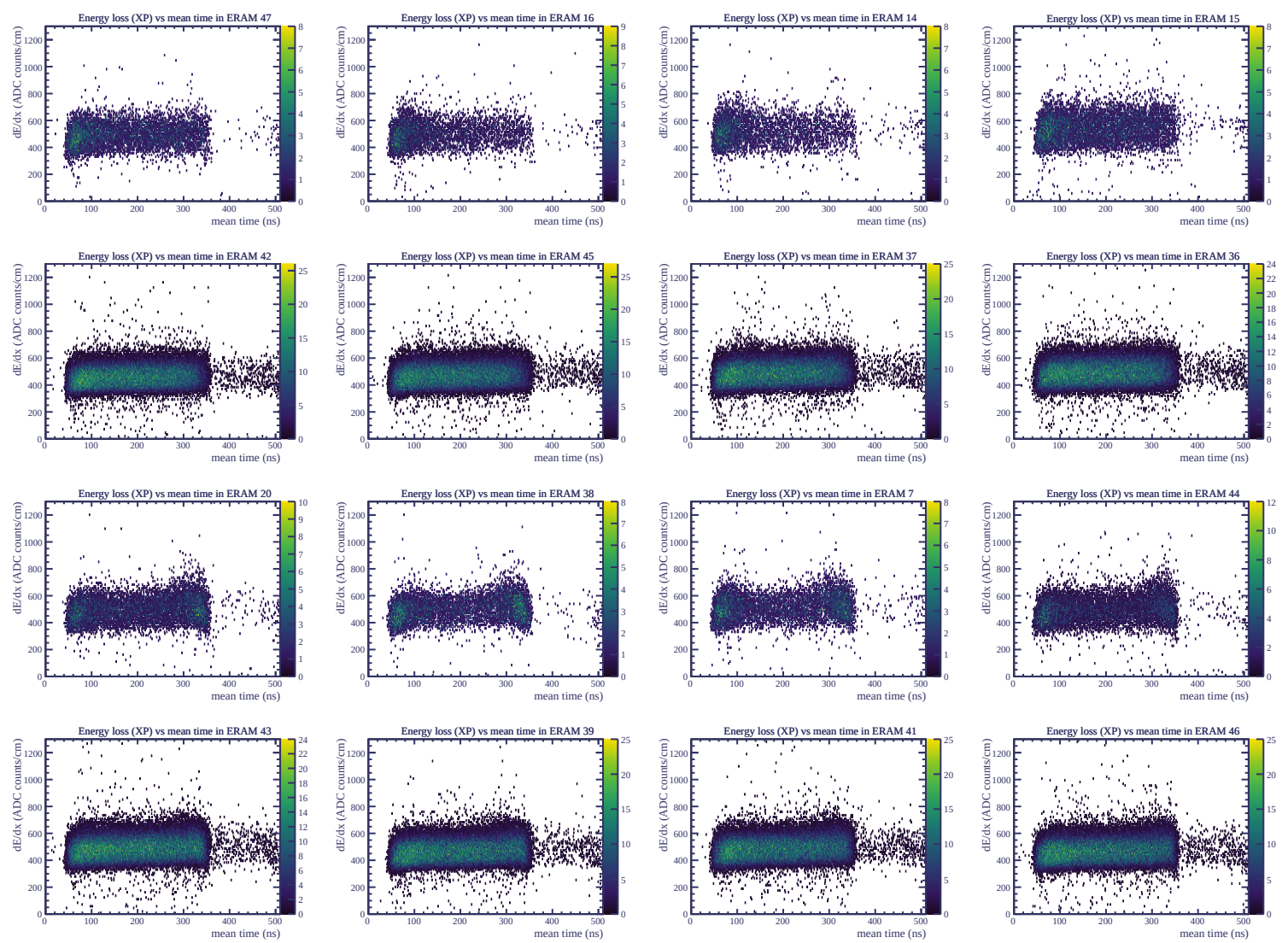
Resolution (%)

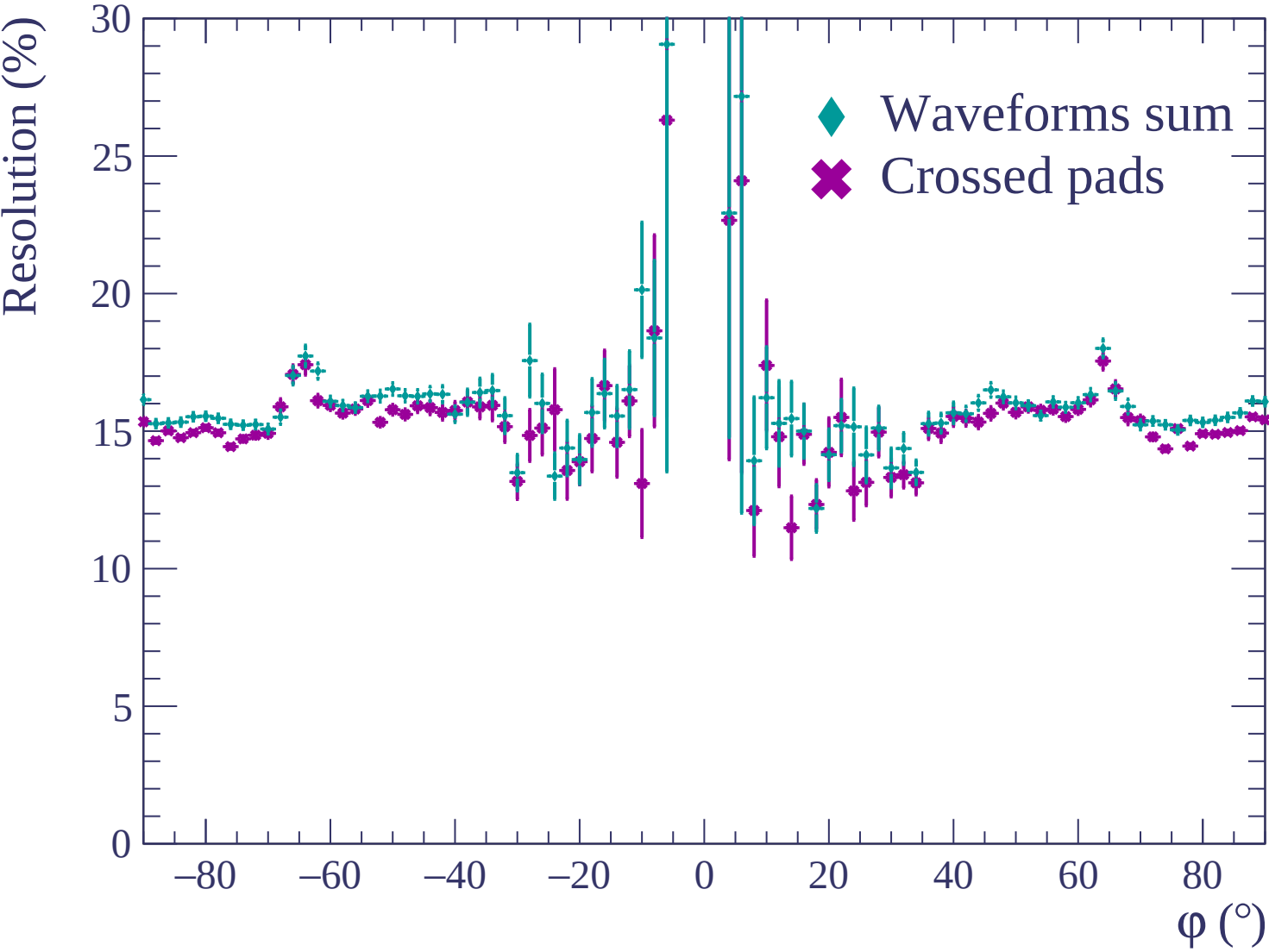


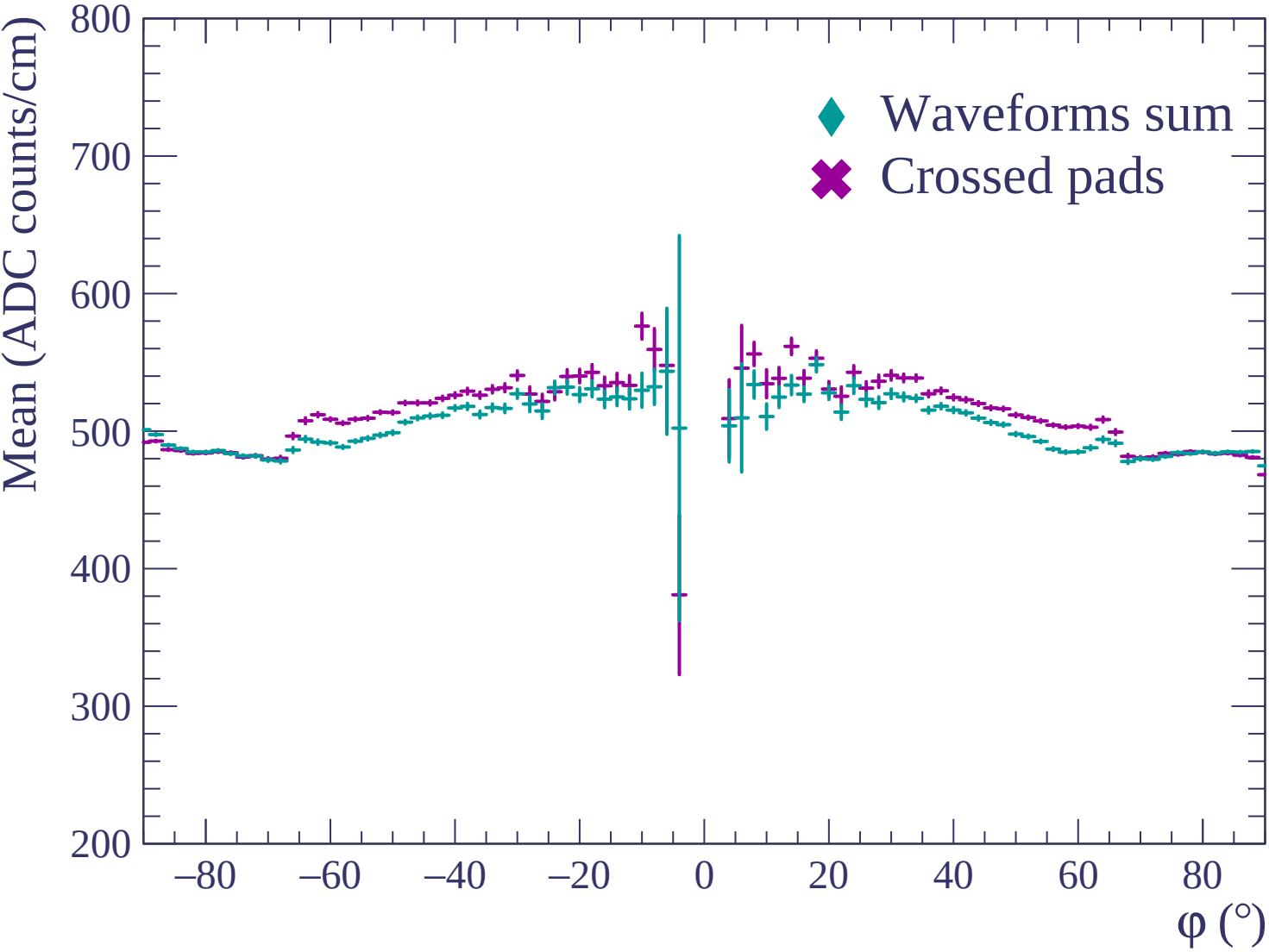


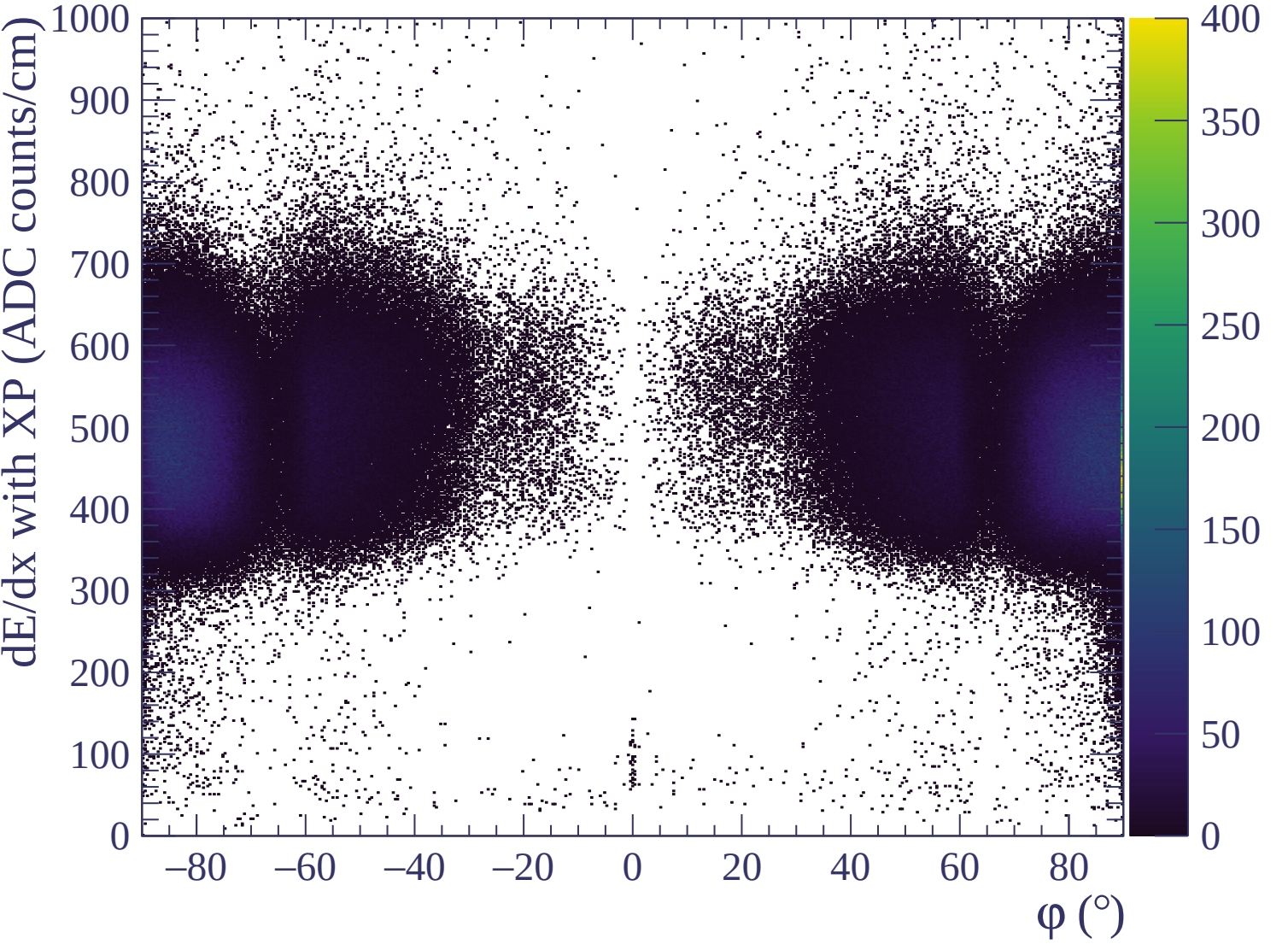


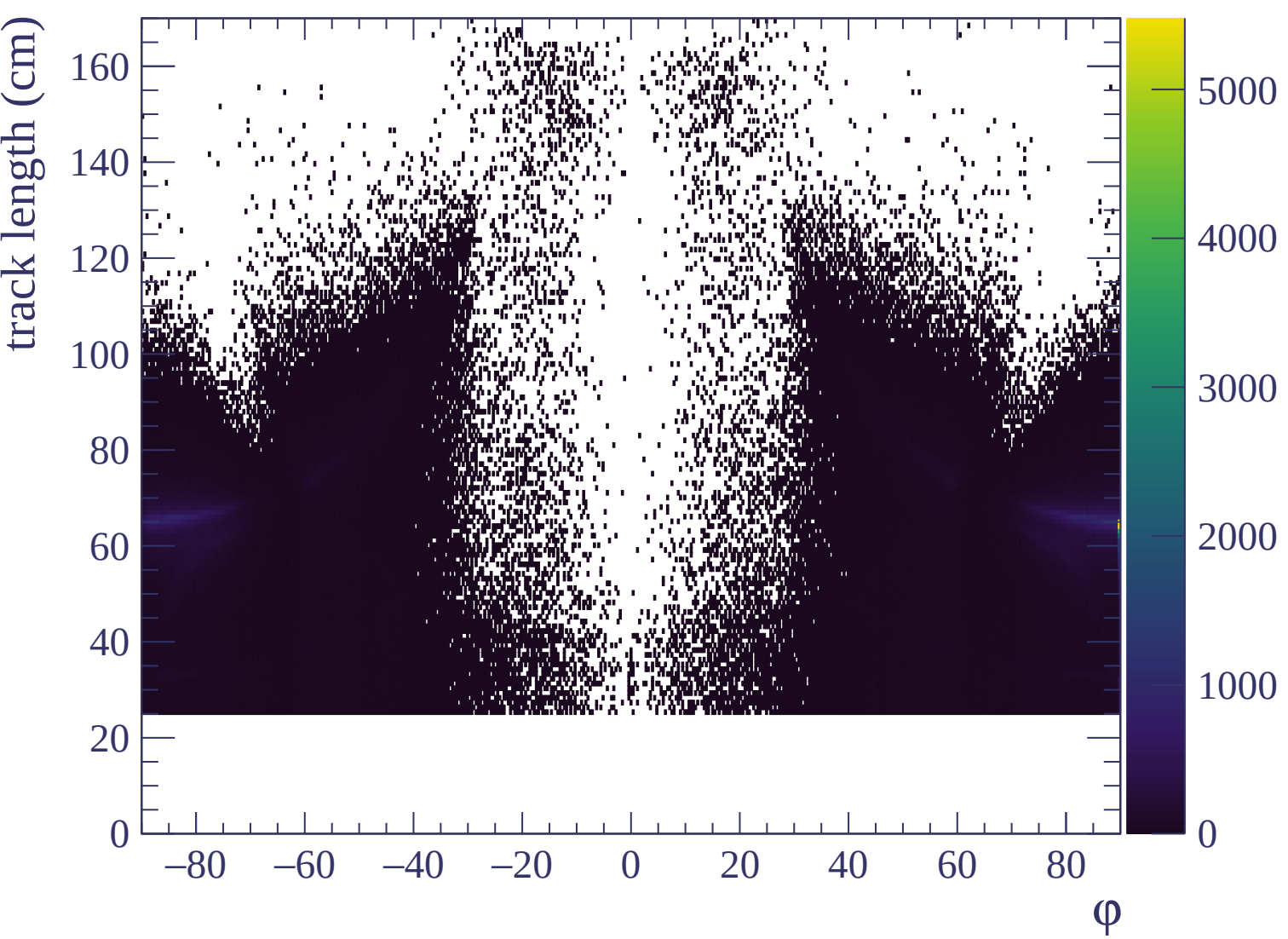


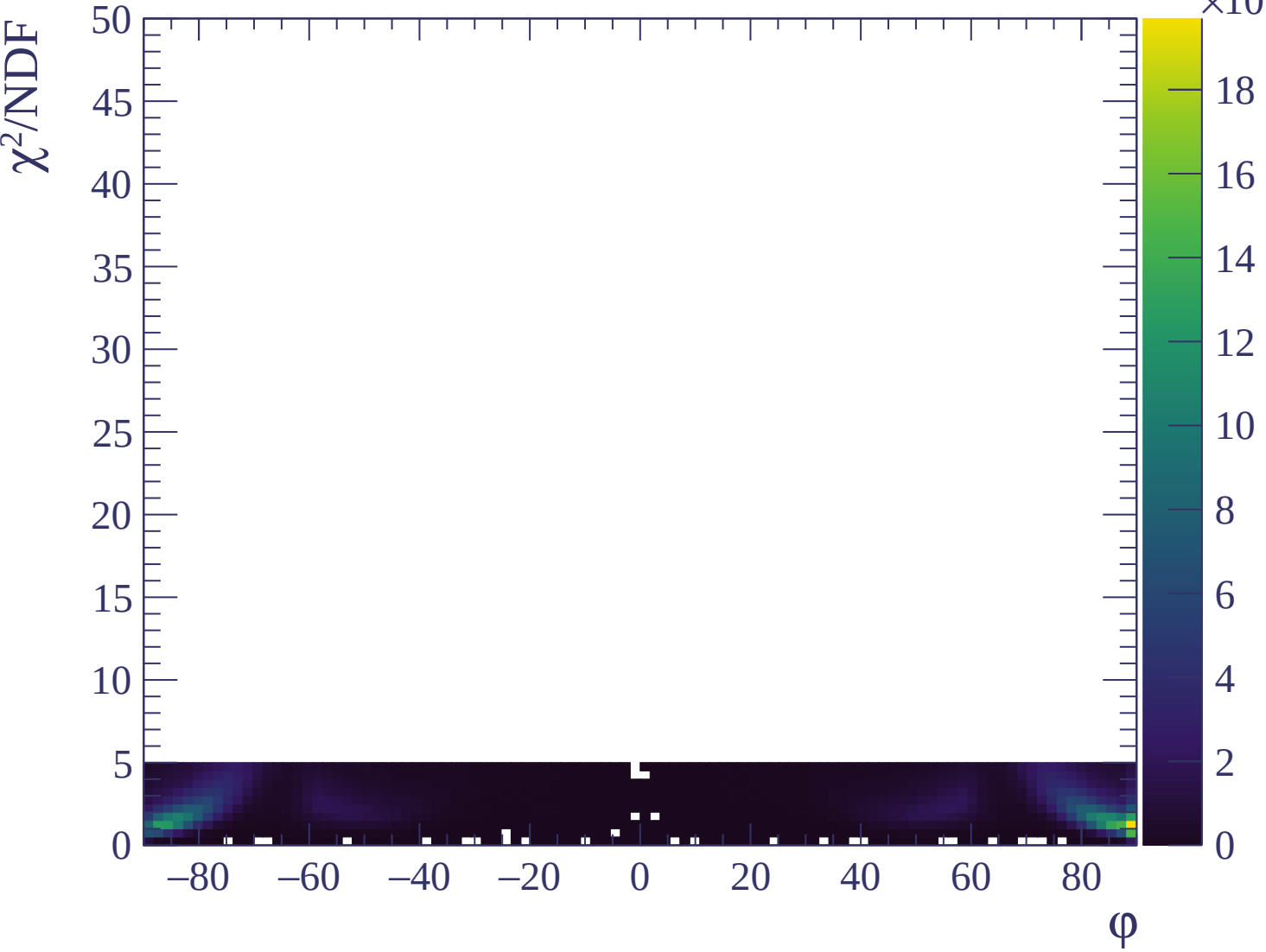


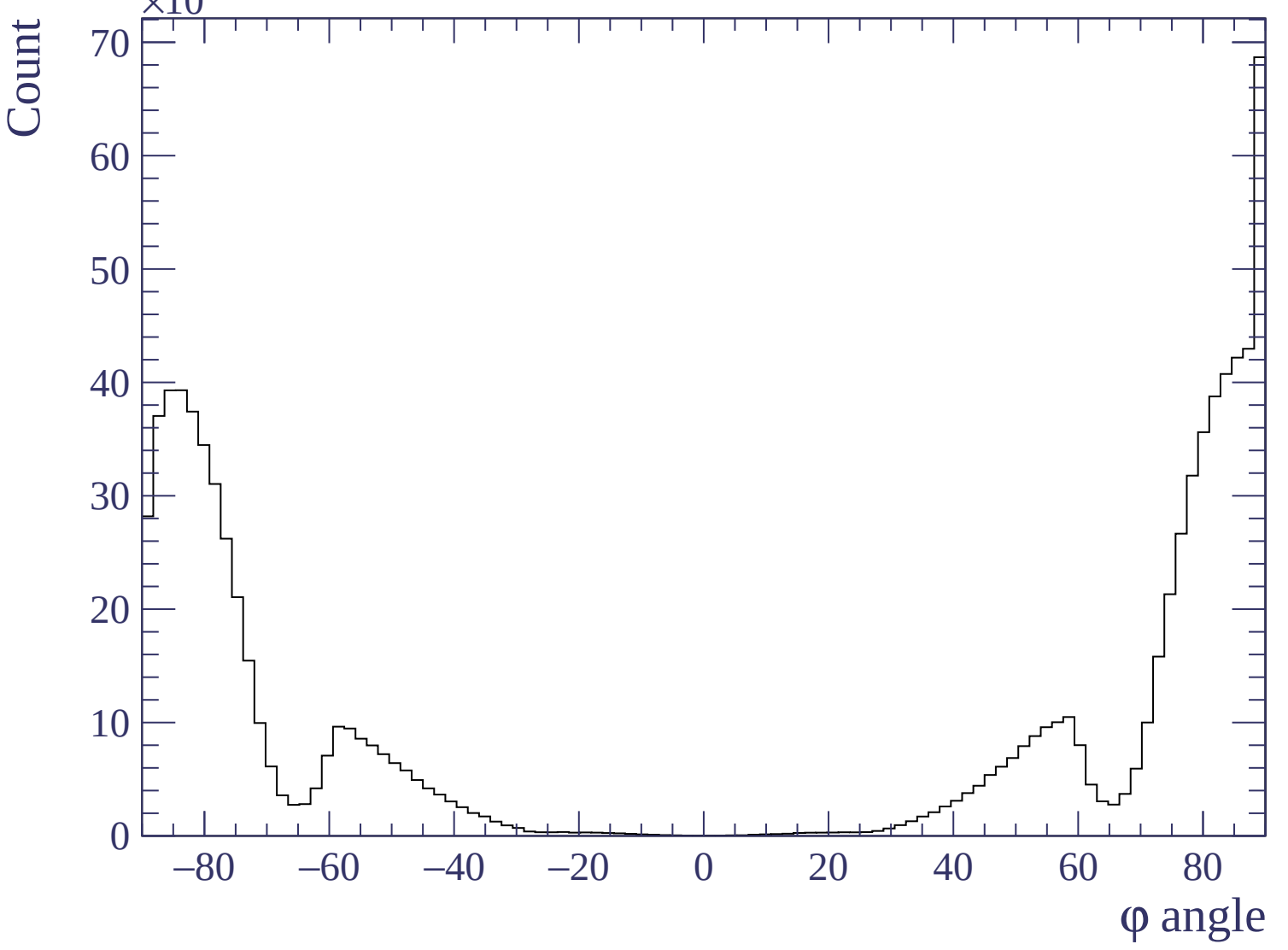


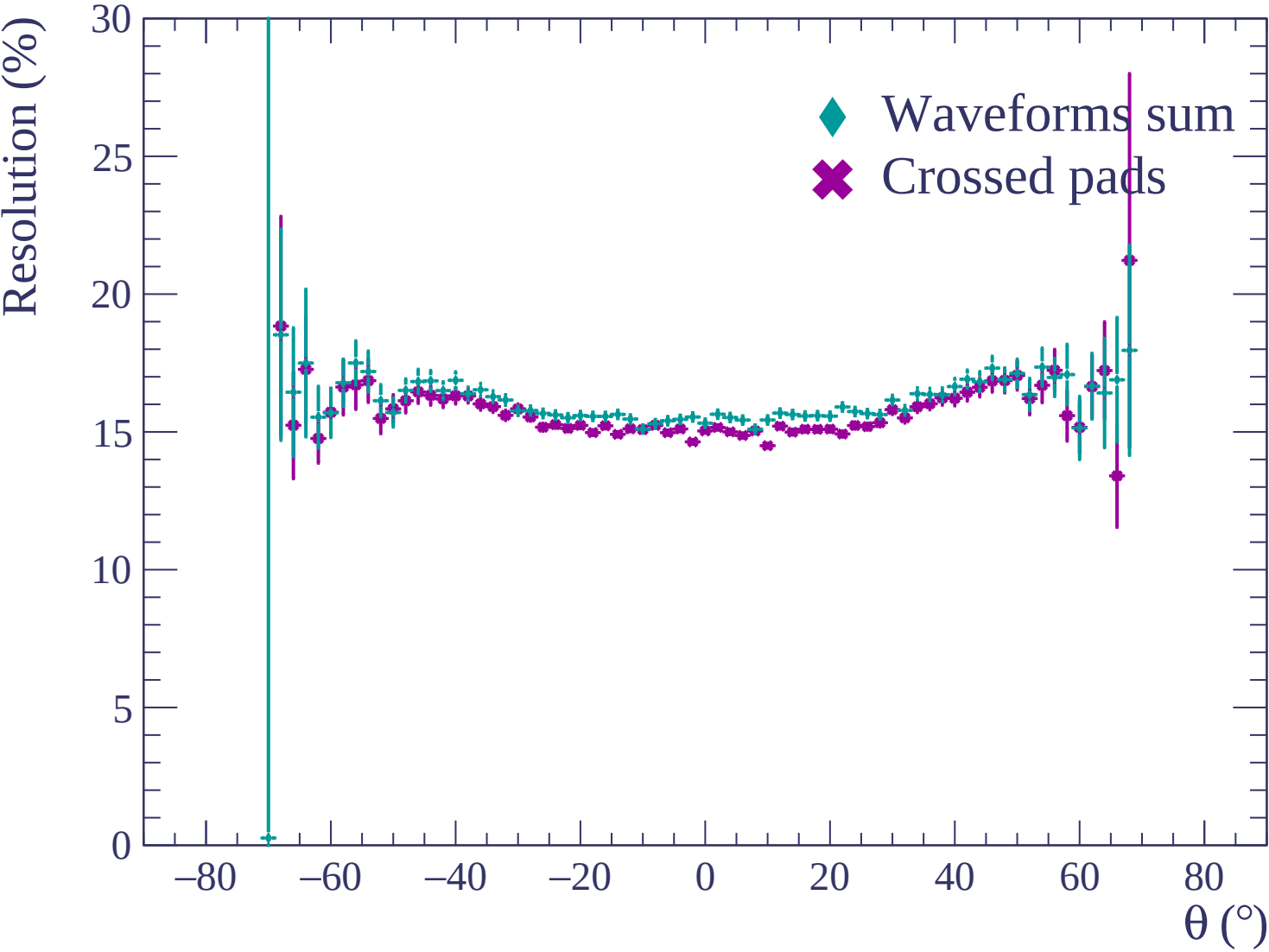


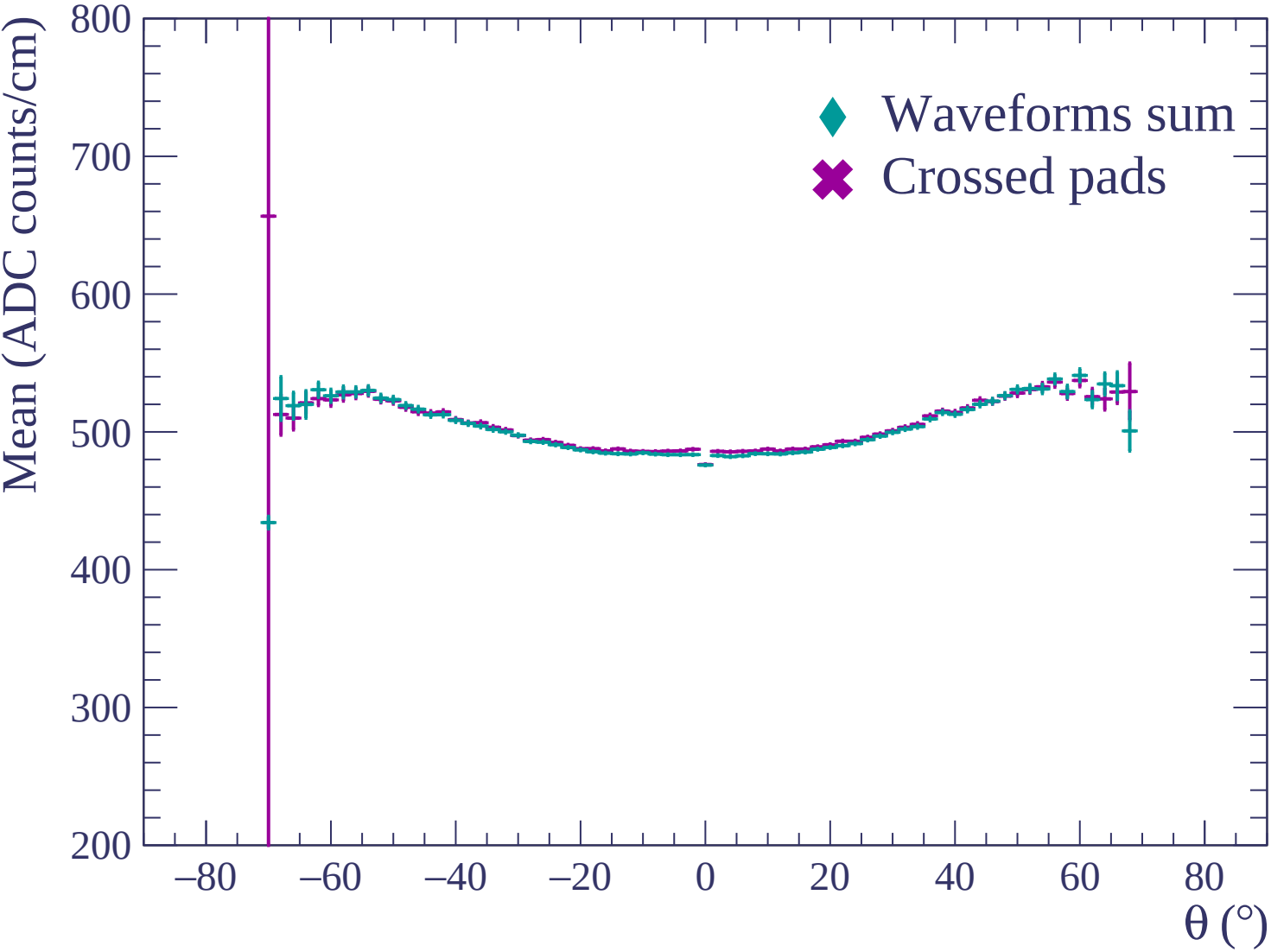


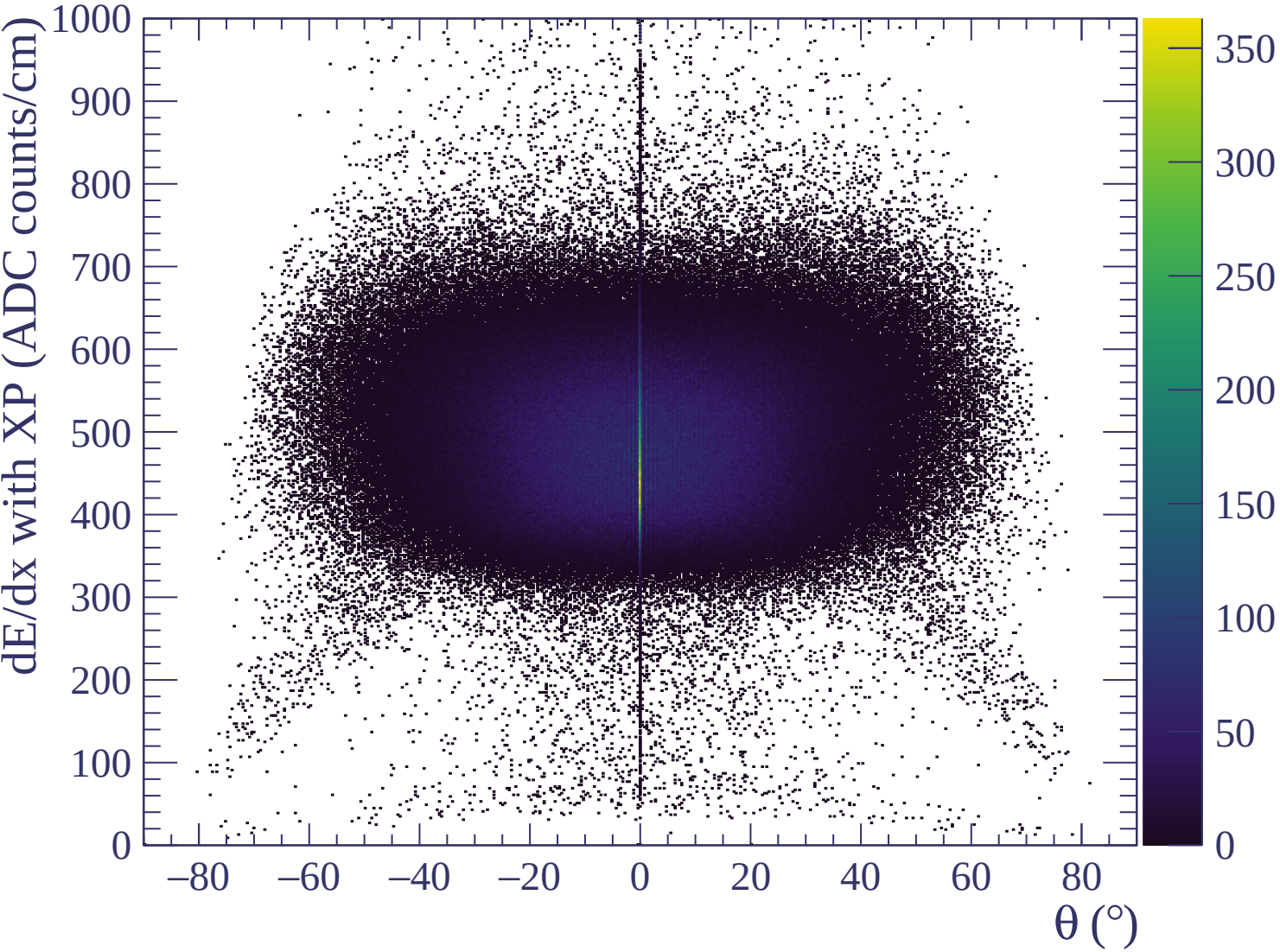


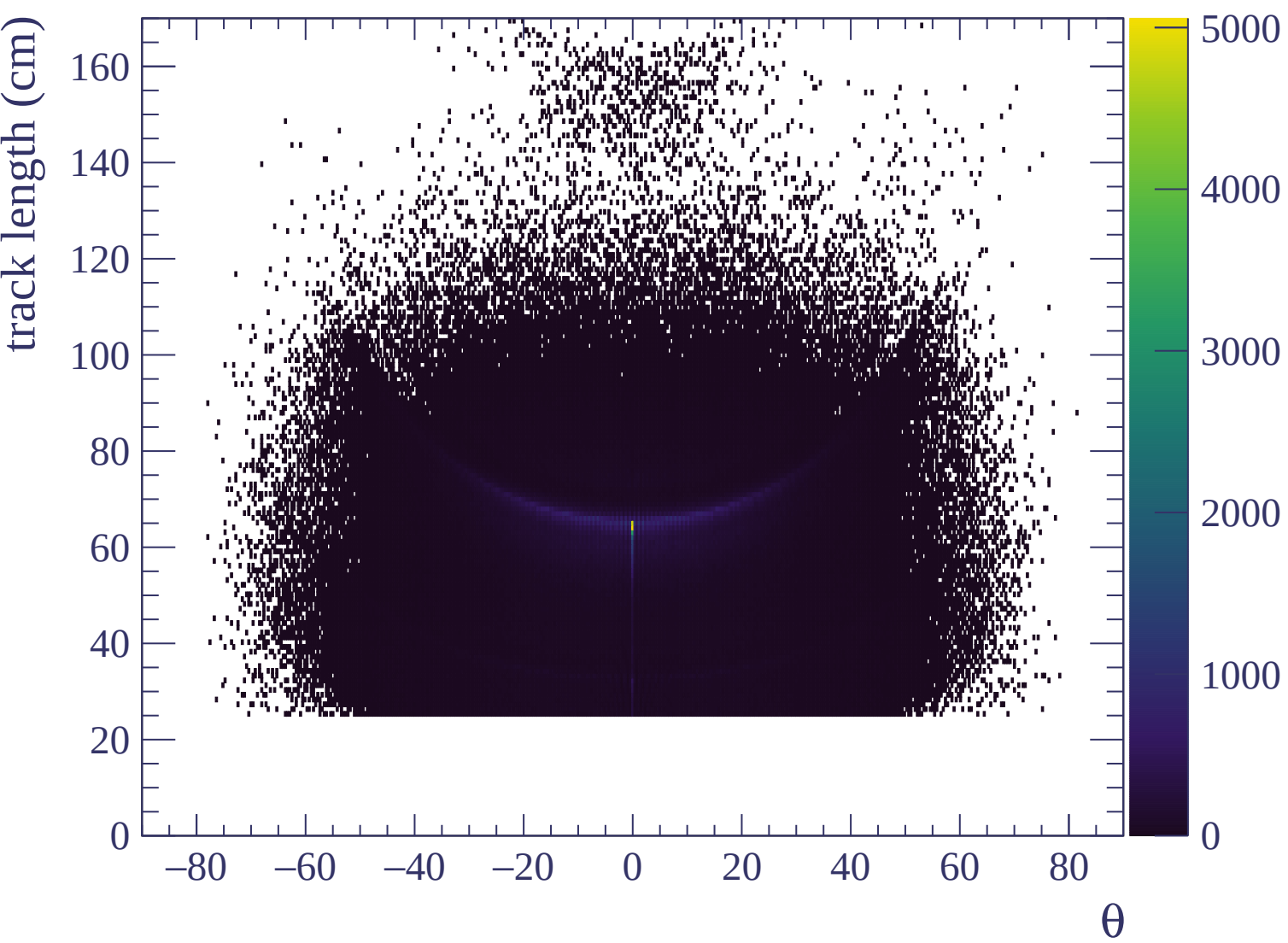


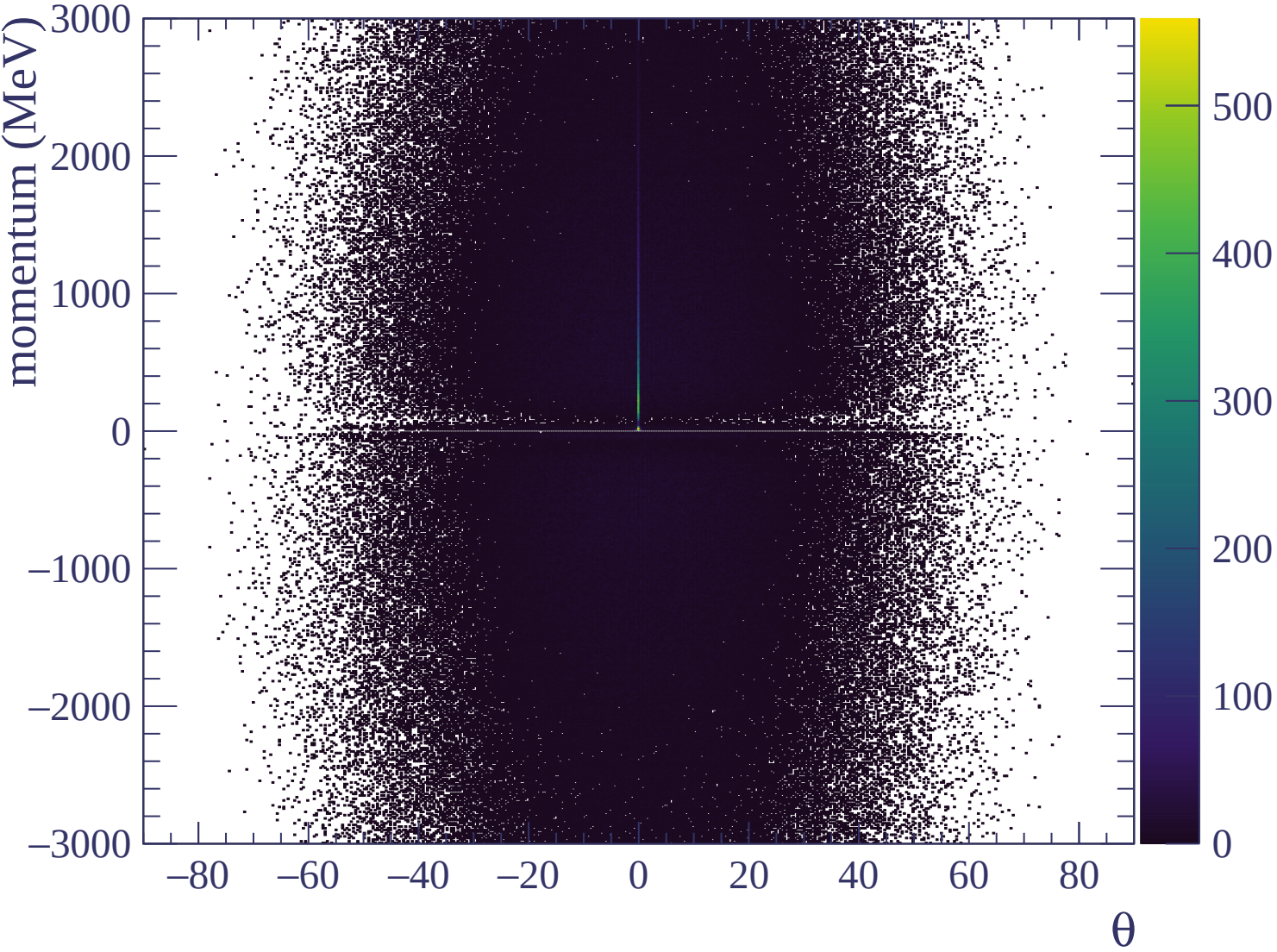


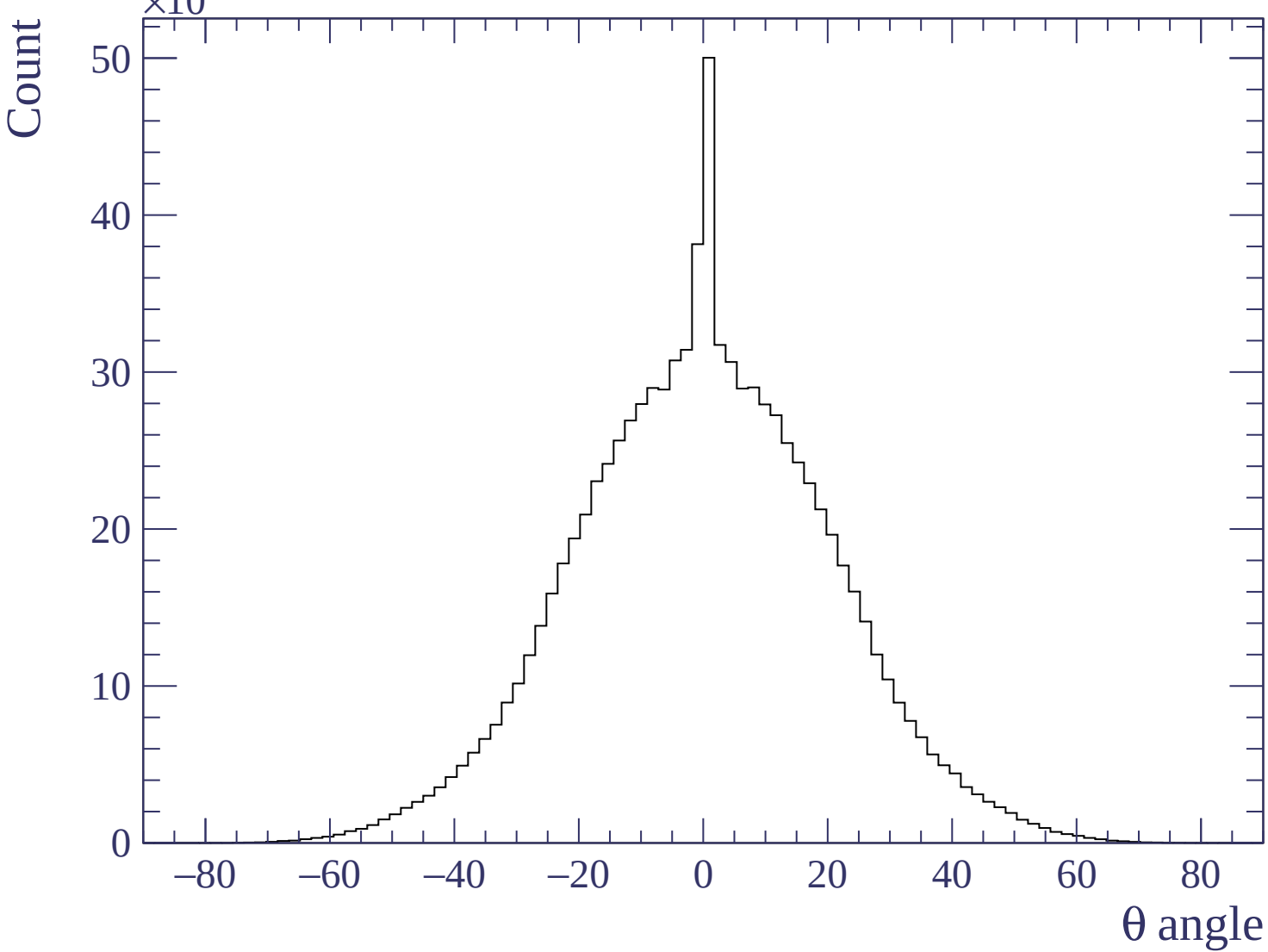


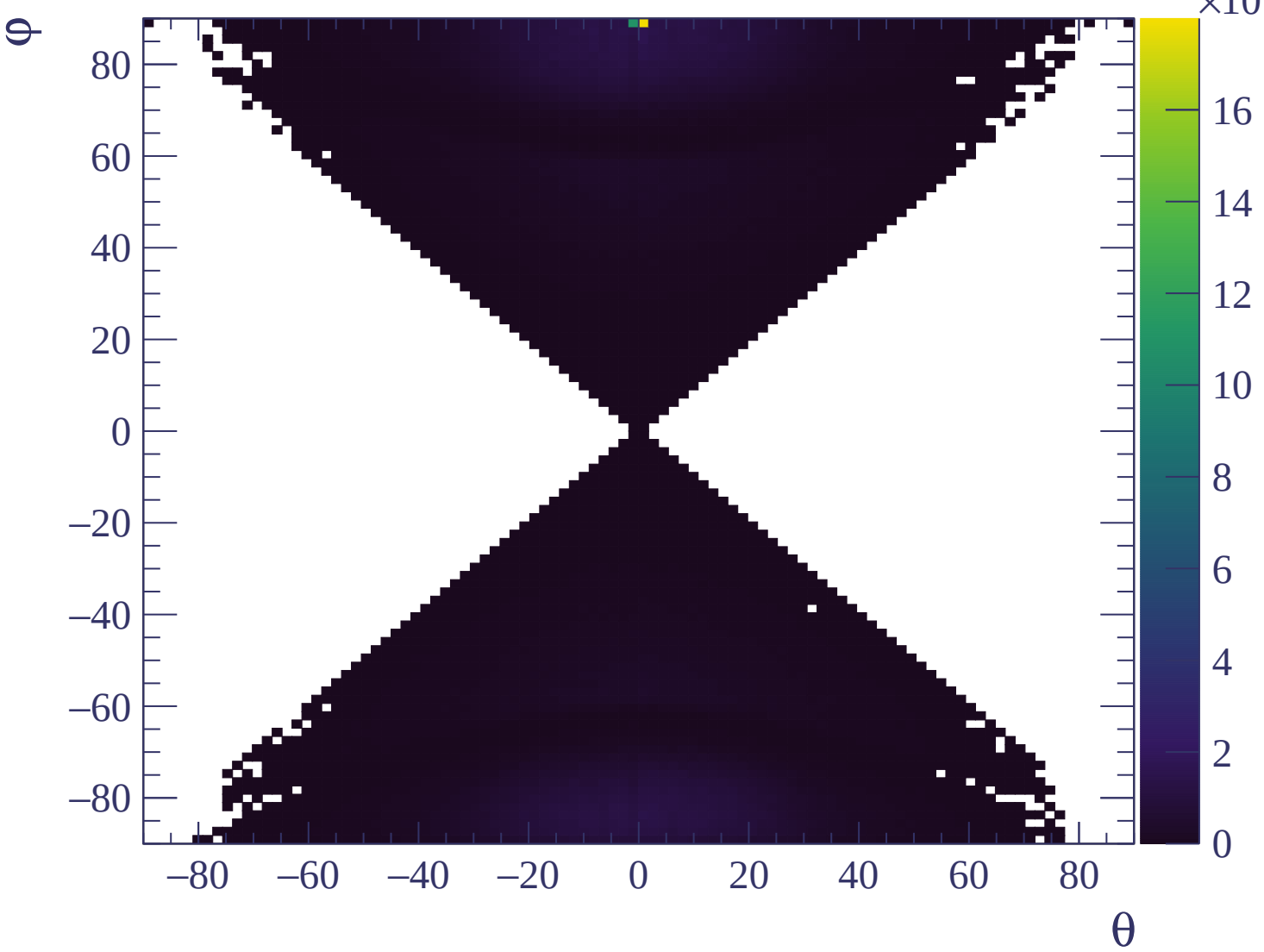


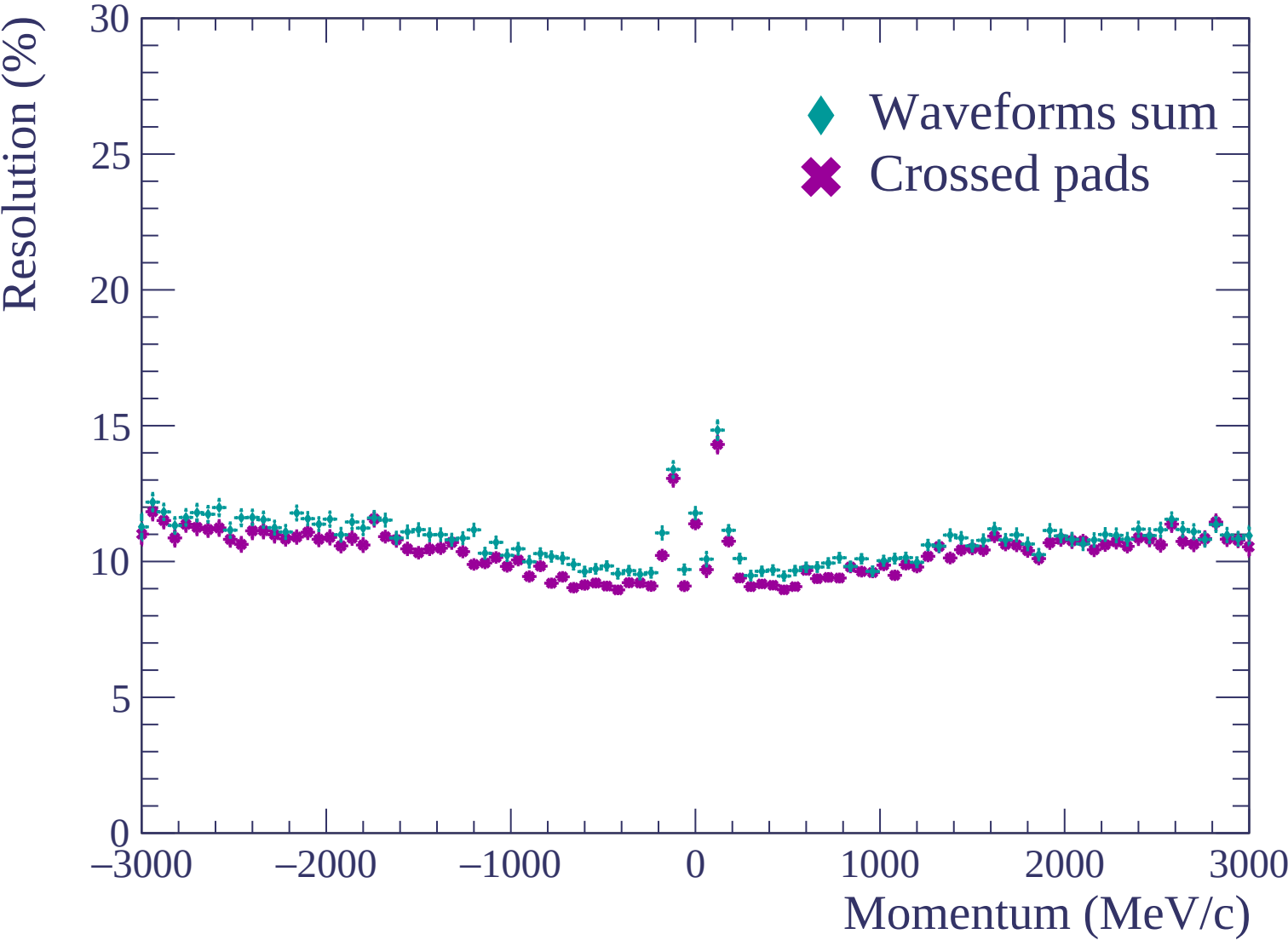


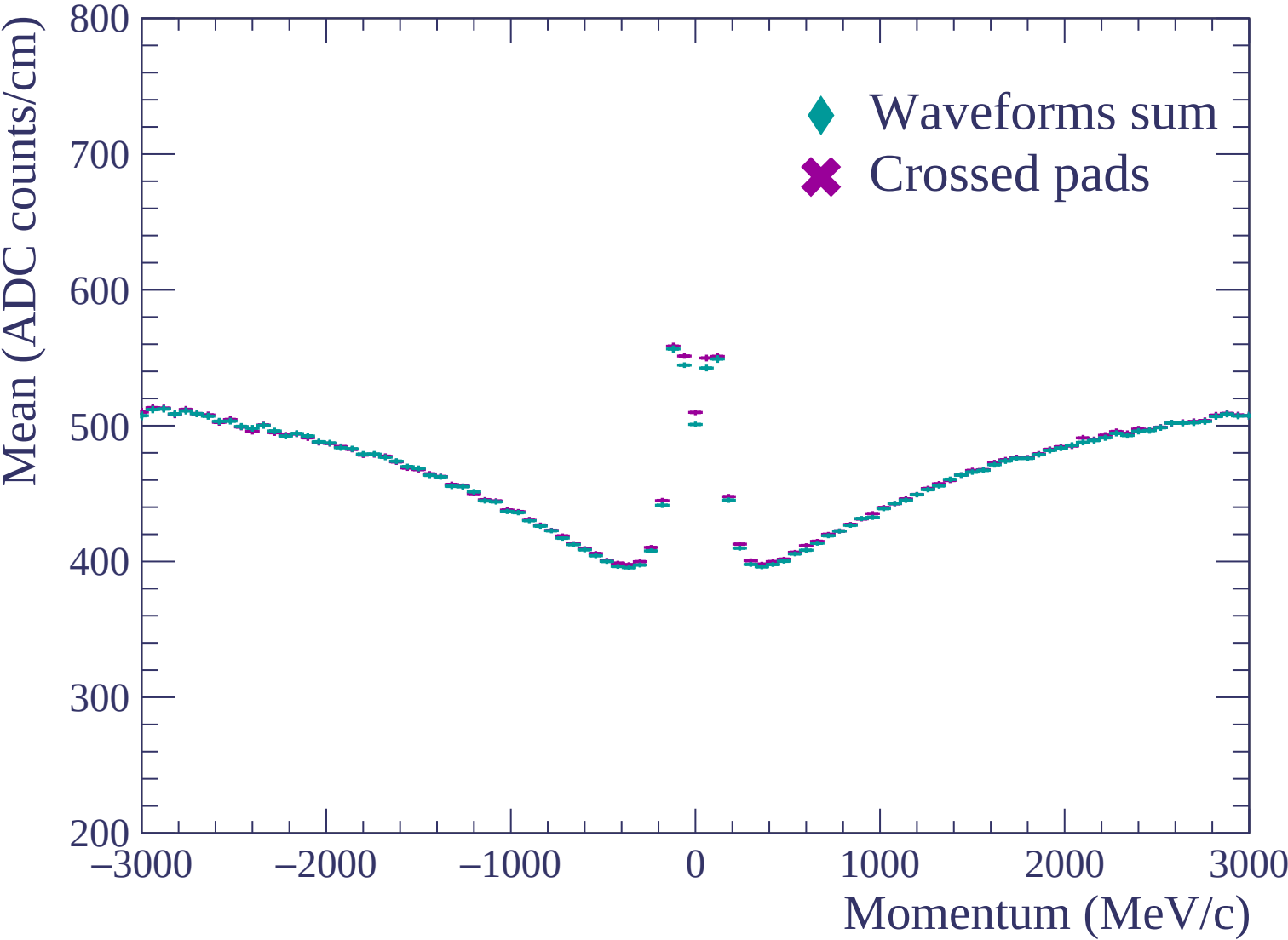


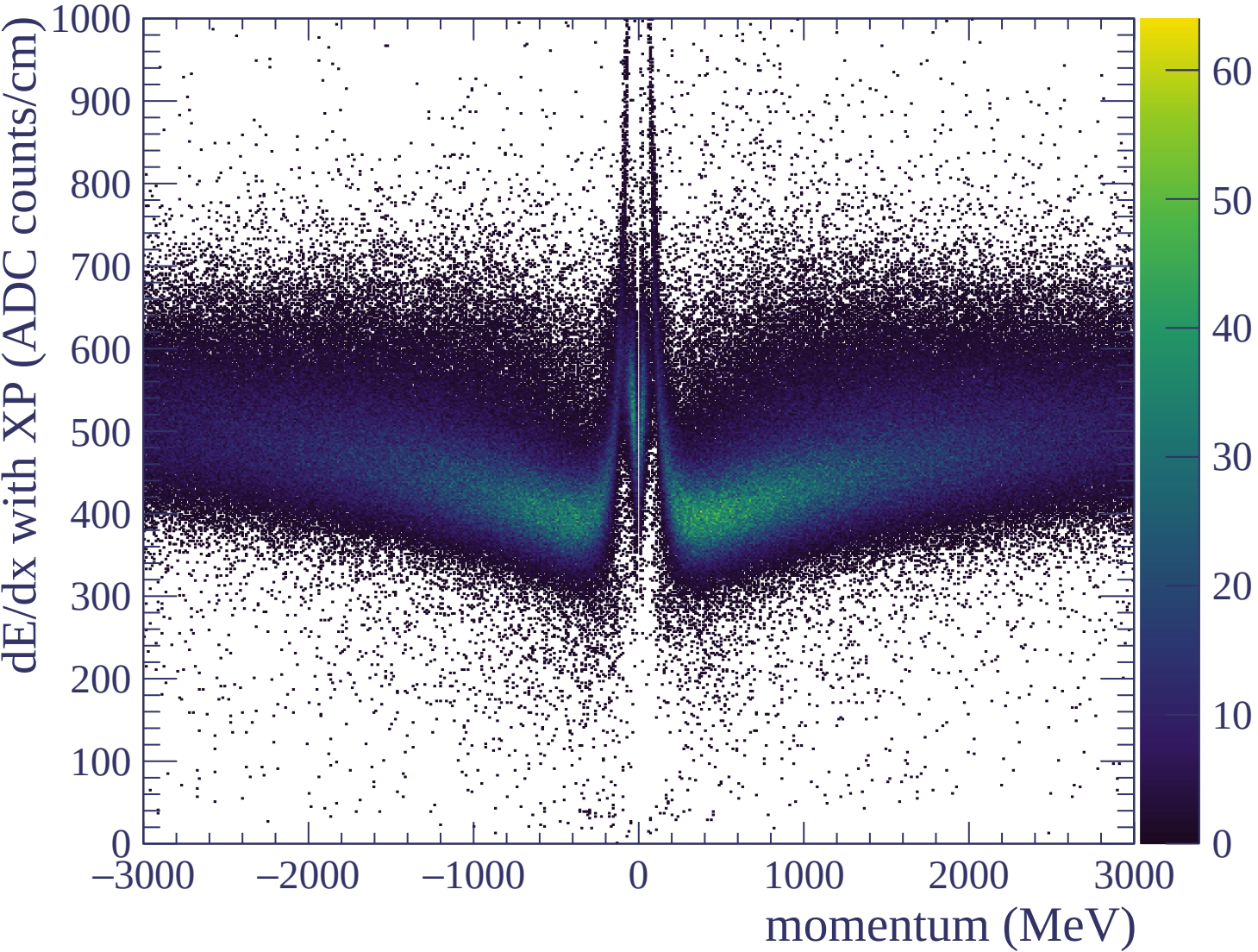


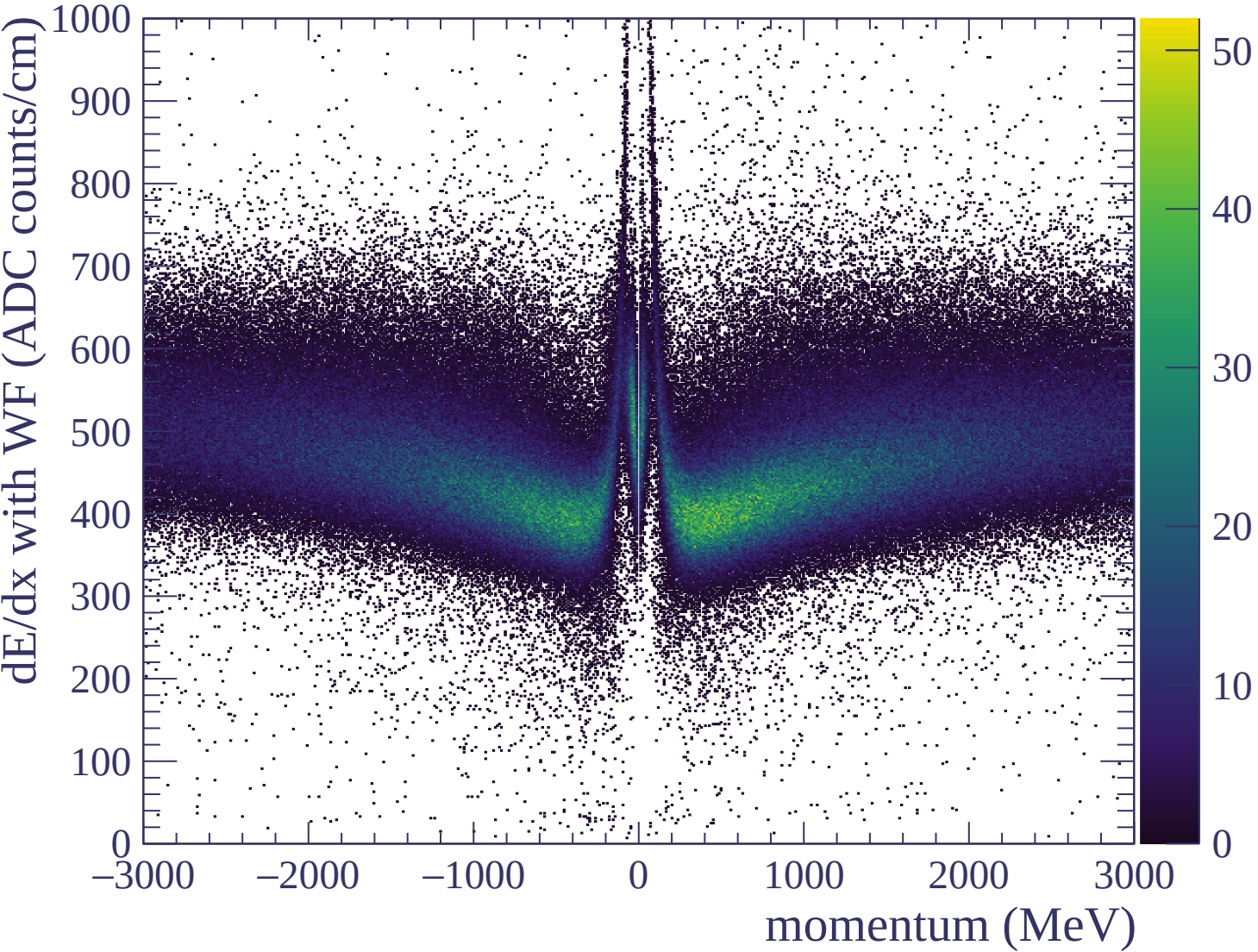


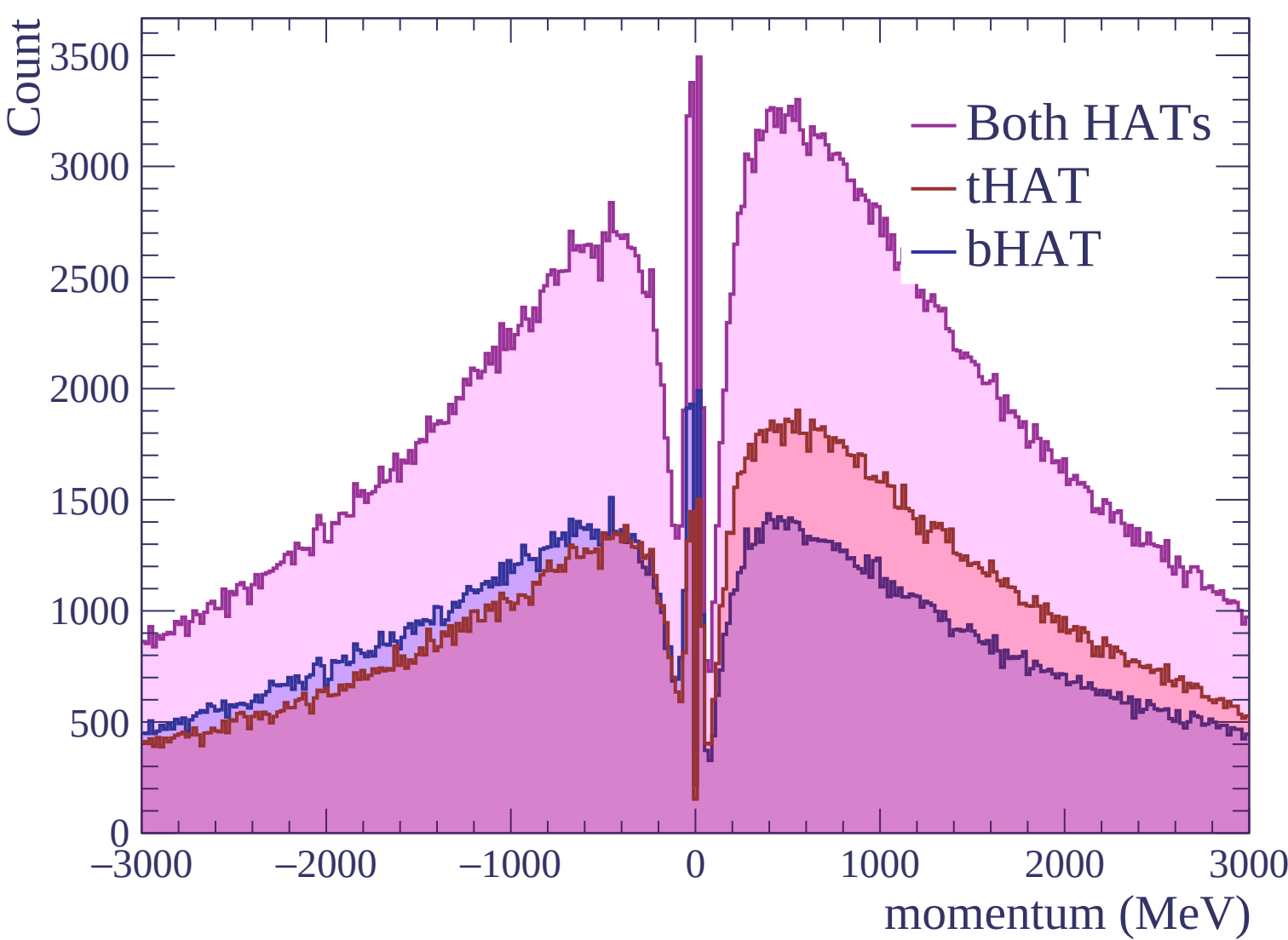


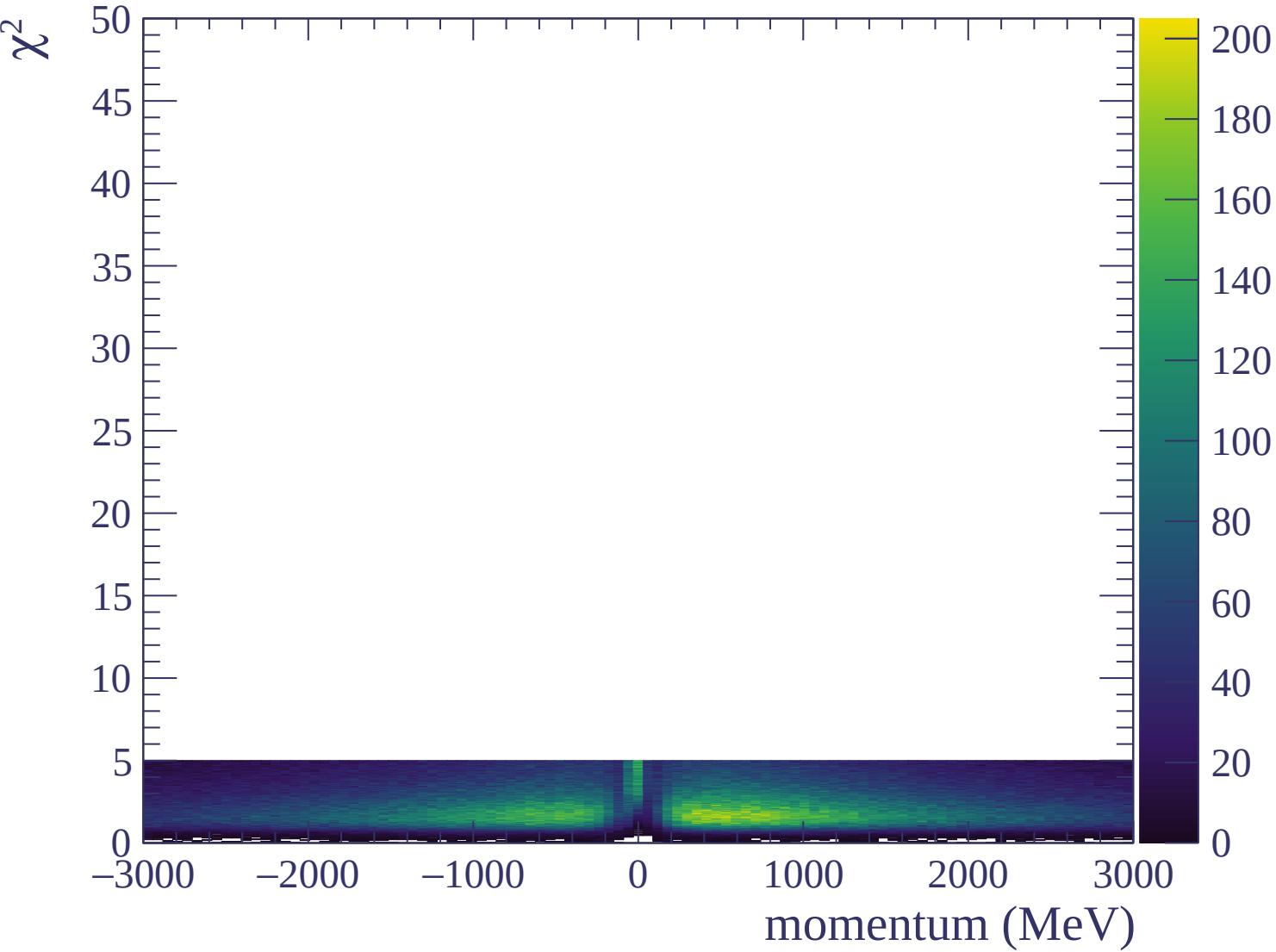


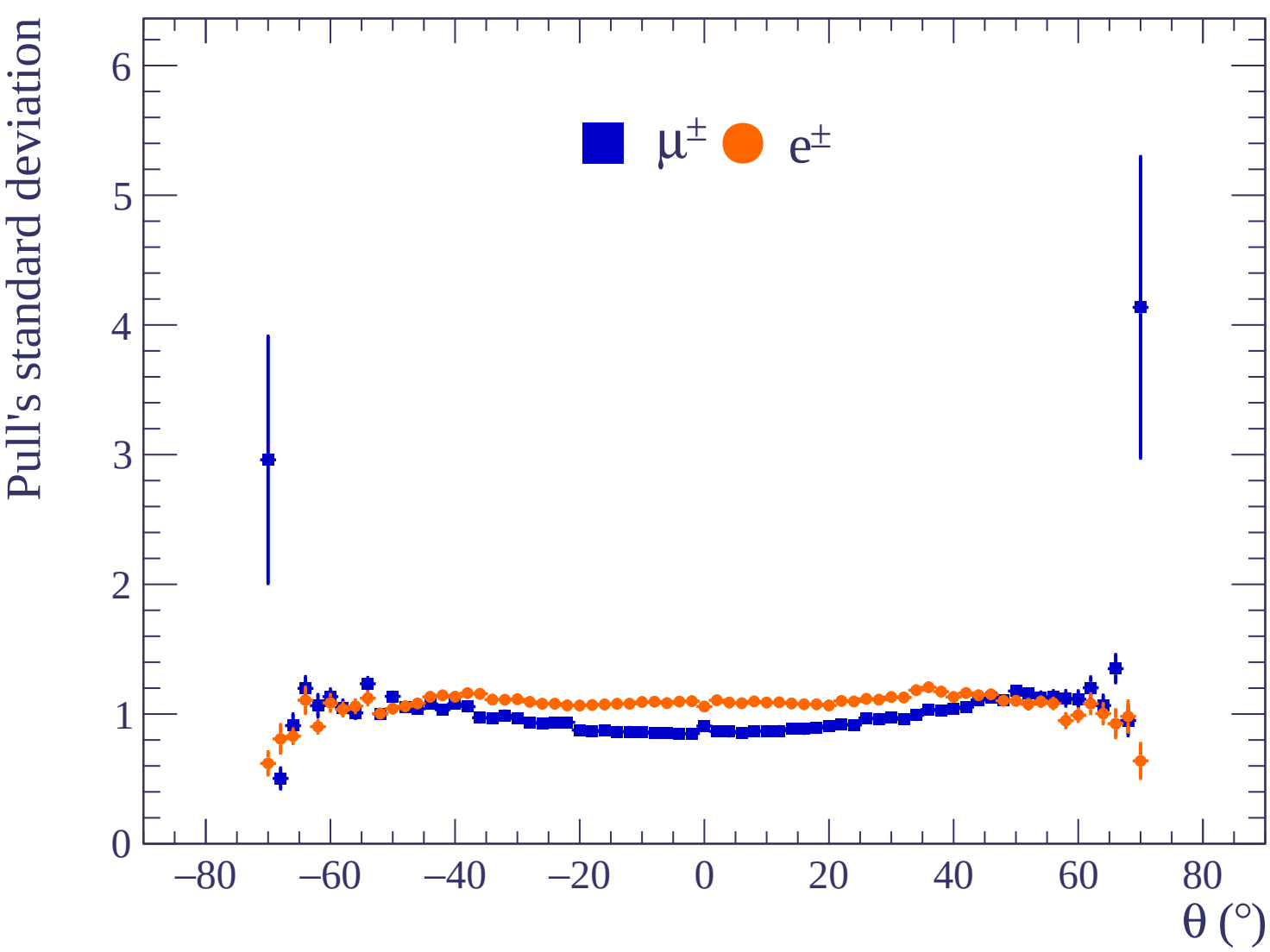


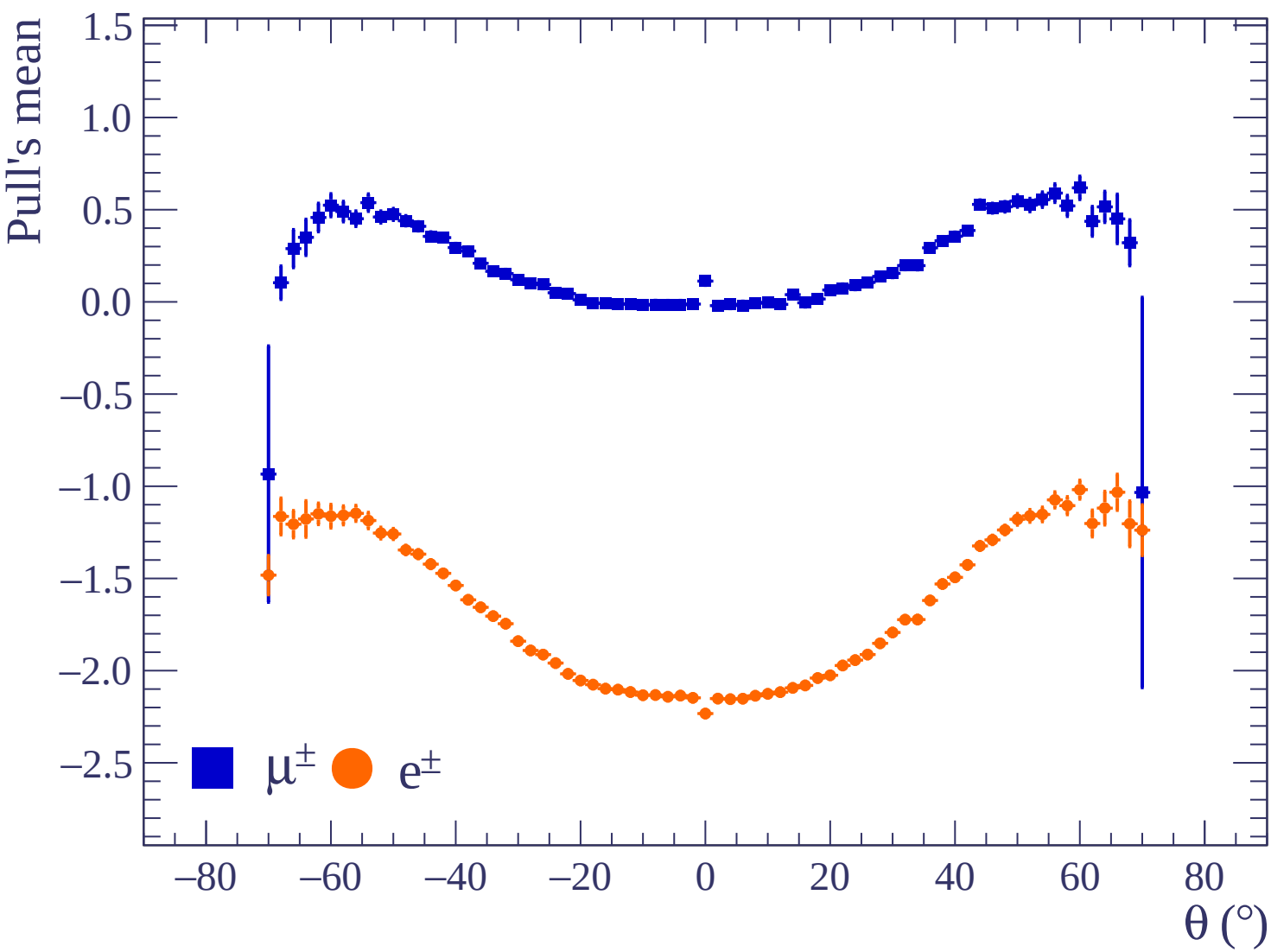


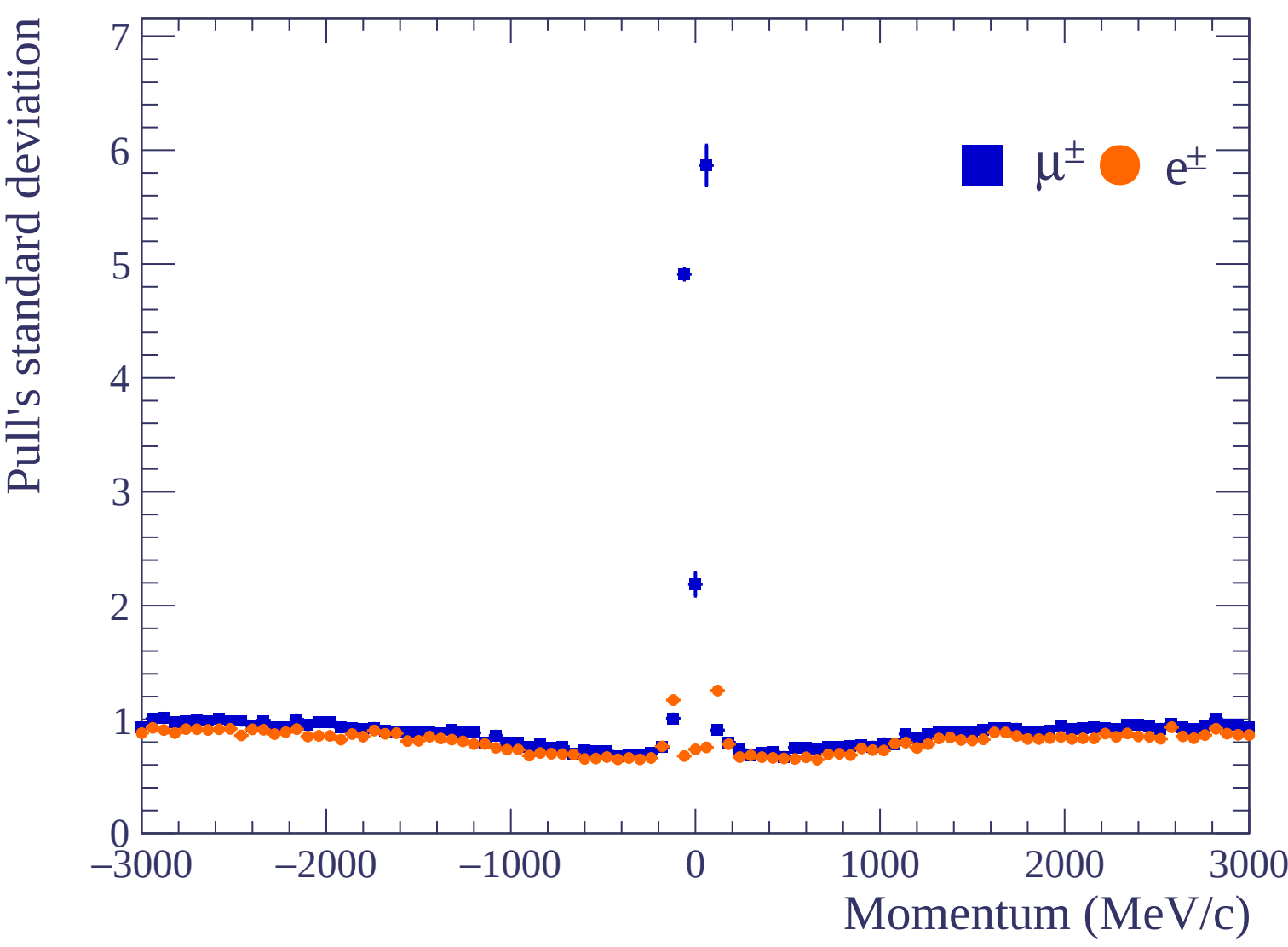


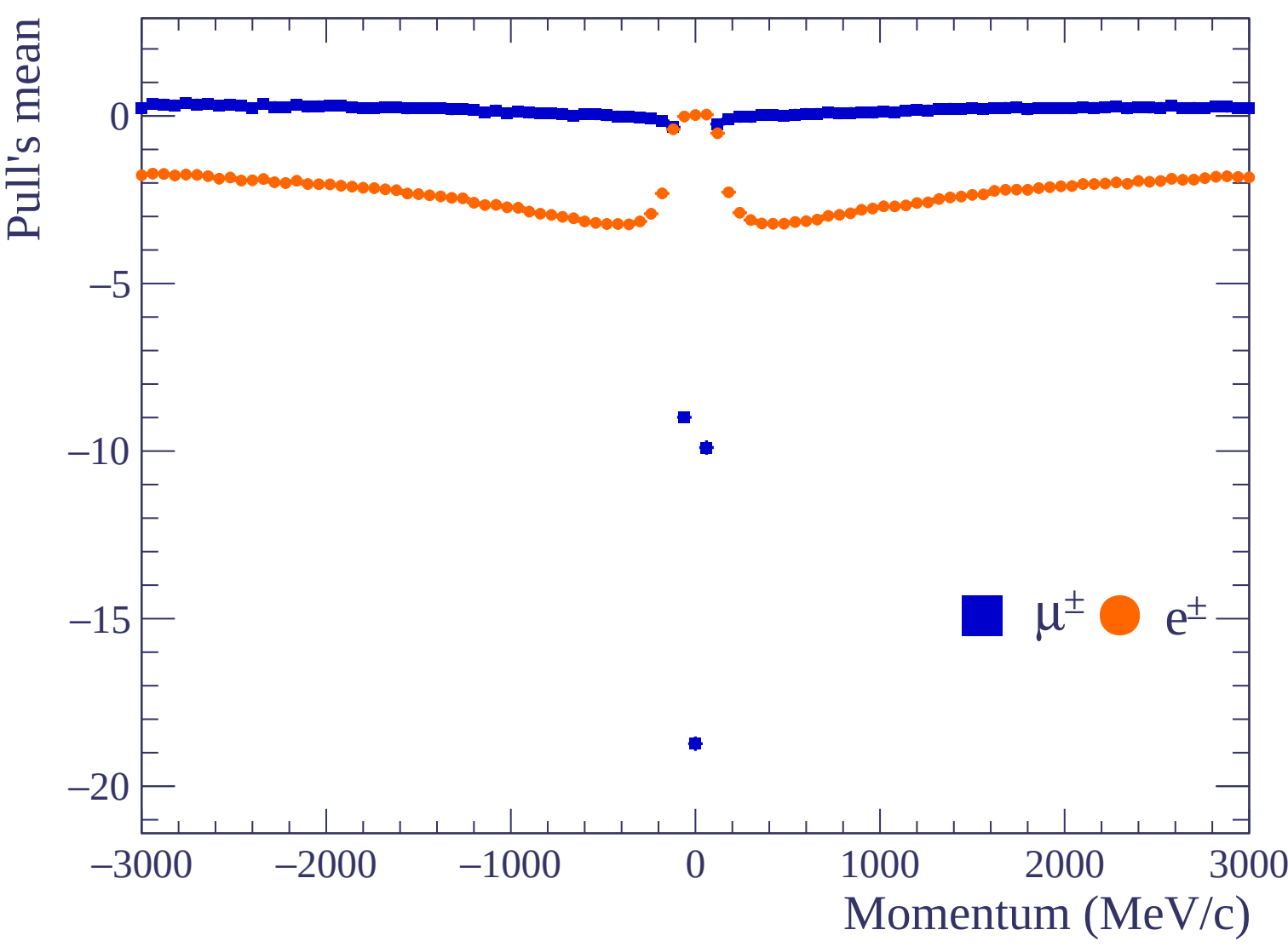


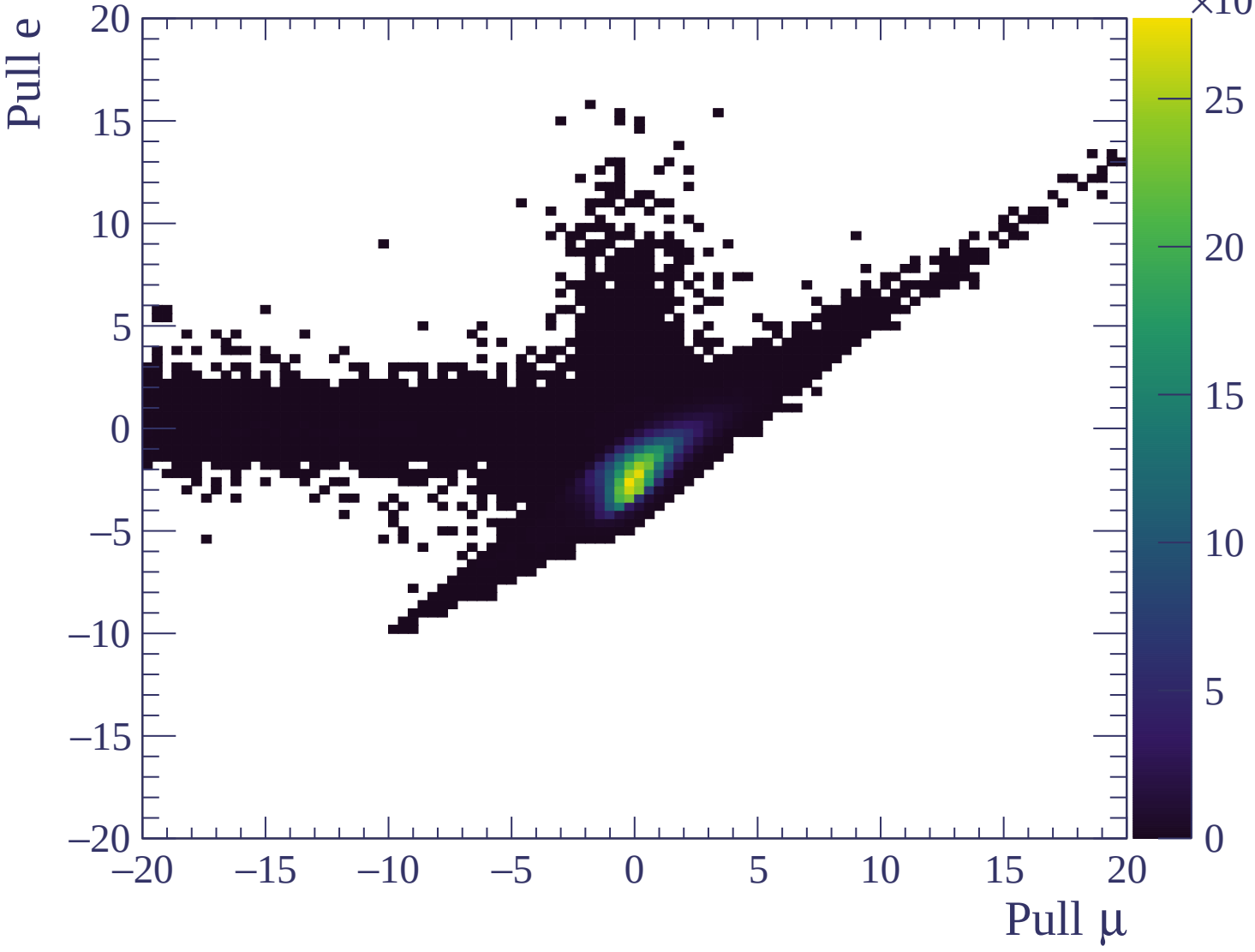


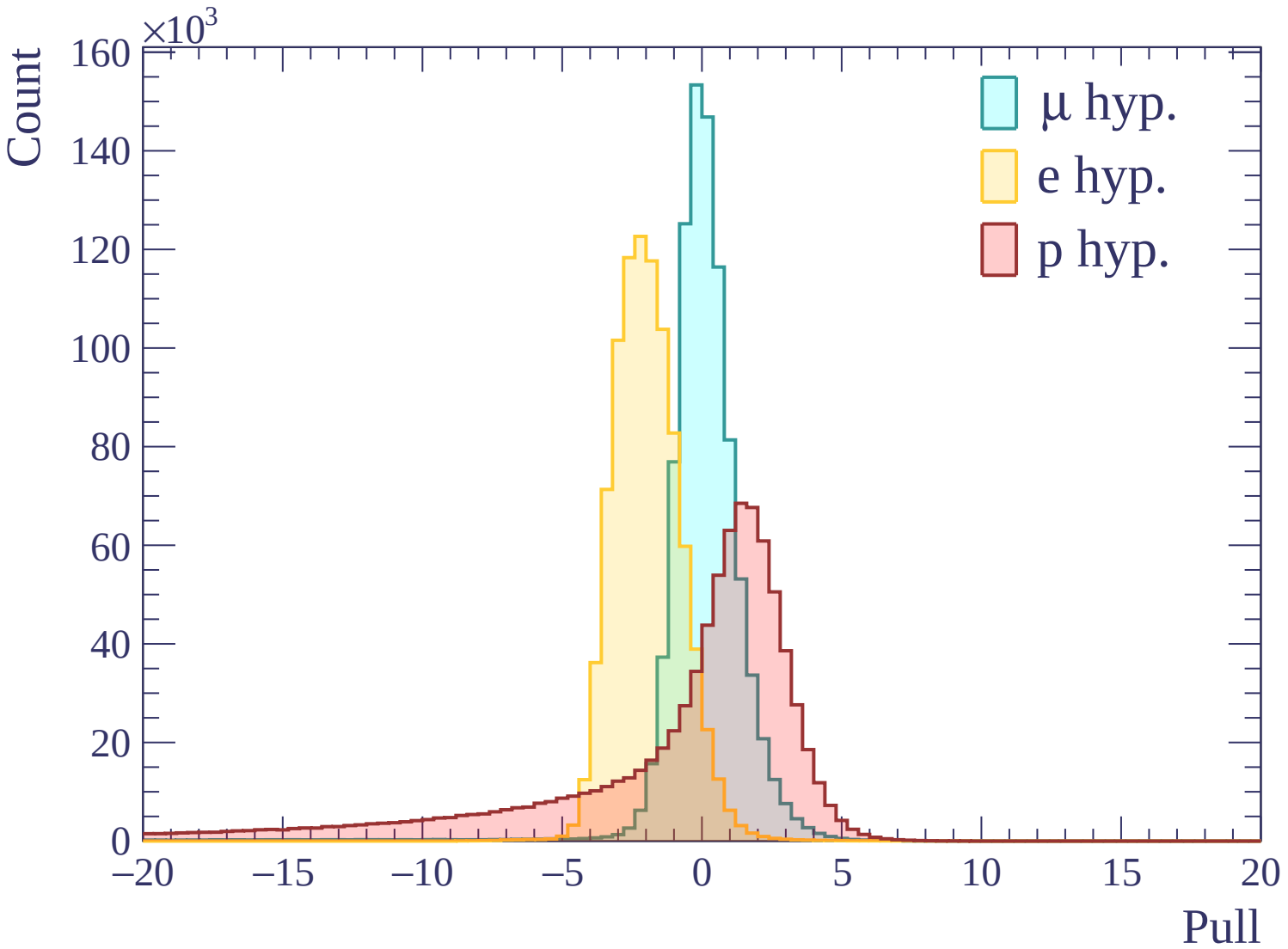


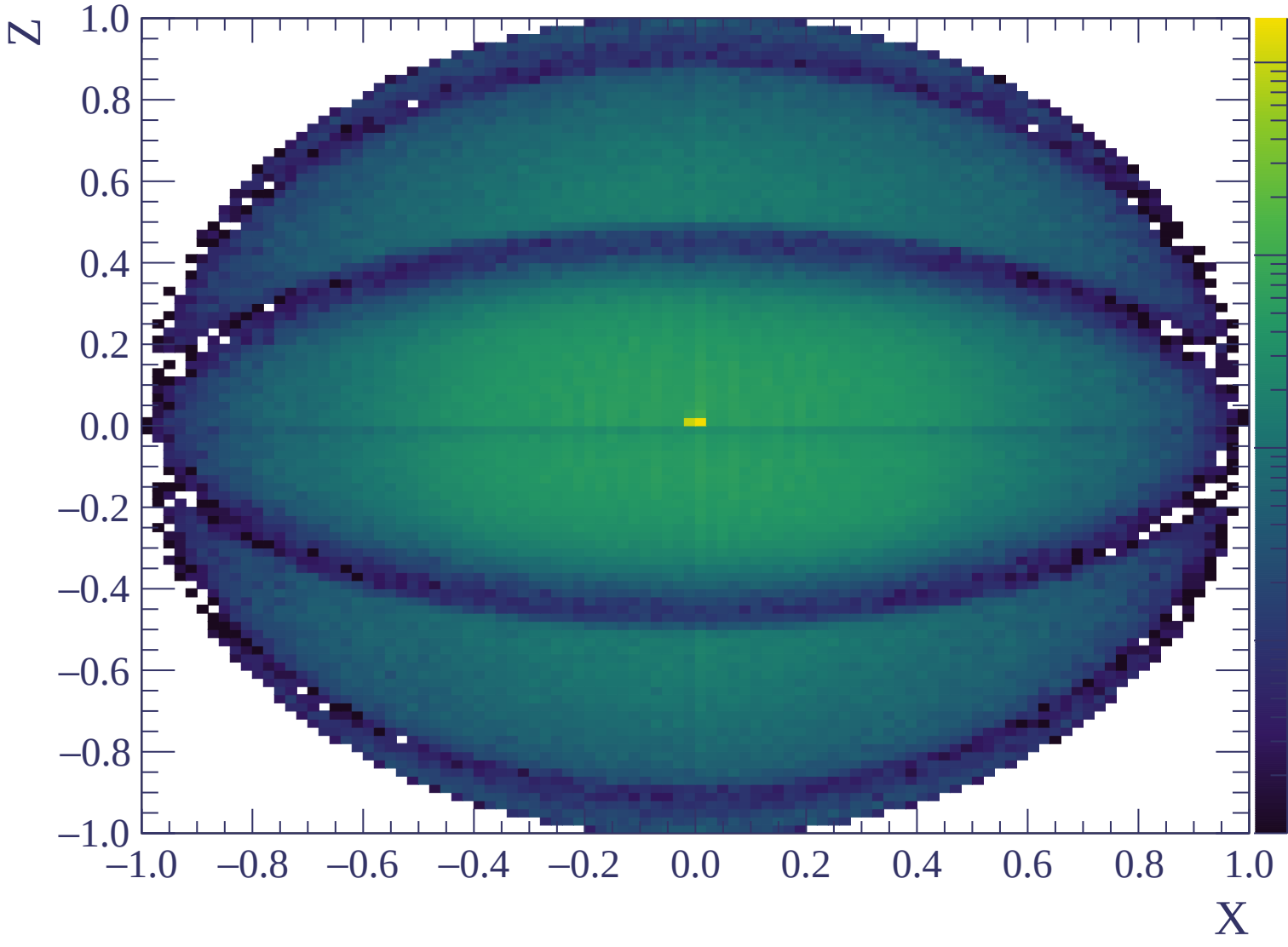


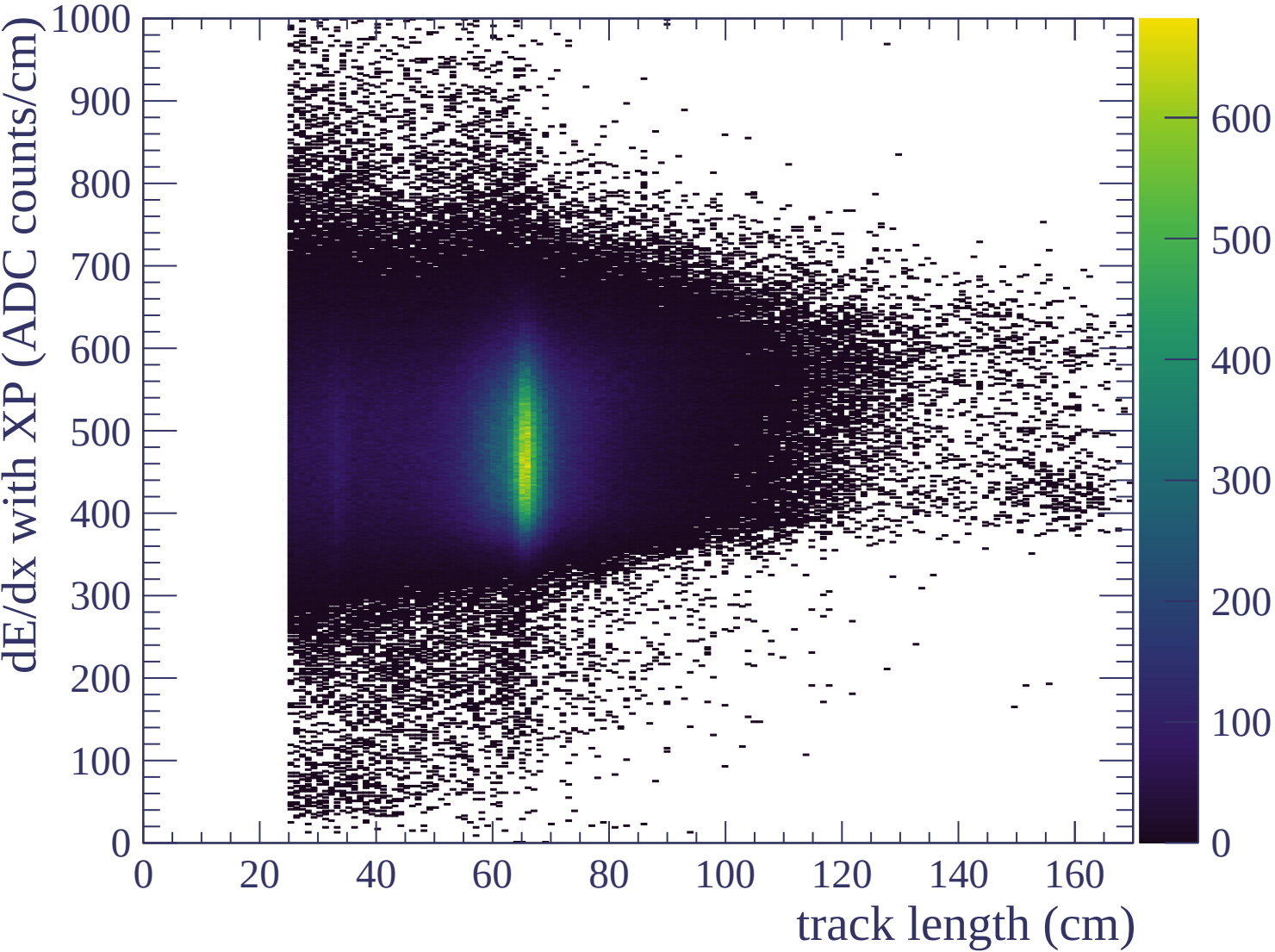


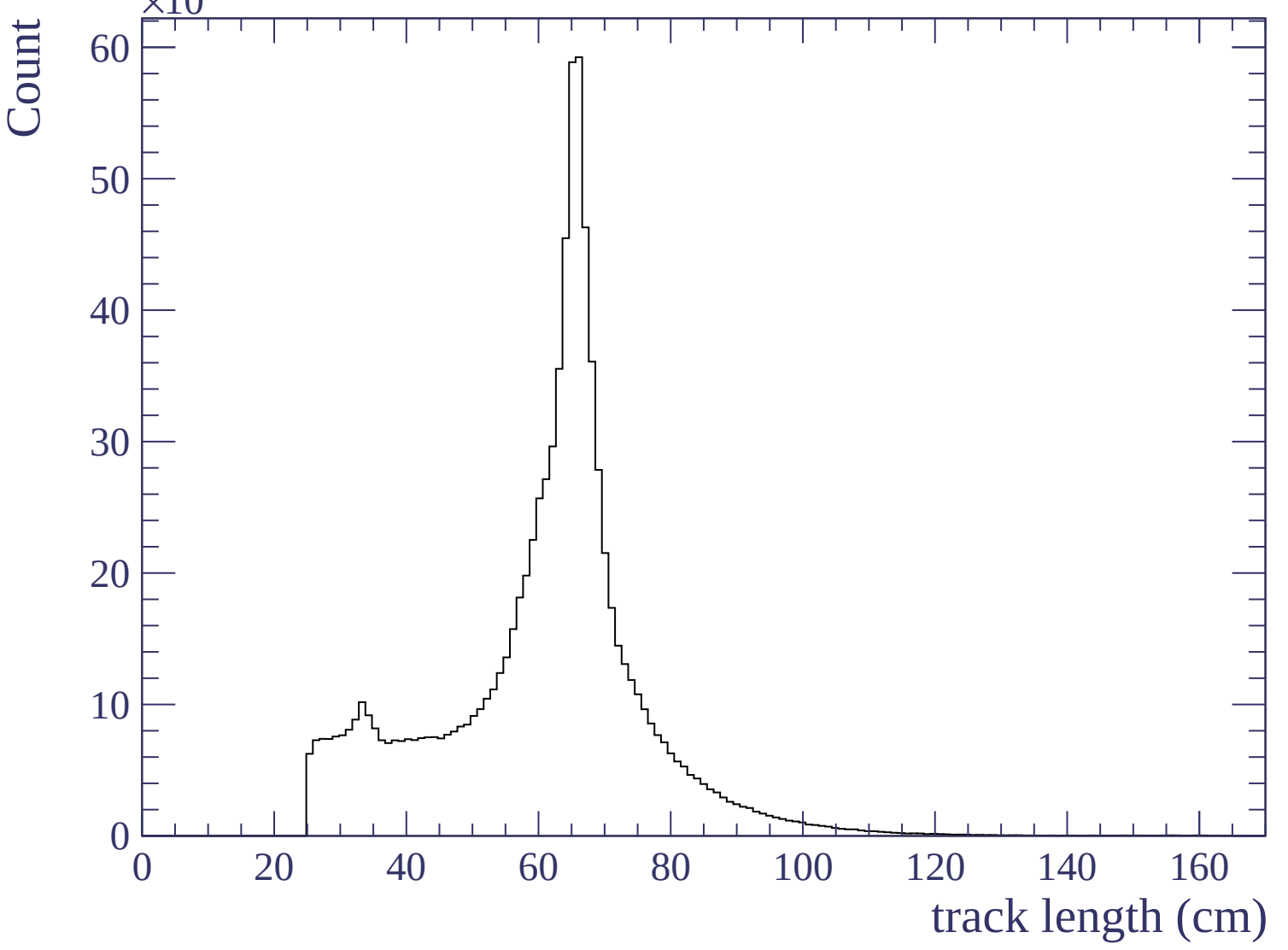


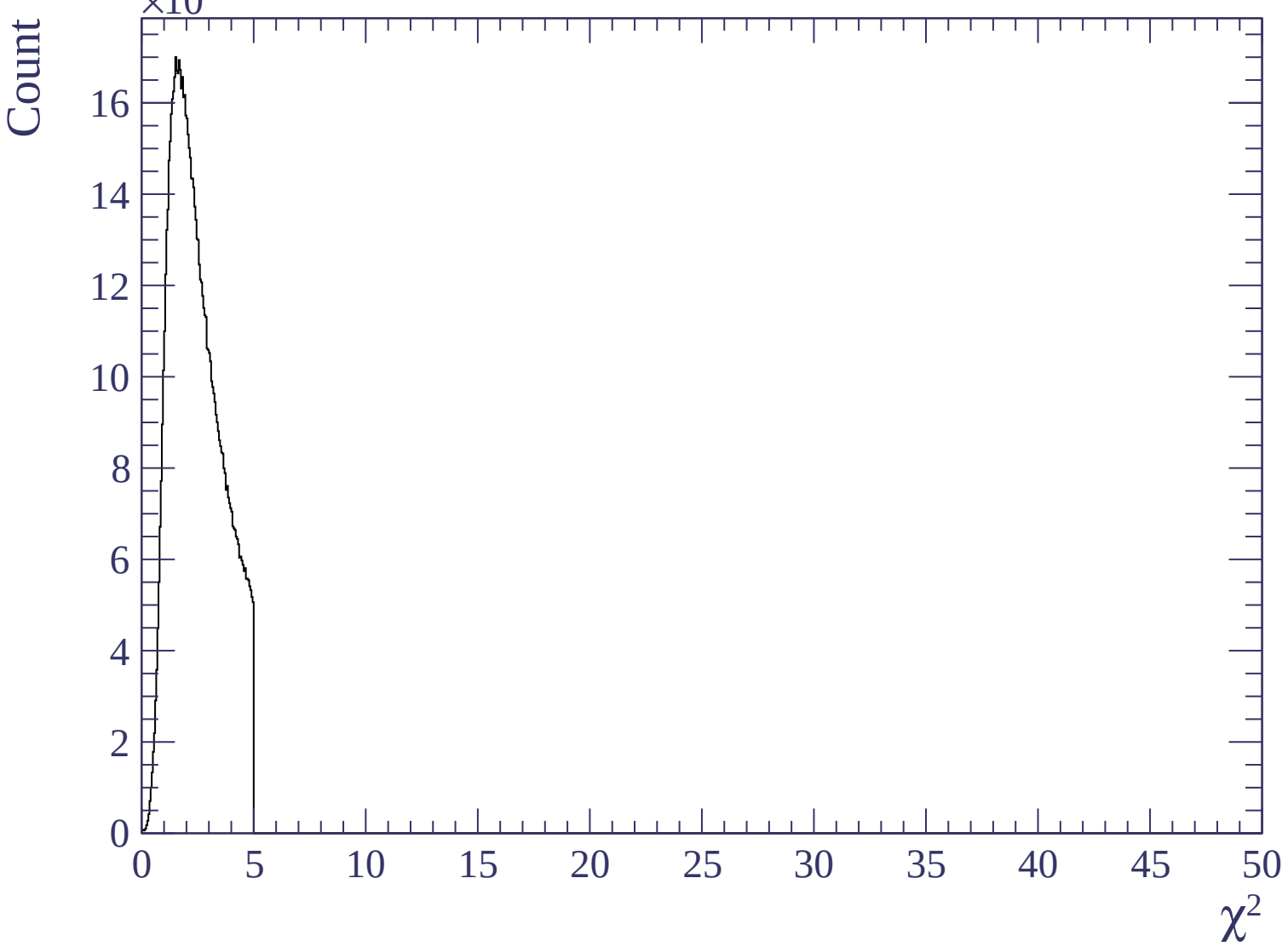


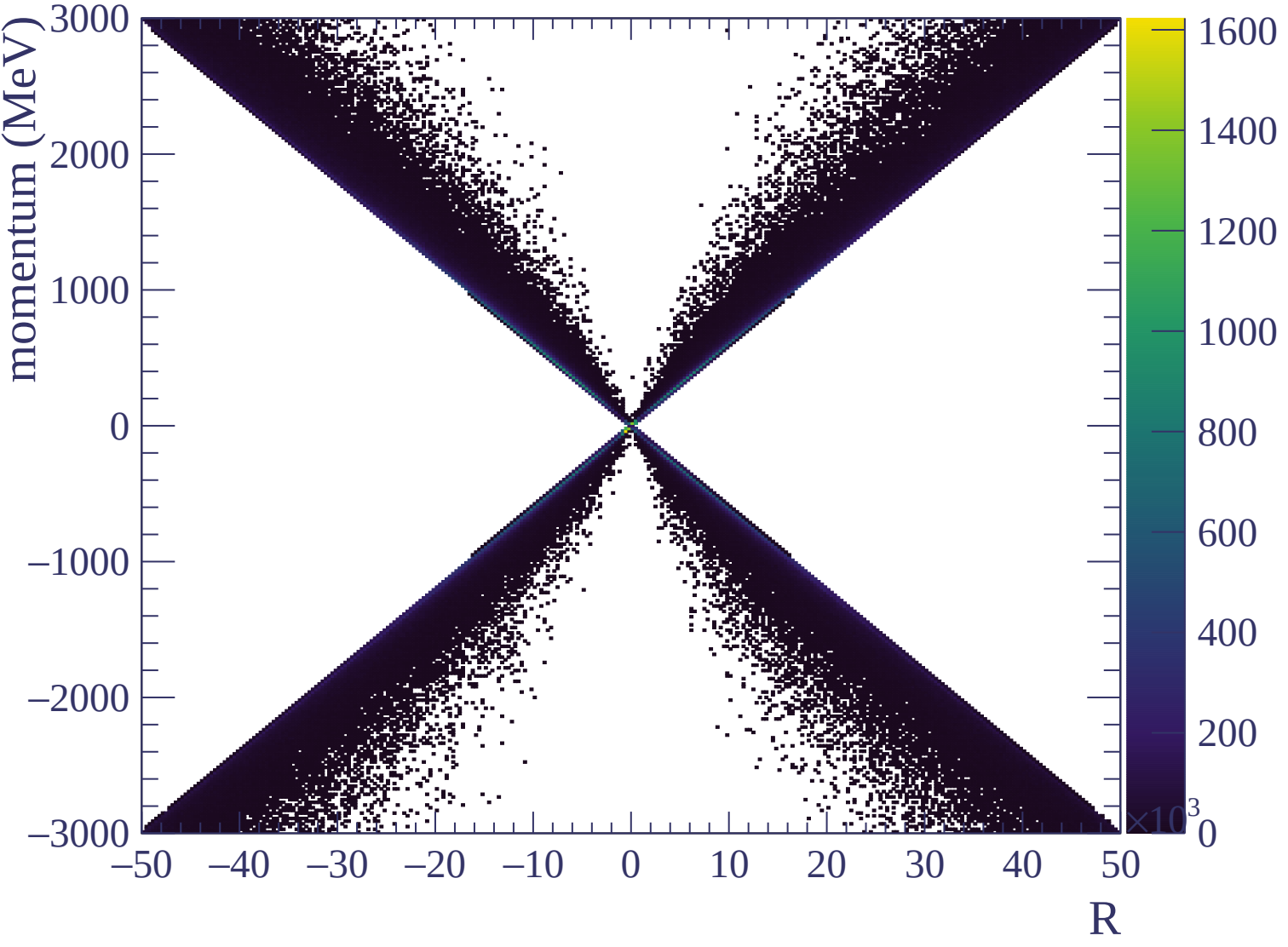


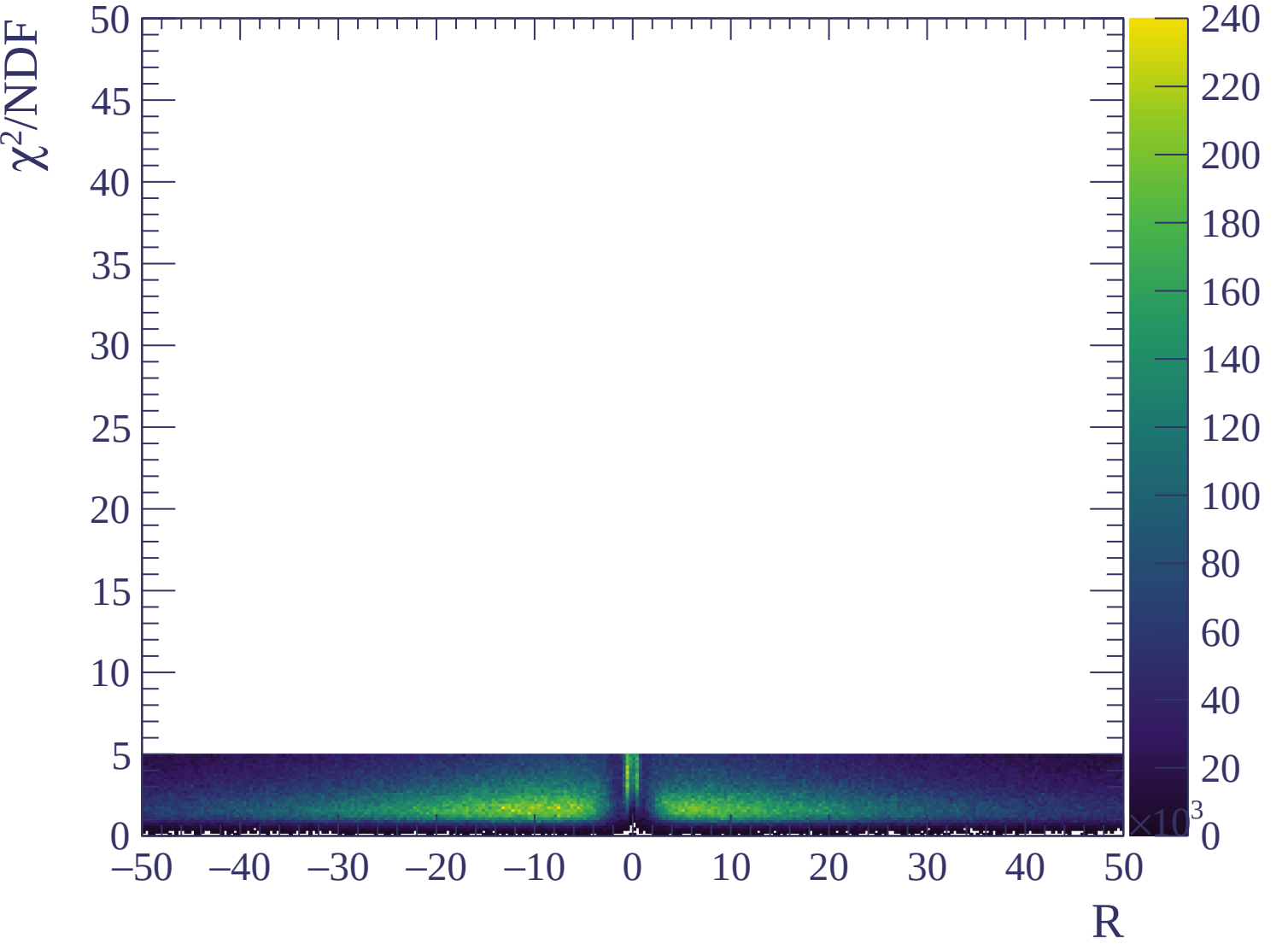






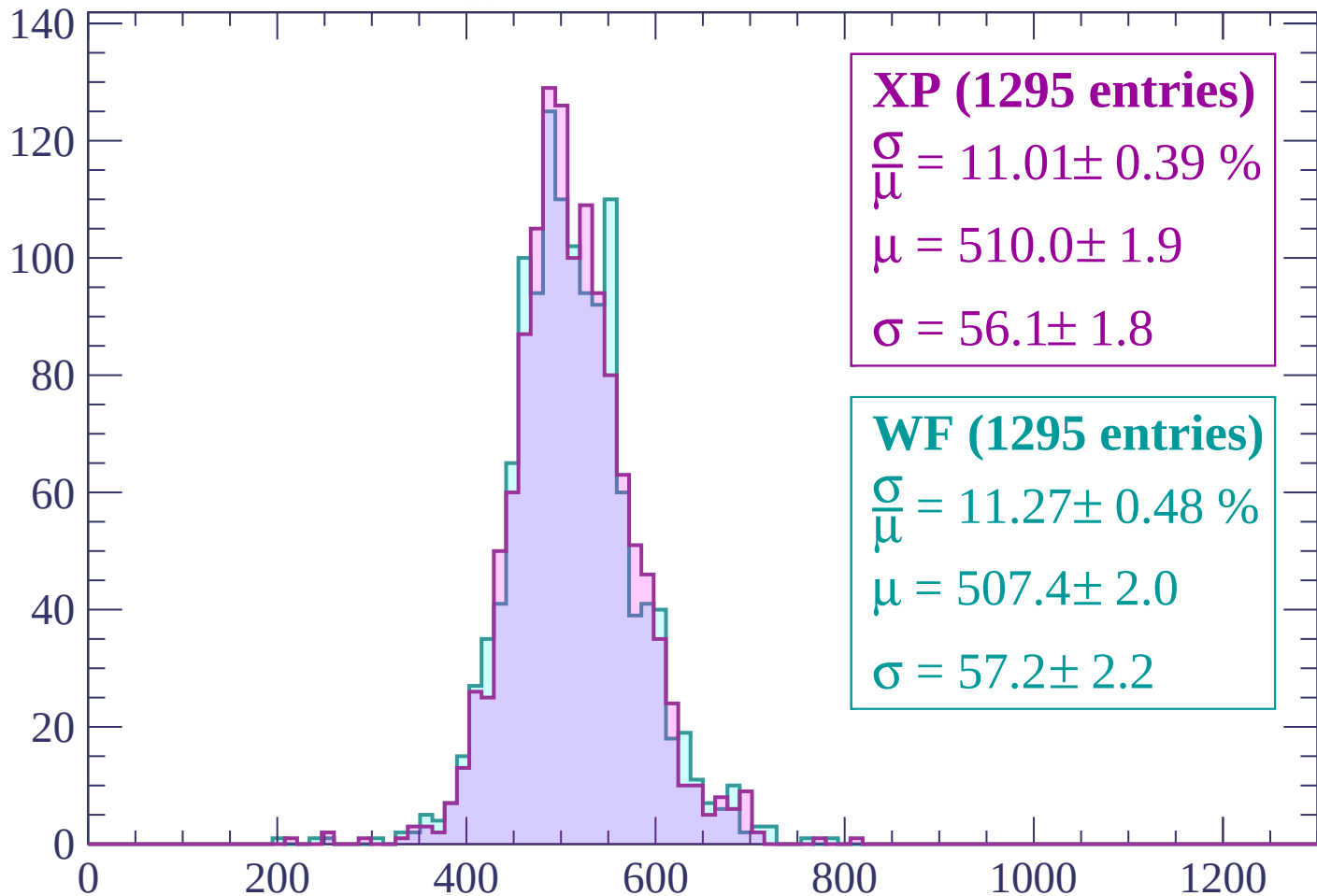






Energy loss | -3000 < p < -2940

Count



XP (1295 entries)

$$\frac{\sigma}{\mu} = 11.01 \pm 0.39 \%$$

$$\mu = 510.0 \pm 1.9$$

$$\sigma = 56.1 \pm 1.8$$

WF (1295 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.48 \%$$

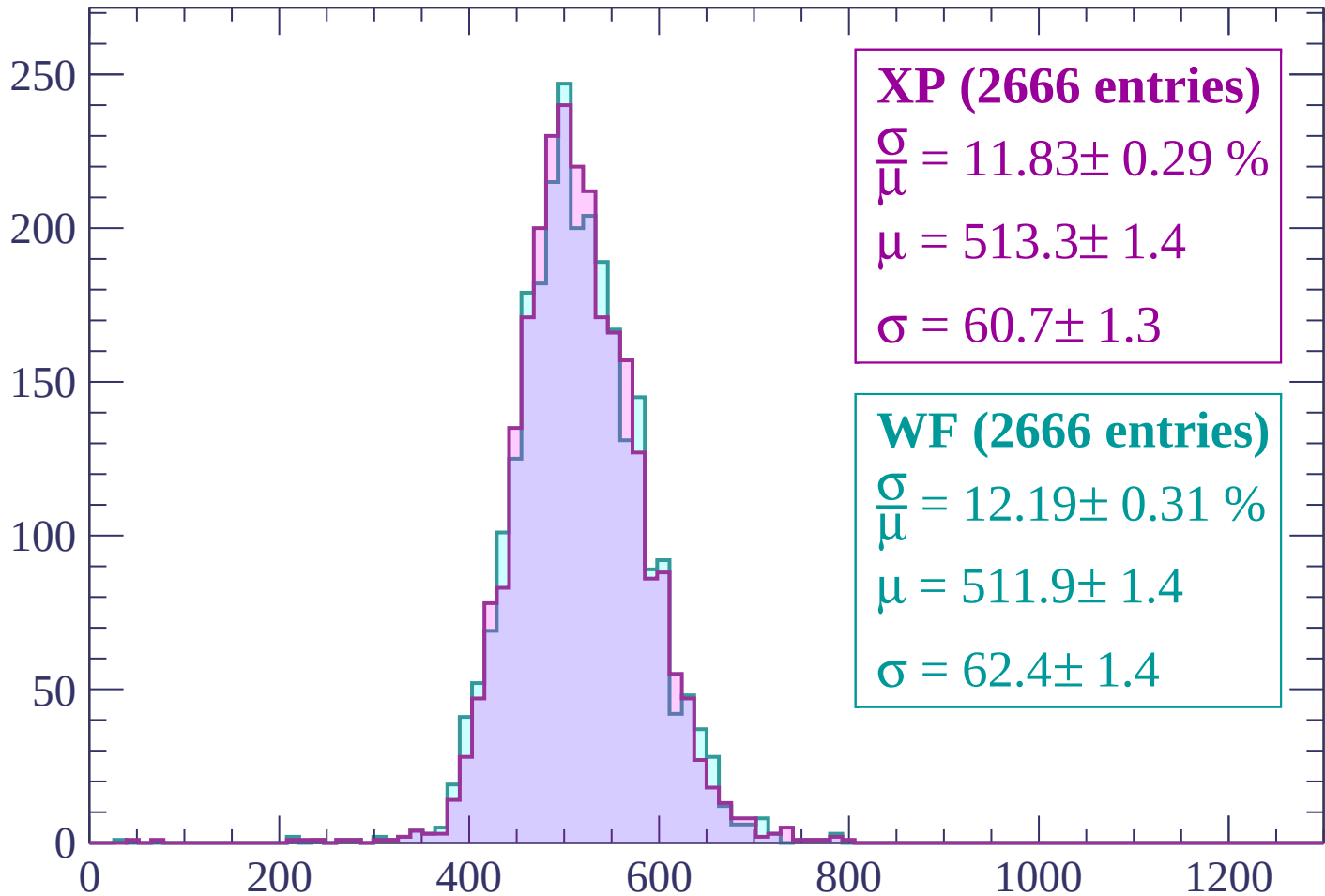
$$\mu = 507.4 \pm 2.0$$

$$\sigma = 57.2 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $-2940 < p < -2880$

Count



XP (2666 entries)

$\frac{\sigma}{\mu} = 11.83 \pm 0.29 \%$

$\mu = 513.3 \pm 1.4$

$\sigma = 60.7 \pm 1.3$

WF (2666 entries)

$\frac{\sigma}{\mu} = 12.19 \pm 0.31 \%$

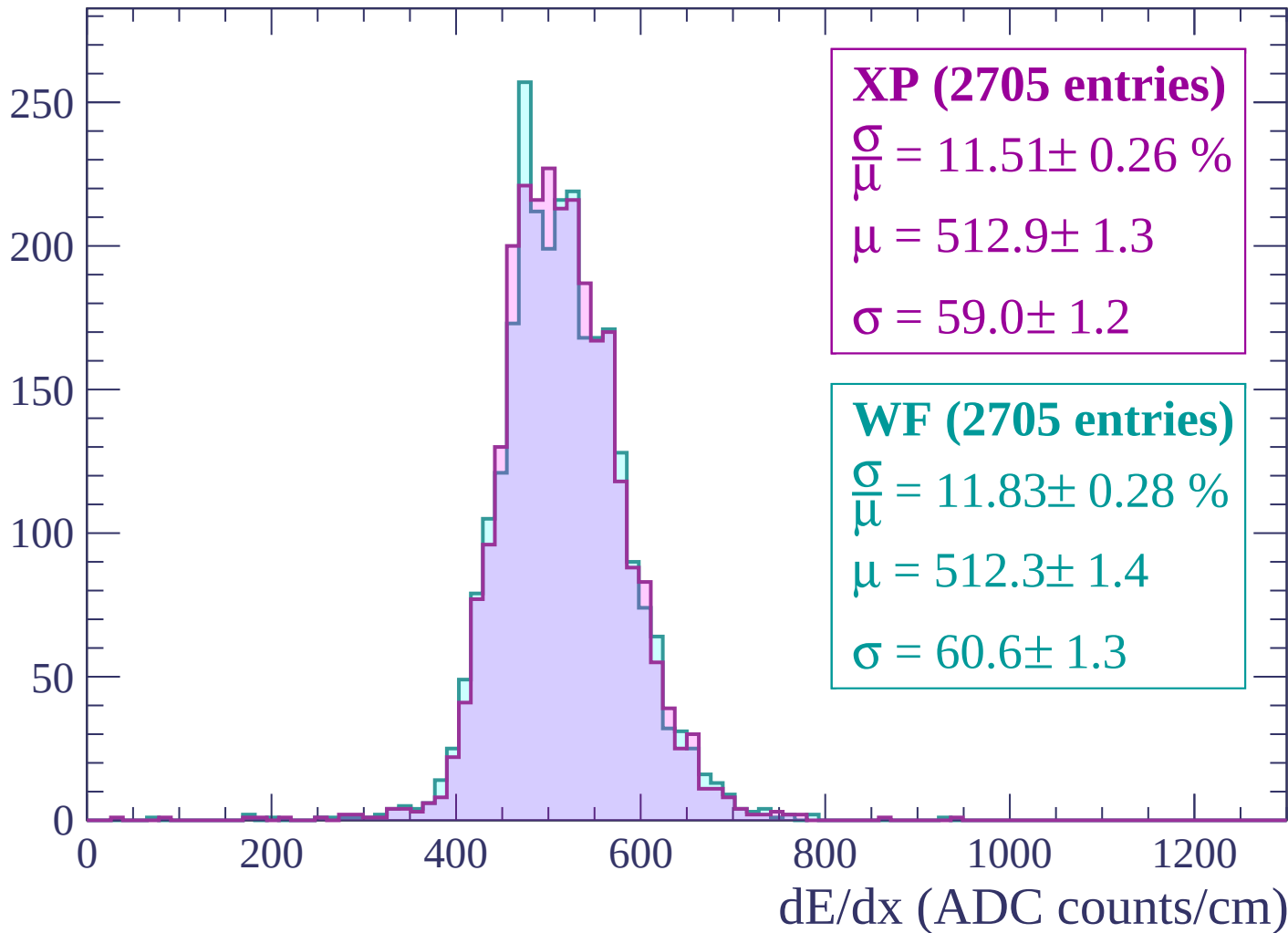
$\mu = 511.9 \pm 1.4$

$\sigma = 62.4 \pm 1.4$

dE/dx (ADC counts/cm)

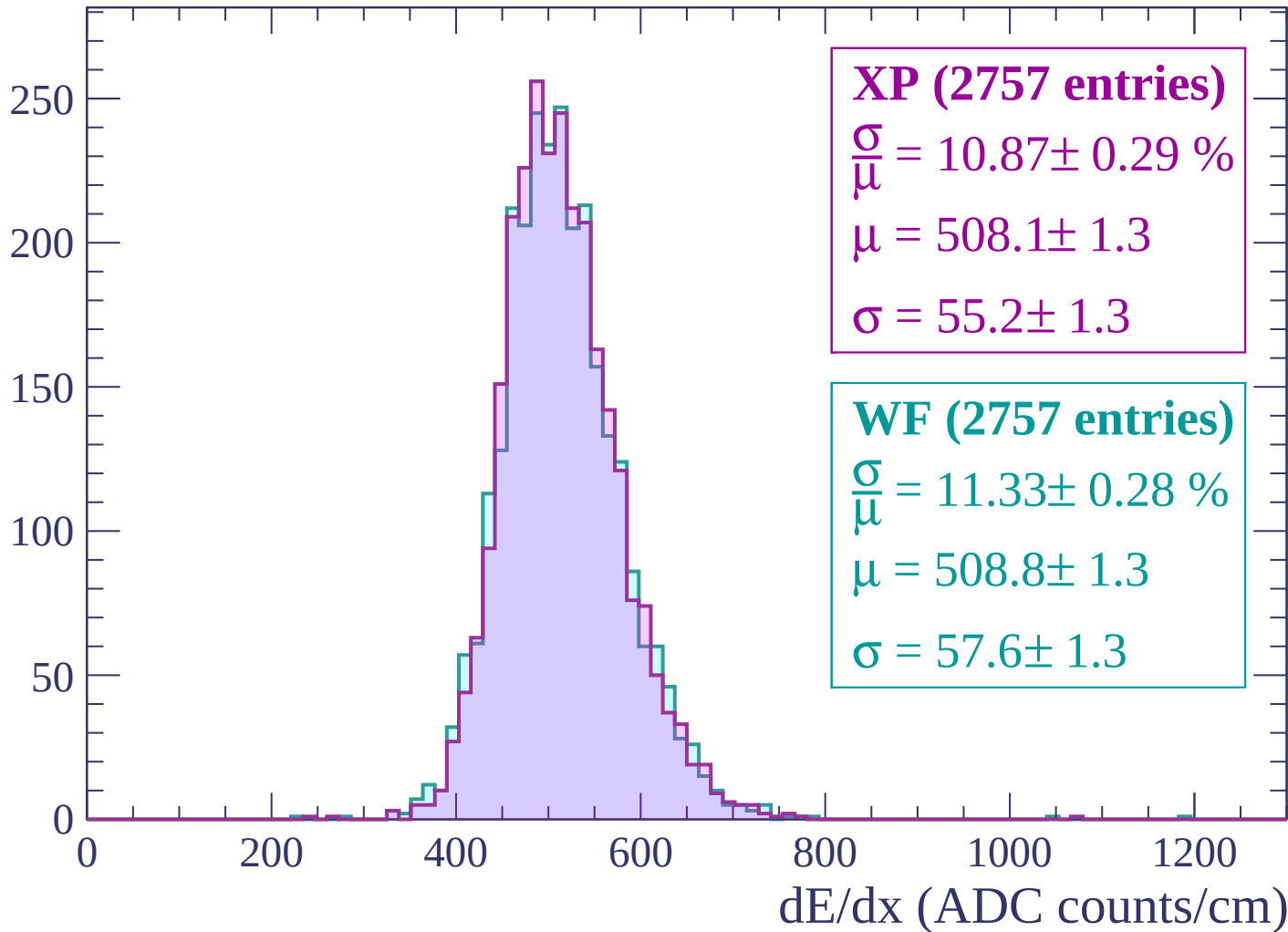
Energy loss | $-2880 < p < -2820$

Count



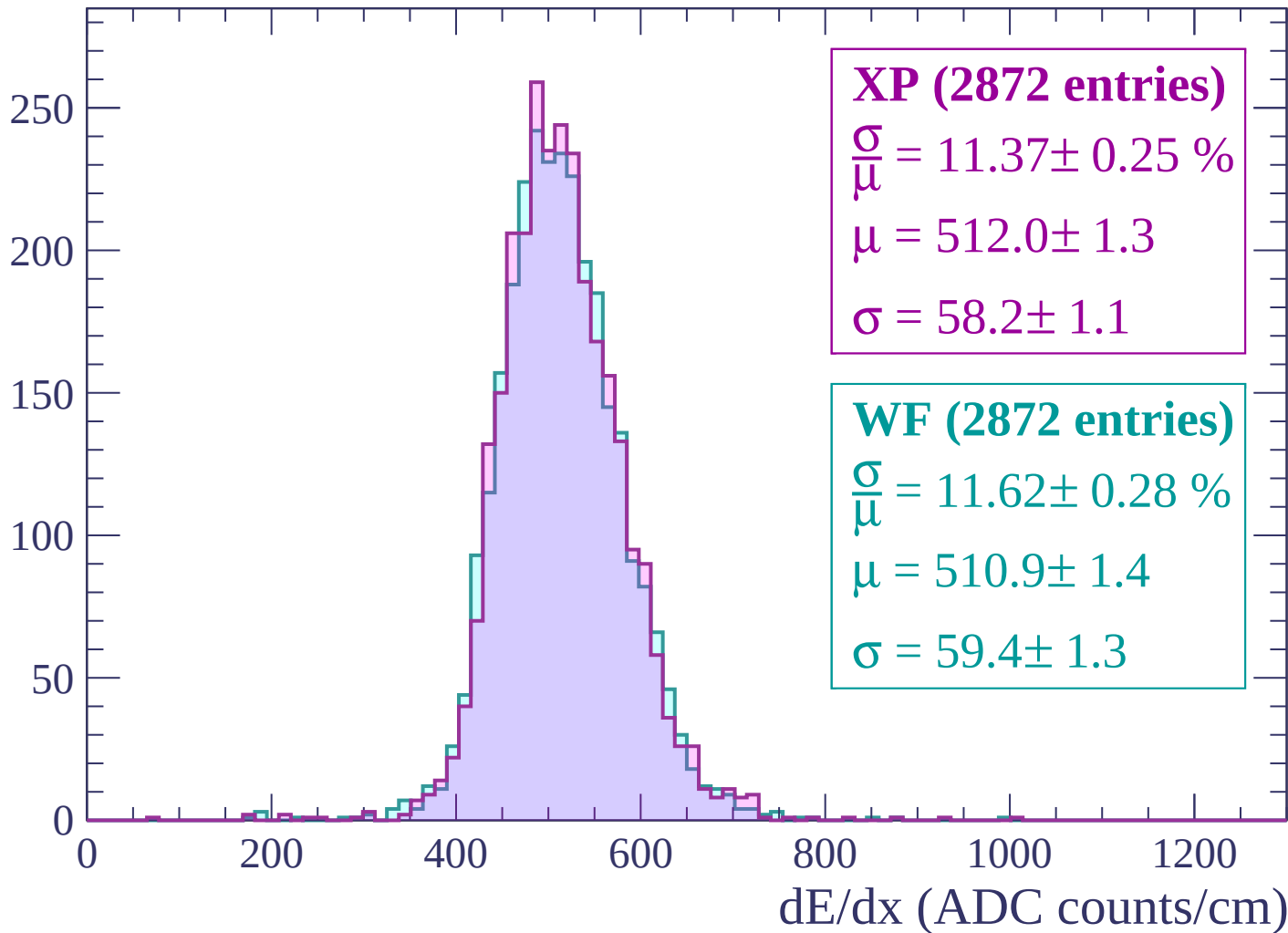
Energy loss | $-2820 < p < -2760$

Count



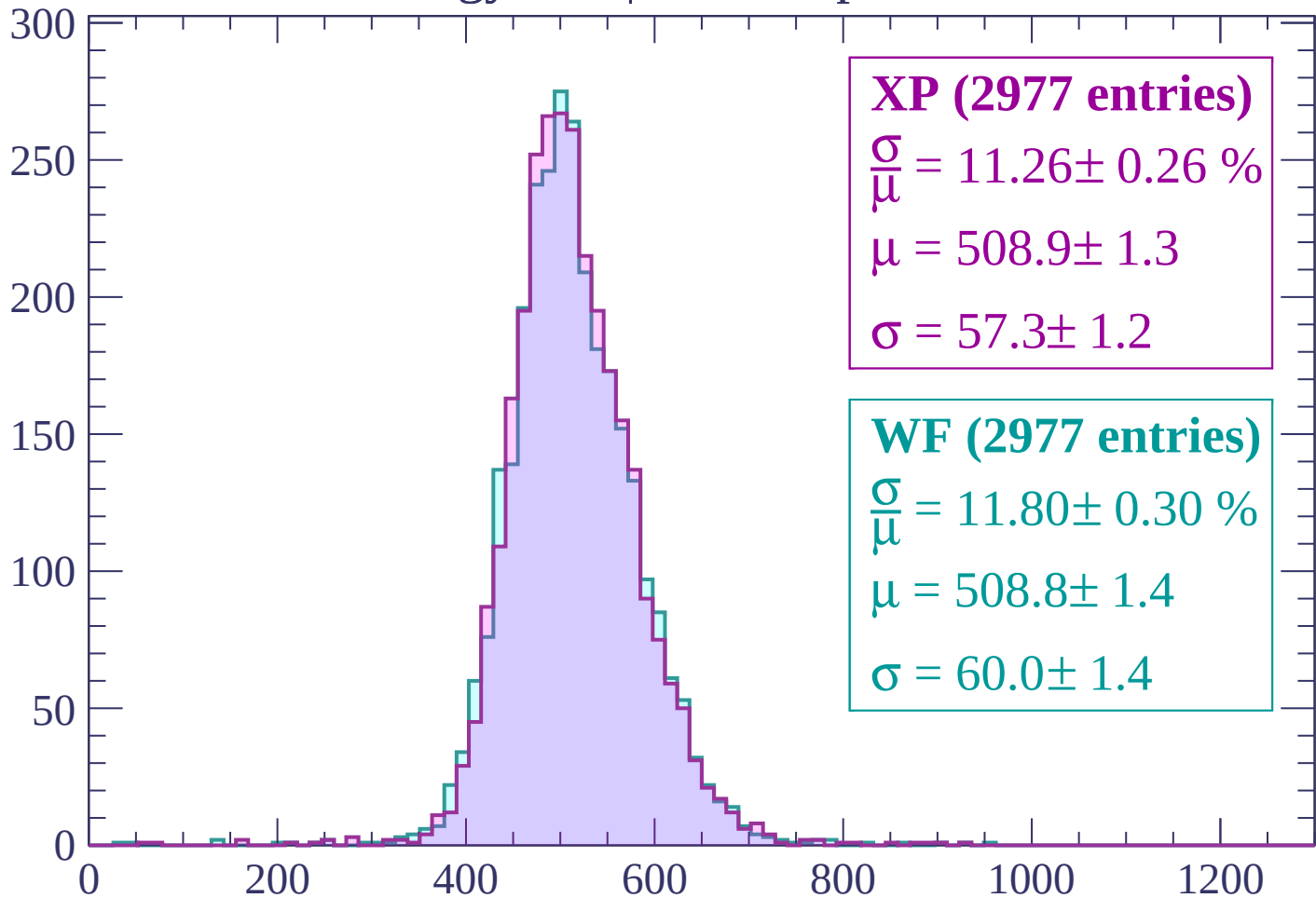
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (2977 entries)

$$\frac{\sigma}{\mu} = 11.26 \pm 0.26 \%$$

$$\mu = 508.9 \pm 1.3$$

$$\sigma = 57.3 \pm 1.2$$

WF (2977 entries)

$$\frac{\sigma}{\mu} = 11.80 \pm 0.30 \%$$

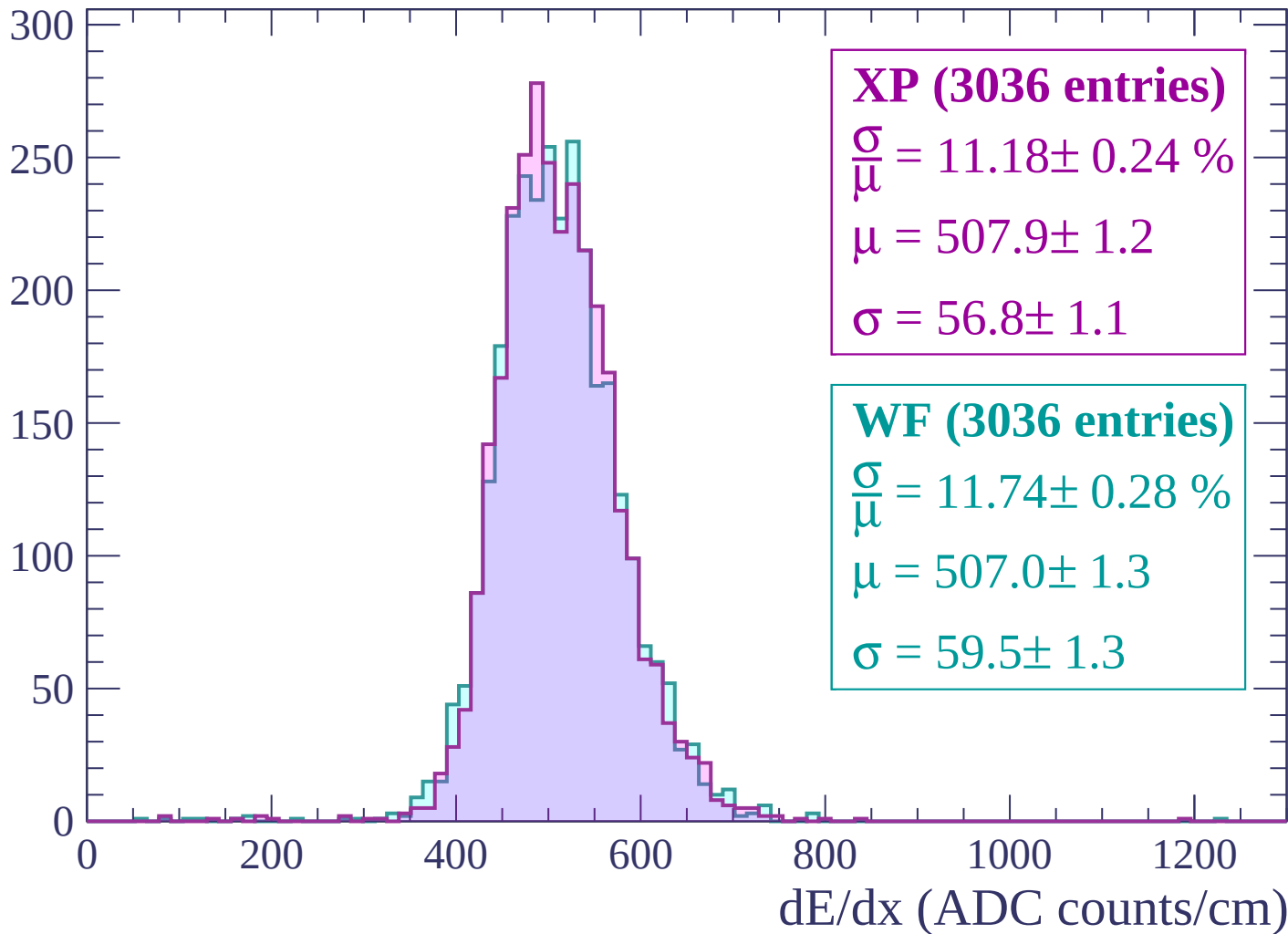
$$\mu = 508.8 \pm 1.4$$

$$\sigma = 60.0 \pm 1.4$$

dE/dx (ADC counts/cm)

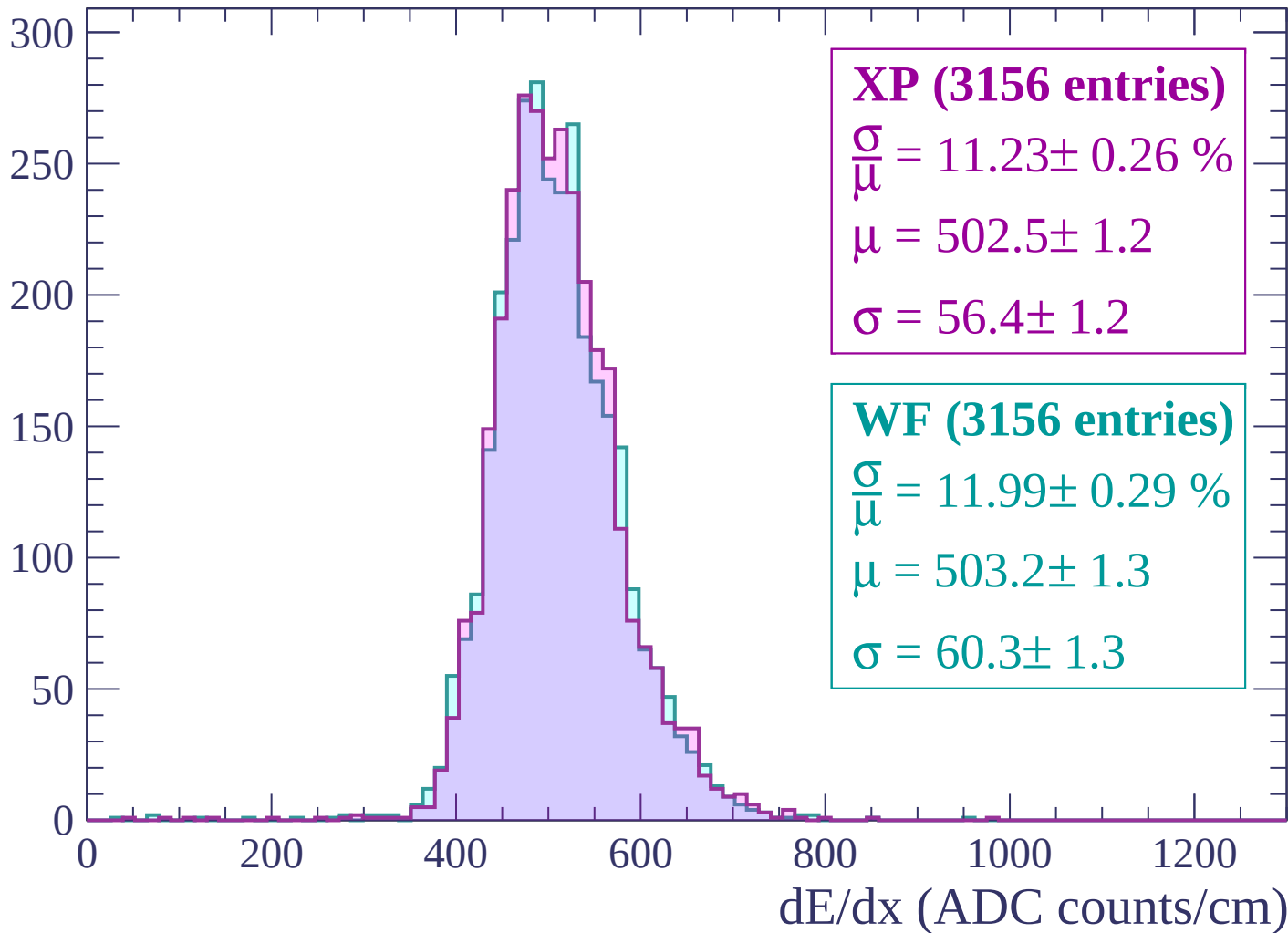
Energy loss | $-2640 < p < -2580$

Count



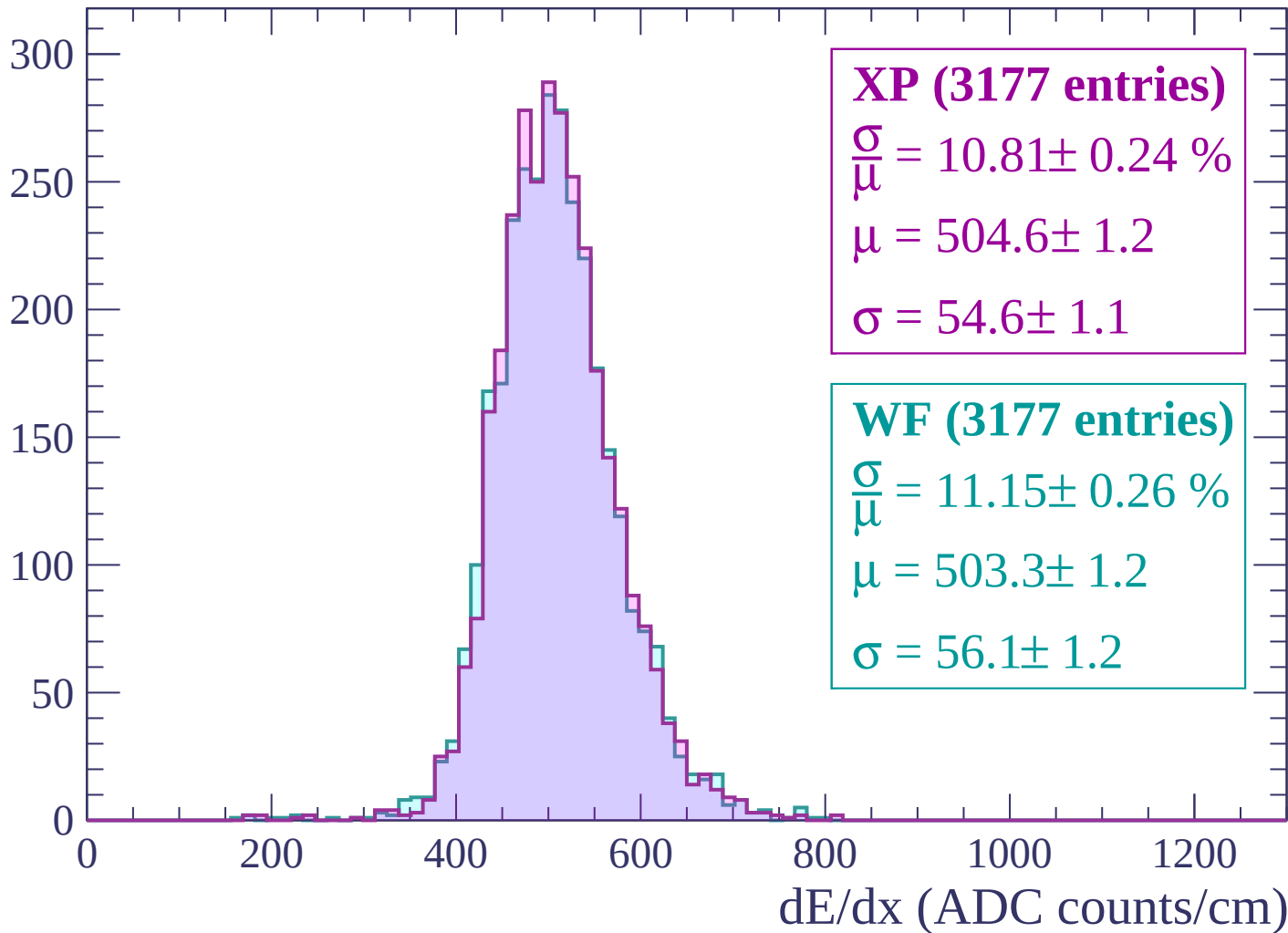
Energy loss | -2580 < p < -2520

Count



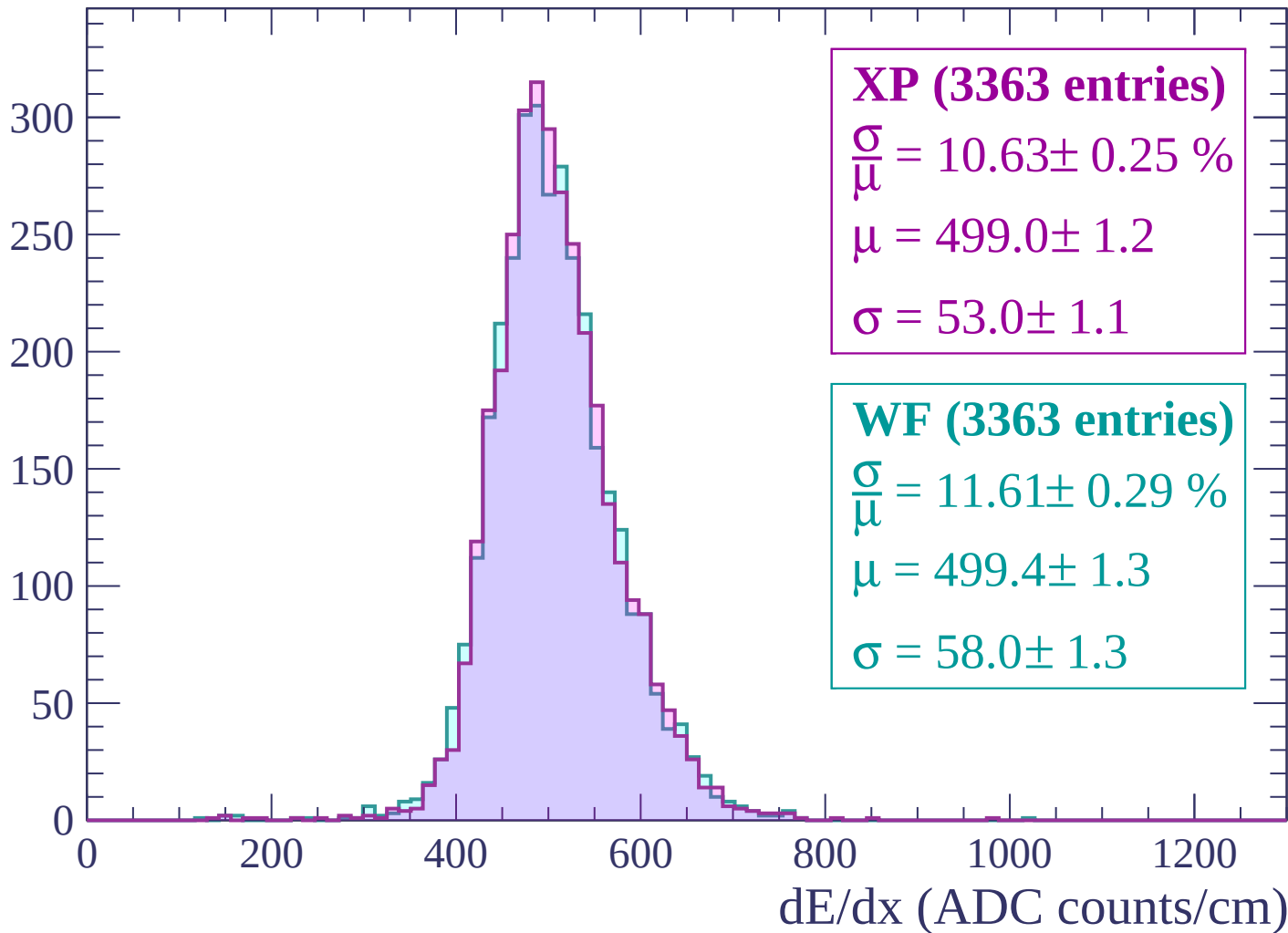
Energy loss | -2520 < p < -2460

Count



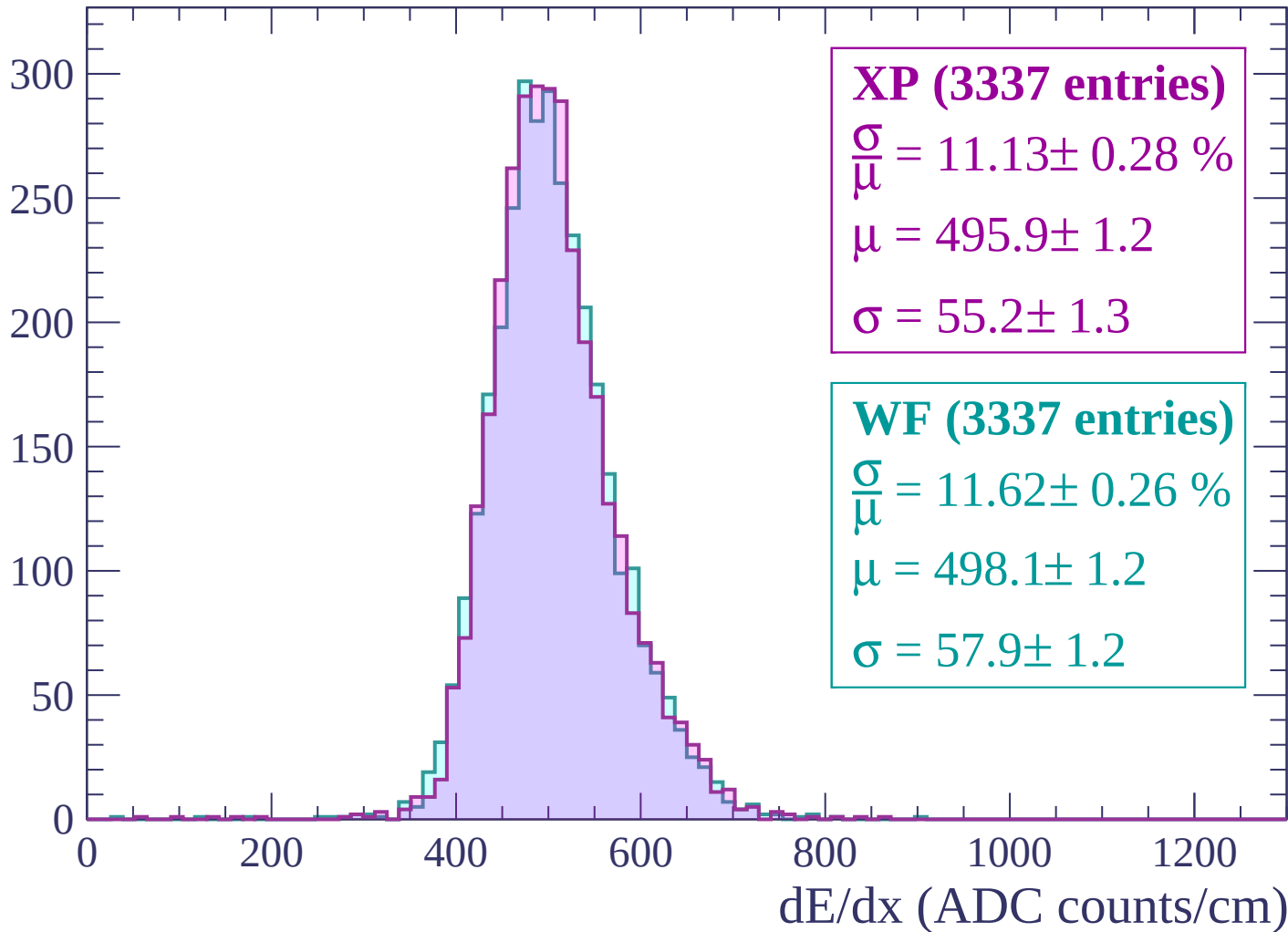
Energy loss | $-2460 < p < -2400$

Count



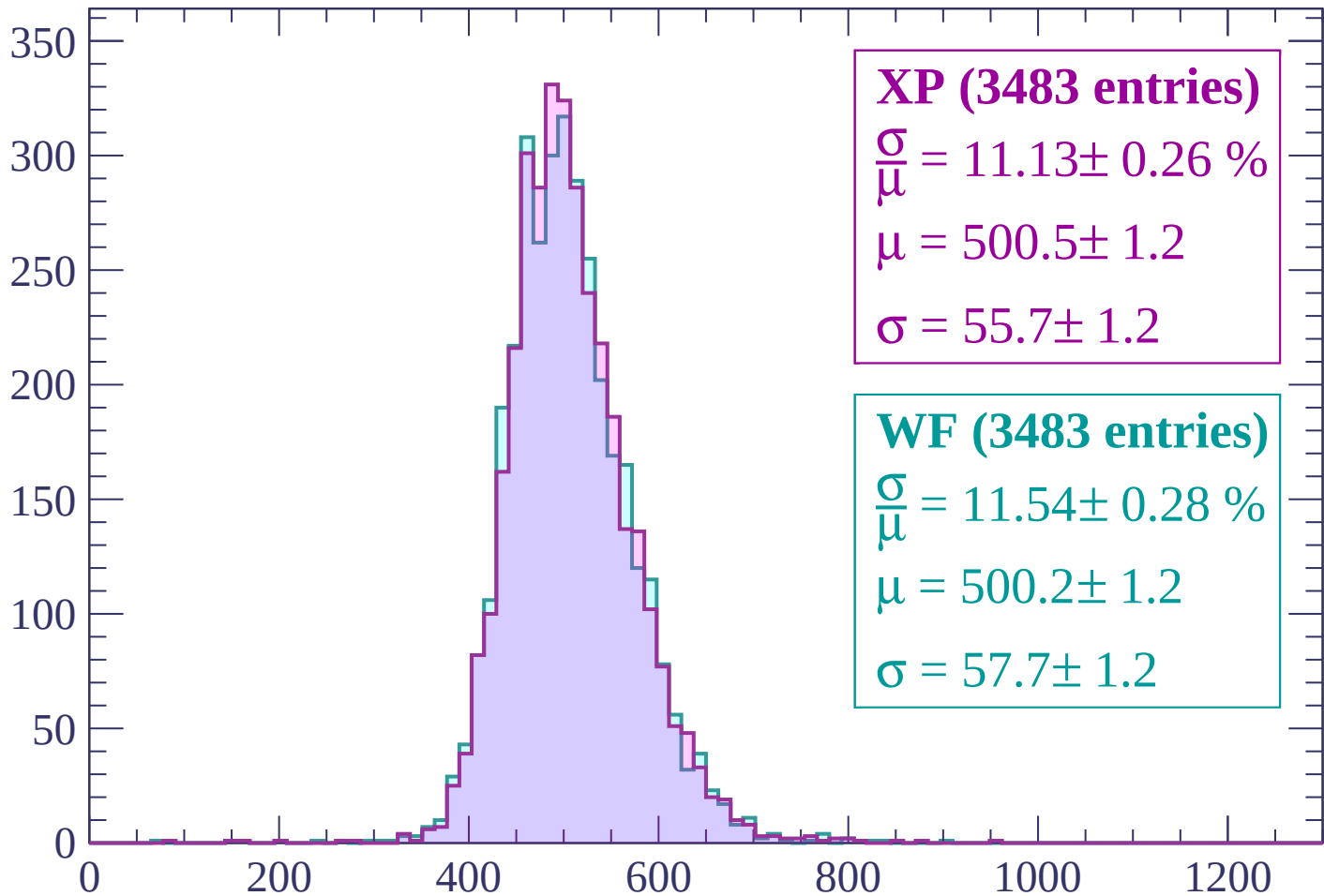
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (3483 entries)

$$\frac{\sigma}{\mu} = 11.13 \pm 0.26 \%$$

$$\mu = 500.5 \pm 1.2$$

$$\sigma = 55.7 \pm 1.2$$

WF (3483 entries)

$$\frac{\sigma}{\mu} = 11.54 \pm 0.28 \%$$

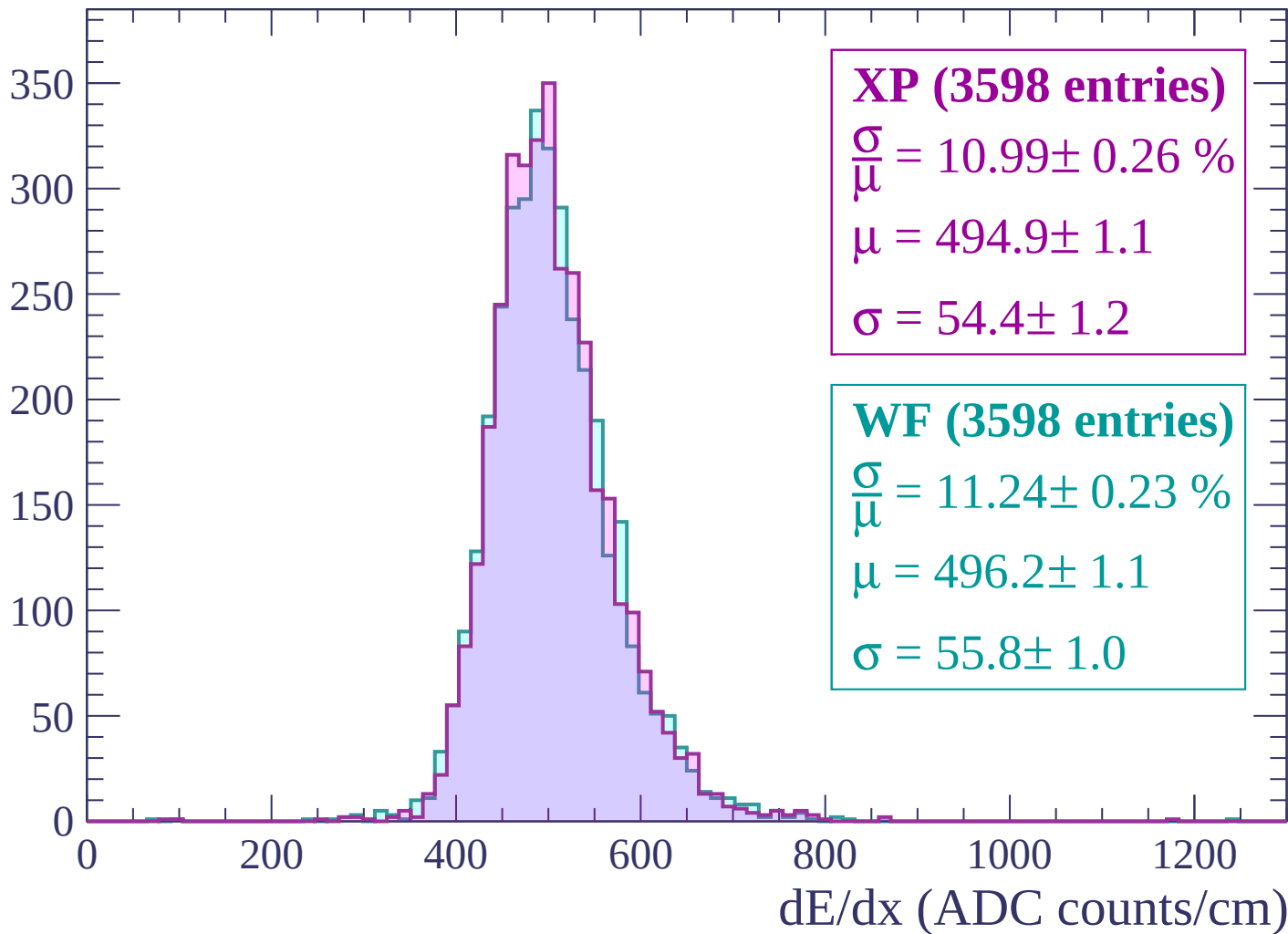
$$\mu = 500.2 \pm 1.2$$

$$\sigma = 57.7 \pm 1.2$$

dE/dx (ADC counts/cm)

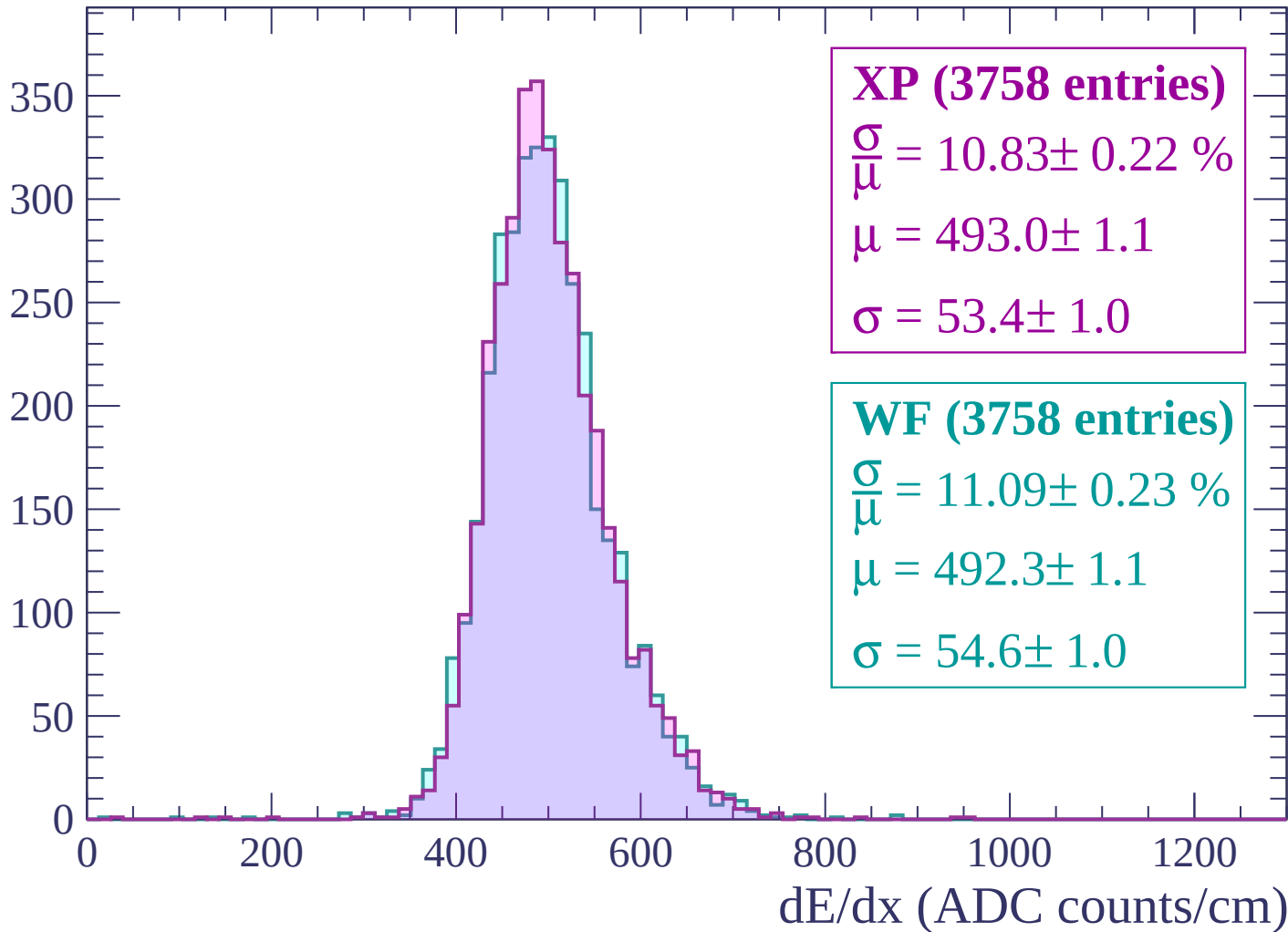
Energy loss | $-2280 < p < -2220$

Count



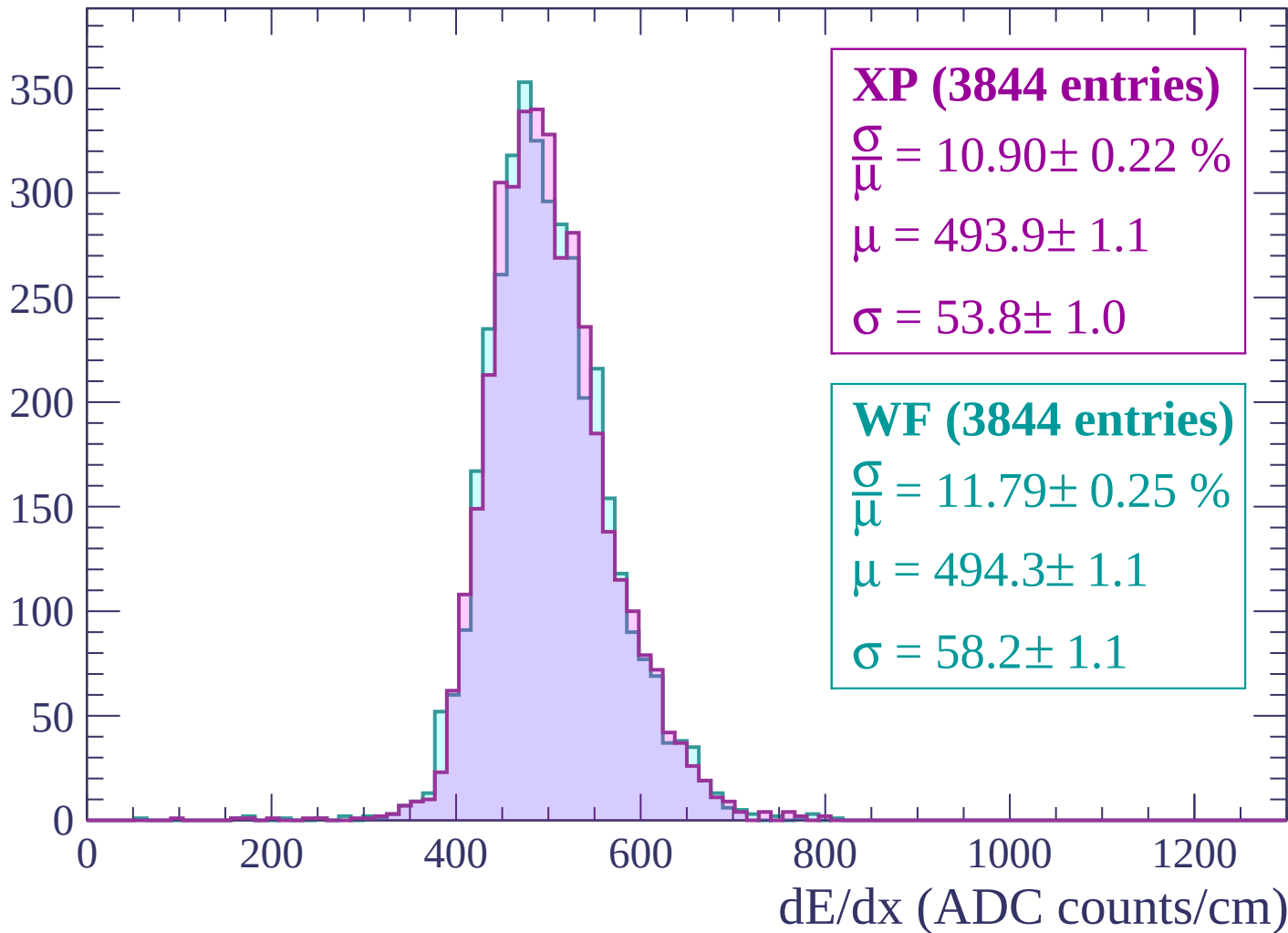
Energy loss | -2220 < p < -2160

Count



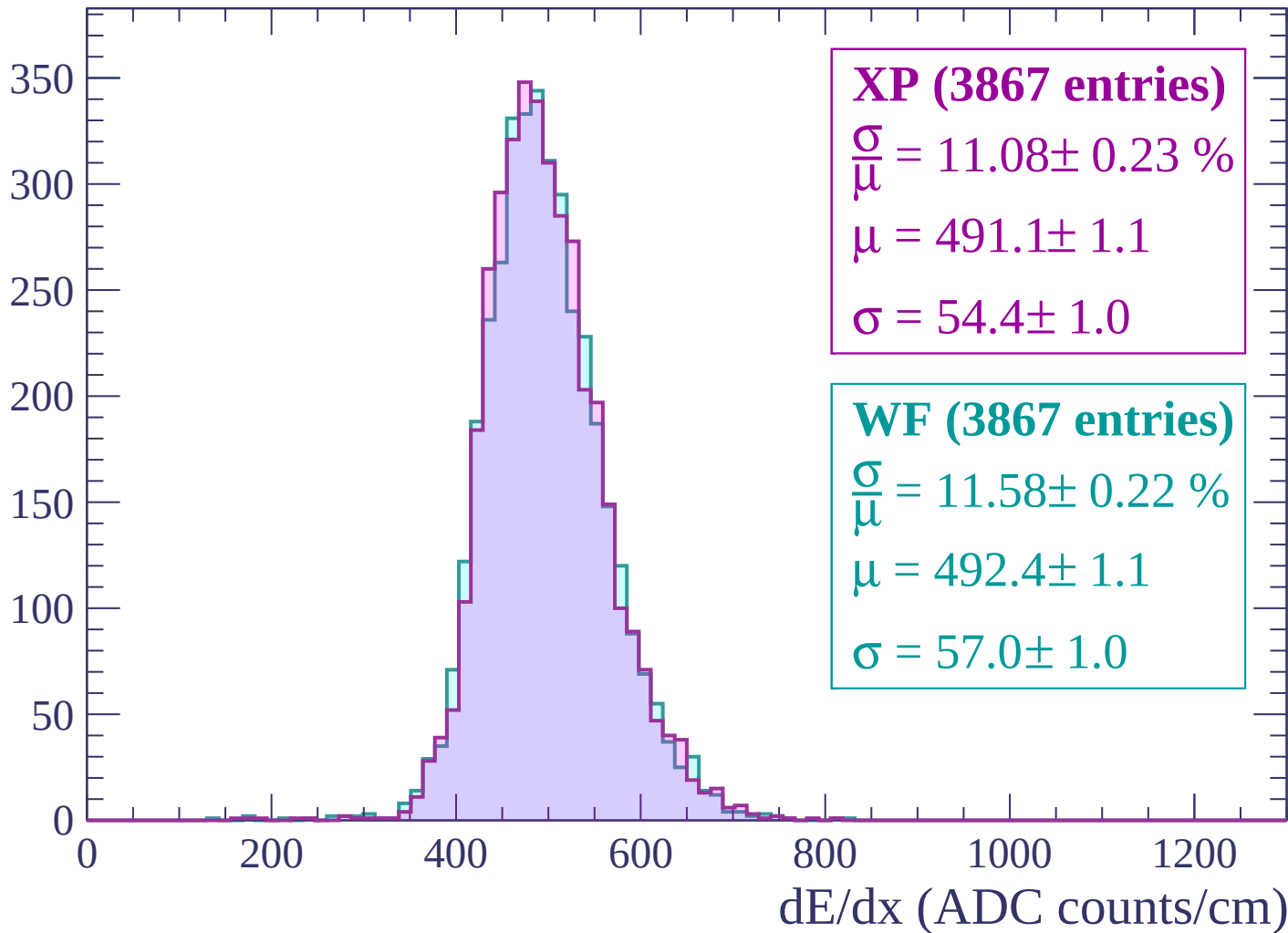
Energy loss | -2160 < p < -2100

Count



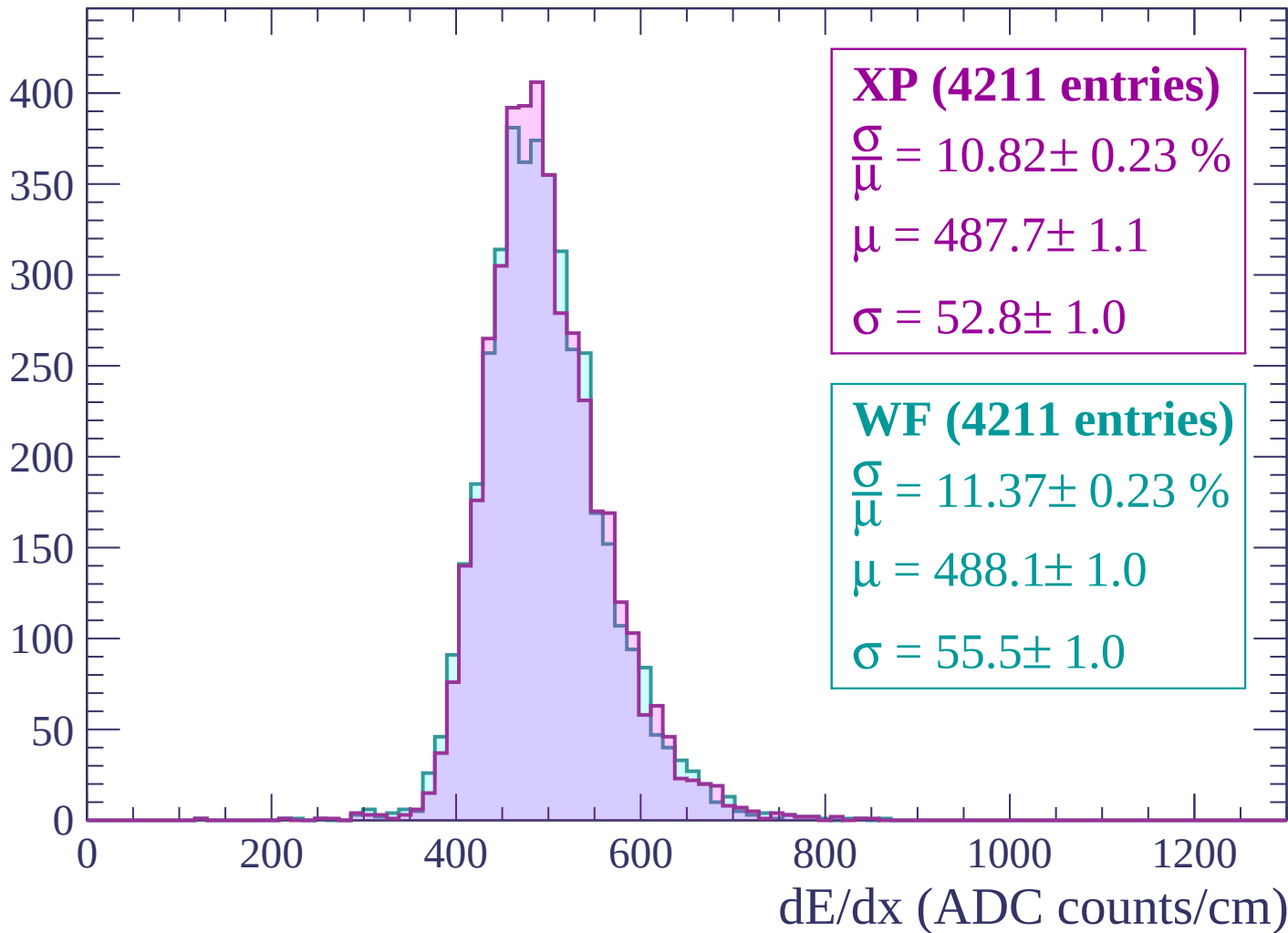
Energy loss | -2100 < p < -2040

Count



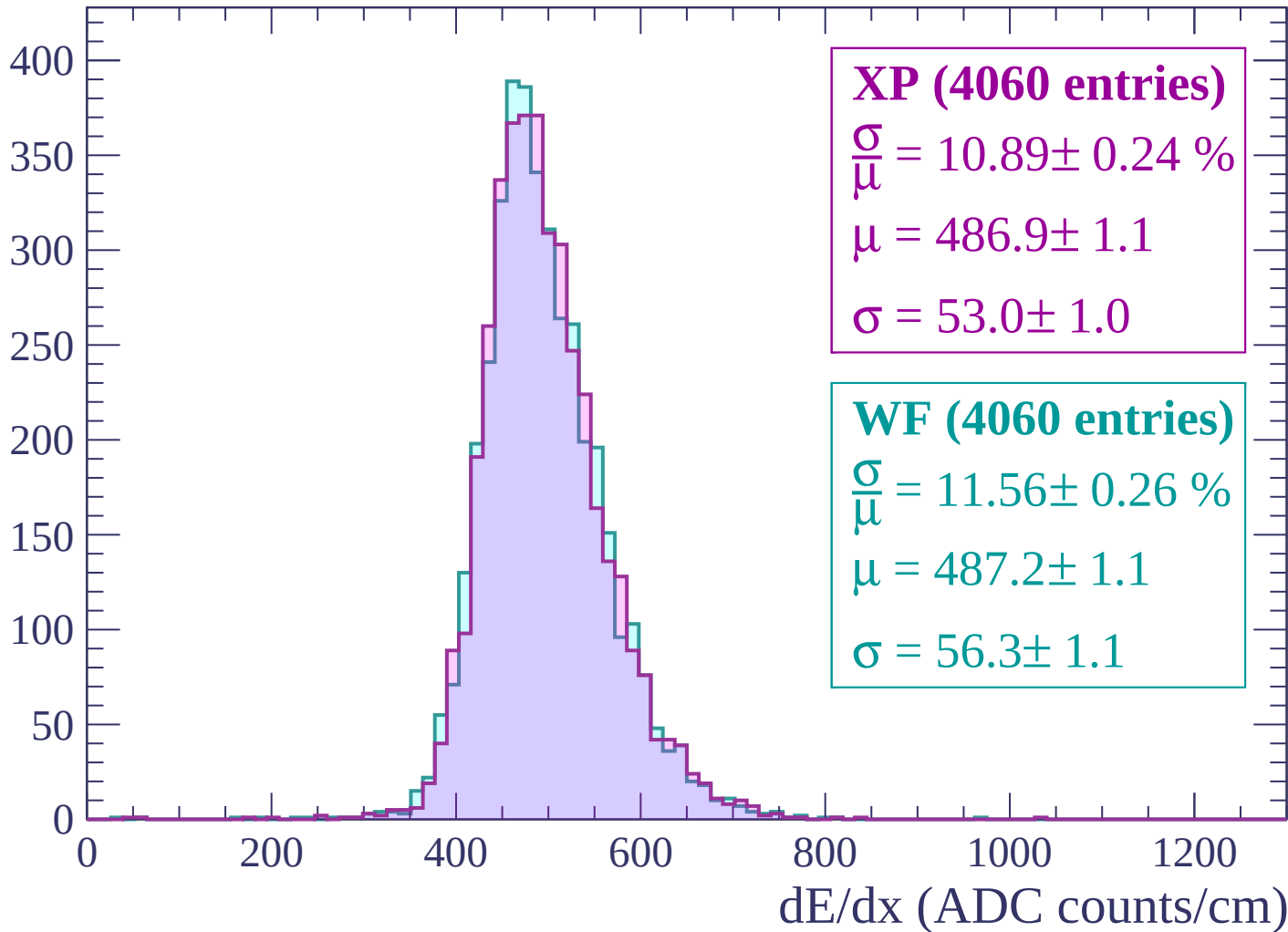
Energy loss | -2040 < p < -1980

Count



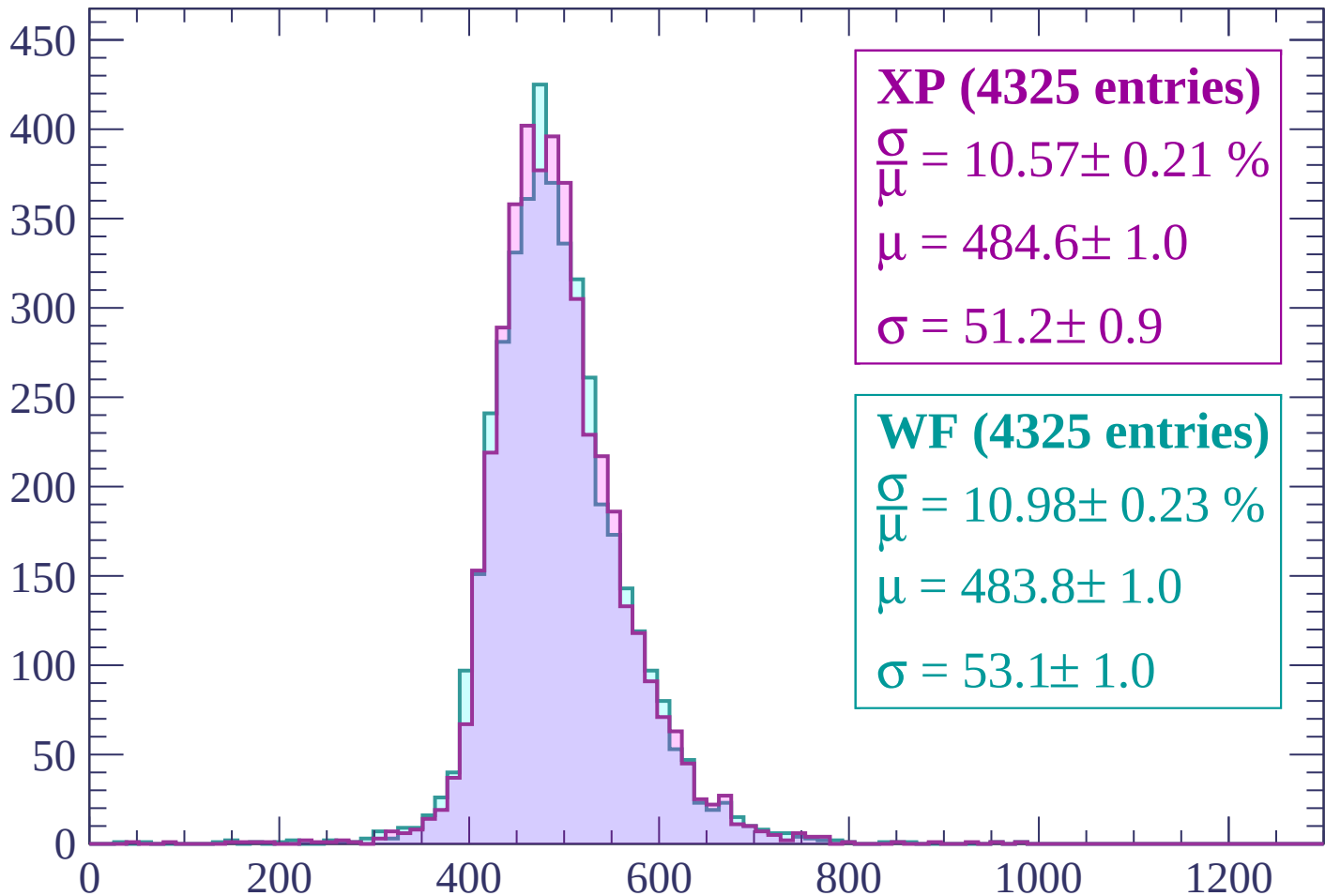
Energy loss | -1980 < p < -1920

Count



Energy loss | -1920 < p < -1860

Count



XP (4325 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.21 \%$$

$$\mu = 484.6 \pm 1.0$$

$$\sigma = 51.2 \pm 0.9$$

WF (4325 entries)

$$\frac{\sigma}{\mu} = 10.98 \pm 0.23 \%$$

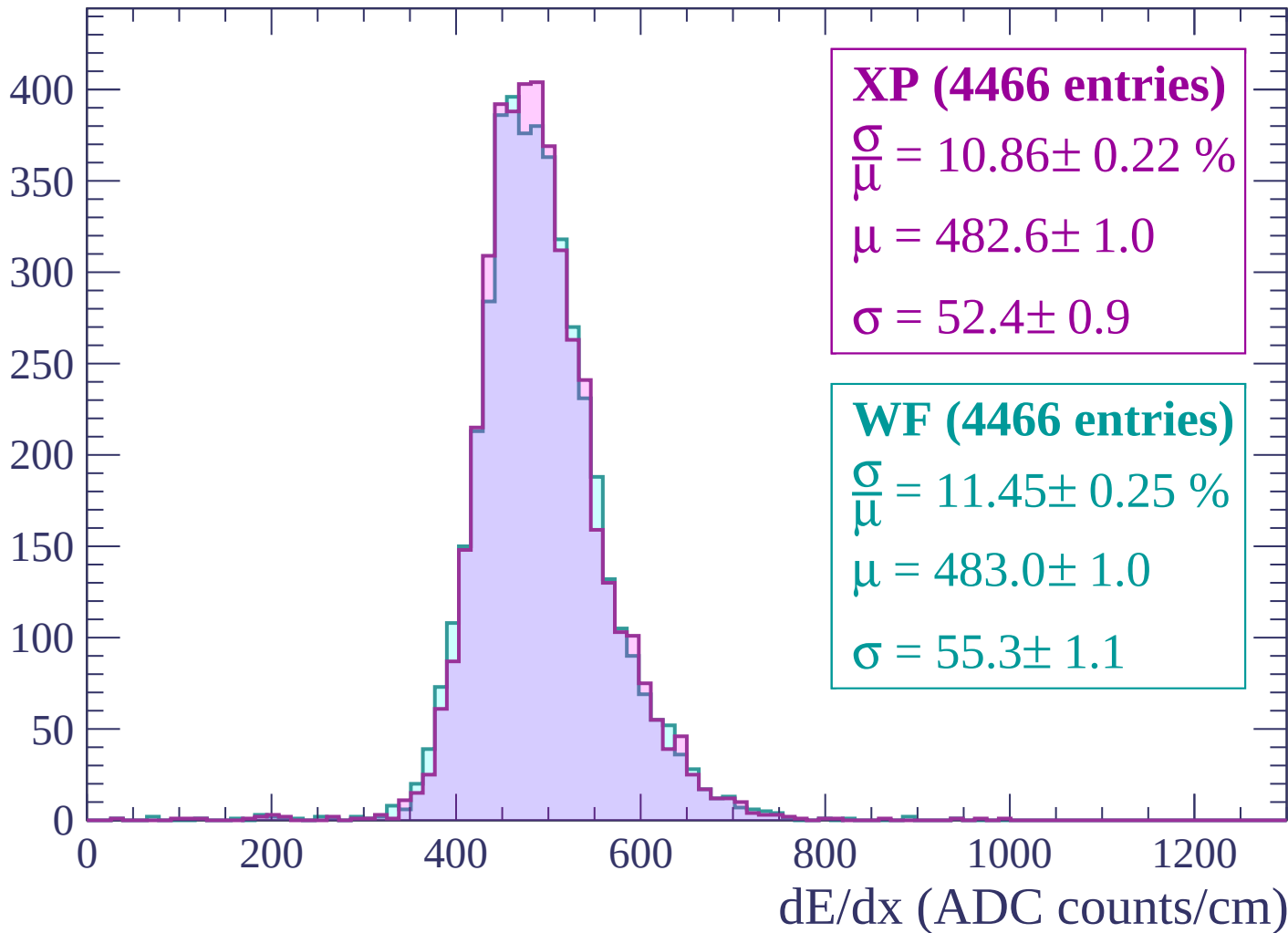
$$\mu = 483.8 \pm 1.0$$

$$\sigma = 53.1 \pm 1.0$$

dE/dx (ADC counts/cm)

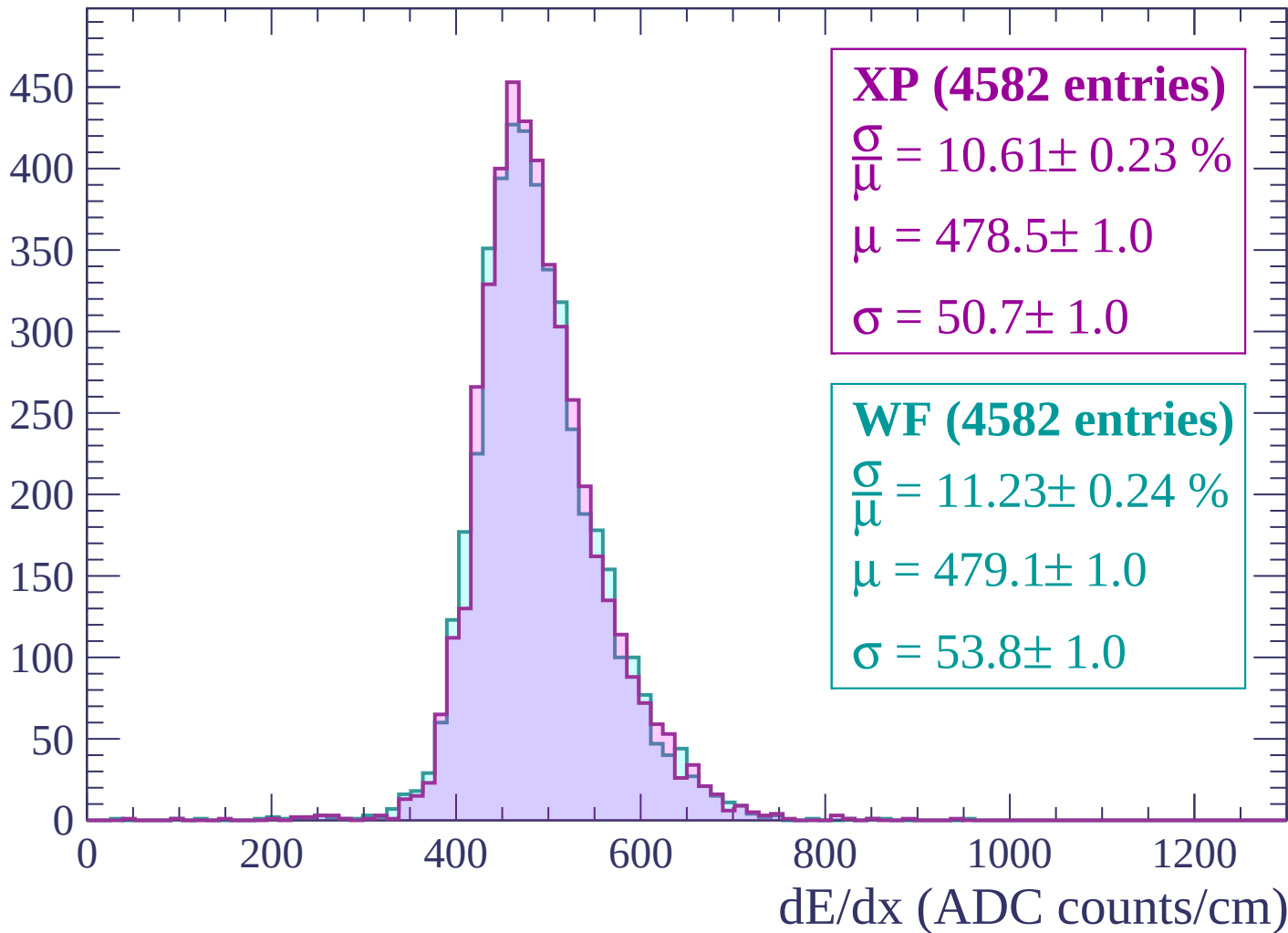
Energy loss | -1860 < p < -1800

Count



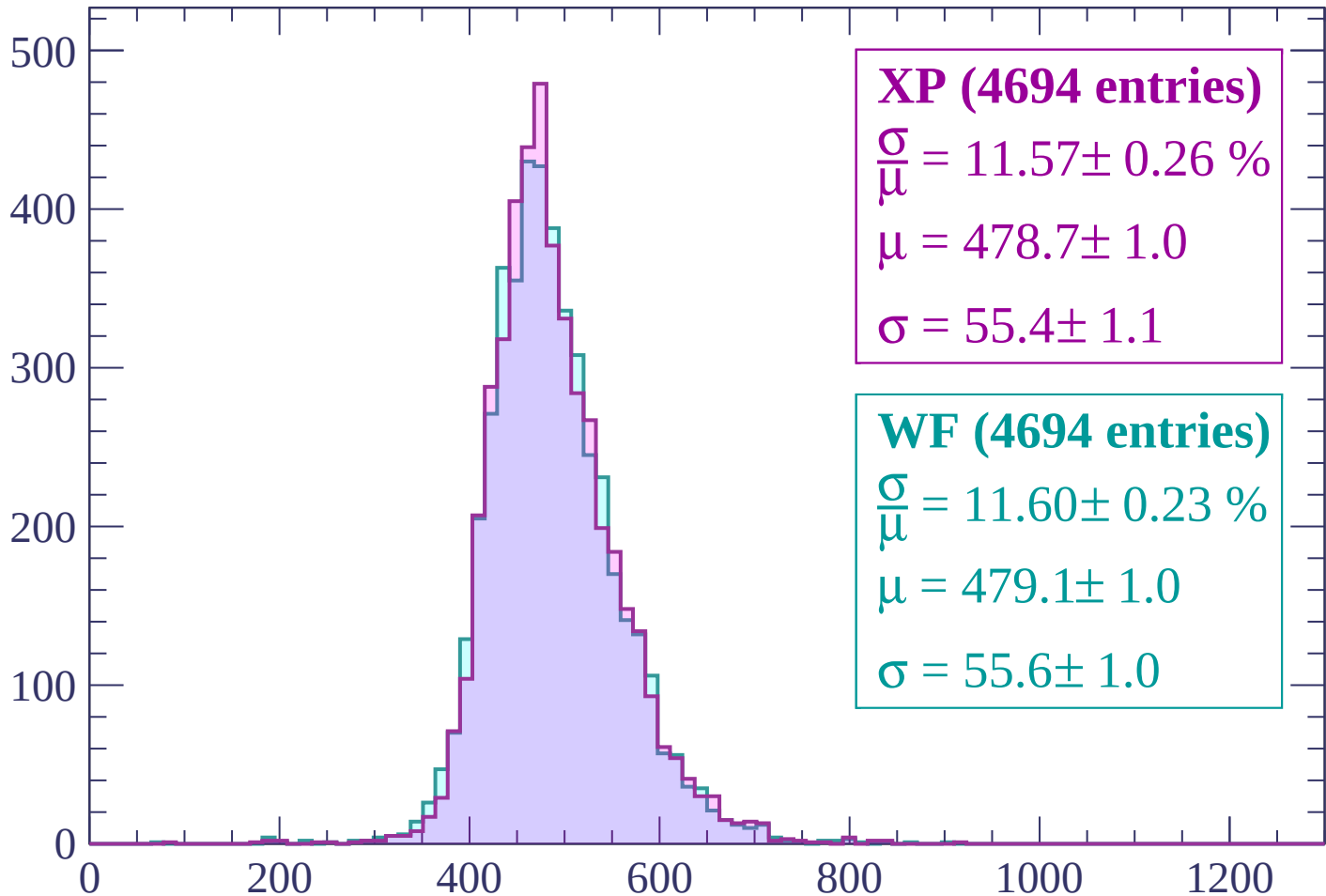
Energy loss | $-1800 < p < -1740$

Count



Energy loss | $-1740 < p < -1680$

Count



XP (4694 entries)

$\frac{\sigma}{\mu} = 11.57 \pm 0.26 \%$

$\mu = 478.7 \pm 1.0$

$\sigma = 55.4 \pm 1.1$

WF (4694 entries)

$\frac{\sigma}{\mu} = 11.60 \pm 0.23 \%$

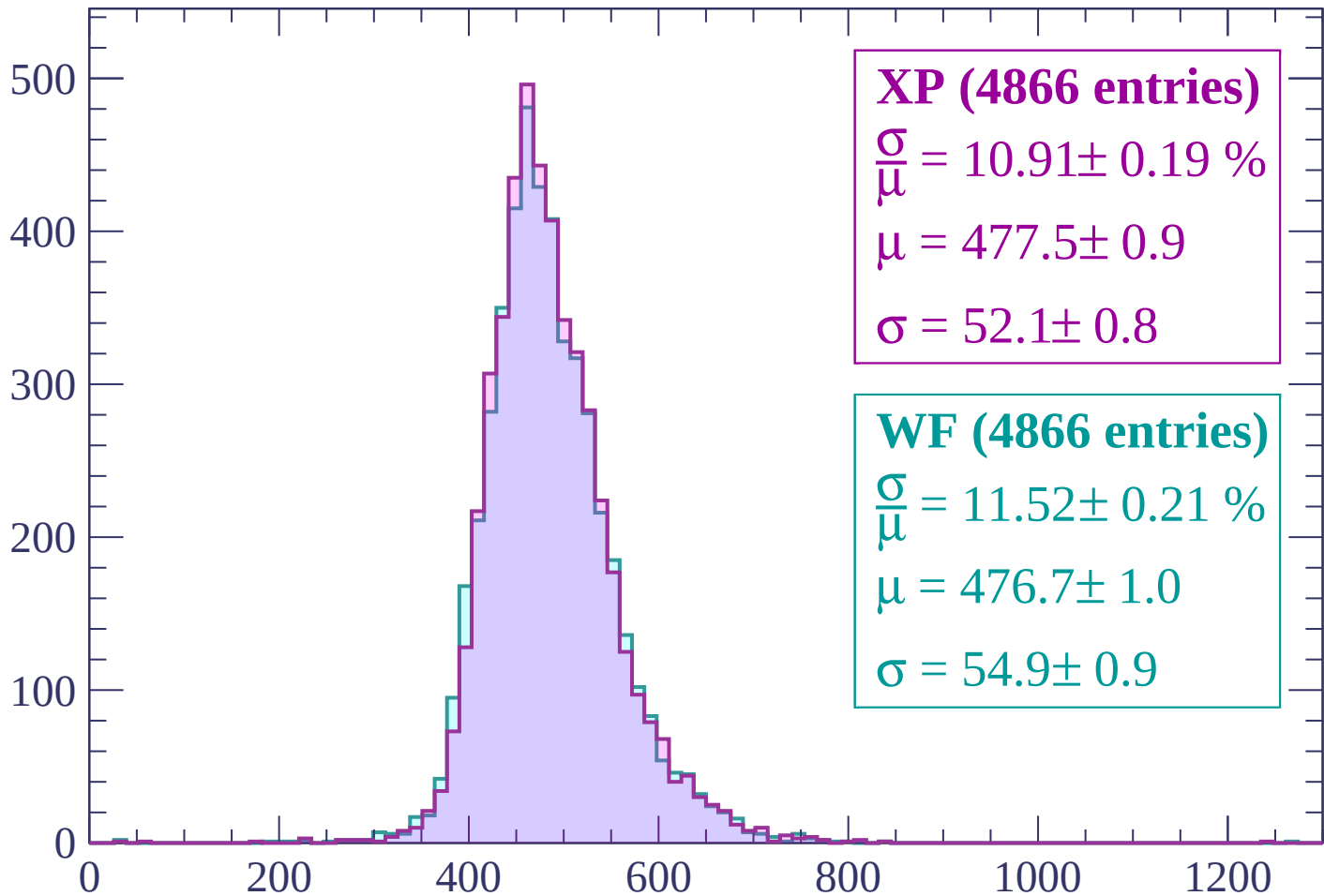
$\mu = 479.1 \pm 1.0$

$\sigma = 55.6 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP (4866 entries)

$\frac{\sigma}{\mu} = 10.91 \pm 0.19 \%$

$\mu = 477.5 \pm 0.9$

$\sigma = 52.1 \pm 0.8$

WF (4866 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.21 \%$

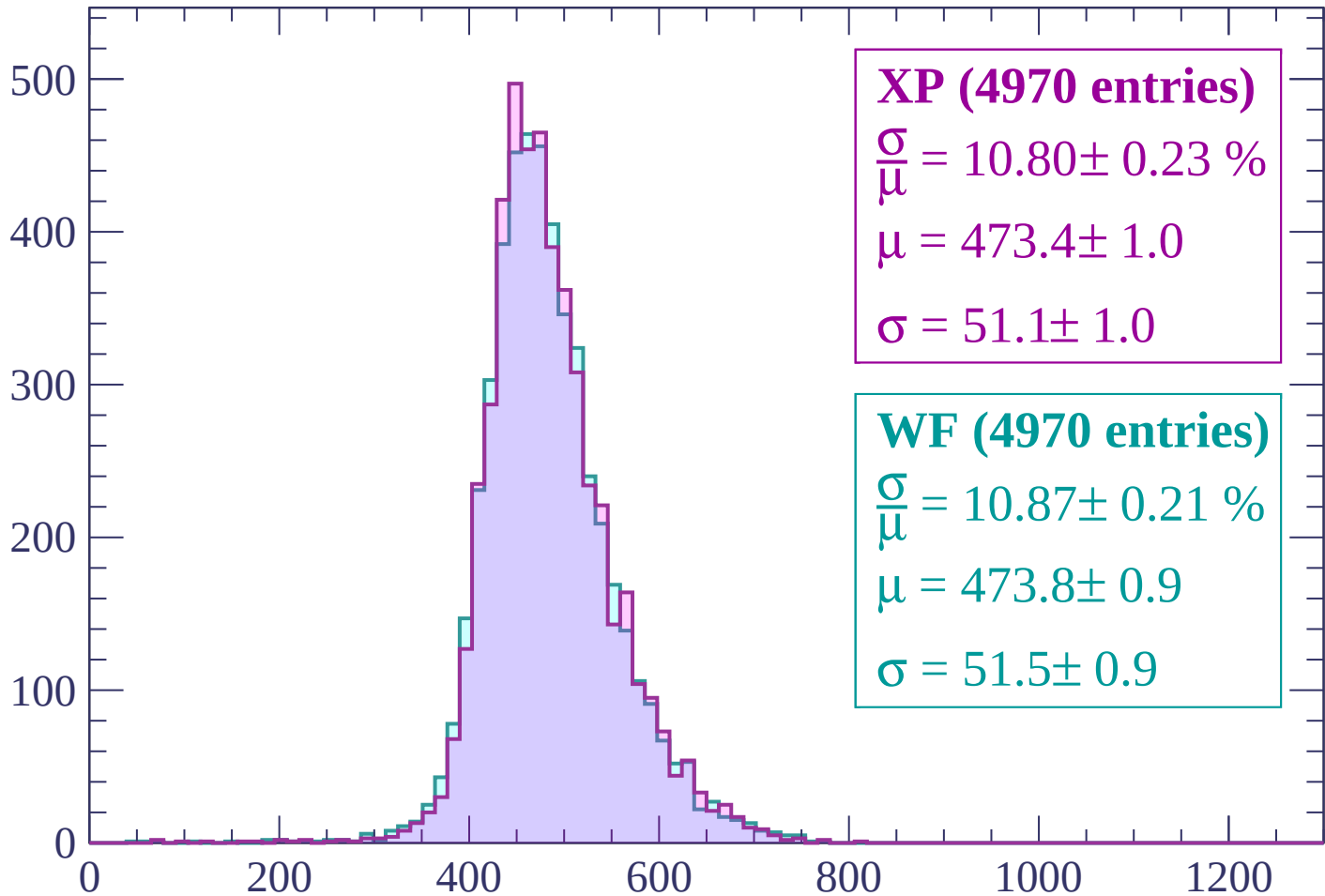
$\mu = 476.7 \pm 1.0$

$\sigma = 54.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count



XP (4970 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.23 \%$

$\mu = 473.4 \pm 1.0$

$\sigma = 51.1 \pm 1.0$

WF (4970 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.21 \%$

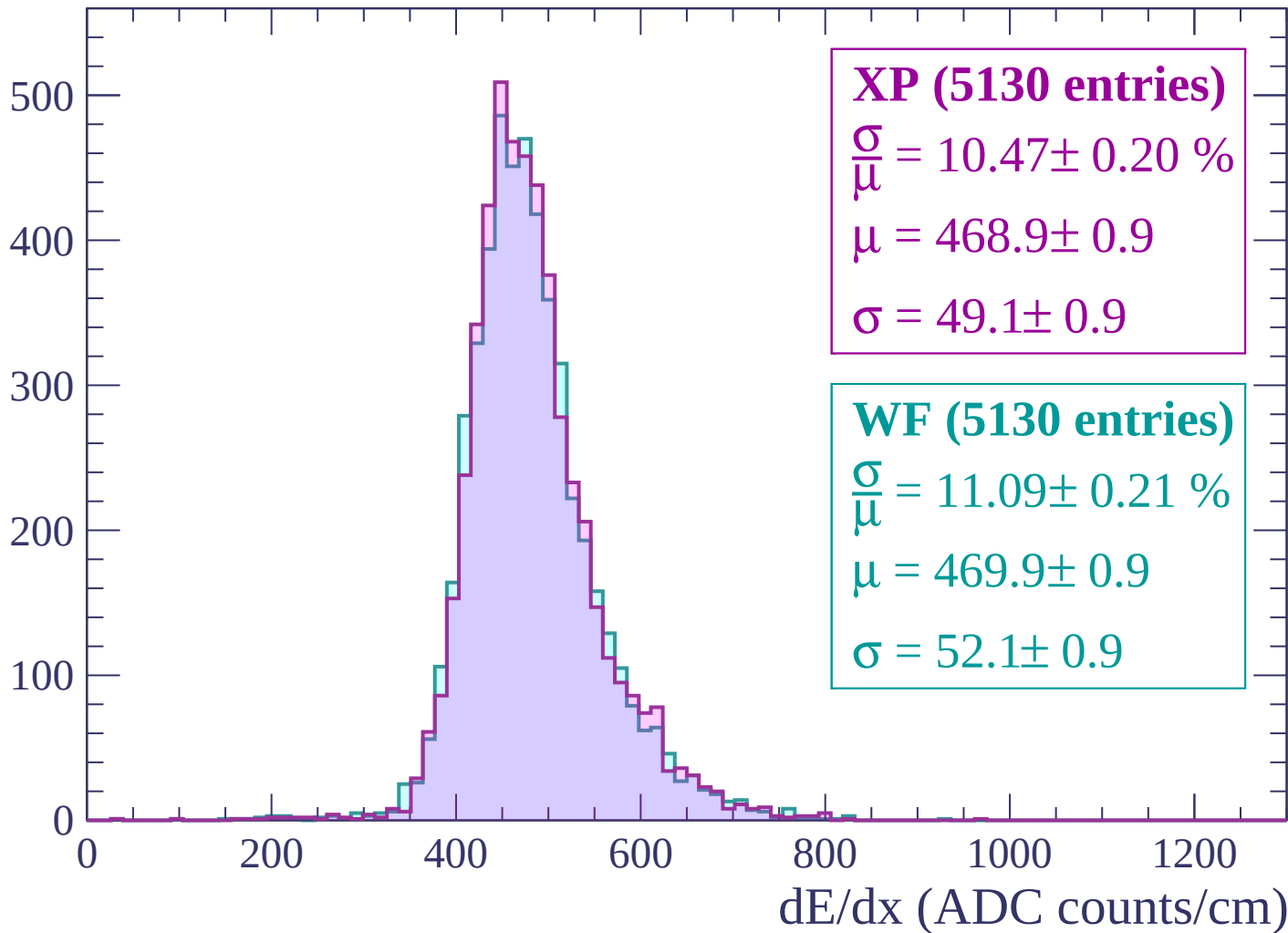
$\mu = 473.8 \pm 0.9$

$\sigma = 51.5 \pm 0.9$

dE/dx (ADC counts/cm)

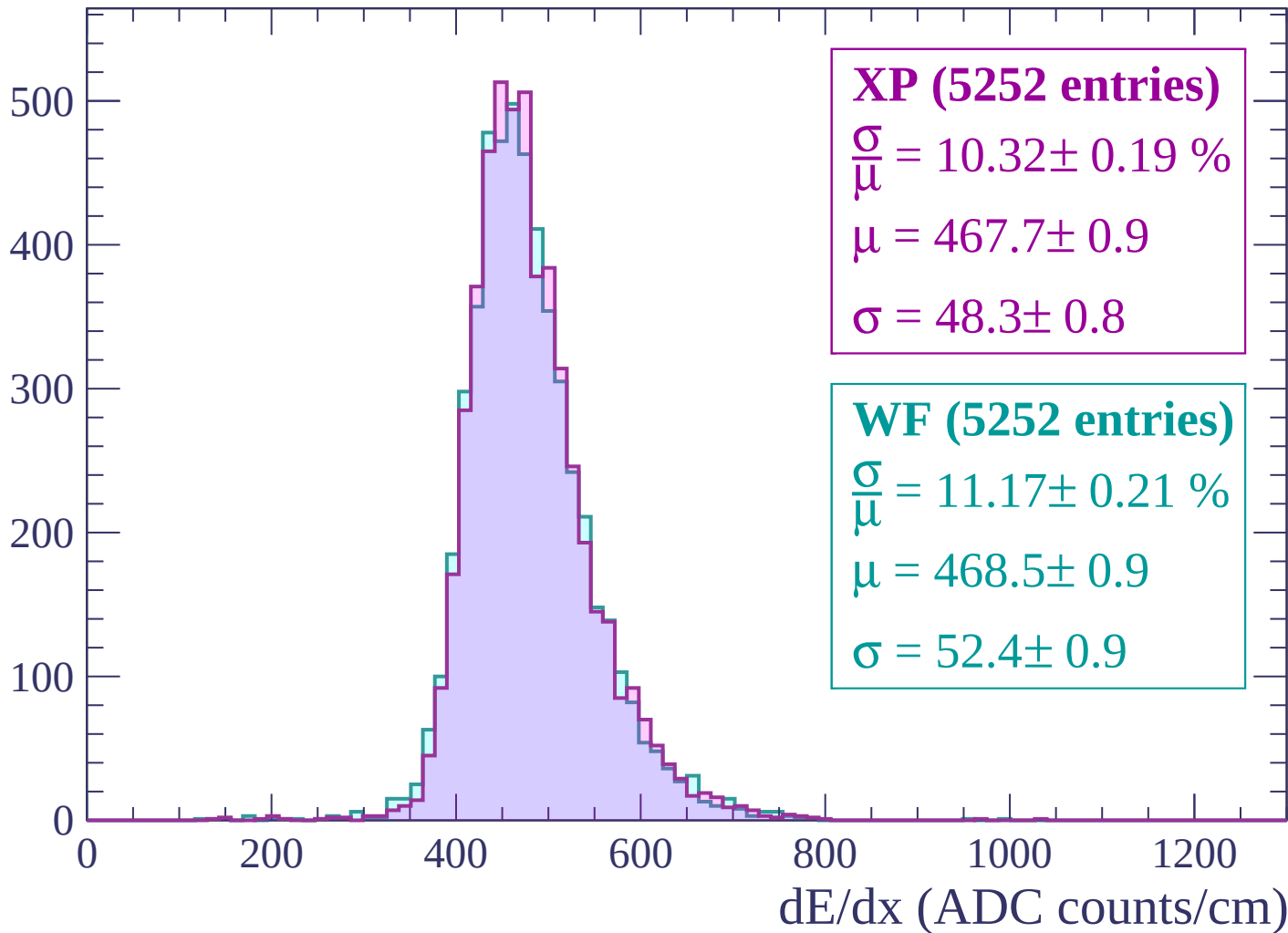
Energy loss | $-1560 < p < -1500$

Count



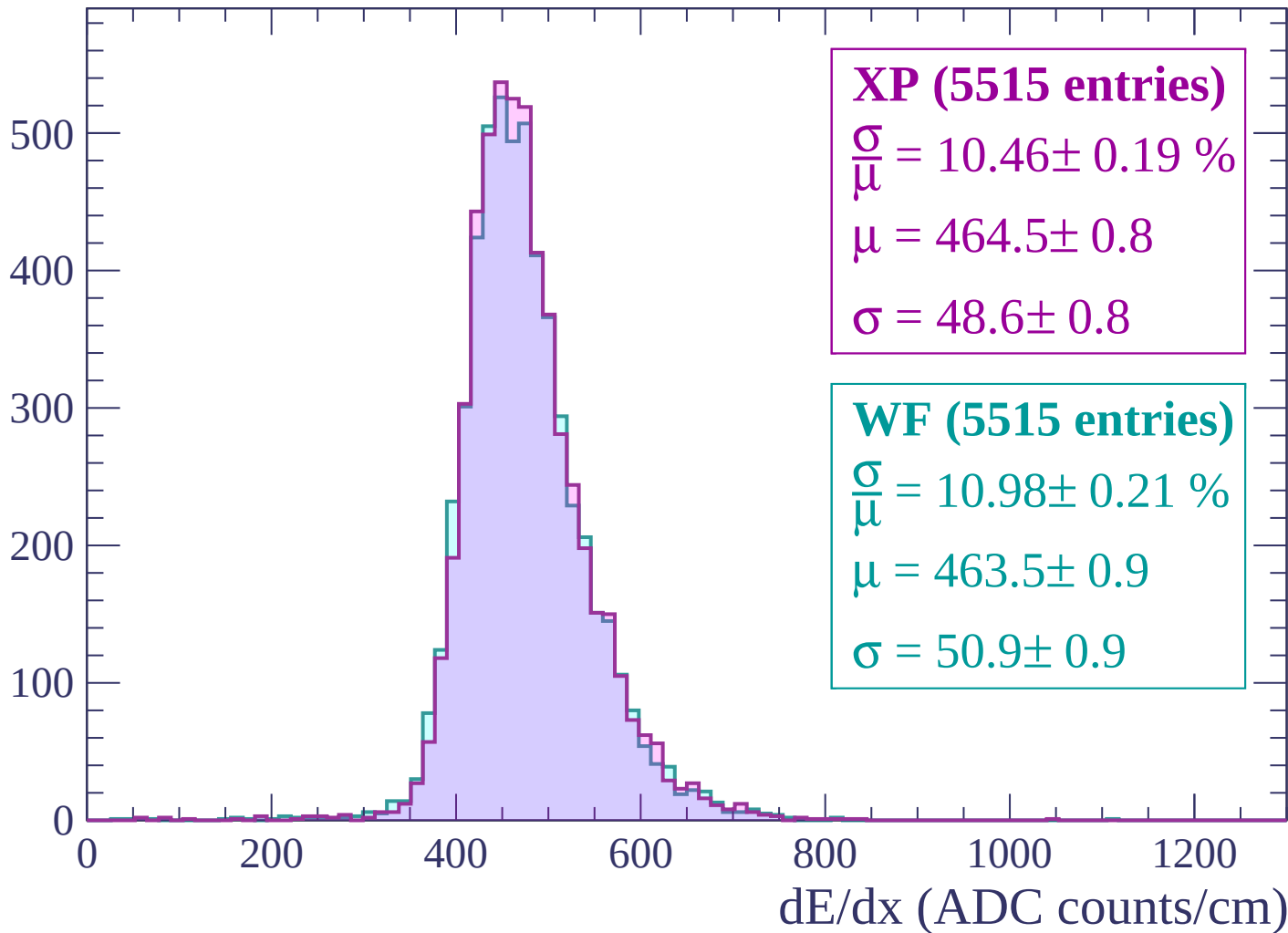
Energy loss | $-1500 < p < -1440$

Count

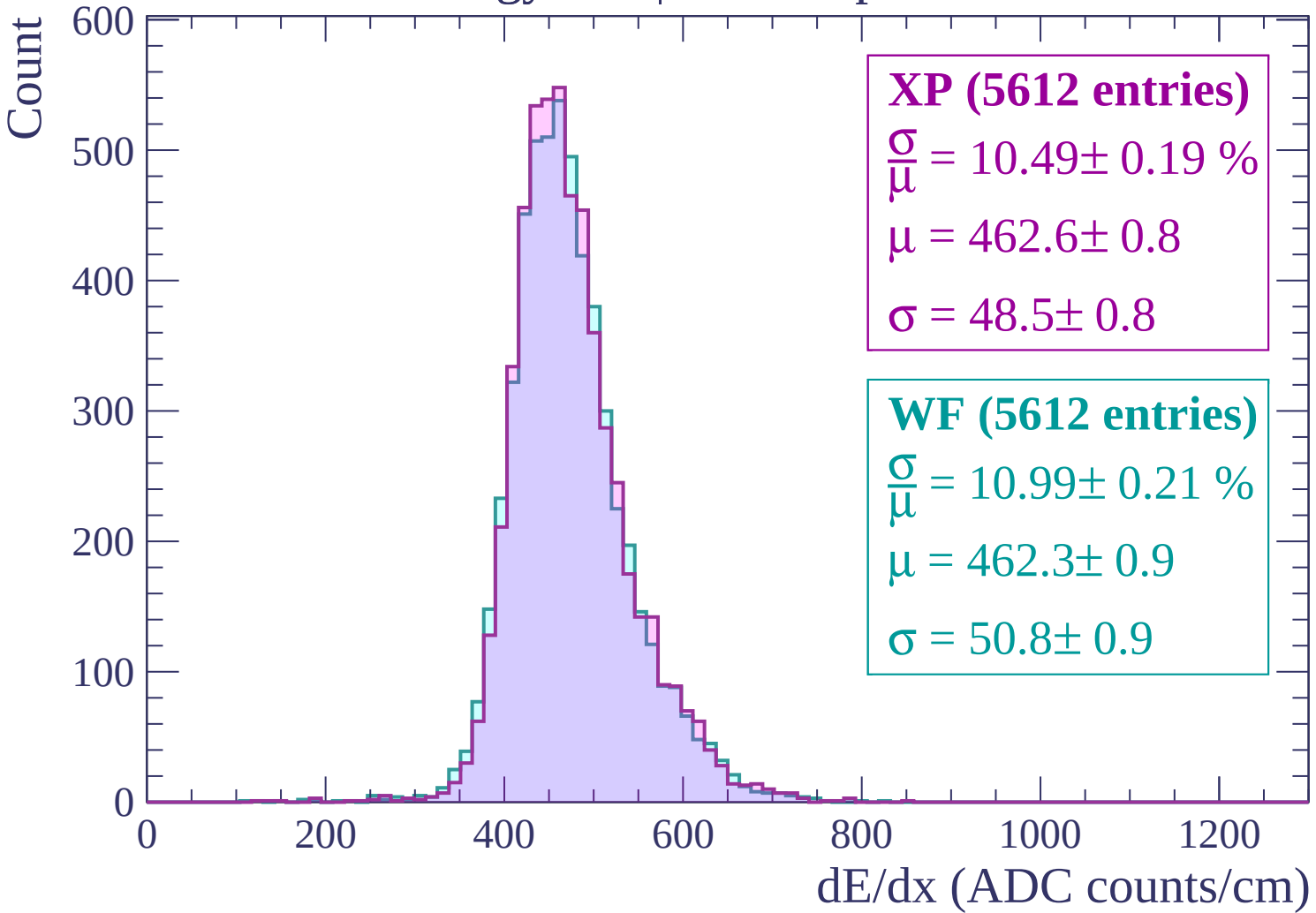


Energy loss | $-1440 < p < -1380$

Count

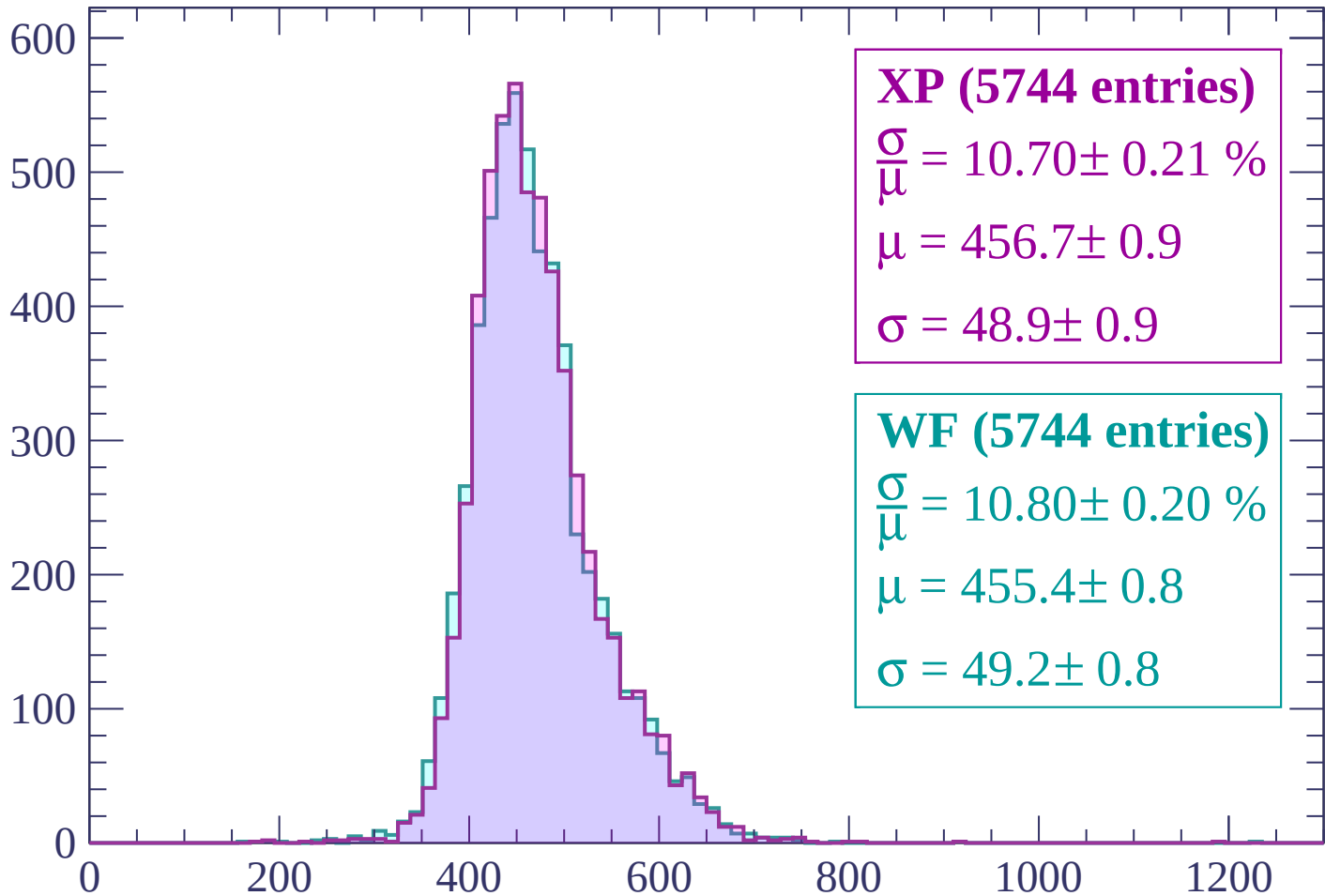


Energy loss | -1380 < p < -1320



Energy loss | $-1320 < p < -1260$

Count



XP (5744 entries)

$\frac{\sigma}{\mu} = 10.70 \pm 0.21 \%$

$\mu = 456.7 \pm 0.9$

$\sigma = 48.9 \pm 0.9$

WF (5744 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.20 \%$

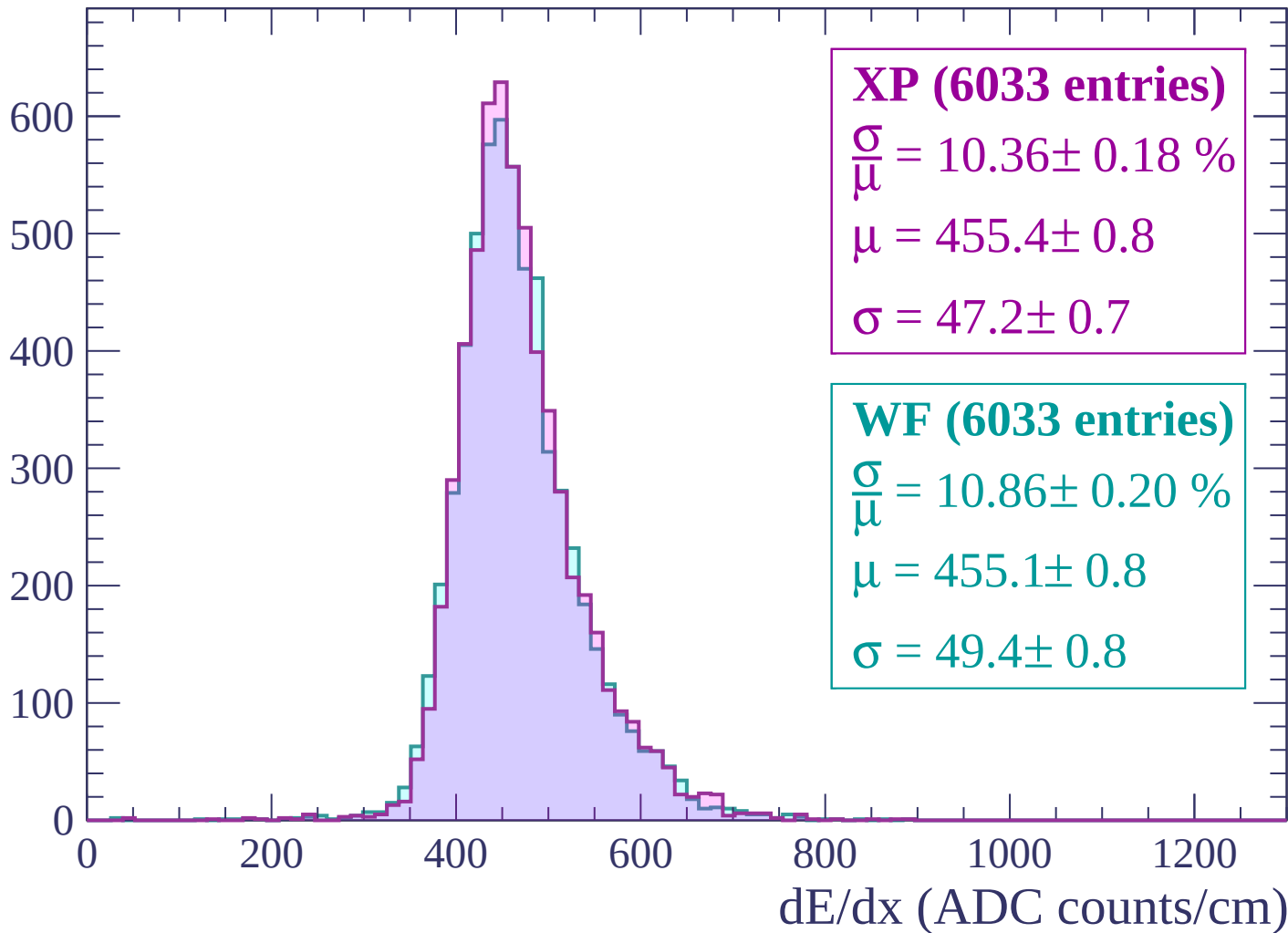
$\mu = 455.4 \pm 0.8$

$\sigma = 49.2 \pm 0.8$

dE/dx (ADC counts/cm)

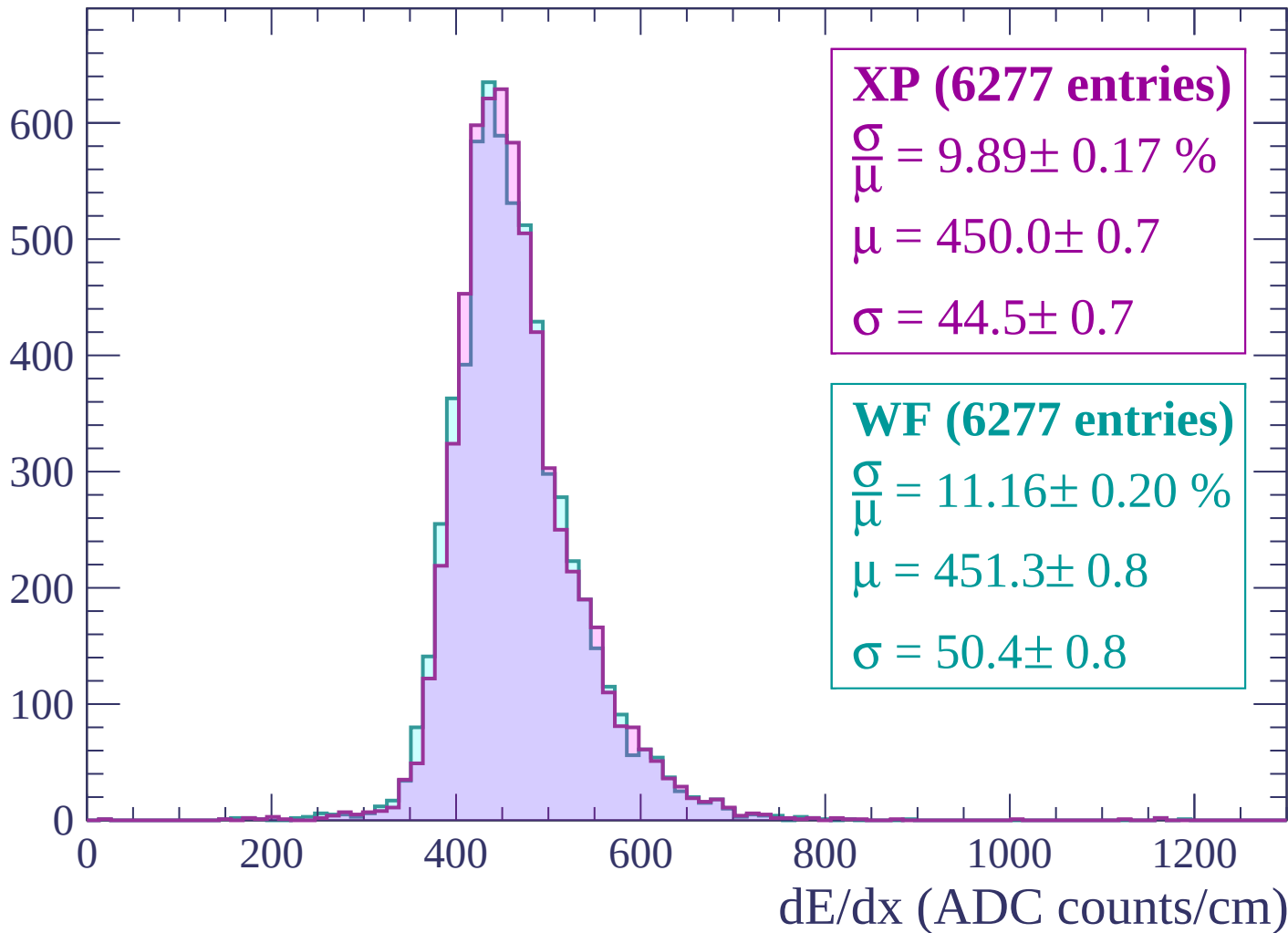
Energy loss | $-1260 < p < -1200$

Count



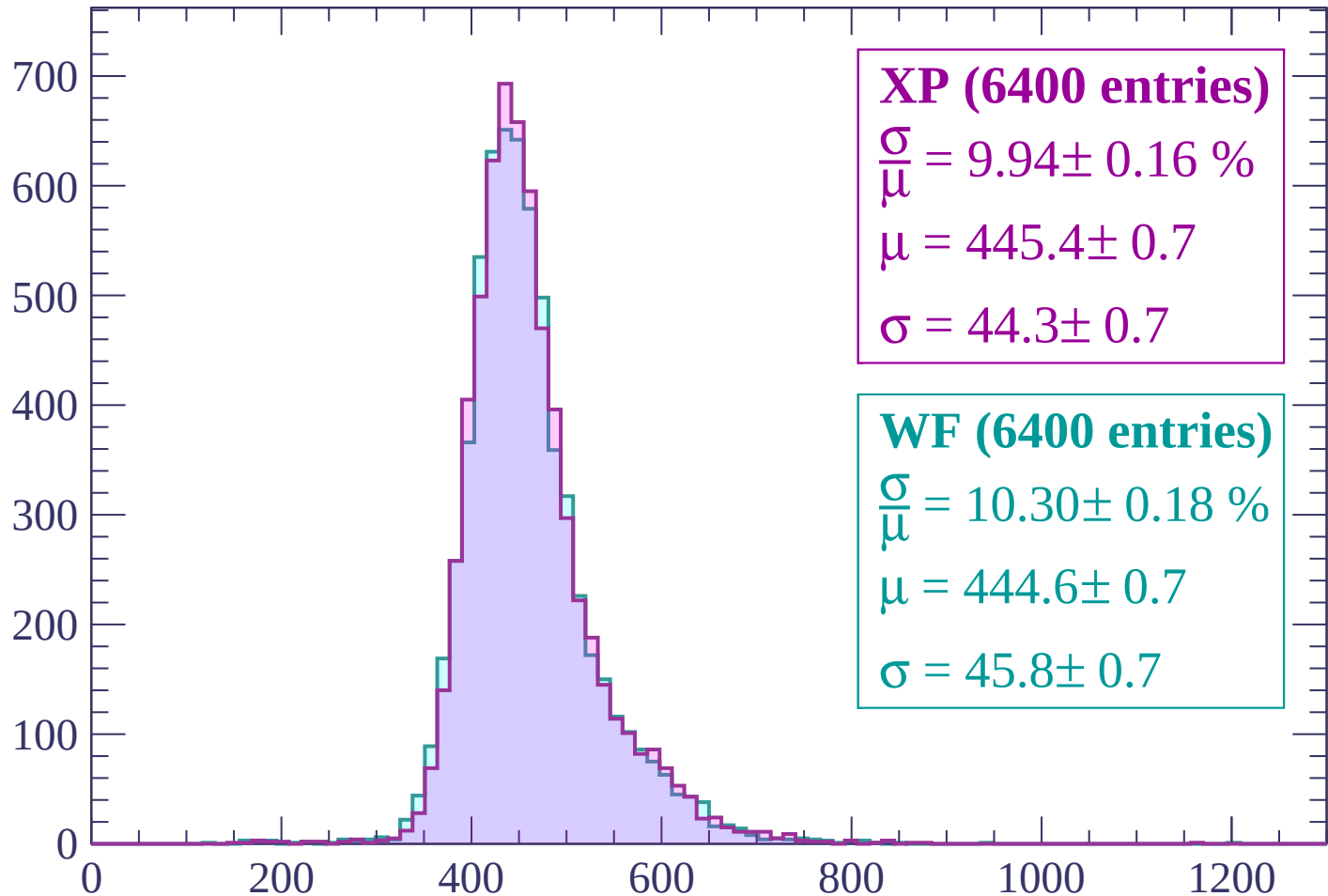
Energy loss | -1200 < p < -1140

Count



Energy loss | $-1140 < p < -1080$

Count



XP (6400 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.16 \%$$

$$\mu = 445.4 \pm 0.7$$

$$\sigma = 44.3 \pm 0.7$$

WF (6400 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.18 \%$$

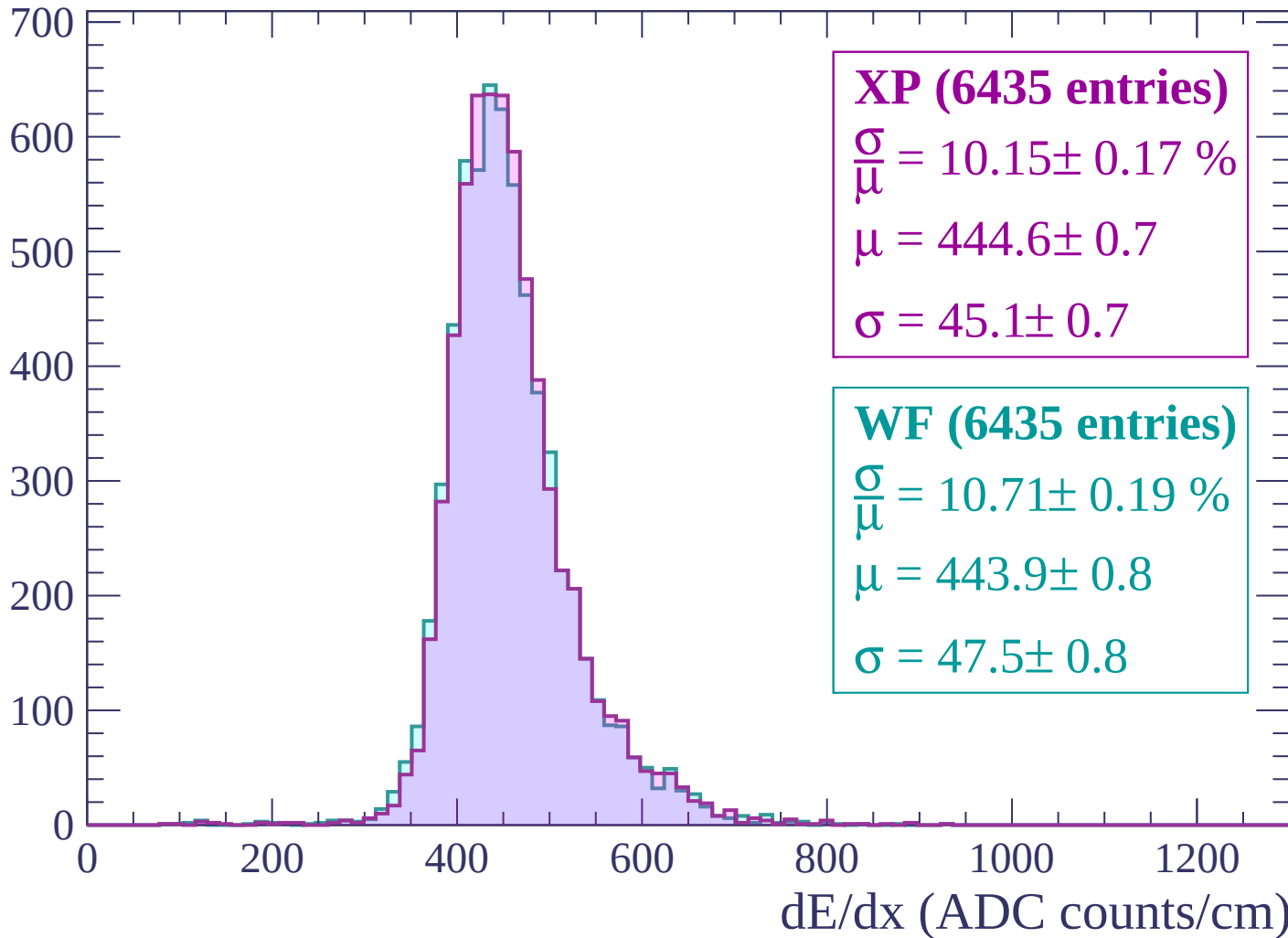
$$\mu = 444.6 \pm 0.7$$

$$\sigma = 45.8 \pm 0.7$$

dE/dx (ADC counts/cm)

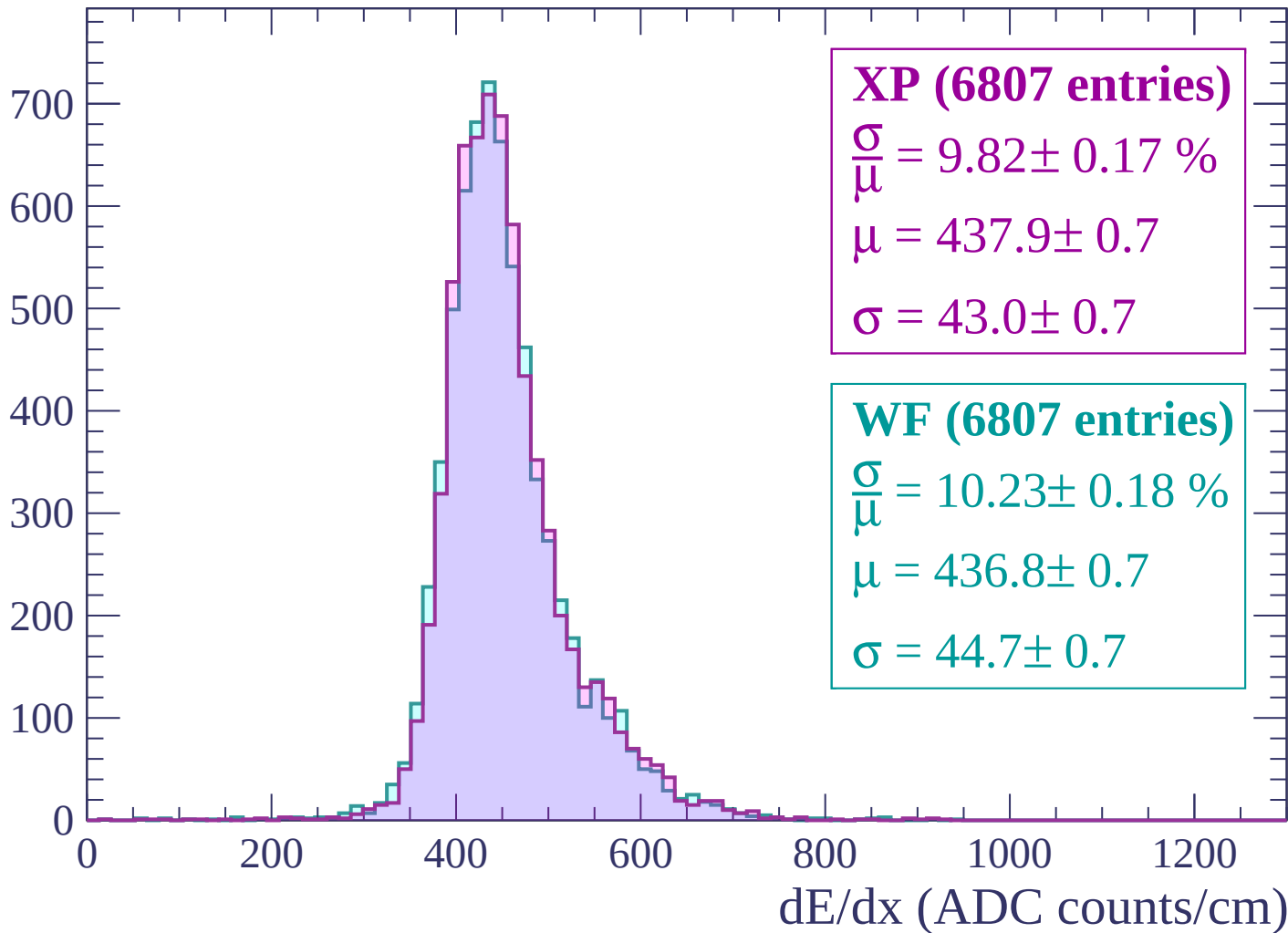
Energy loss | $-1080 < p < -1020$

Count



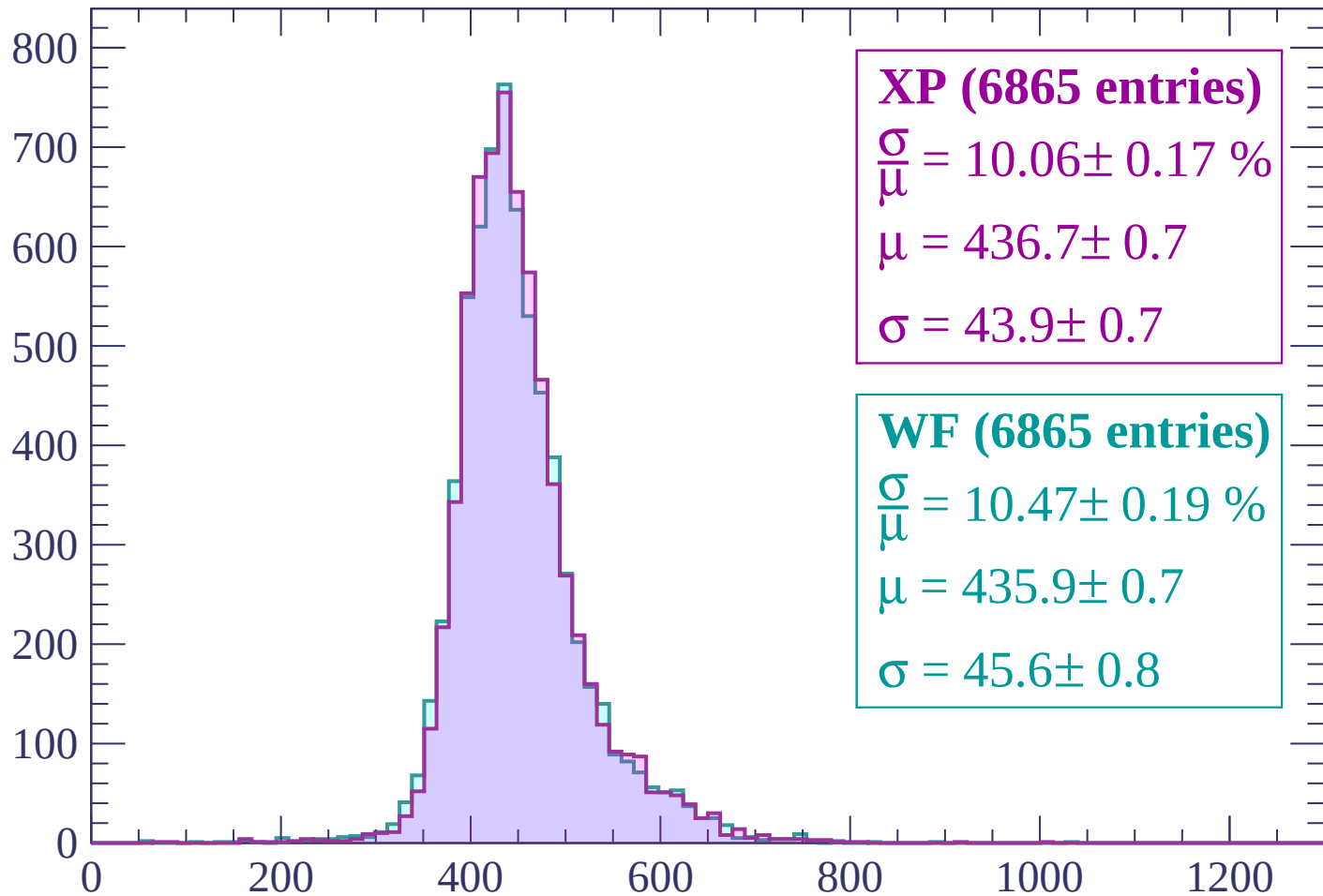
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (6865 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.17 \%$$

$$\mu = 436.7 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

WF (6865 entries)

$$\frac{\sigma}{\mu} = 10.47 \pm 0.19 \%$$

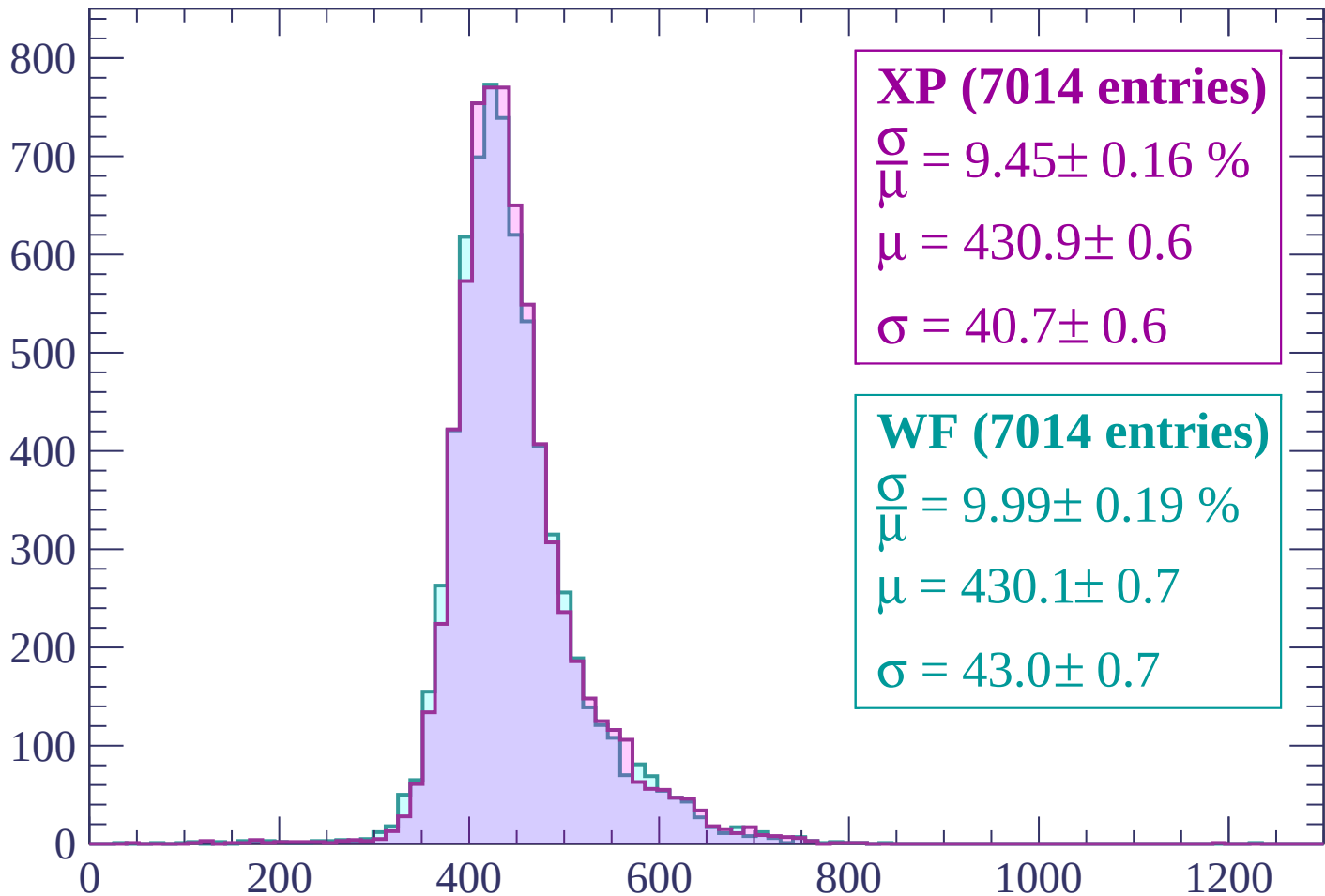
$$\mu = 435.9 \pm 0.7$$

$$\sigma = 45.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (7014 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.16 \%$$

$$\mu = 430.9 \pm 0.6$$

$$\sigma = 40.7 \pm 0.6$$

WF (7014 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.19 \%$$

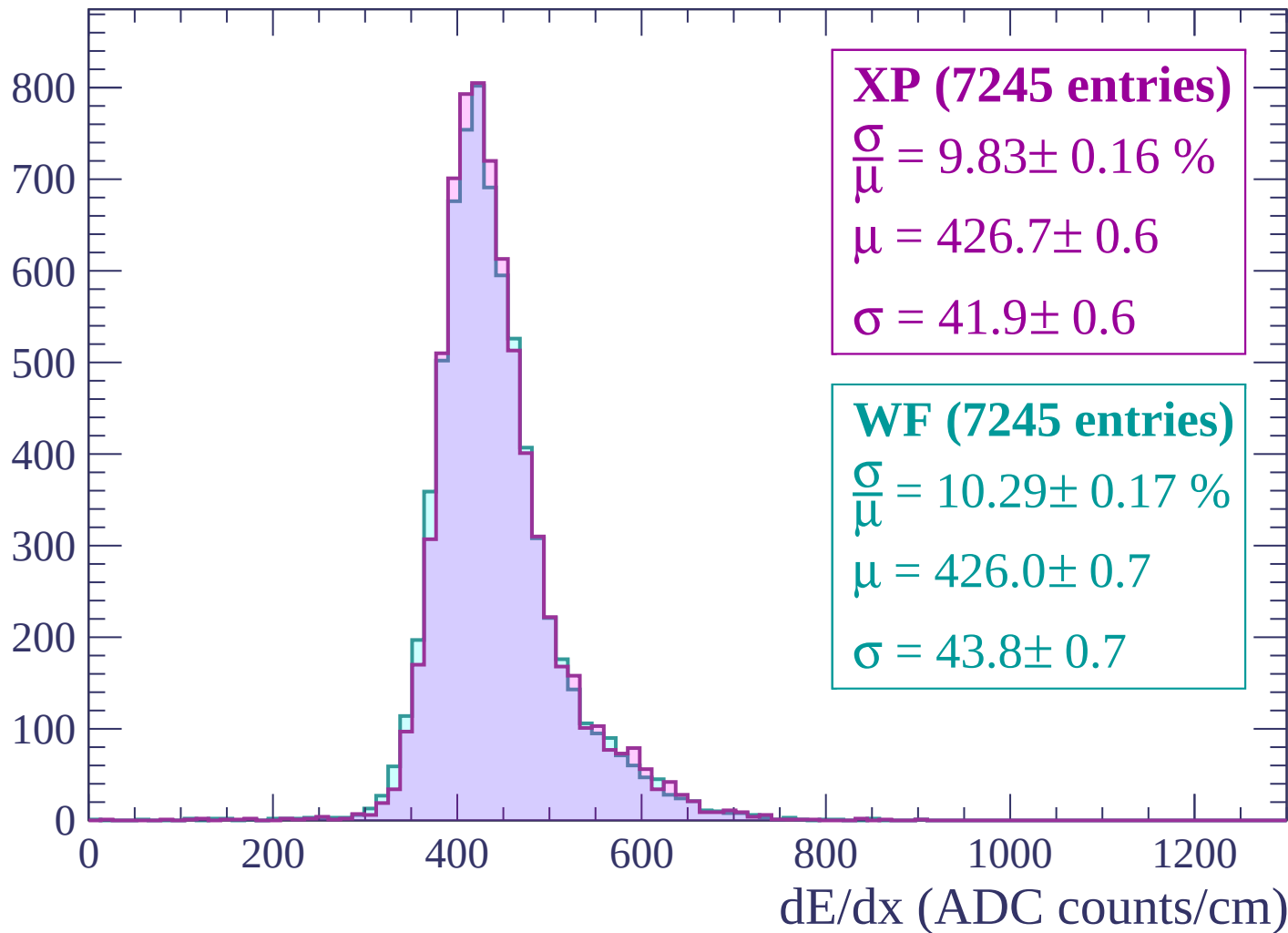
$$\mu = 430.1 \pm 0.7$$

$$\sigma = 43.0 \pm 0.7$$

dE/dx (ADC counts/cm)

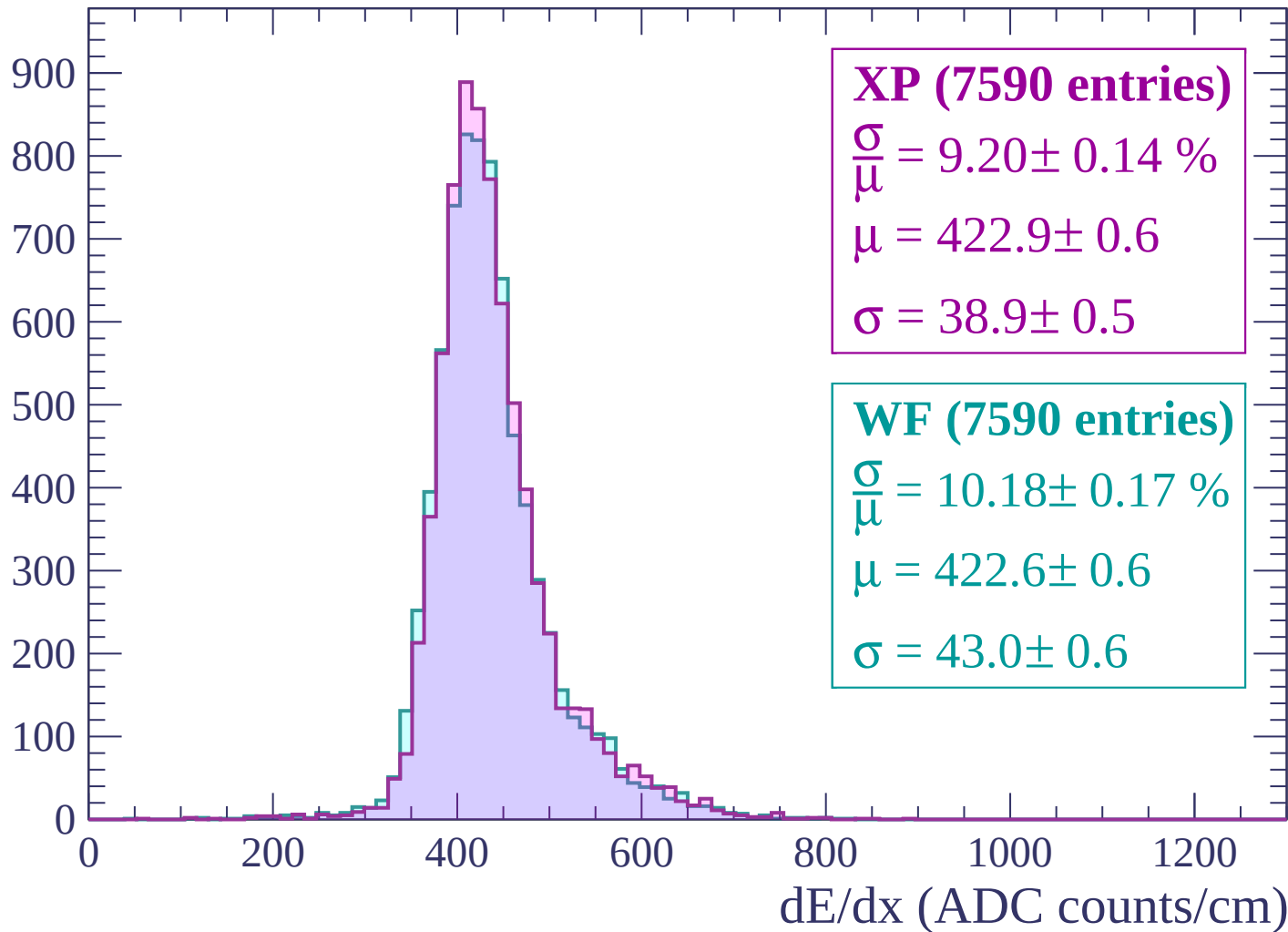
Energy loss | $-840 < p < -780$

Count



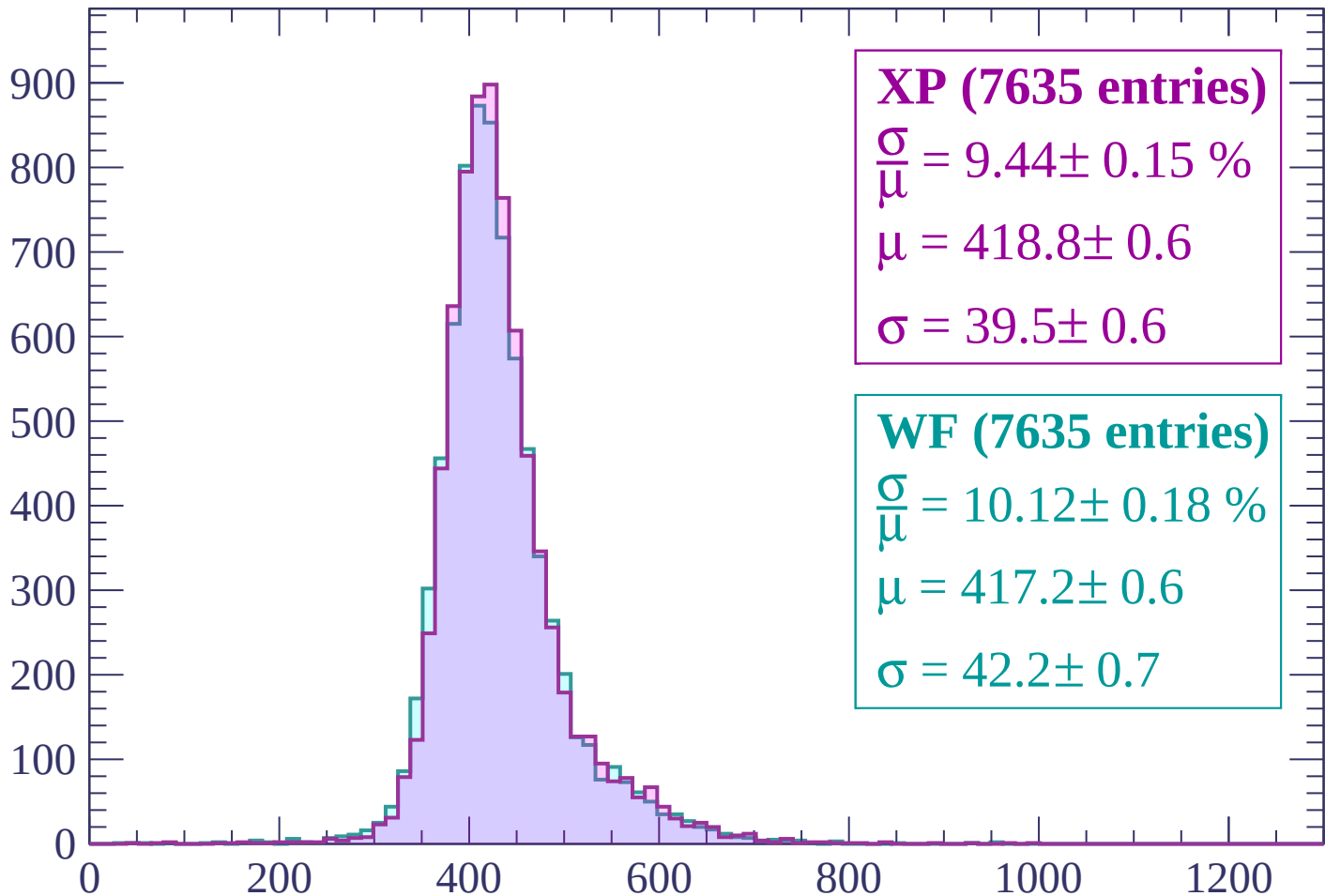
Energy loss | $-780 < p < -720$

Count



Energy loss | $-720 < p < -660$

Count



XP (7635 entries)

$$\frac{\sigma}{\mu} = 9.44 \pm 0.15 \%$$

$$\mu = 418.8 \pm 0.6$$

$$\sigma = 39.5 \pm 0.6$$

WF (7635 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.18 \%$$

$$\mu = 417.2 \pm 0.6$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7994 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.15 \%$

$\mu = 413.1 \pm 0.5$

$\sigma = 37.3 \pm 0.6$

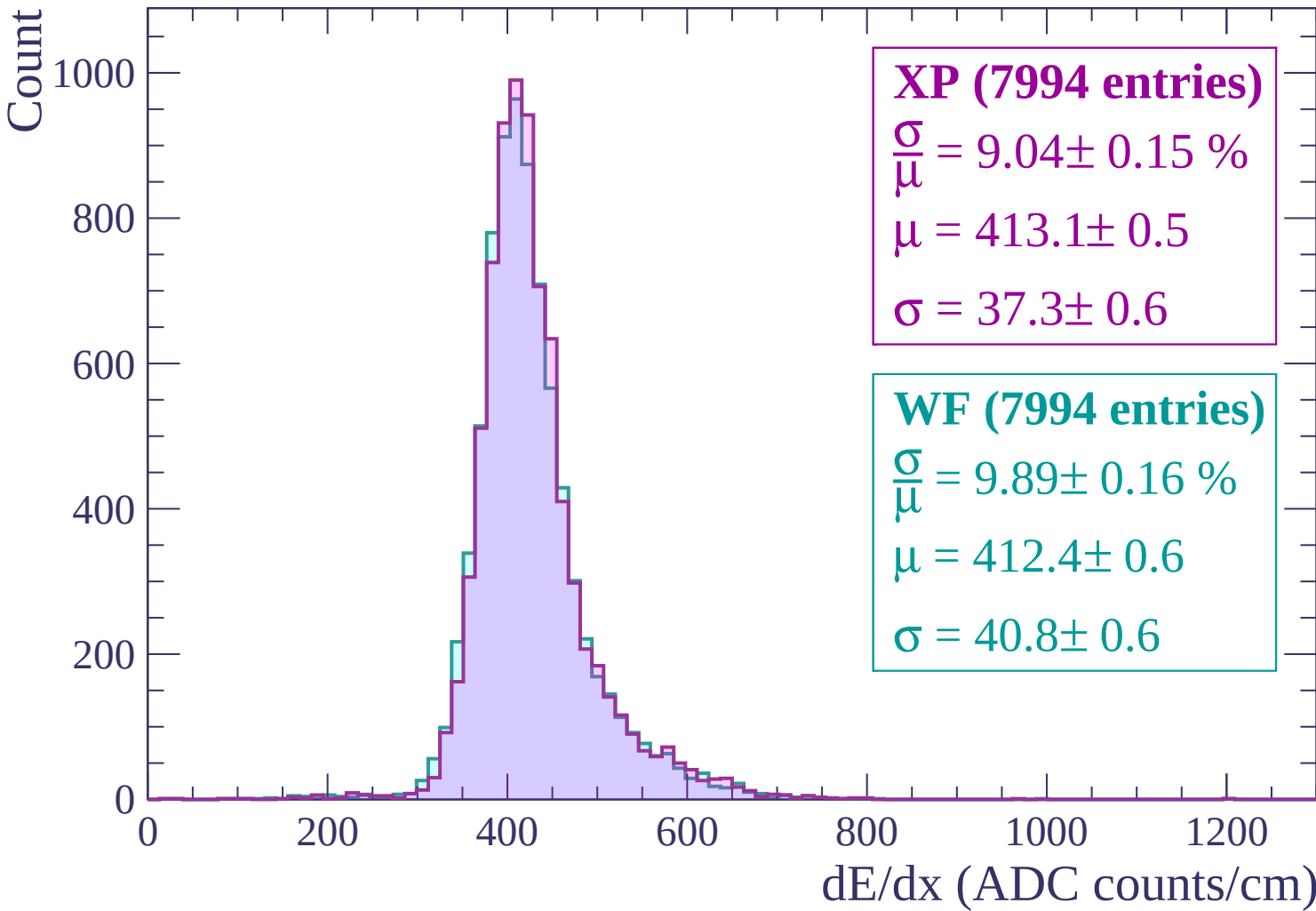
WF (7994 entries)

$\frac{\sigma}{\mu} = 9.89 \pm 0.16 \%$

$\mu = 412.4 \pm 0.6$

$\sigma = 40.8 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $-600 < p < -540$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8001 entries)

$\frac{\sigma}{\mu} = 9.14 \pm 0.15 \%$

$\mu = 409.4 \pm 0.5$

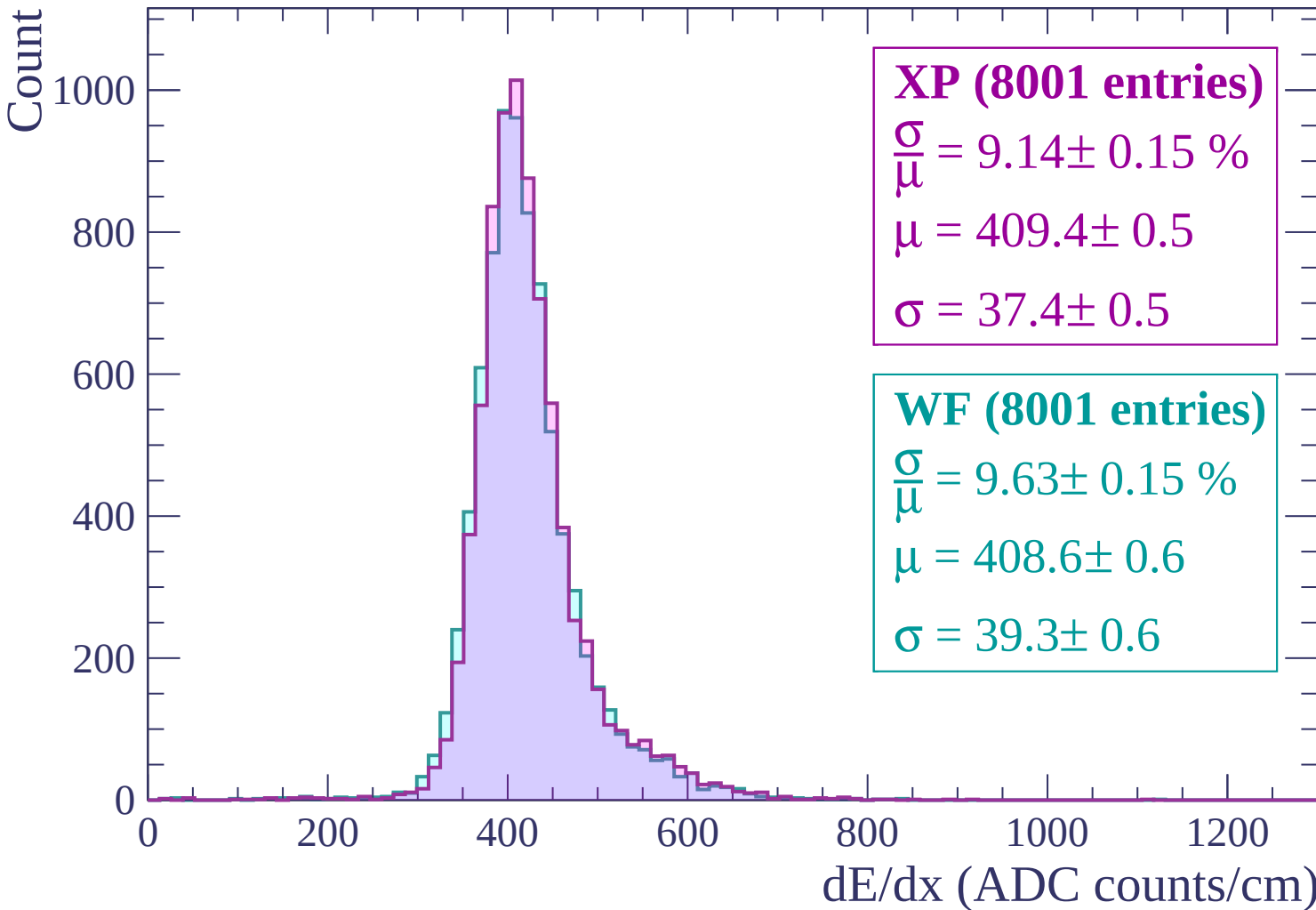
$\sigma = 37.4 \pm 0.5$

WF (8001 entries)

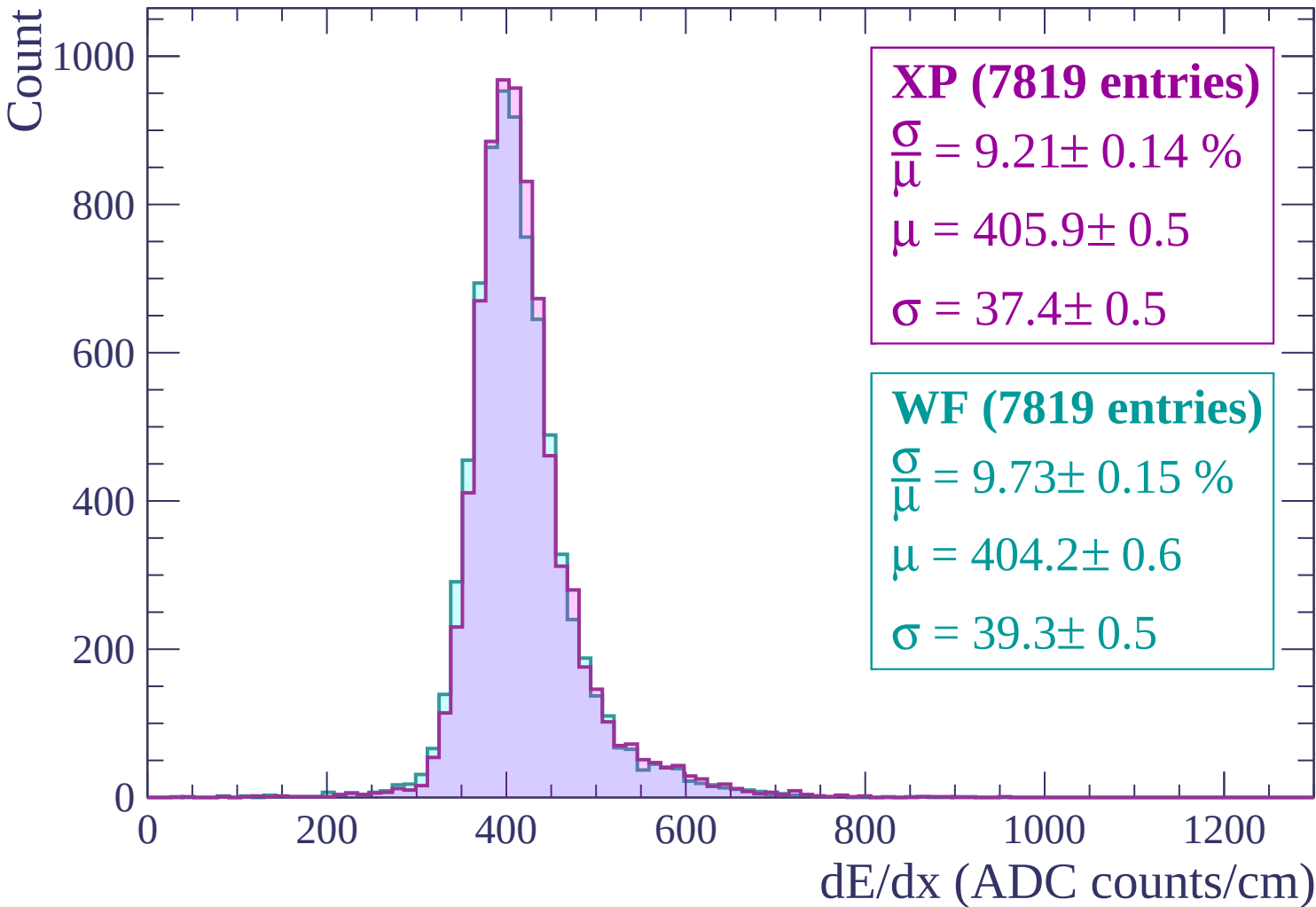
$\frac{\sigma}{\mu} = 9.63 \pm 0.15 \%$

$\mu = 408.6 \pm 0.6$

$\sigma = 39.3 \pm 0.6$



Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8208 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.13 \%$$

$$\mu = 400.9 \pm 0.5$$

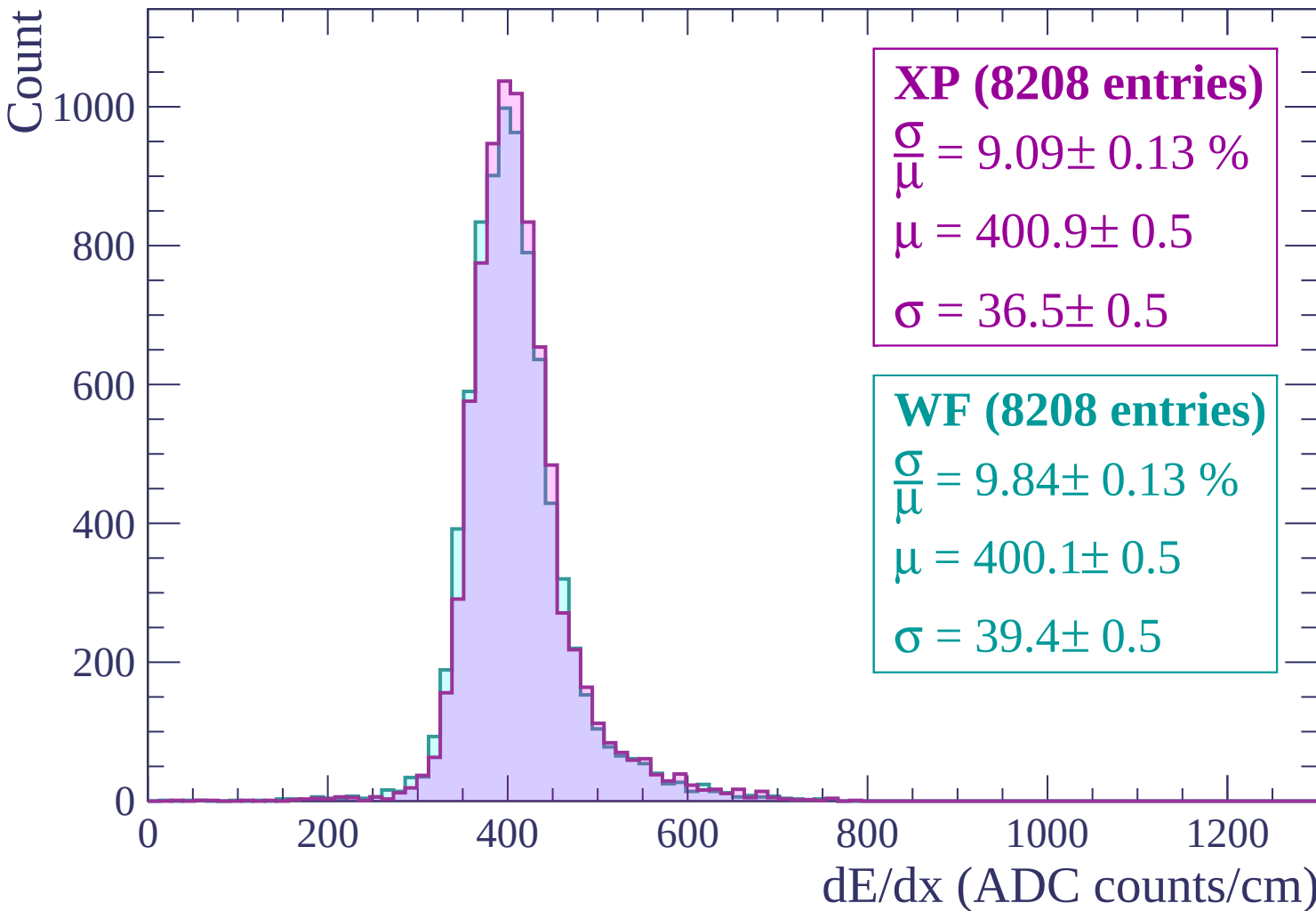
$$\sigma = 36.5 \pm 0.5$$

WF (8208 entries)

$$\frac{\sigma}{\mu} = 9.84 \pm 0.13 \%$$

$$\mu = 400.1 \pm 0.5$$

$$\sigma = 39.4 \pm 0.5$$



Energy loss | $-420 < p < -360$

Count

1000

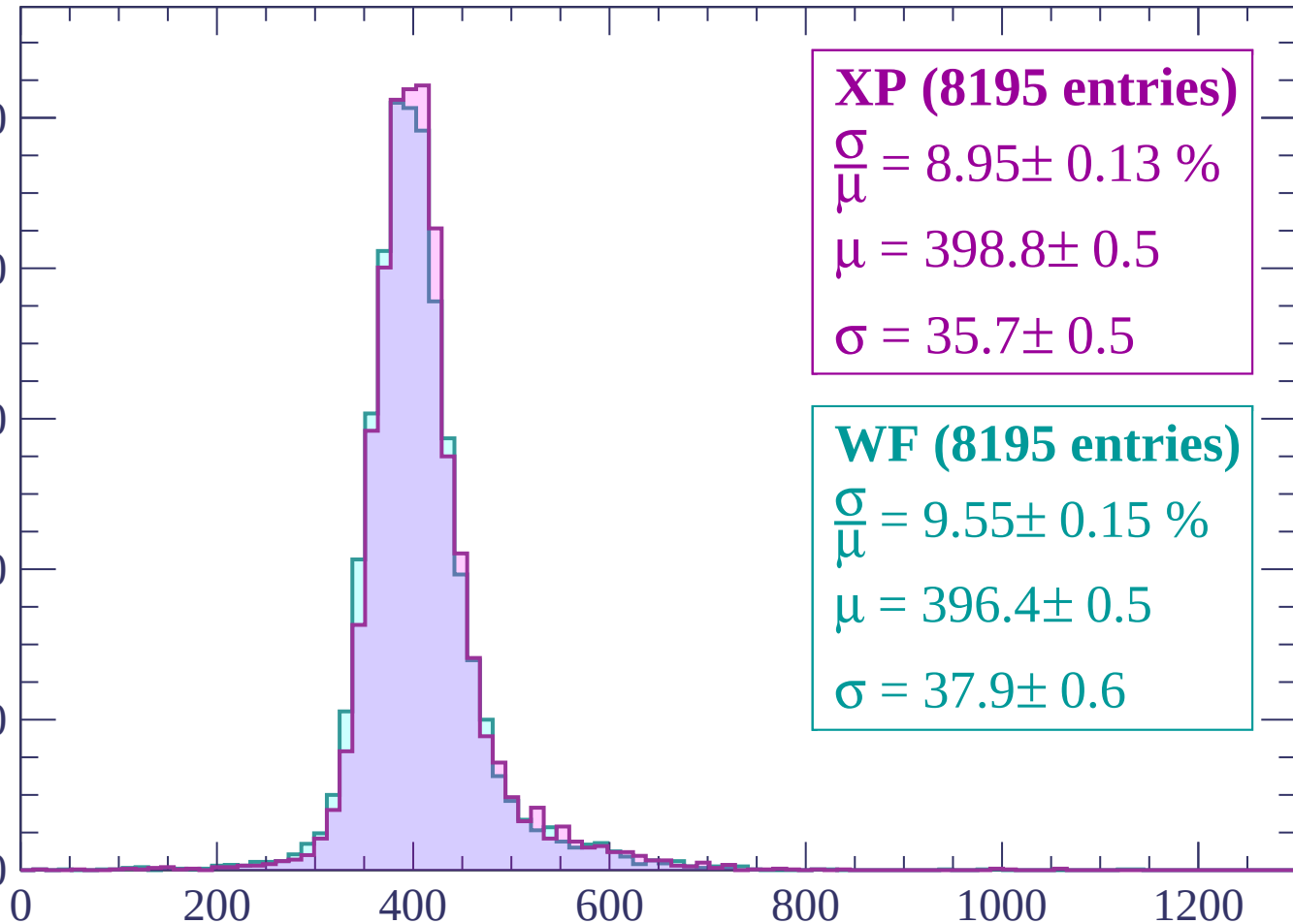
800

600

400

200

0



XP (8195 entries)

$\frac{\sigma}{\mu} = 8.95 \pm 0.13 \%$

$\mu = 398.8 \pm 0.5$

$\sigma = 35.7 \pm 0.5$

WF (8195 entries)

$\frac{\sigma}{\mu} = 9.55 \pm 0.15 \%$

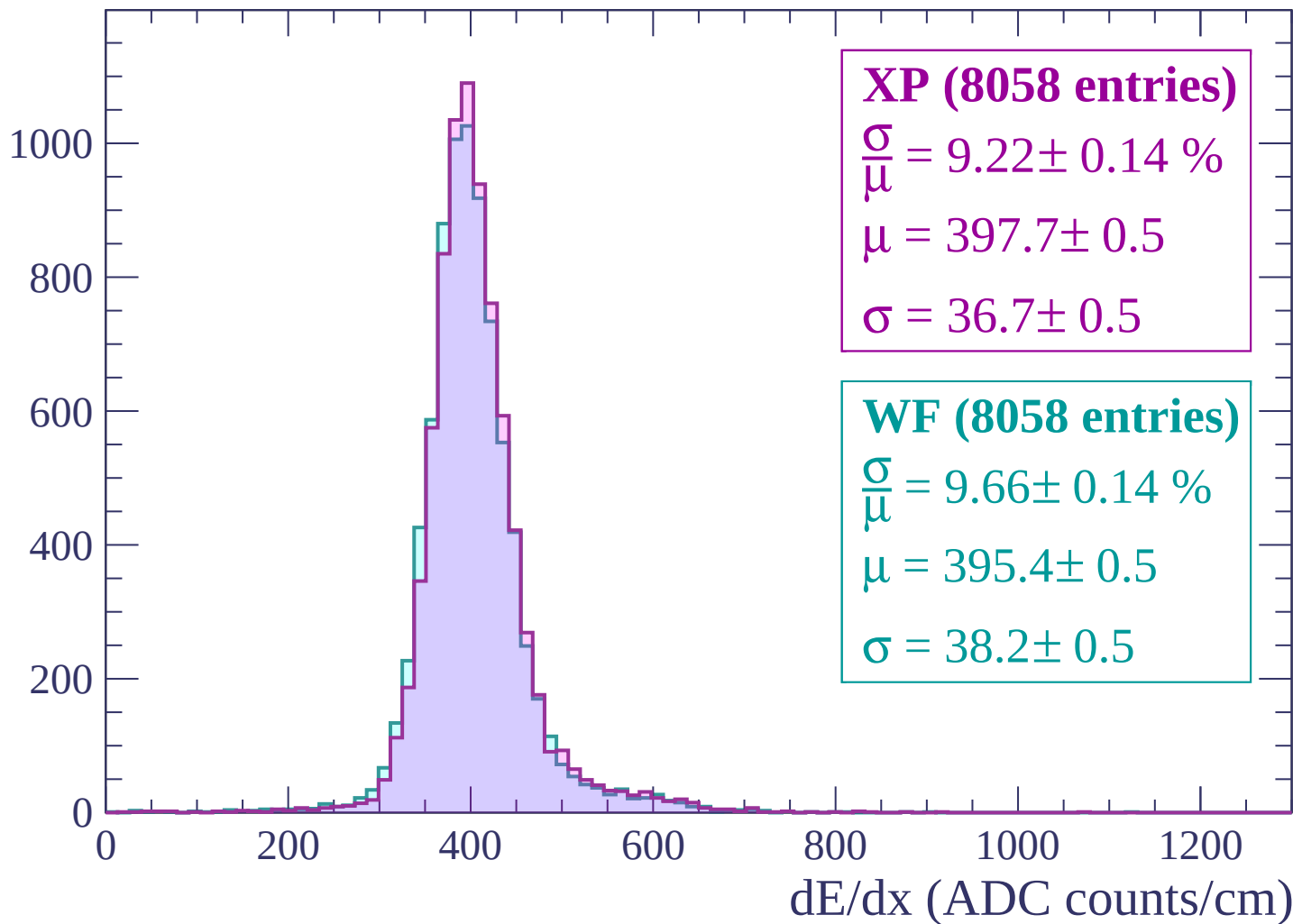
$\mu = 396.4 \pm 0.5$

$\sigma = 37.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count

1000

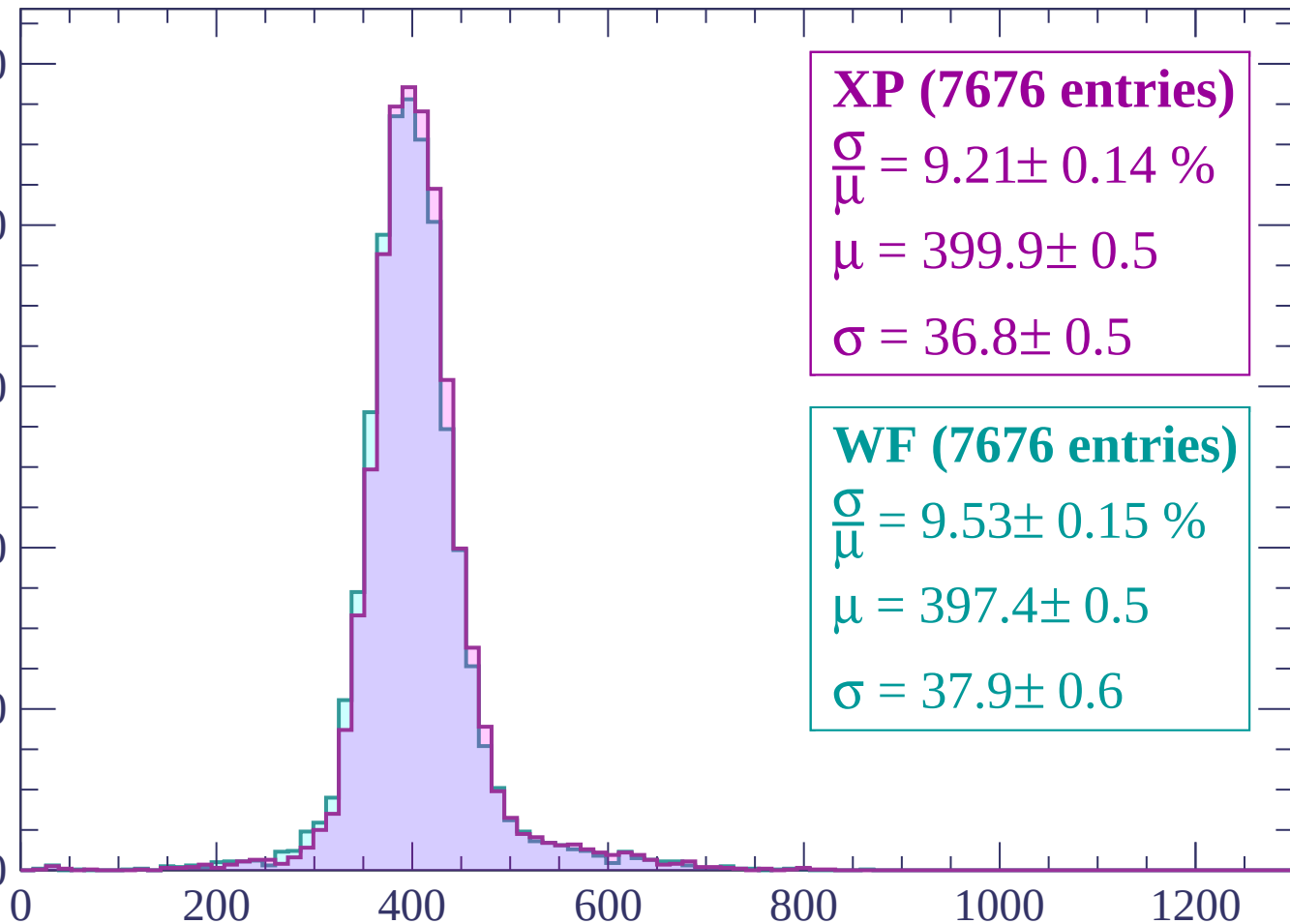
800

600

400

200

0



XP (7676 entries)

$\frac{\sigma}{\mu} = 9.21 \pm 0.14 \%$

$\mu = 399.9 \pm 0.5$

$\sigma = 36.8 \pm 0.5$

WF (7676 entries)

$\frac{\sigma}{\mu} = 9.53 \pm 0.15 \%$

$\mu = 397.4 \pm 0.5$

$\sigma = 37.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count

1000

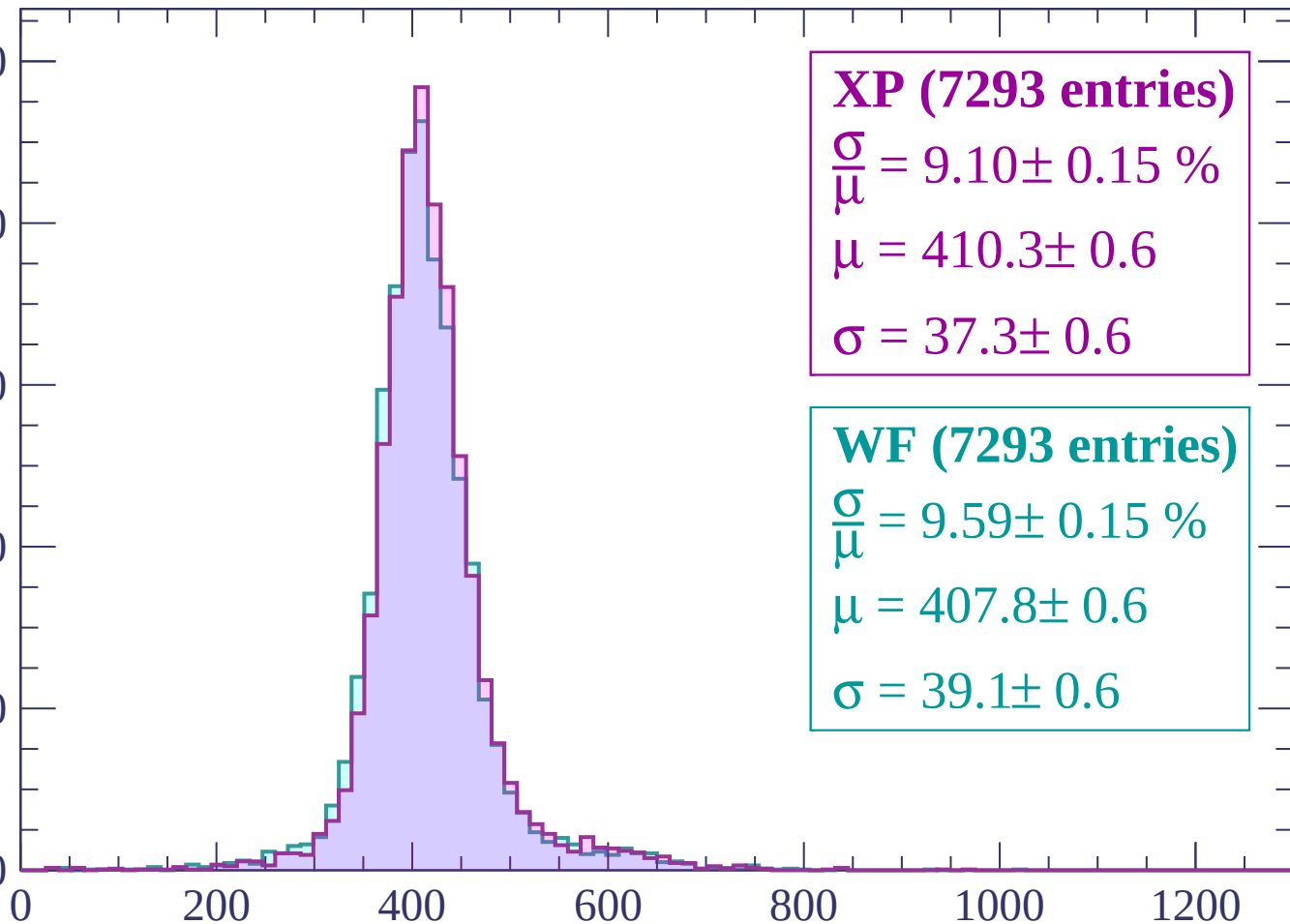
800

600

400

200

0



XP (7293 entries)

$\frac{\sigma}{\mu} = 9.10 \pm 0.15 \%$

$\mu = 410.3 \pm 0.6$

$\sigma = 37.3 \pm 0.6$

WF (7293 entries)

$\frac{\sigma}{\mu} = 9.59 \pm 0.15 \%$

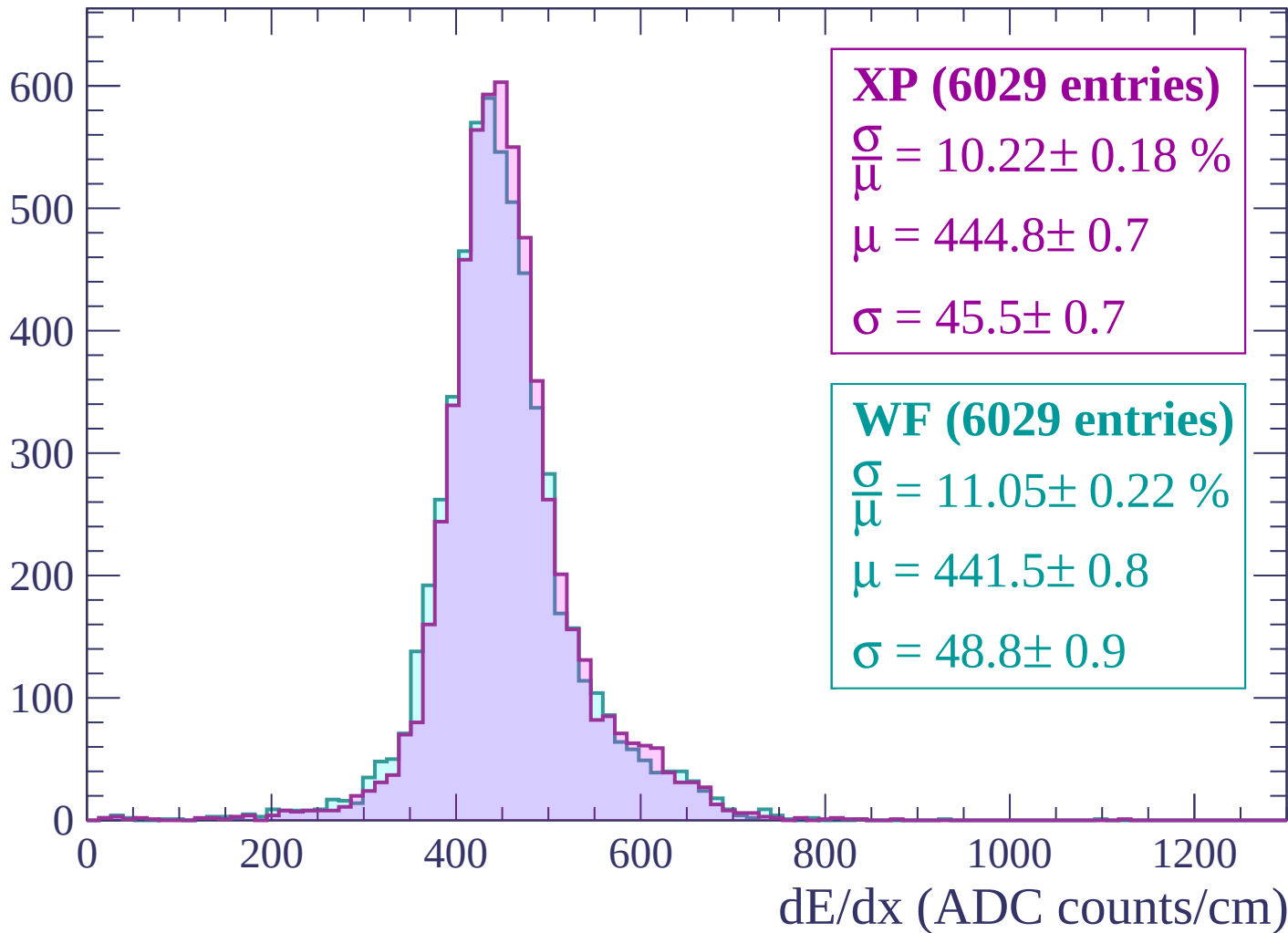
$\mu = 407.8 \pm 0.6$

$\sigma = 39.1 \pm 0.6$

dE/dx (ADC counts/cm)

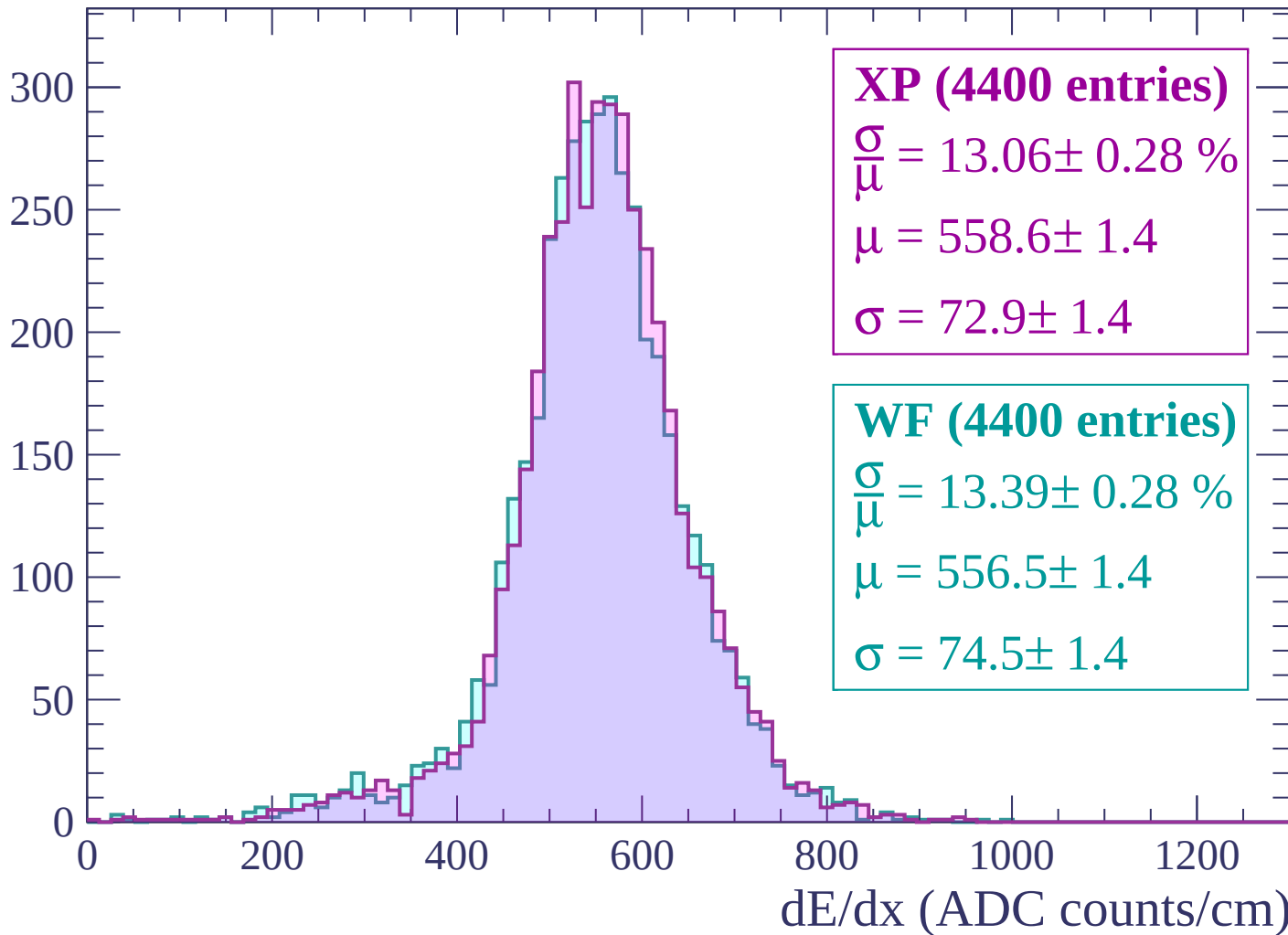
Energy loss | -180 < p < -120

Count



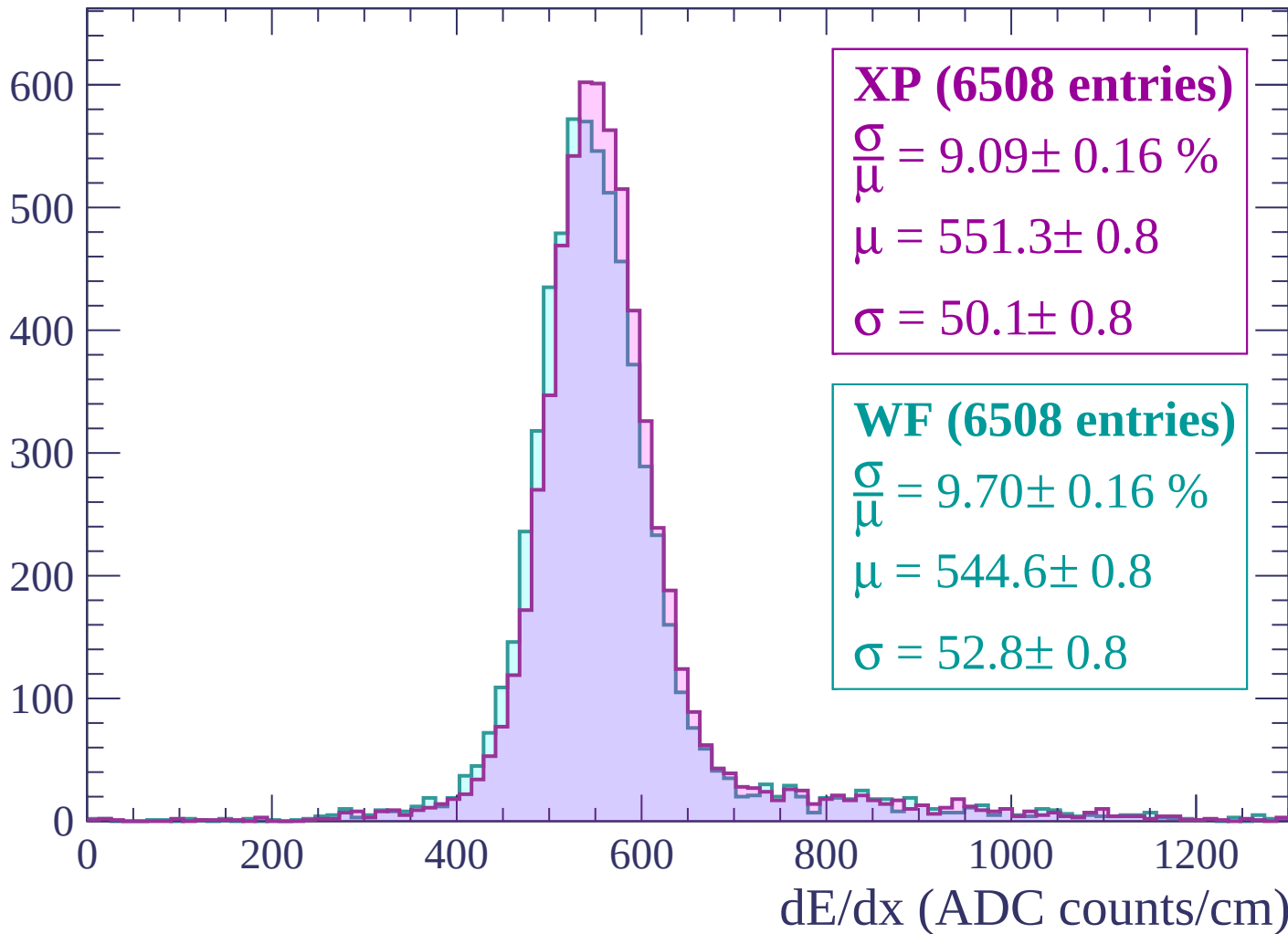
Energy loss | $-120 < p < -60$

Count



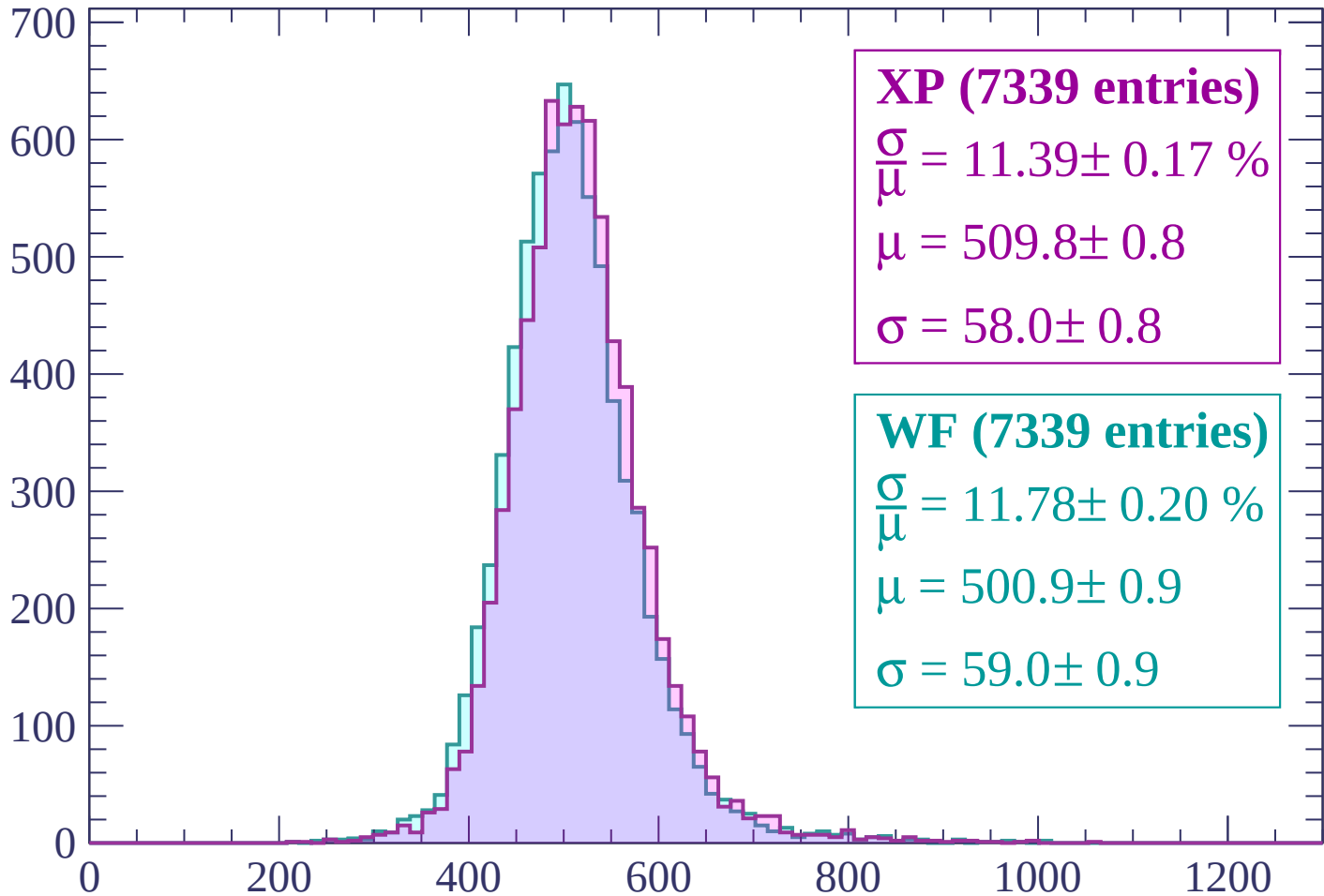
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP (7339 entries)

$\frac{\sigma}{\mu} = 11.39 \pm 0.17 \%$

$\mu = 509.8 \pm 0.8$

$\sigma = 58.0 \pm 0.8$

WF (7339 entries)

$\frac{\sigma}{\mu} = 11.78 \pm 0.20 \%$

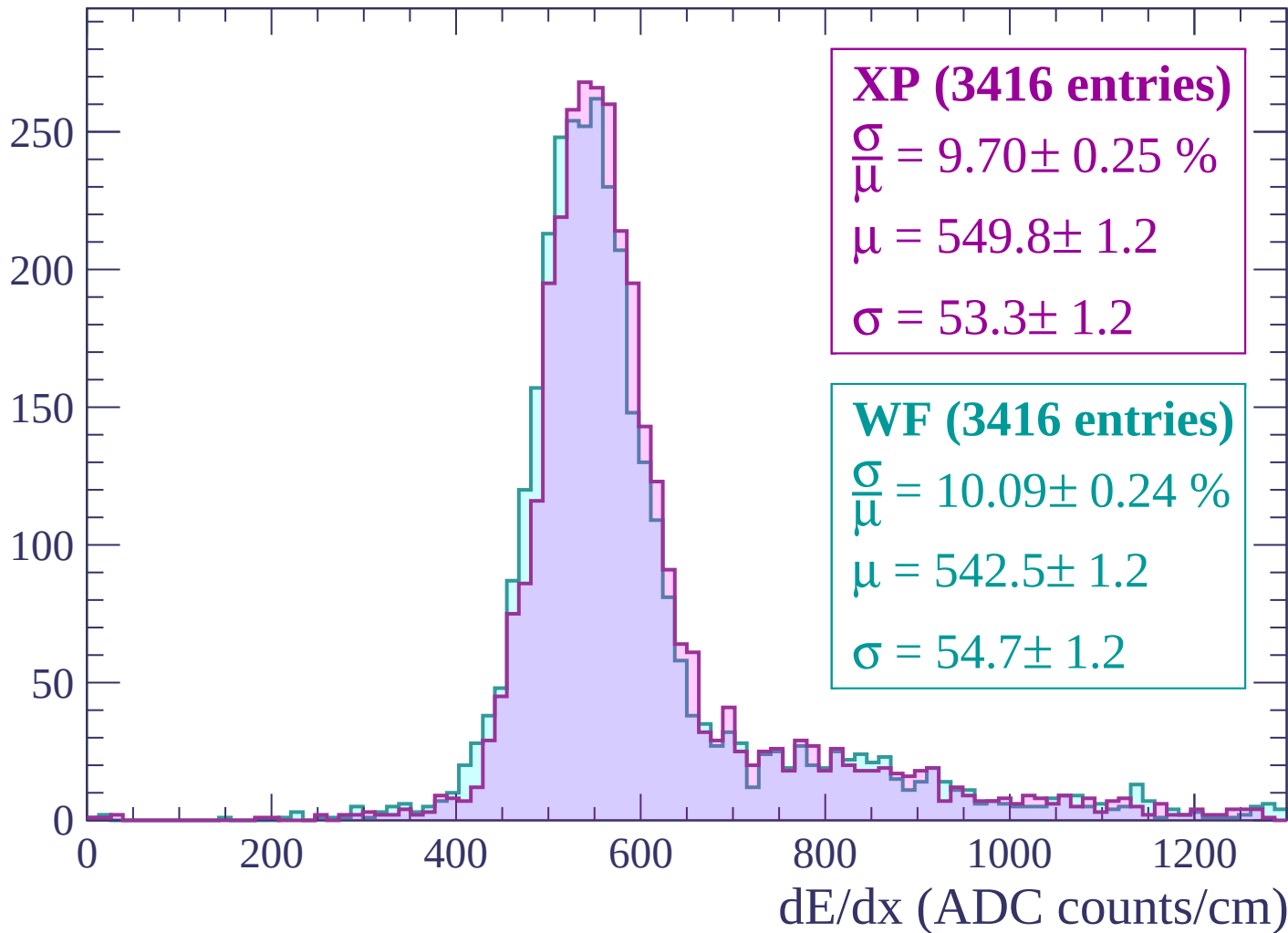
$\mu = 500.9 \pm 0.9$

$\sigma = 59.0 \pm 0.9$

dE/dx (ADC counts/cm)

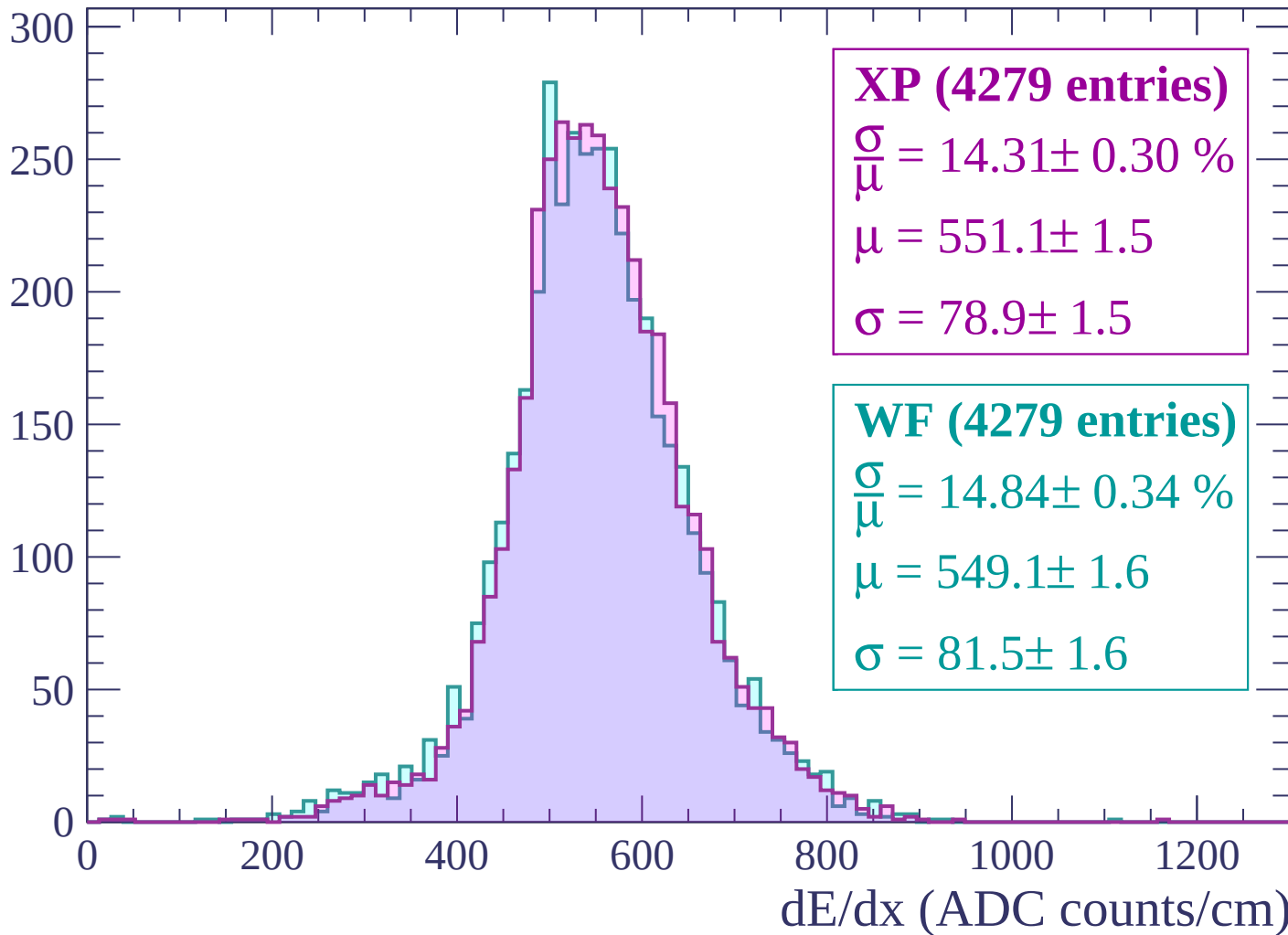
Energy loss | $60 < p < 120$

Count



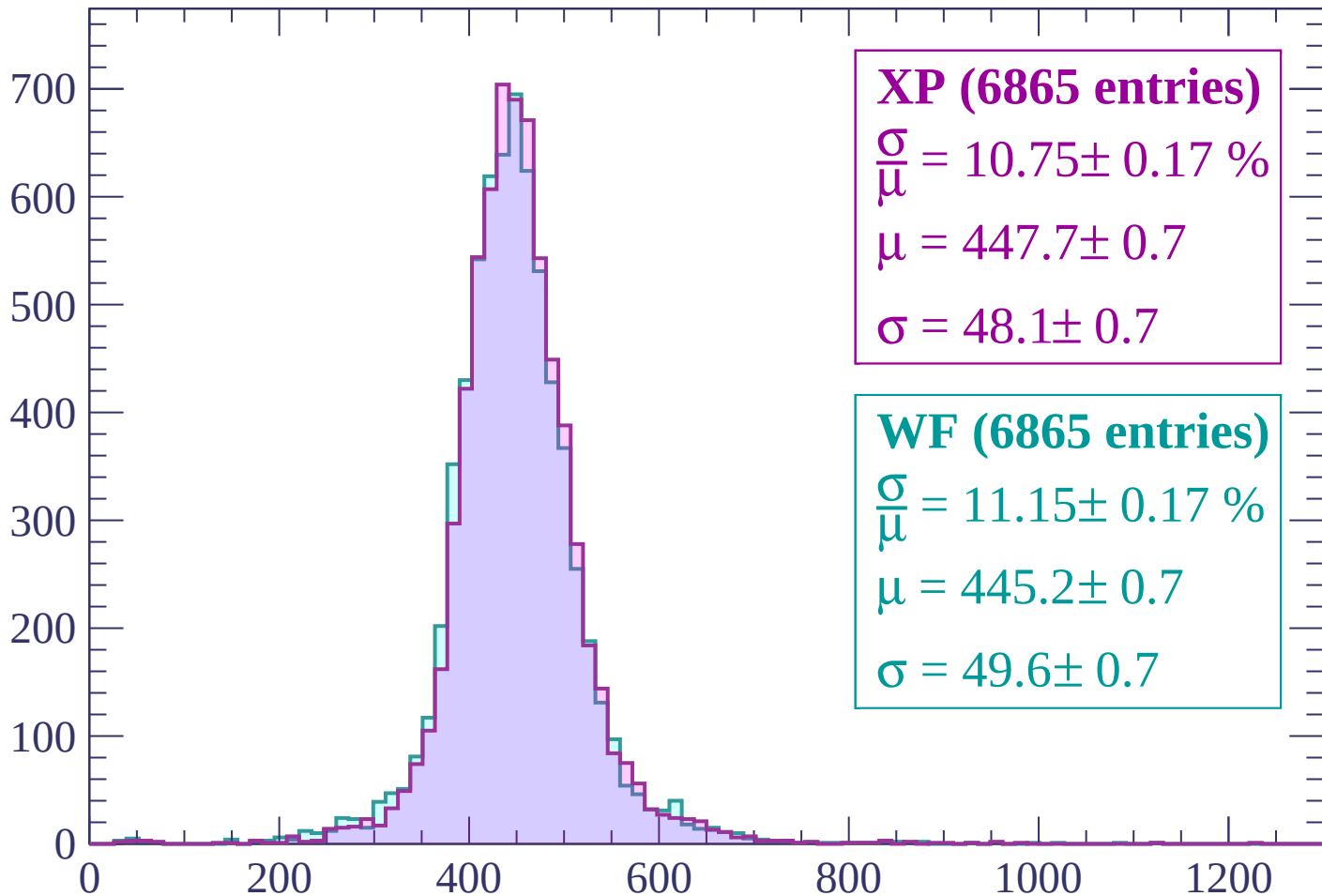
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (6865 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.17 \%$

$\mu = 447.7 \pm 0.7$

$\sigma = 48.1 \pm 0.7$

WF (6865 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.17 \%$

$\mu = 445.2 \pm 0.7$

$\sigma = 49.6 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8375 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.13 \%$

$\mu = 412.8 \pm 0.5$

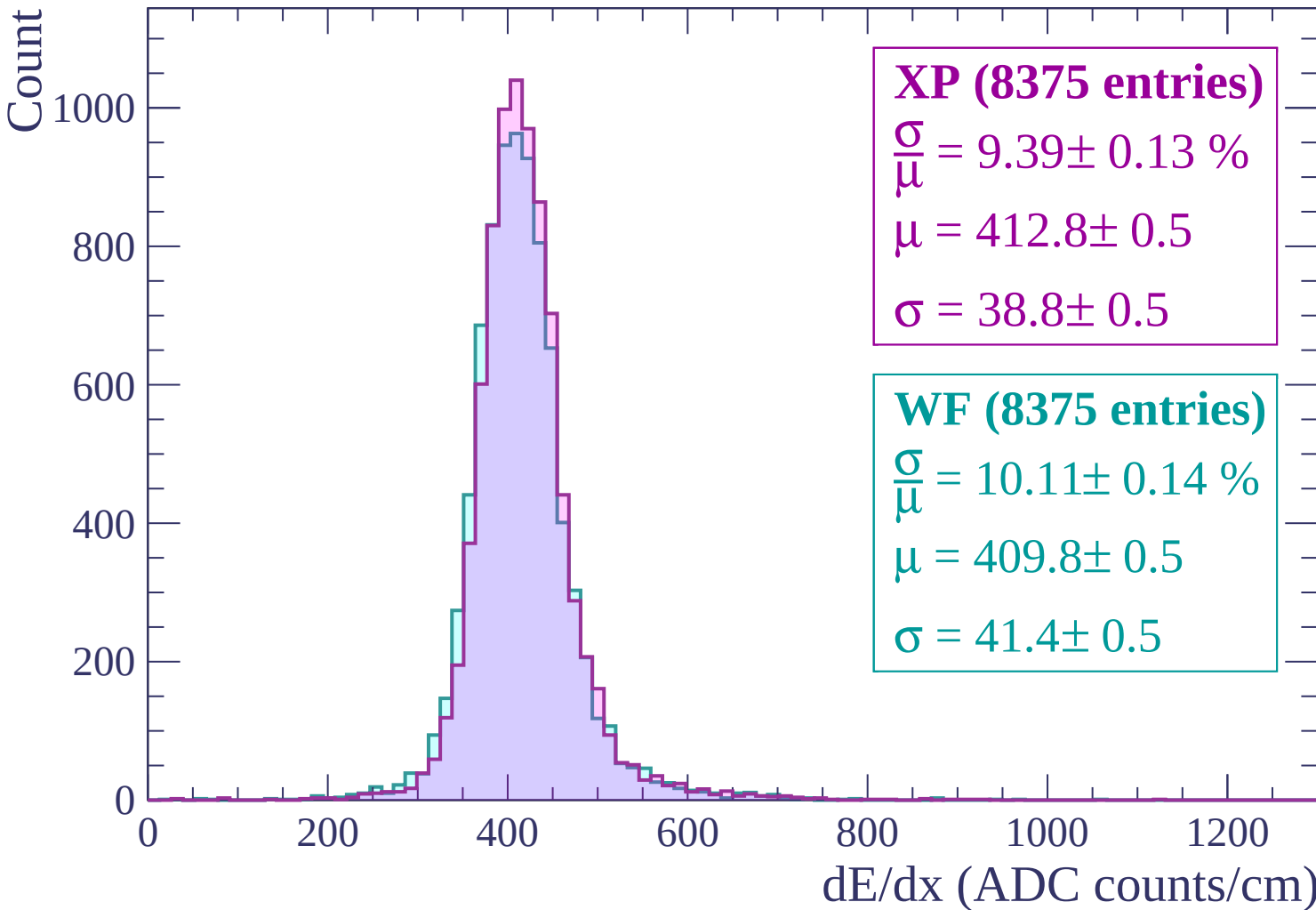
$\sigma = 38.8 \pm 0.5$

WF (8375 entries)

$\frac{\sigma}{\mu} = 10.11 \pm 0.14 \%$

$\mu = 409.8 \pm 0.5$

$\sigma = 41.4 \pm 0.5$



Energy loss | $300 < p < 360$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9170 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.13 \%$

$\mu = 400.6 \pm 0.5$

$\sigma = 36.3 \pm 0.5$

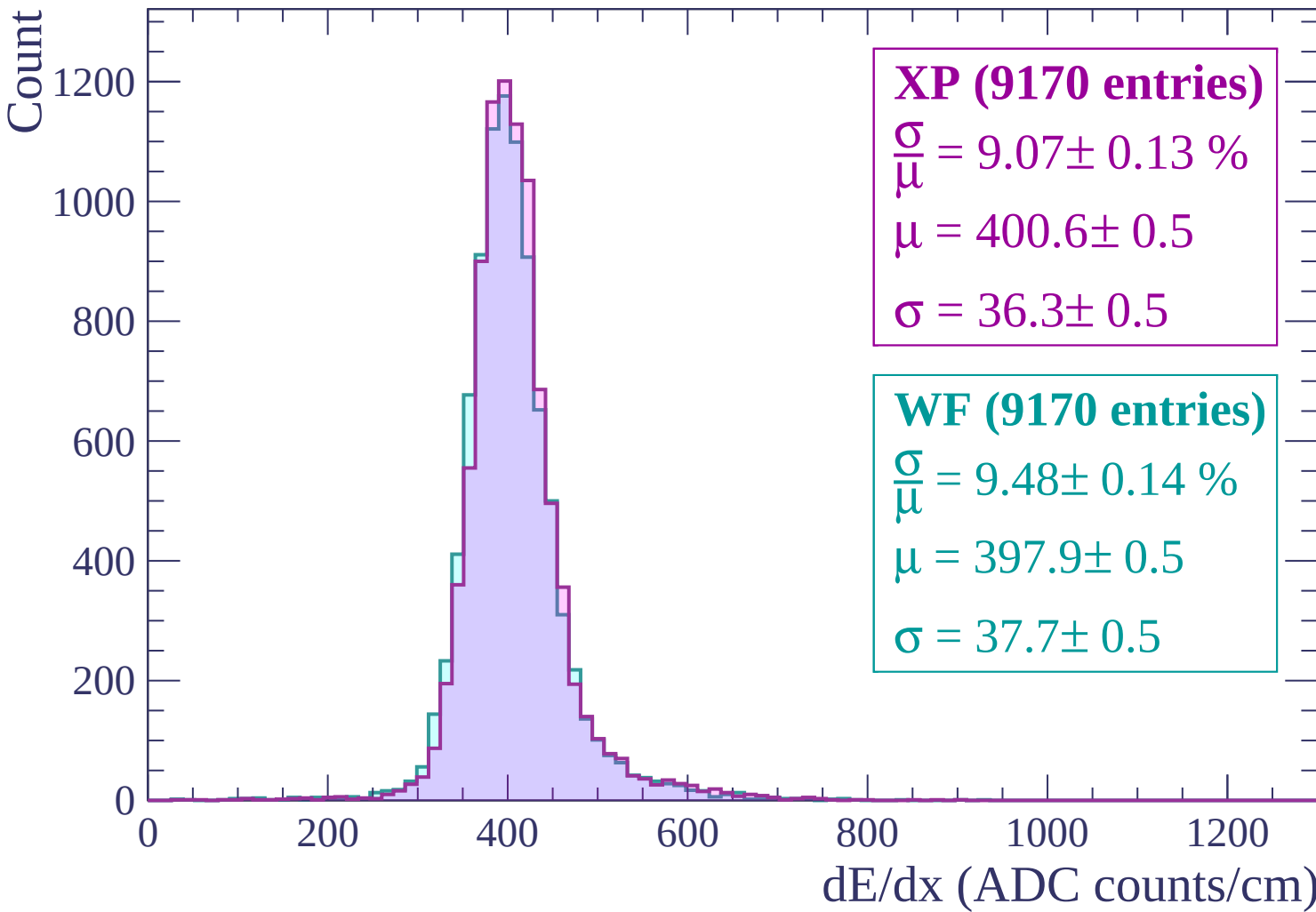
WF (9170 entries)

$\frac{\sigma}{\mu} = 9.48 \pm 0.14 \%$

$\mu = 397.9 \pm 0.5$

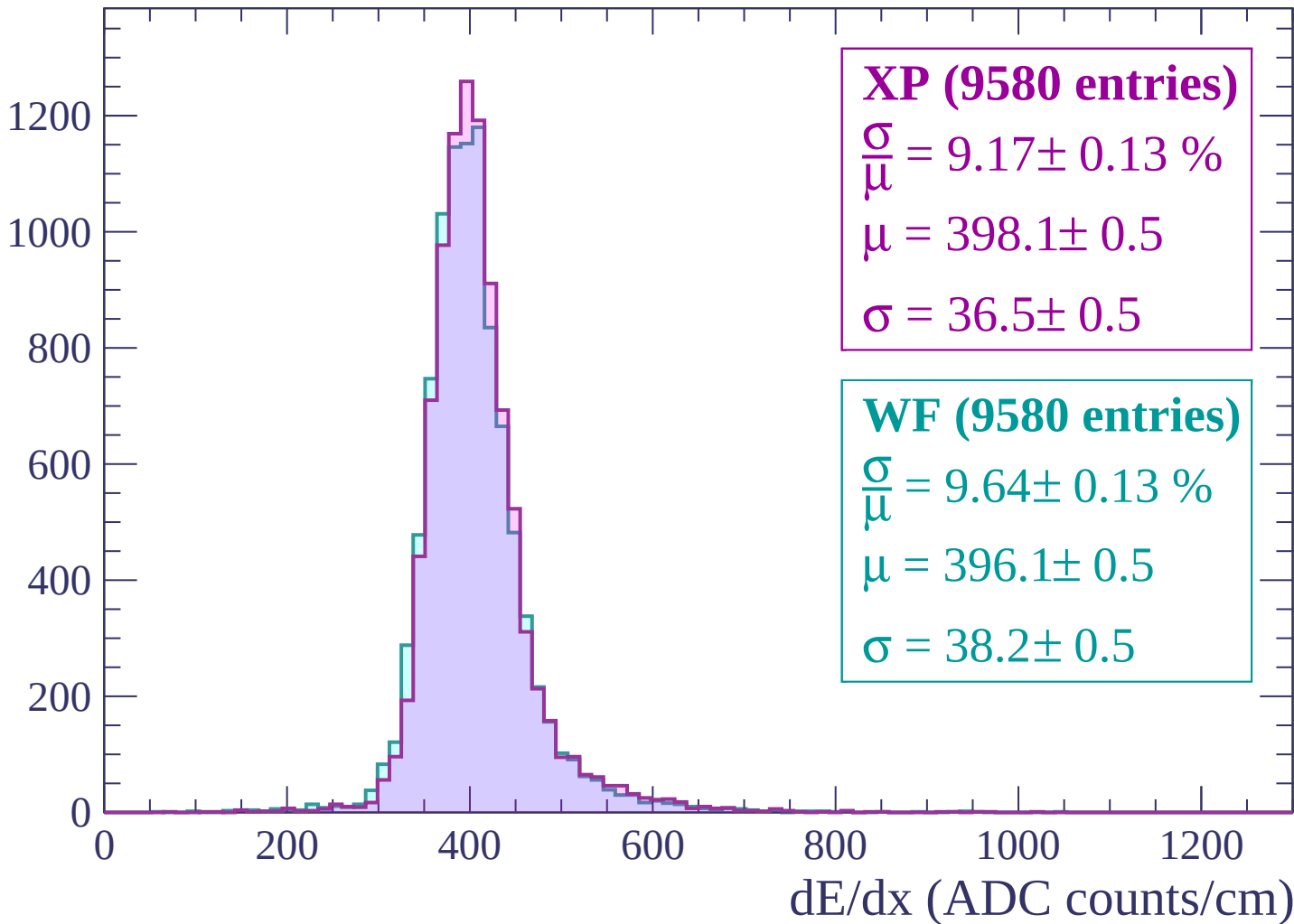
$\sigma = 37.7 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$

Count



Energy loss | $420 < p < 480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (9744 entries)

$$\frac{\sigma}{\mu} = 9.13 \pm 0.13 \%$$

$$\mu = 399.9 \pm 0.5$$

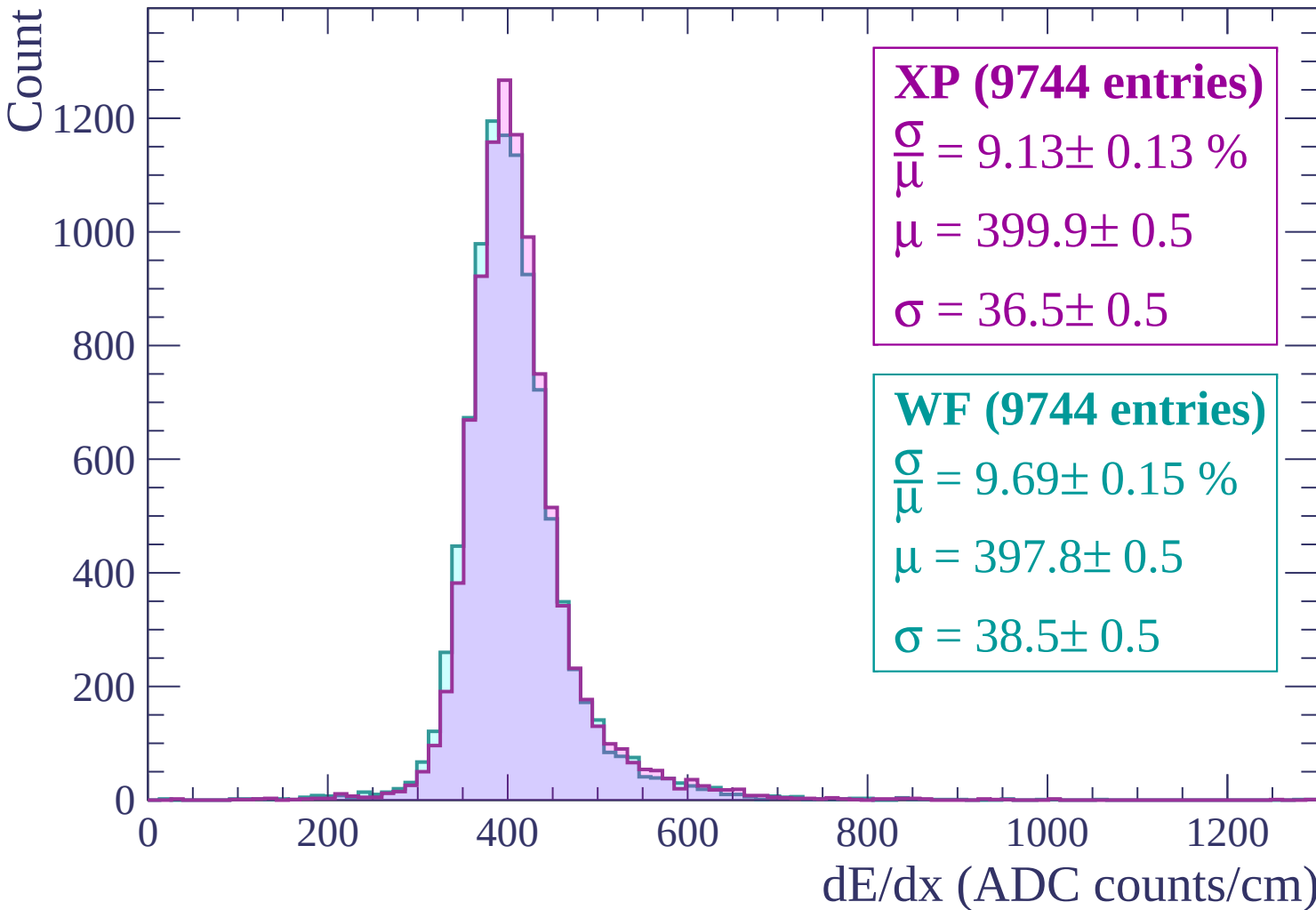
$$\sigma = 36.5 \pm 0.5$$

WF (9744 entries)

$$\frac{\sigma}{\mu} = 9.69 \pm 0.15 \%$$

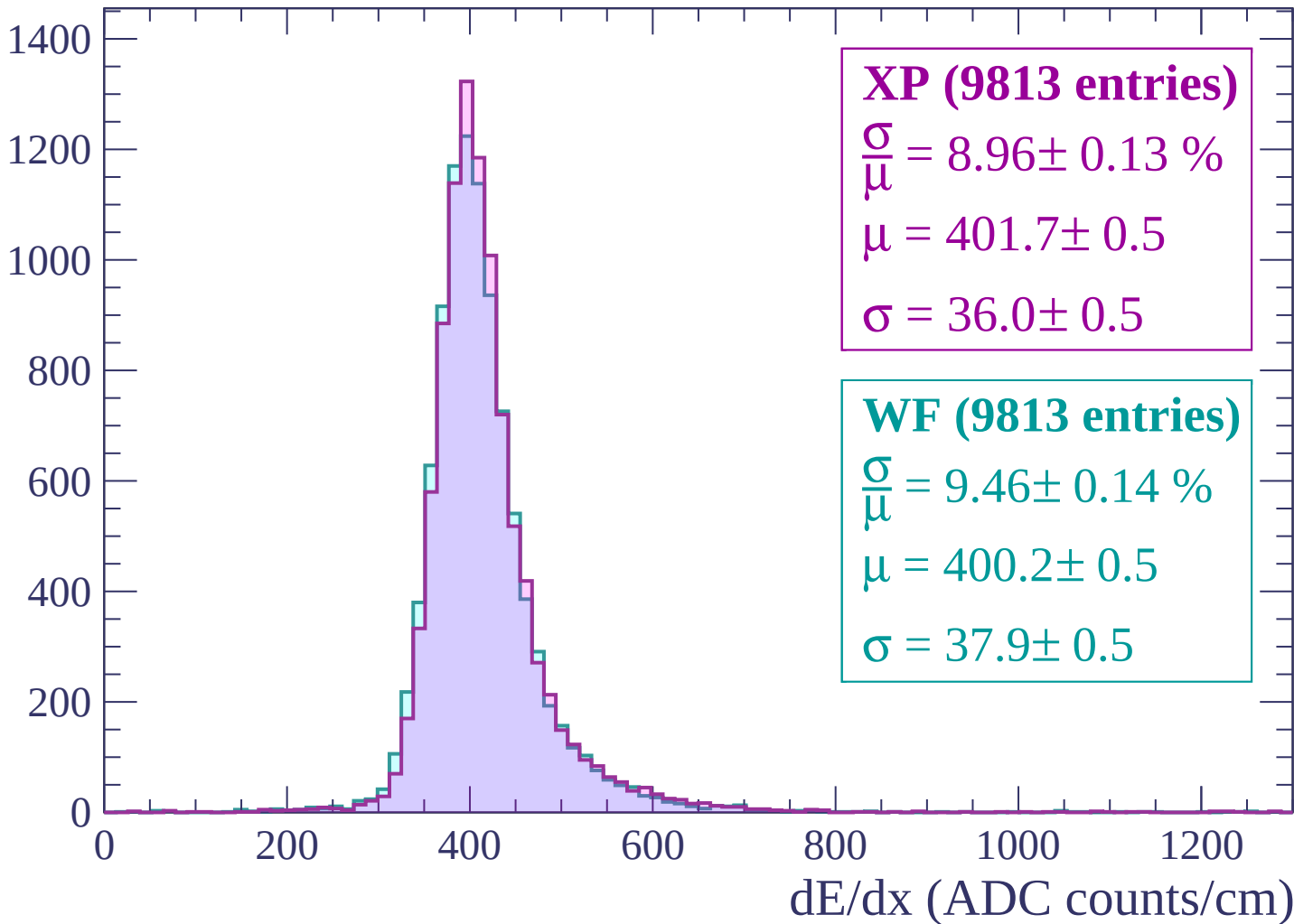
$$\mu = 397.8 \pm 0.5$$

$$\sigma = 38.5 \pm 0.5$$



Energy loss | $480 < p < 540$

Count



Energy loss | $540 < p < 600$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (9776 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.12 \%$

$\mu = 406.8 \pm 0.5$

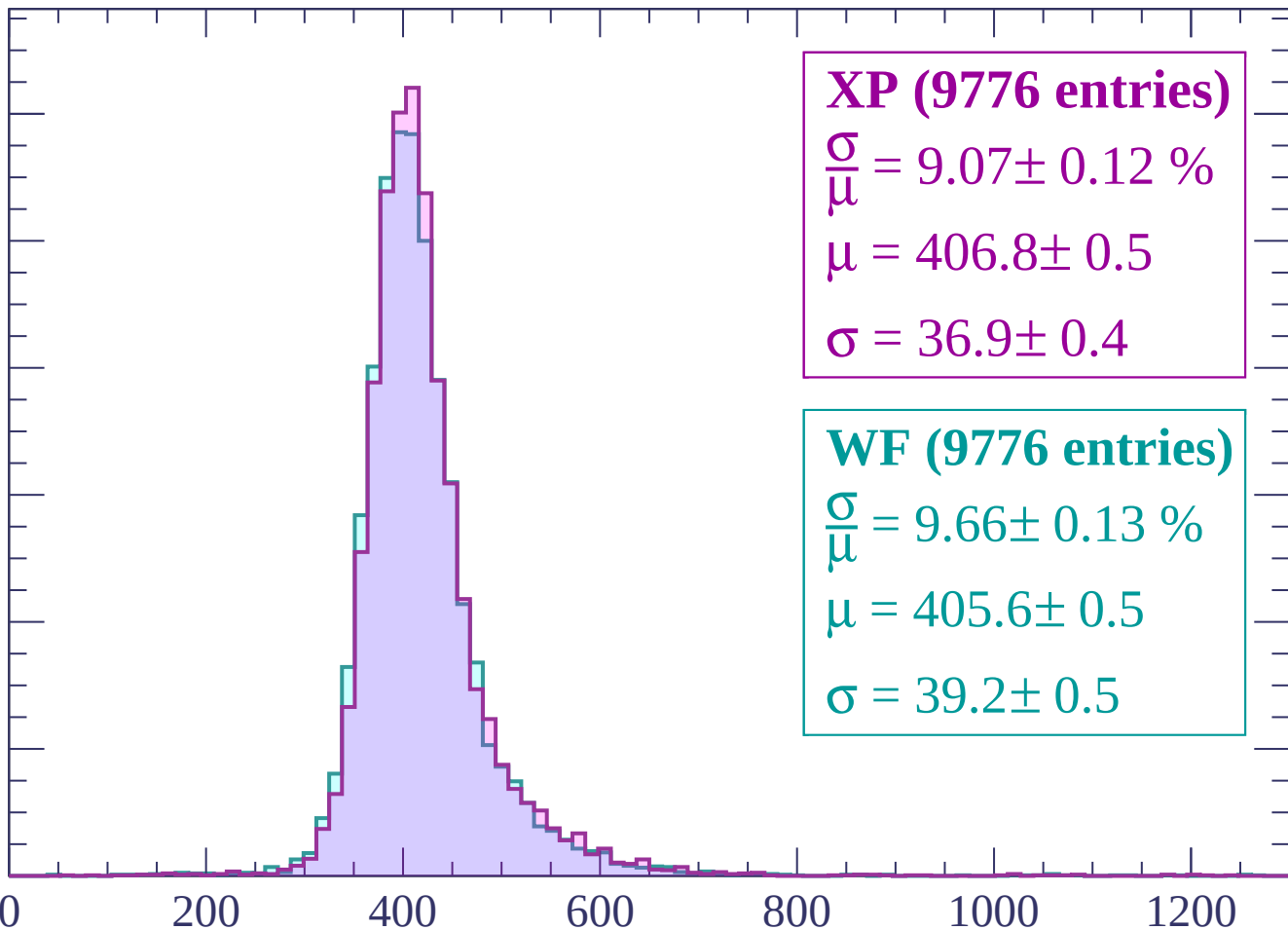
$\sigma = 36.9 \pm 0.4$

WF (9776 entries)

$\frac{\sigma}{\mu} = 9.66 \pm 0.13 \%$

$\mu = 405.6 \pm 0.5$

$\sigma = 39.2 \pm 0.5$



Energy loss | $600 < p < 660$

Count

1200

1000

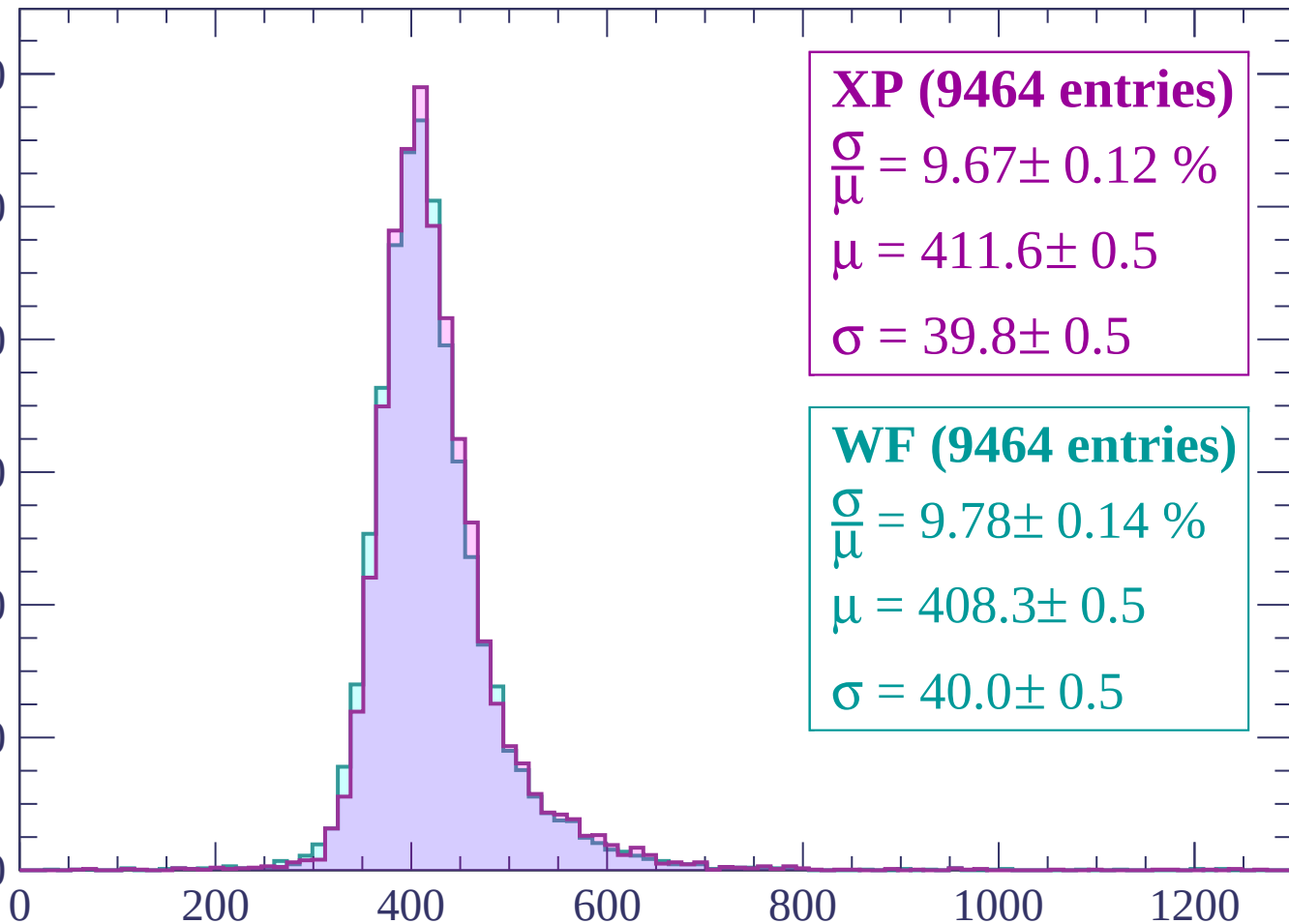
800

600

400

200

0



XP (9464 entries)

$\frac{\sigma}{\mu} = 9.67 \pm 0.12 \%$

$\mu = 411.6 \pm 0.5$

$\sigma = 39.8 \pm 0.5$

WF (9464 entries)

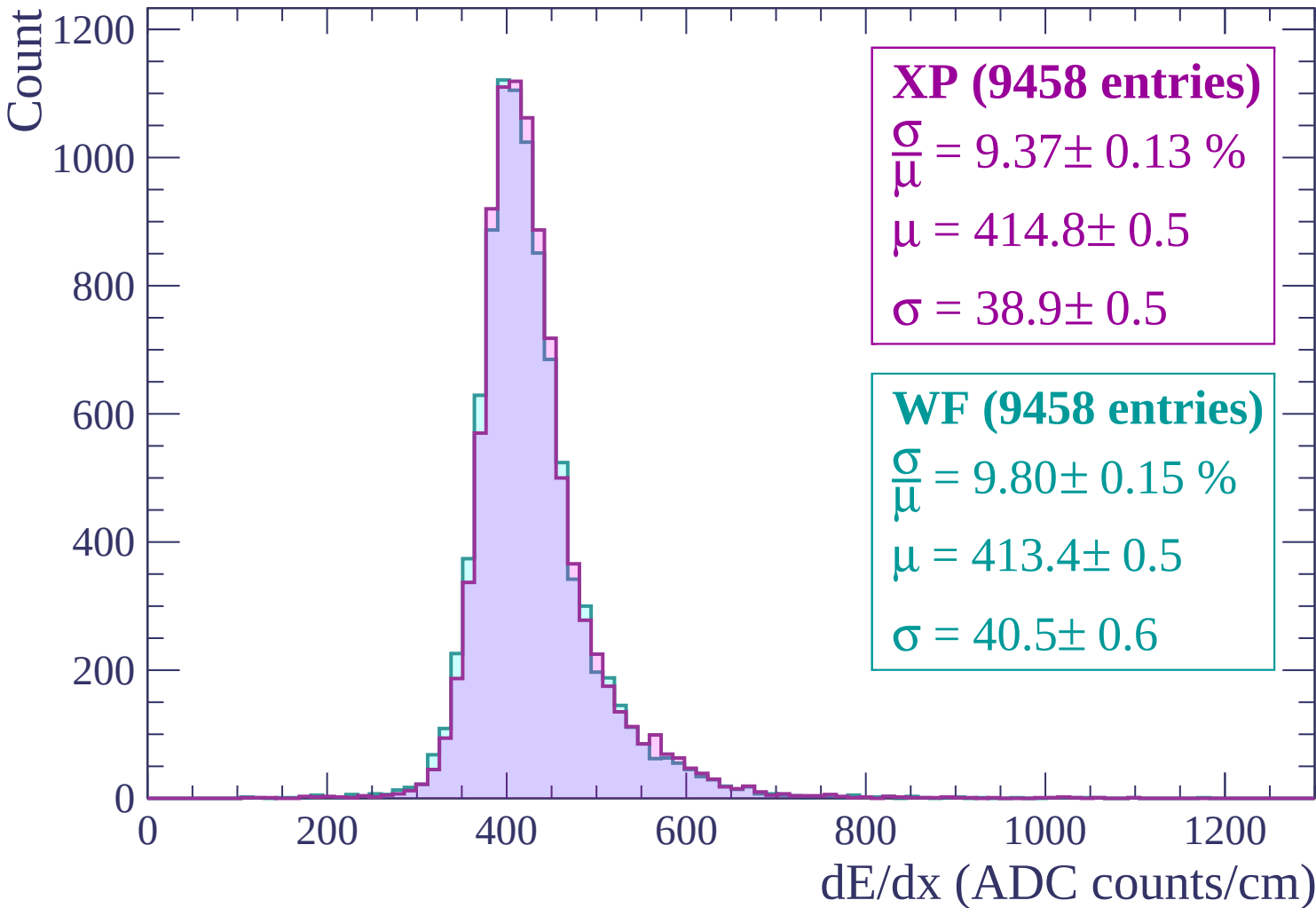
$\frac{\sigma}{\mu} = 9.78 \pm 0.14 \%$

$\mu = 408.3 \pm 0.5$

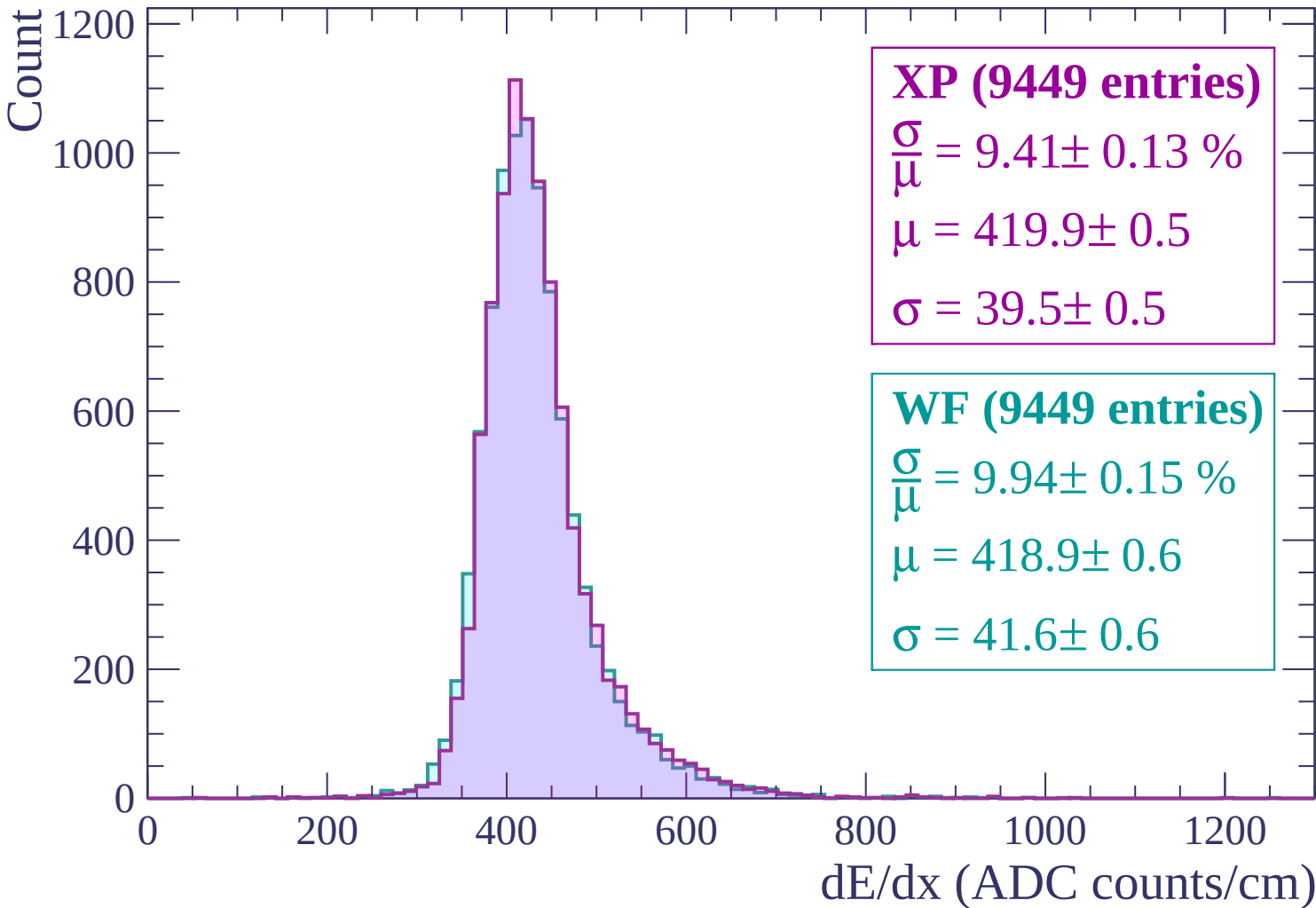
$\sigma = 40.0 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

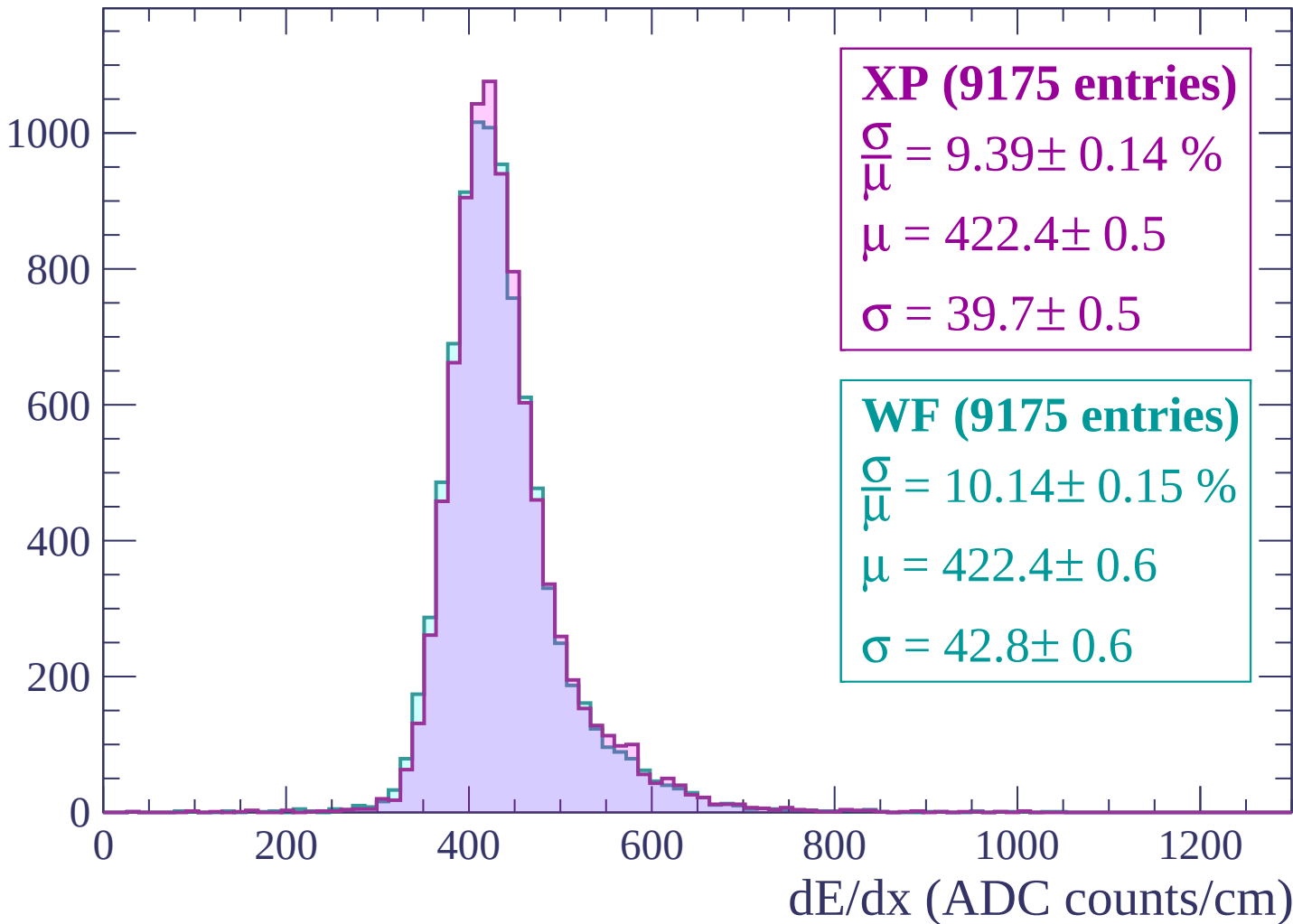


Energy loss | $720 < p < 780$



Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8916 entries)

$$\frac{\sigma}{\mu} = 9.80 \pm 0.16 \%$$

$$\mu = 427.2 \pm 0.6$$

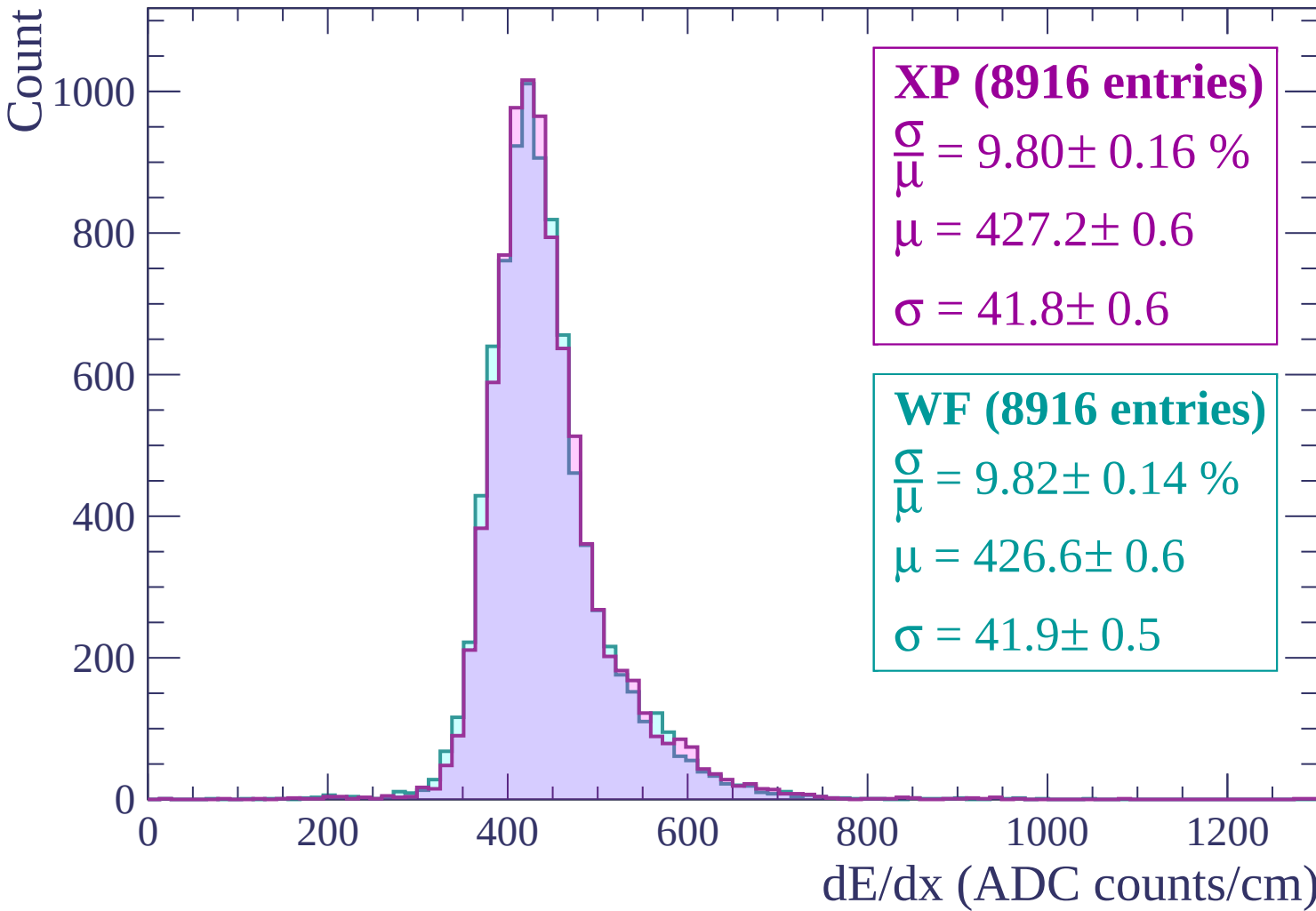
$$\sigma = 41.8 \pm 0.6$$

WF (8916 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.14 \%$$

$$\mu = 426.6 \pm 0.6$$

$$\sigma = 41.9 \pm 0.5$$



Energy loss | $900 < p < 960$

Count

1000

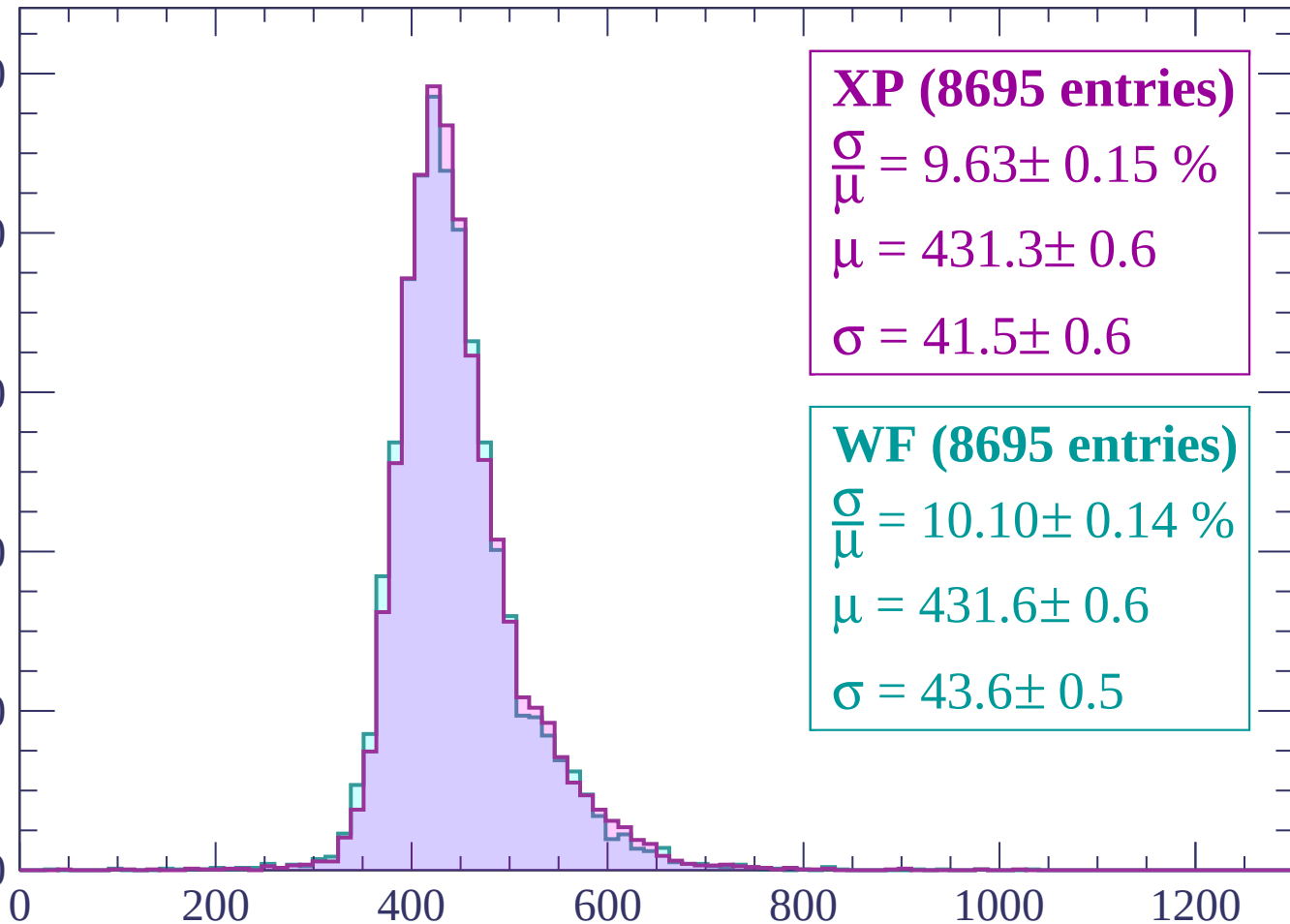
800

600

400

200

0



XP (8695 entries)

$\frac{\sigma}{\mu} = 9.63 \pm 0.15 \%$

$\mu = 431.3 \pm 0.6$

$\sigma = 41.5 \pm 0.6$

WF (8695 entries)

$\frac{\sigma}{\mu} = 10.10 \pm 0.14 \%$

$\mu = 431.6 \pm 0.6$

$\sigma = 43.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count

1000

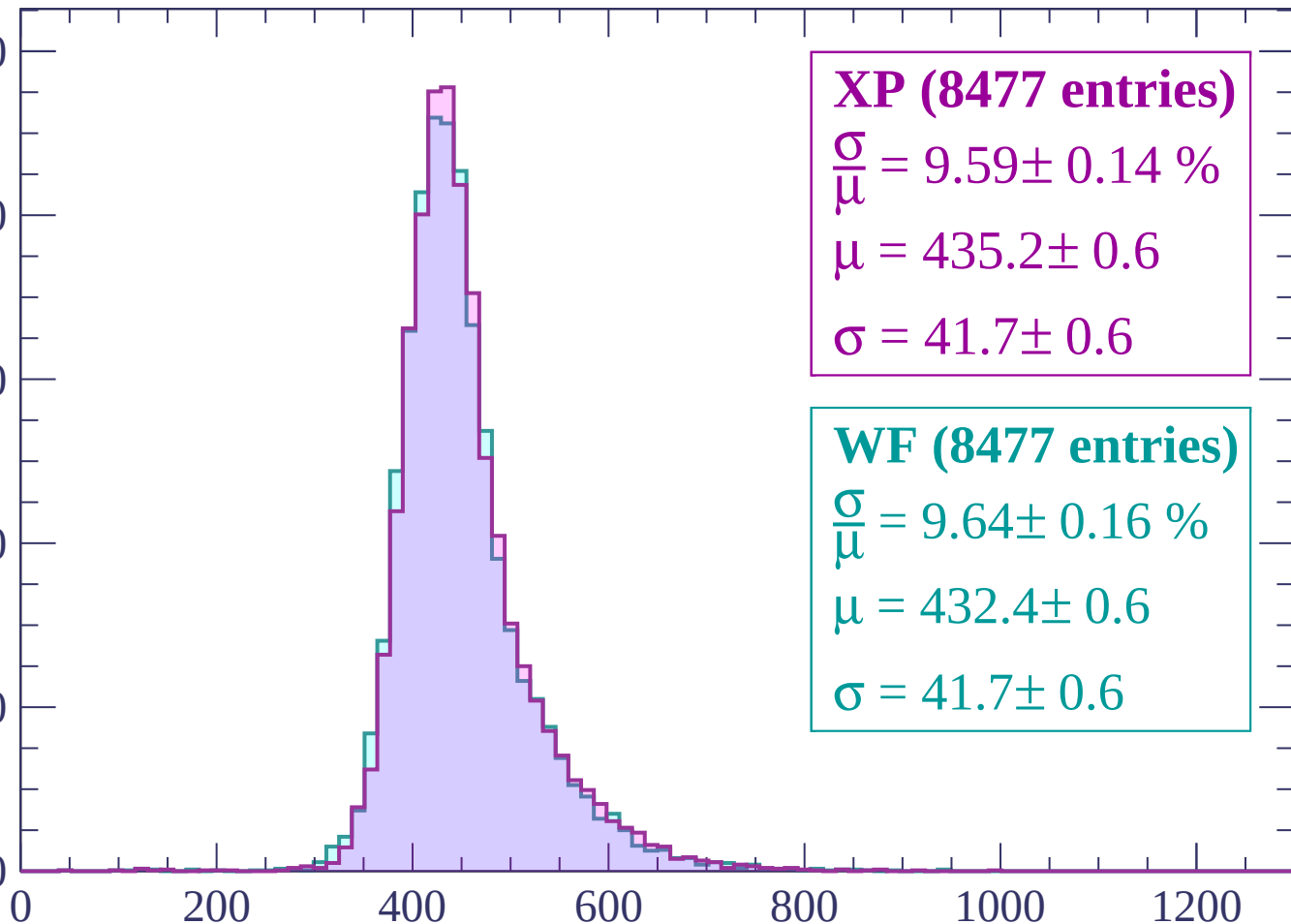
800

600

400

200

0



XP (8477 entries)

$\frac{\sigma}{\mu} = 9.59 \pm 0.14 \%$

$\mu = 435.2 \pm 0.6$

$\sigma = 41.7 \pm 0.6$

WF (8477 entries)

$\frac{\sigma}{\mu} = 9.64 \pm 0.16 \%$

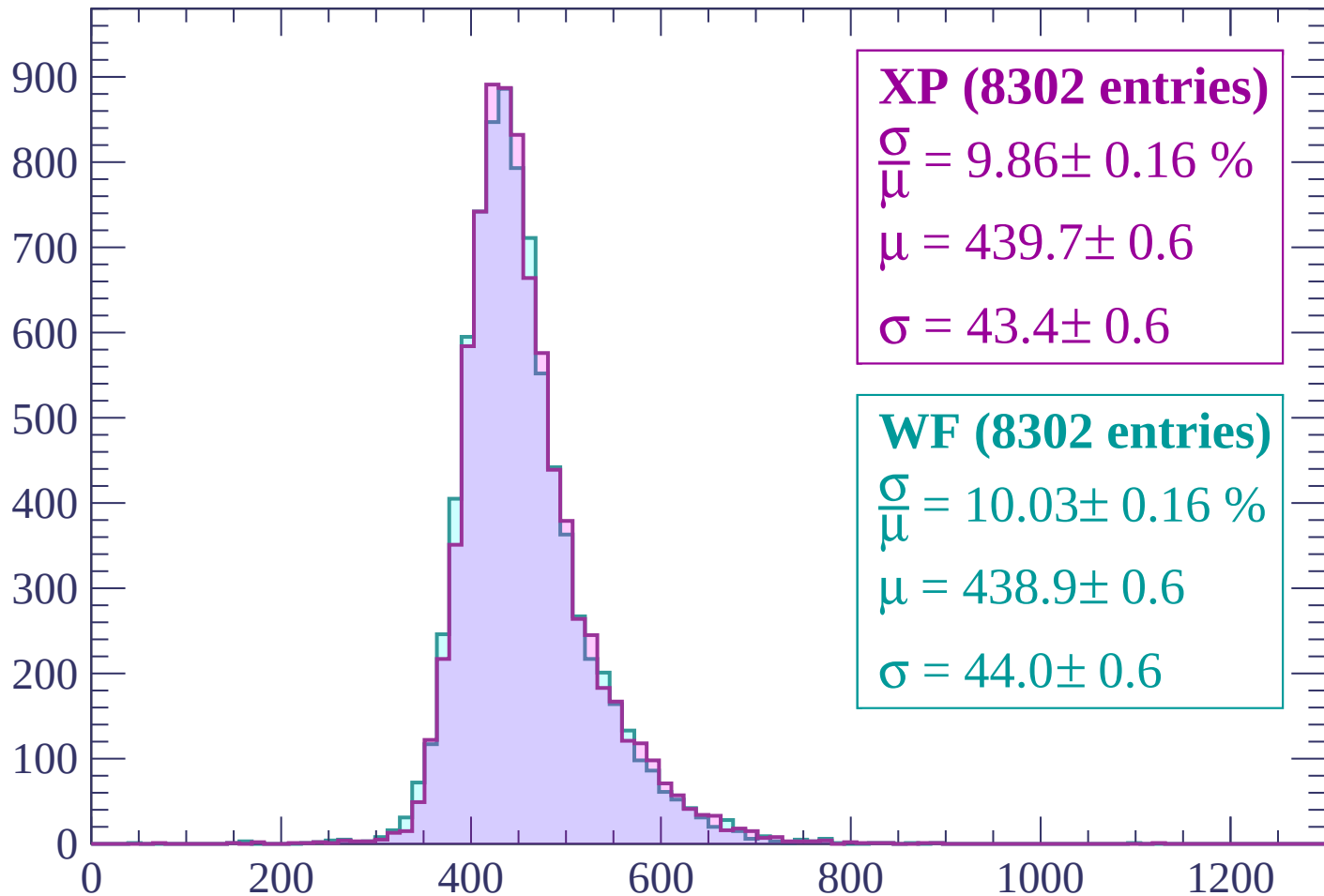
$\mu = 432.4 \pm 0.6$

$\sigma = 41.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (8302 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.16 \%$$

$$\mu = 439.7 \pm 0.6$$

$$\sigma = 43.4 \pm 0.6$$

WF (8302 entries)

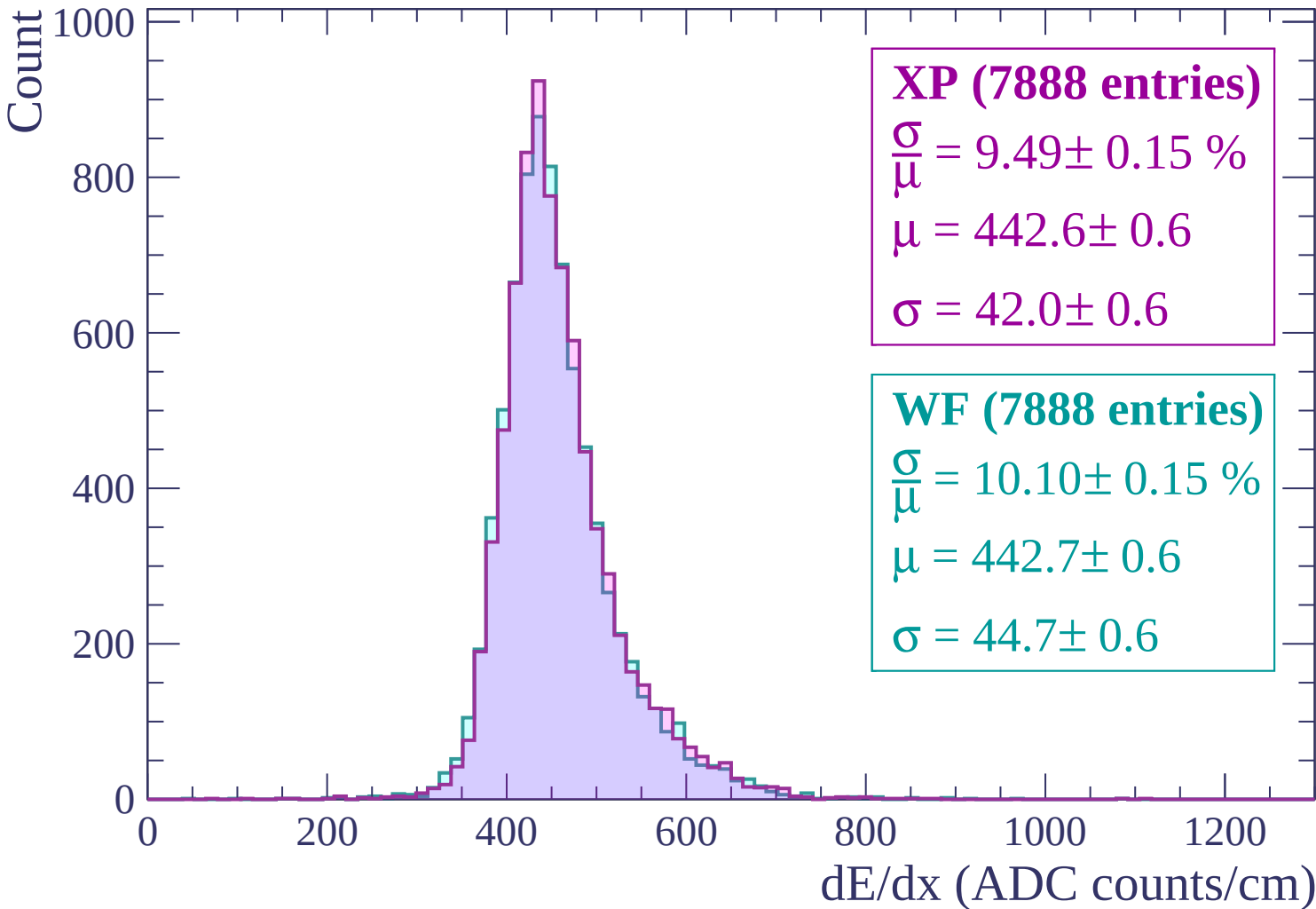
$$\frac{\sigma}{\mu} = 10.03 \pm 0.16 \%$$

$$\mu = 438.9 \pm 0.6$$

$$\sigma = 44.0 \pm 0.6$$

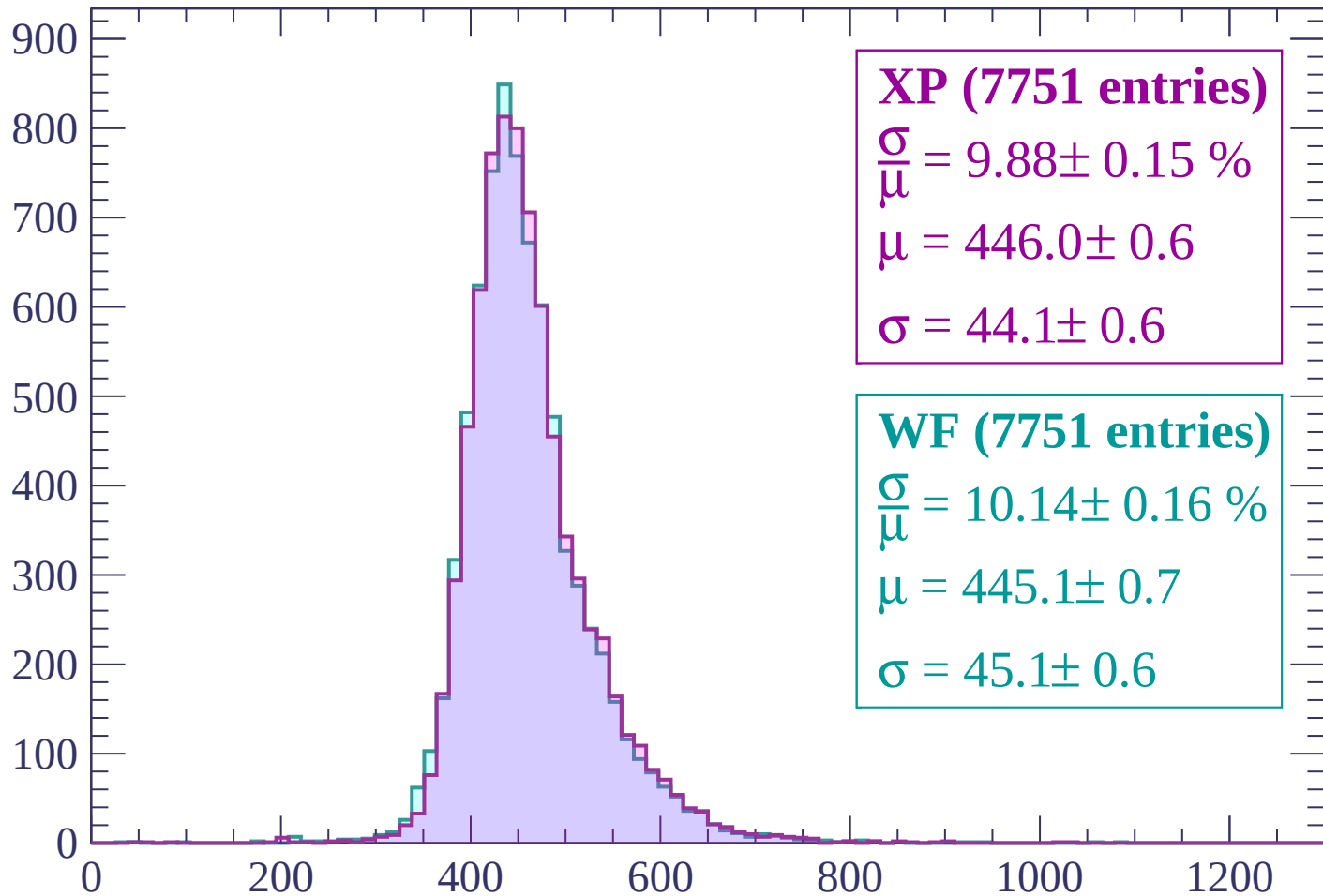
dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$



Energy loss | $1140 < p < 1200$

Count



XP (7751 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.15 \%$$

$$\mu = 446.0 \pm 0.6$$

$$\sigma = 44.1 \pm 0.6$$

WF (7751 entries)

$$\frac{\sigma}{\mu} = 10.14 \pm 0.16 \%$$

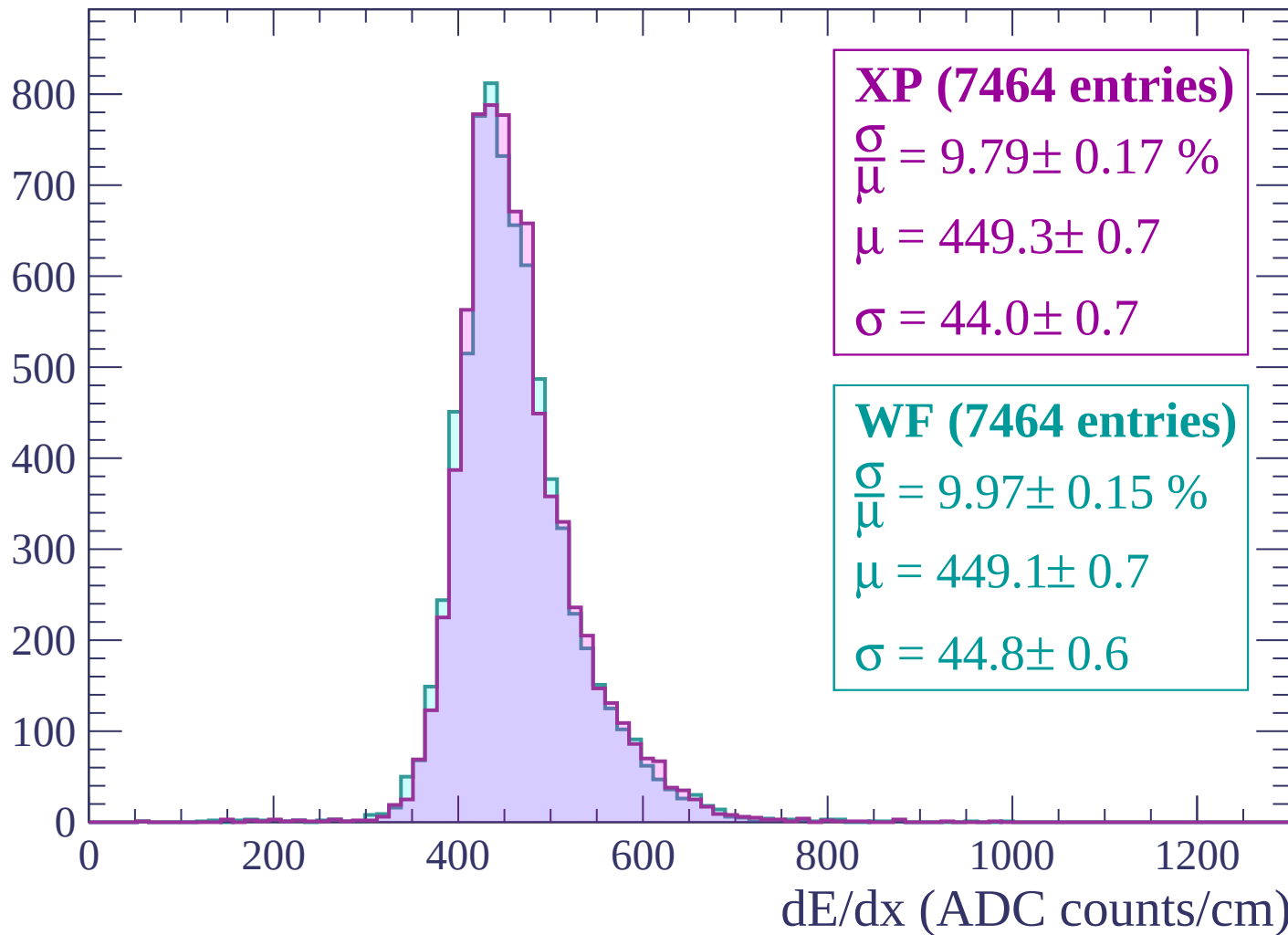
$$\mu = 445.1 \pm 0.7$$

$$\sigma = 45.1 \pm 0.6$$

dE/dx (ADC counts/cm)

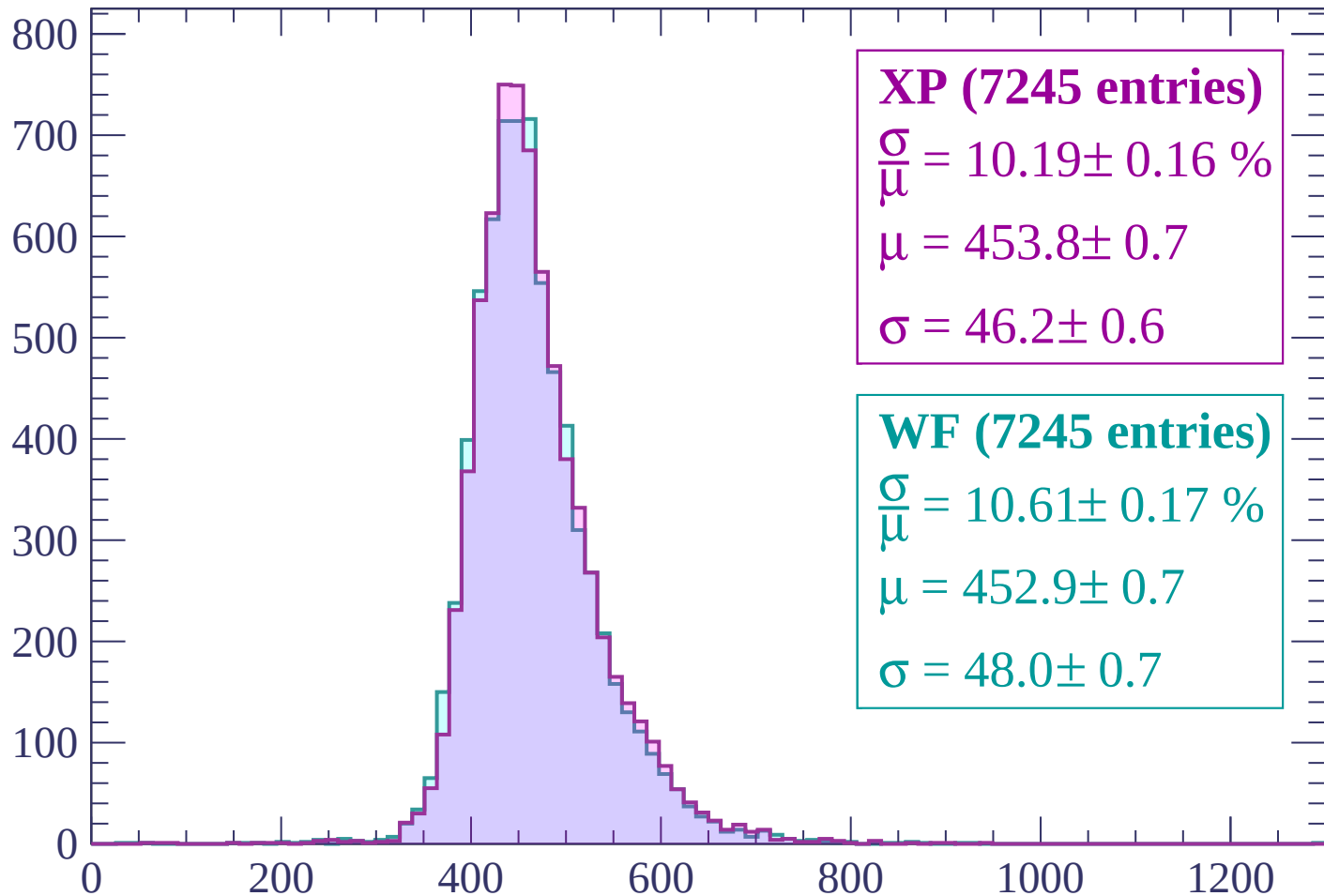
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (7245 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.16 \%$

$\mu = 453.8 \pm 0.7$

$\sigma = 46.2 \pm 0.6$

WF (7245 entries)

$\frac{\sigma}{\mu} = 10.61 \pm 0.17 \%$

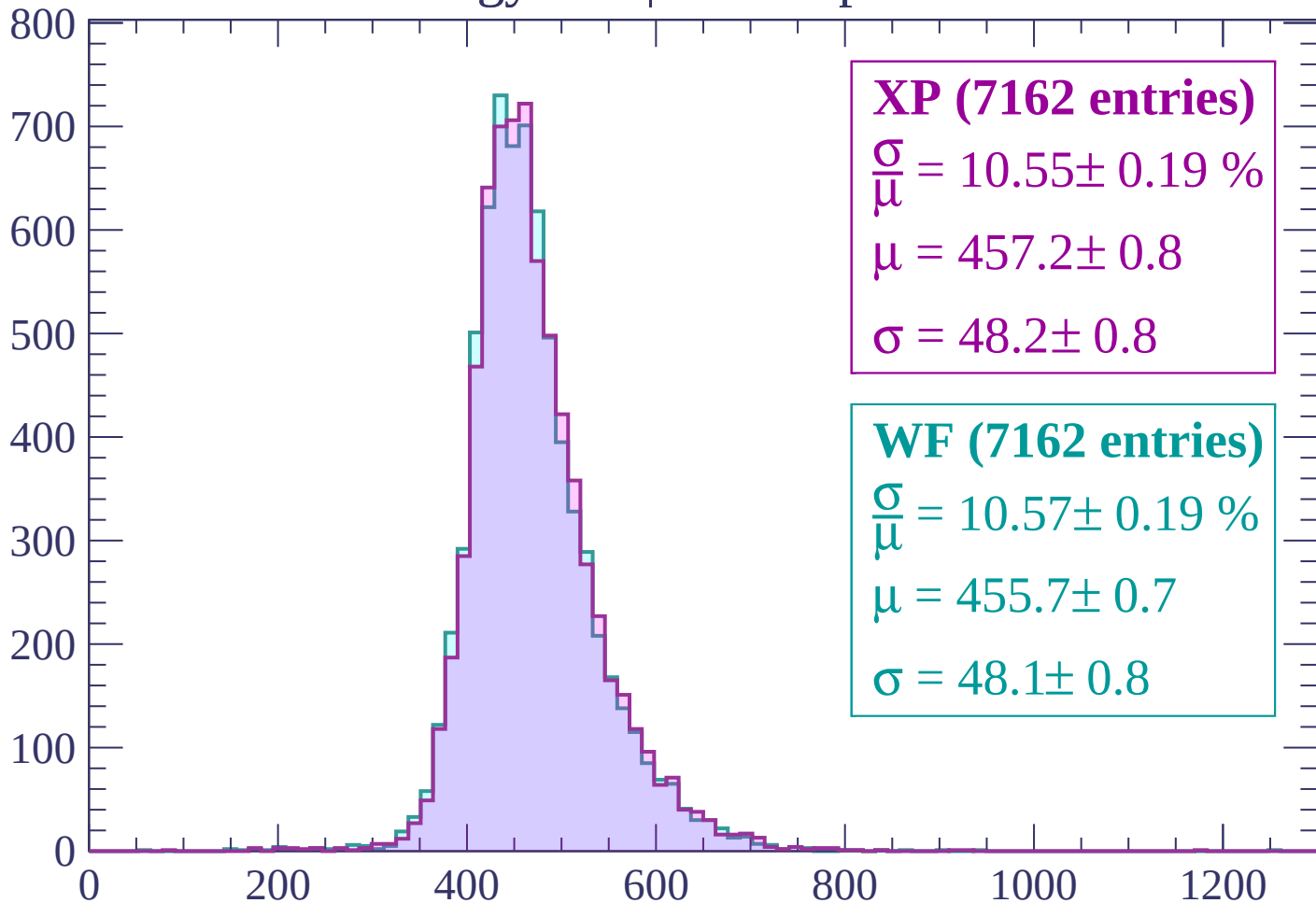
$\mu = 452.9 \pm 0.7$

$\sigma = 48.0 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (7162 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.19 \%$

$\mu = 457.2 \pm 0.8$

$\sigma = 48.2 \pm 0.8$

WF (7162 entries)

$\frac{\sigma}{\mu} = 10.57 \pm 0.19 \%$

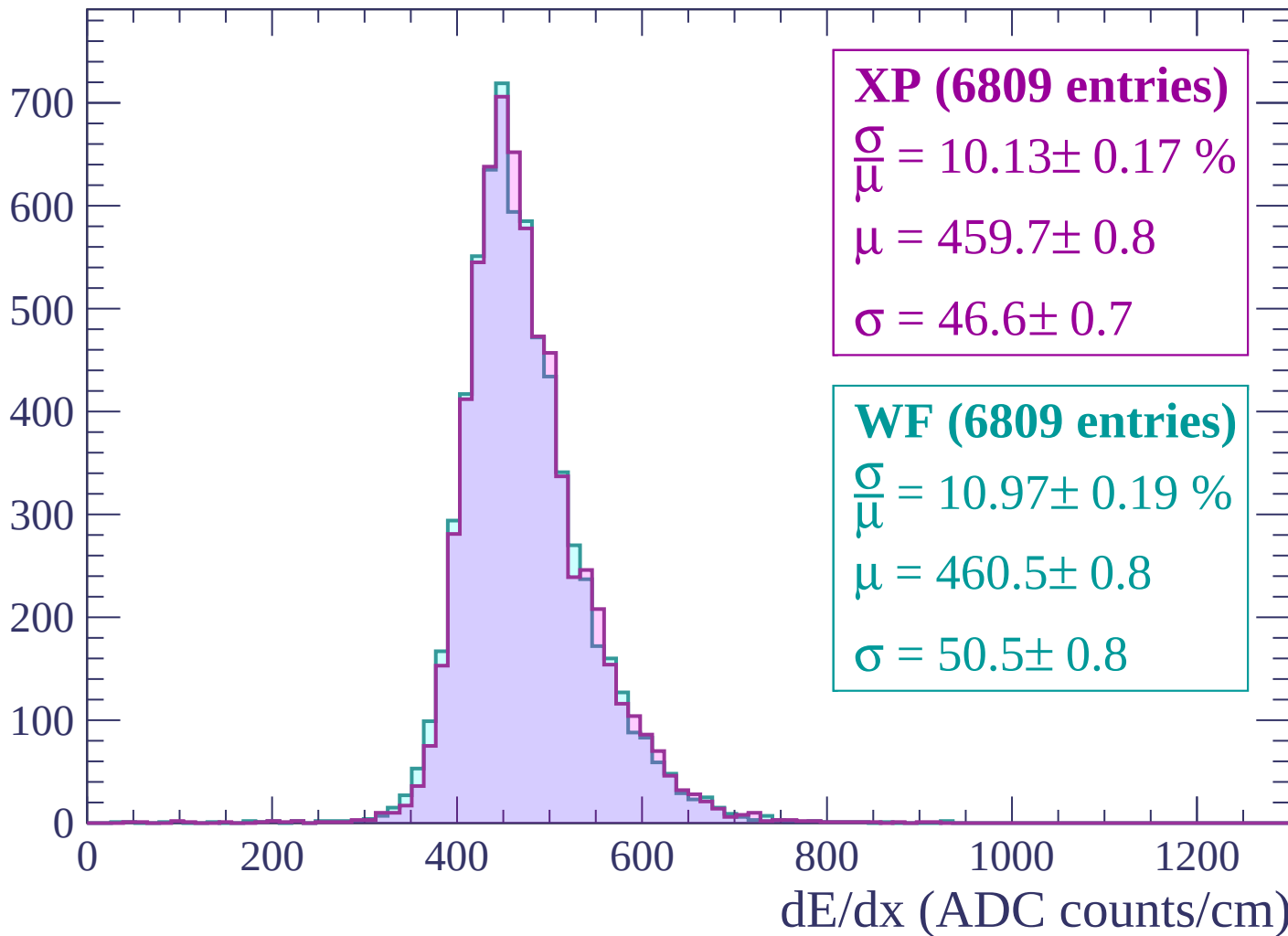
$\mu = 455.7 \pm 0.7$

$\sigma = 48.1 \pm 0.8$

dE/dx (ADC counts/cm)

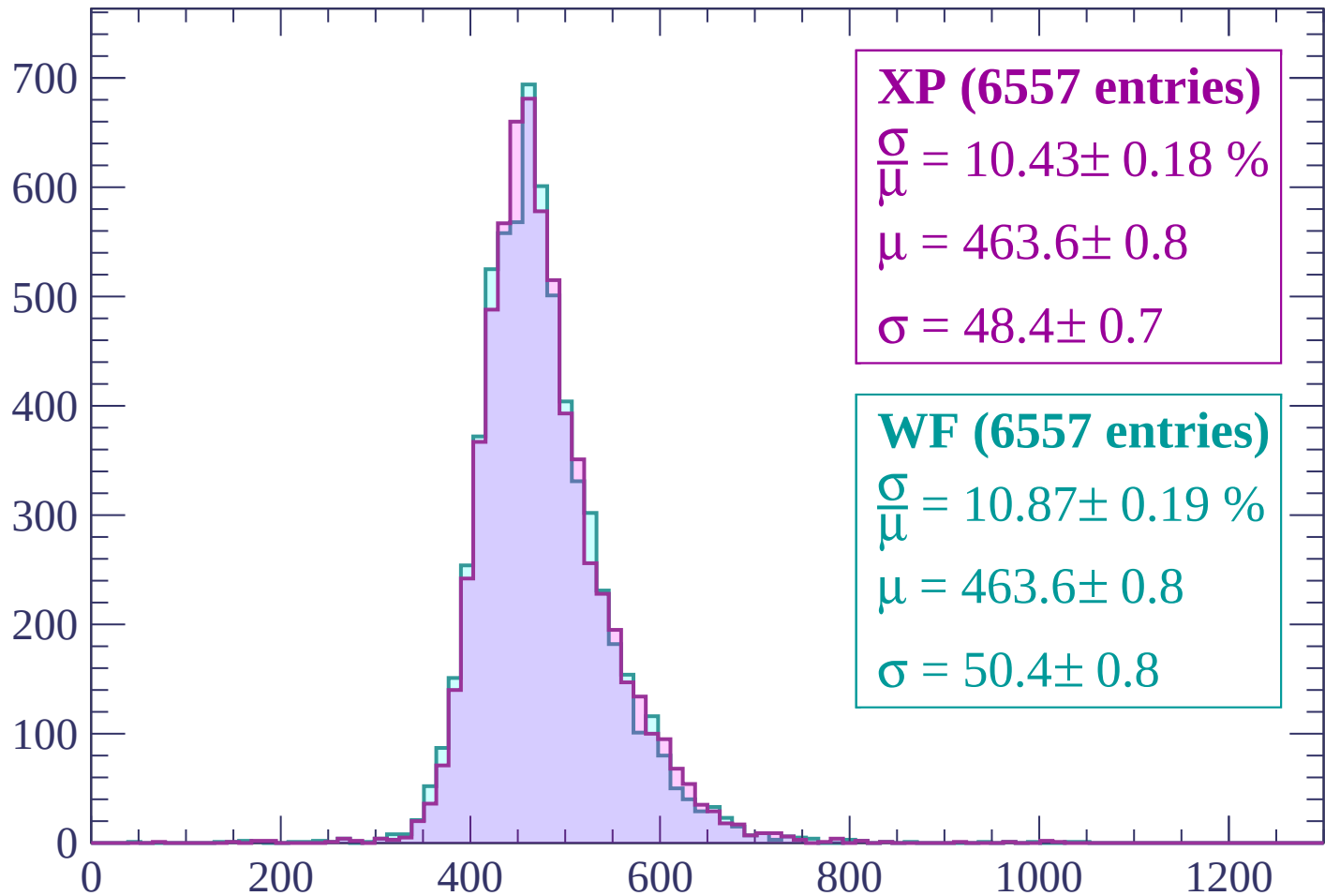
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP (6557 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.18 \%$

$\mu = 463.6 \pm 0.8$

$\sigma = 48.4 \pm 0.7$

WF (6557 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.19 \%$

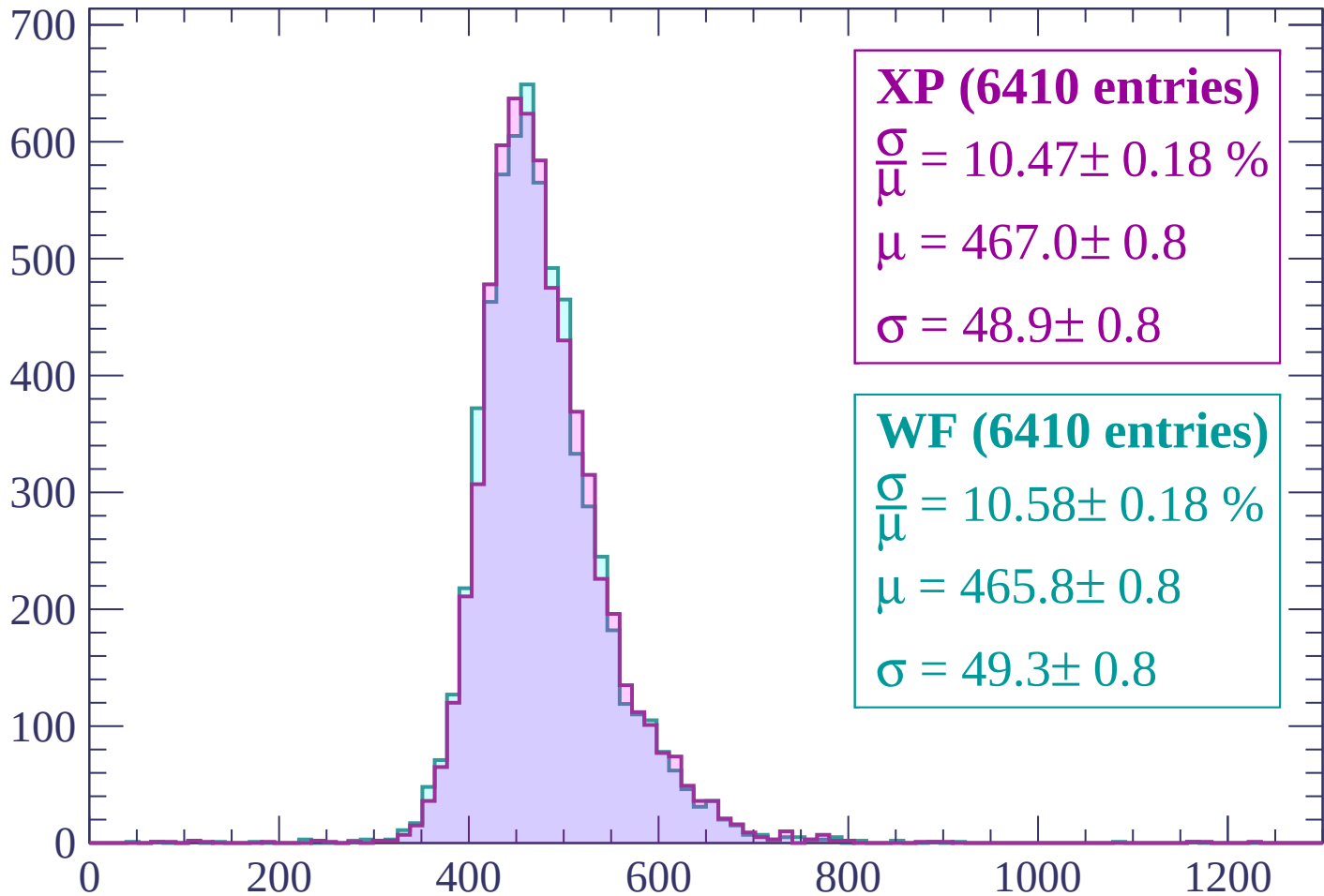
$\mu = 463.6 \pm 0.8$

$\sigma = 50.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (6410 entries)

$\frac{\sigma}{\mu} = 10.47 \pm 0.18 \%$

$\mu = 467.0 \pm 0.8$

$\sigma = 48.9 \pm 0.8$

WF (6410 entries)

$\frac{\sigma}{\mu} = 10.58 \pm 0.18 \%$

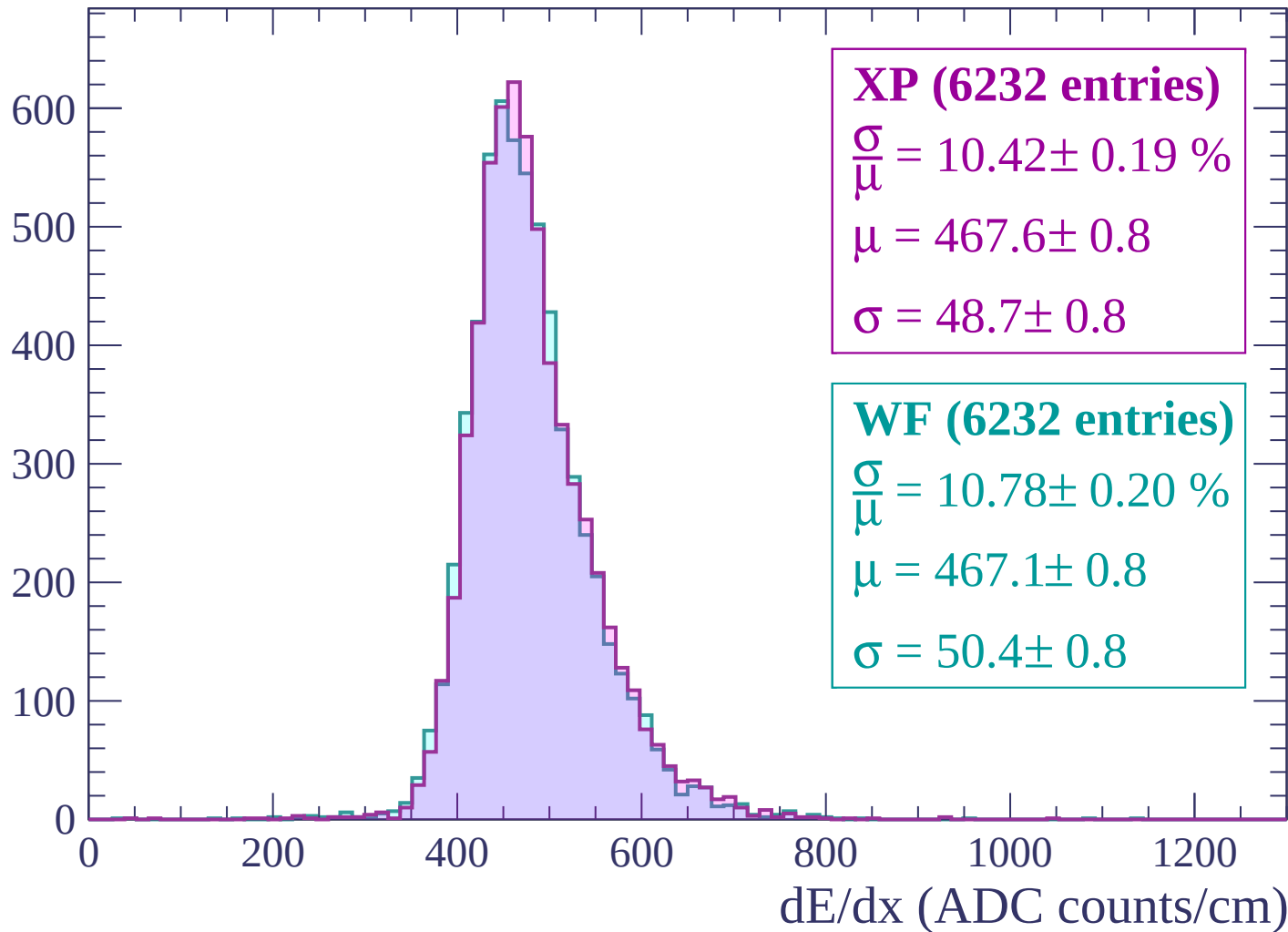
$\mu = 465.8 \pm 0.8$

$\sigma = 49.3 \pm 0.8$

dE/dx (ADC counts/cm)

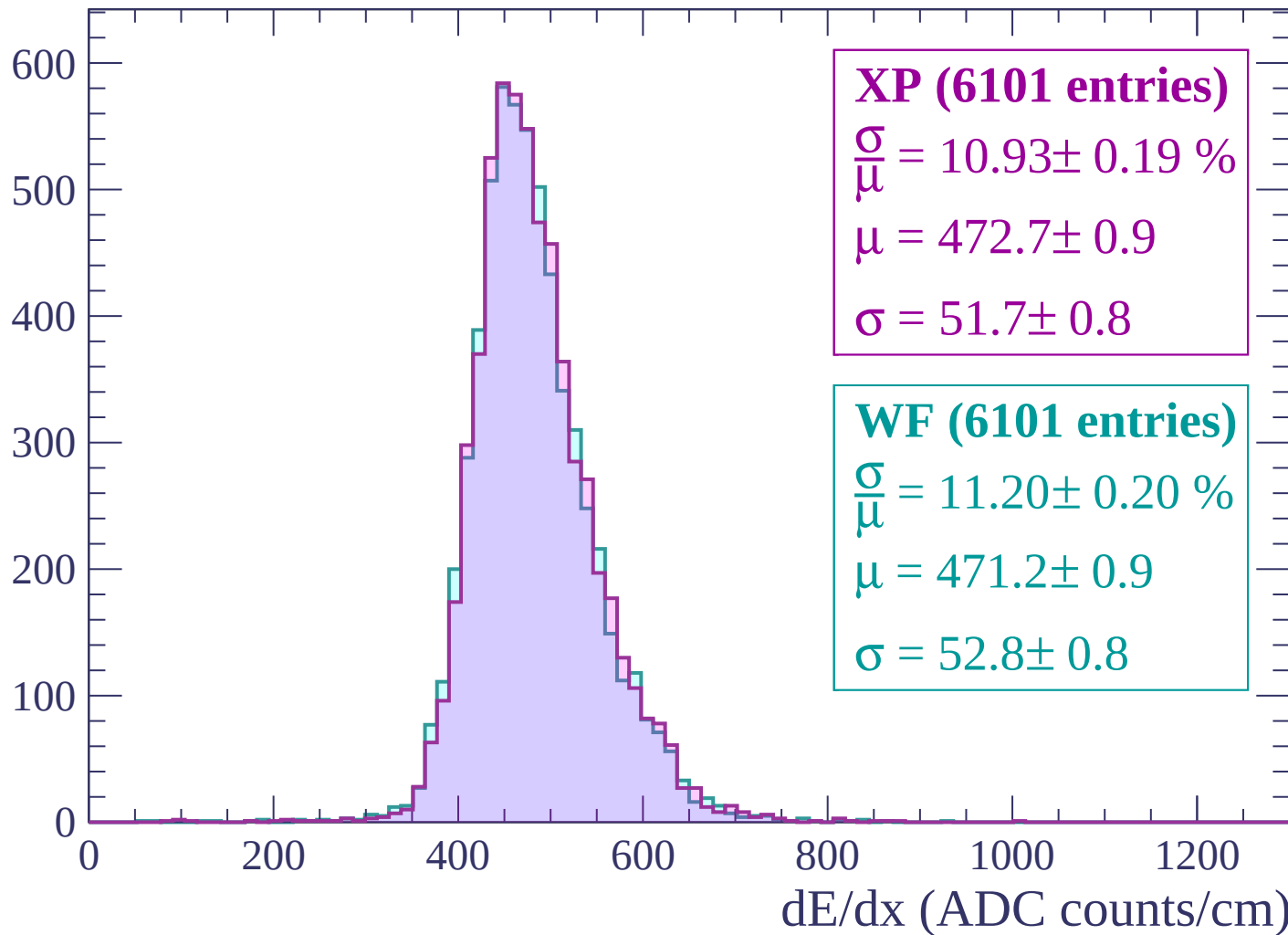
Energy loss | $1560 < p < 1620$

Count



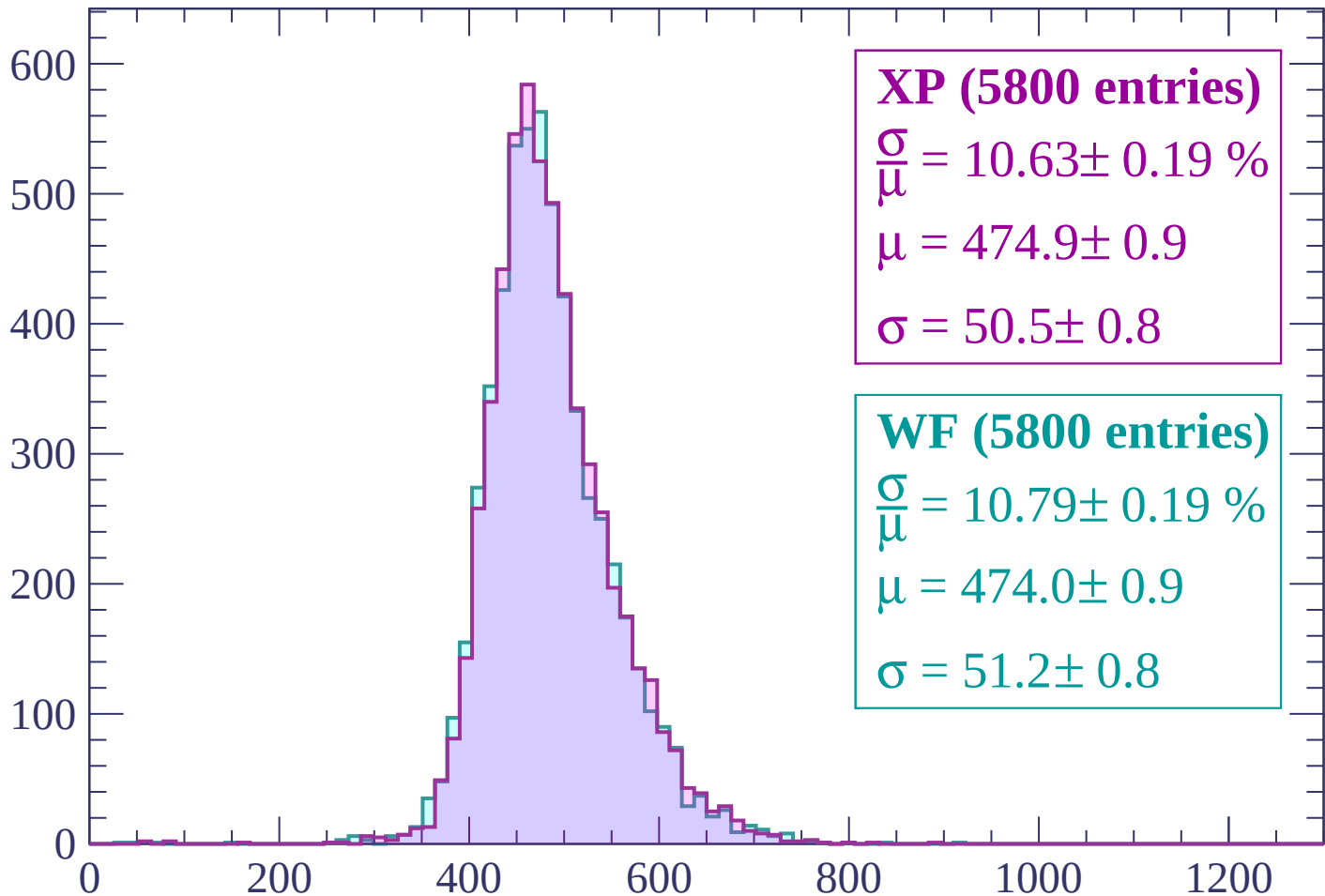
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP (5800 entries)

$\frac{\sigma}{\mu} = 10.63 \pm 0.19 \%$

$\mu = 474.9 \pm 0.9$

$\sigma = 50.5 \pm 0.8$

WF (5800 entries)

$\frac{\sigma}{\mu} = 10.79 \pm 0.19 \%$

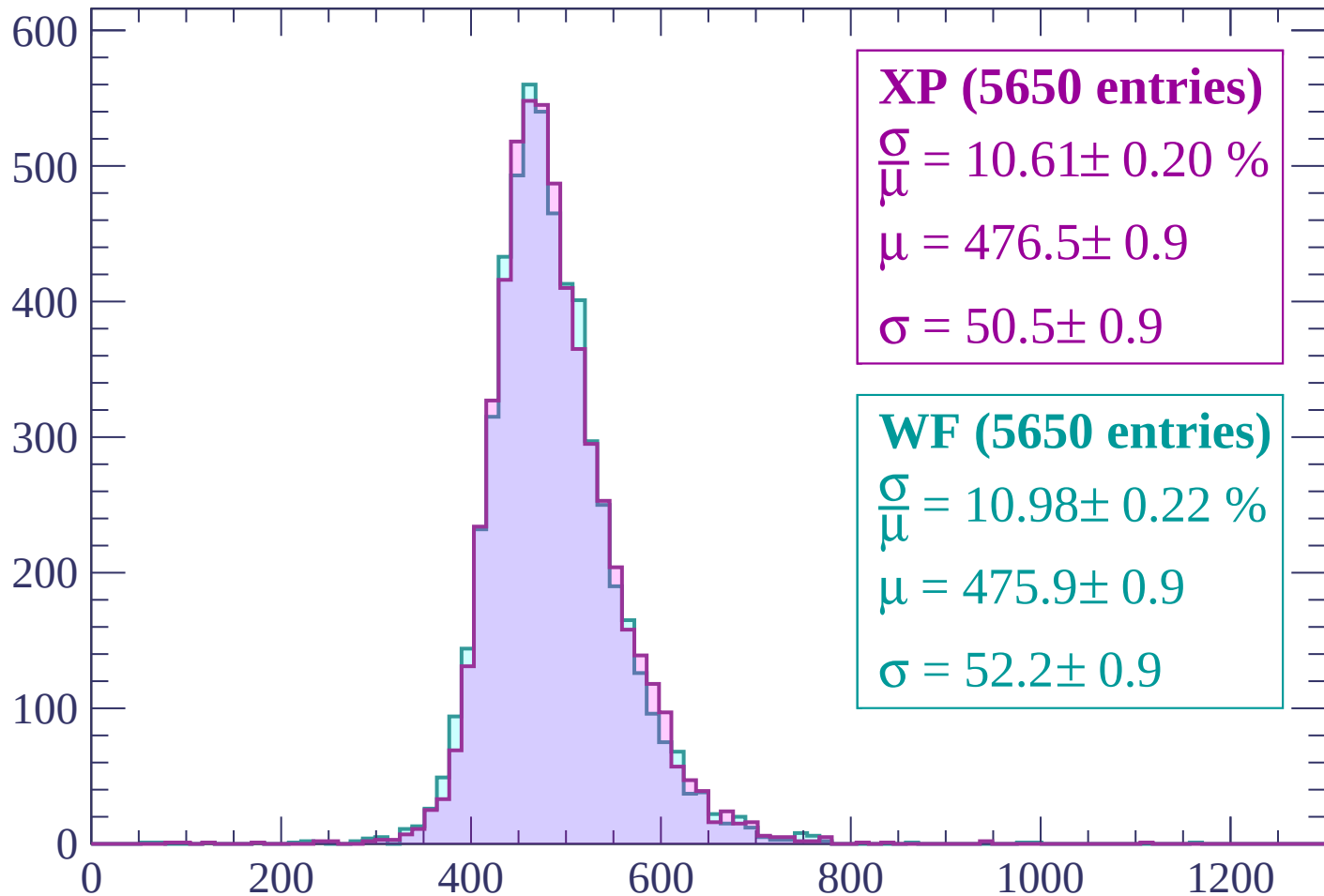
$\mu = 474.0 \pm 0.9$

$\sigma = 51.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (5650 entries)

$$\frac{\sigma}{\mu} = 10.61 \pm 0.20 \%$$

$$\mu = 476.5 \pm 0.9$$

$$\sigma = 50.5 \pm 0.9$$

WF (5650 entries)

$$\frac{\sigma}{\mu} = 10.98 \pm 0.22 \%$$

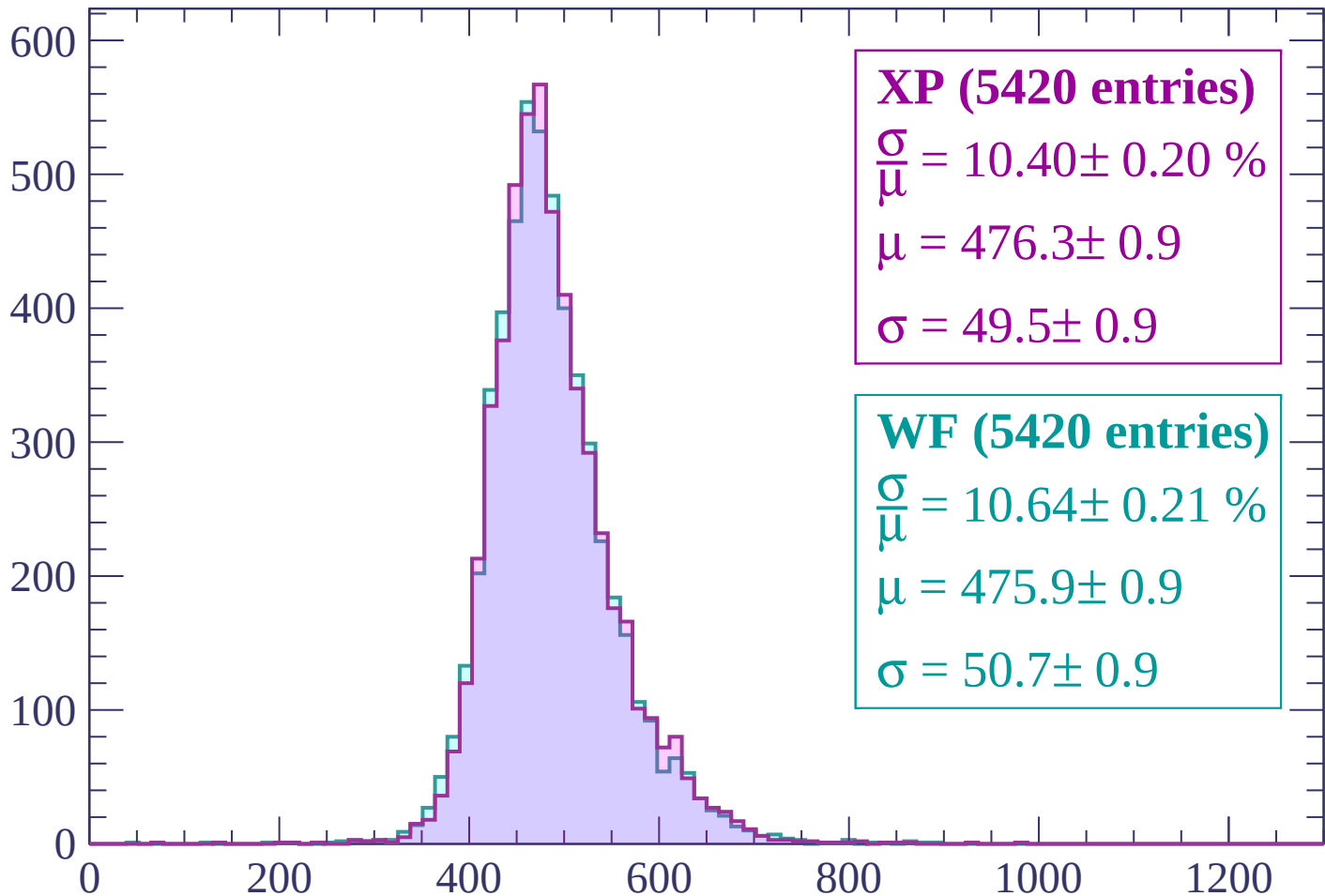
$$\mu = 475.9 \pm 0.9$$

$$\sigma = 52.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (5420 entries)

$\frac{\sigma}{\mu} = 10.40 \pm 0.20 \%$

$\mu = 476.3 \pm 0.9$

$\sigma = 49.5 \pm 0.9$

WF (5420 entries)

$\frac{\sigma}{\mu} = 10.64 \pm 0.21 \%$

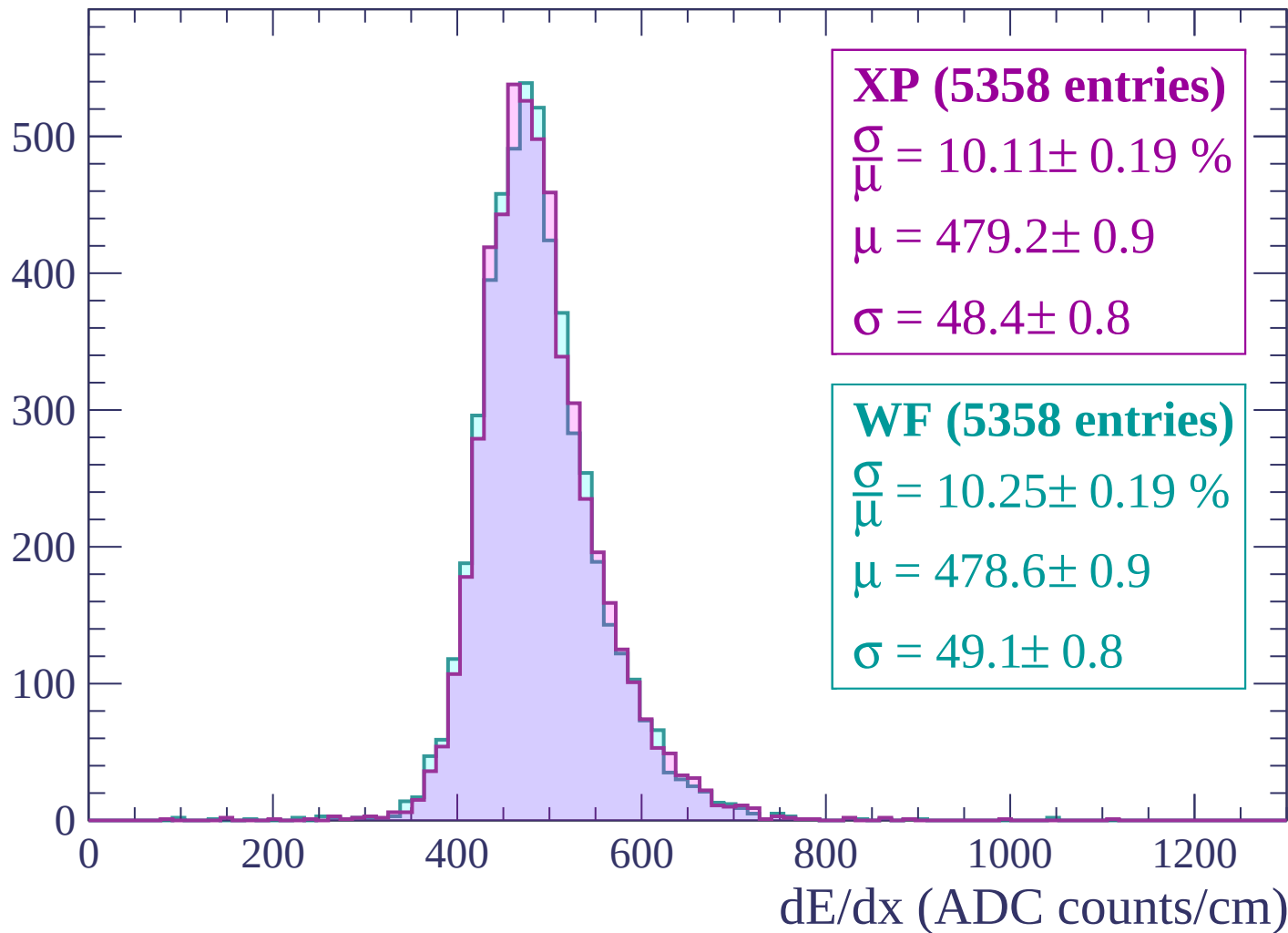
$\mu = 475.9 \pm 0.9$

$\sigma = 50.7 \pm 0.9$

dE/dx (ADC counts/cm)

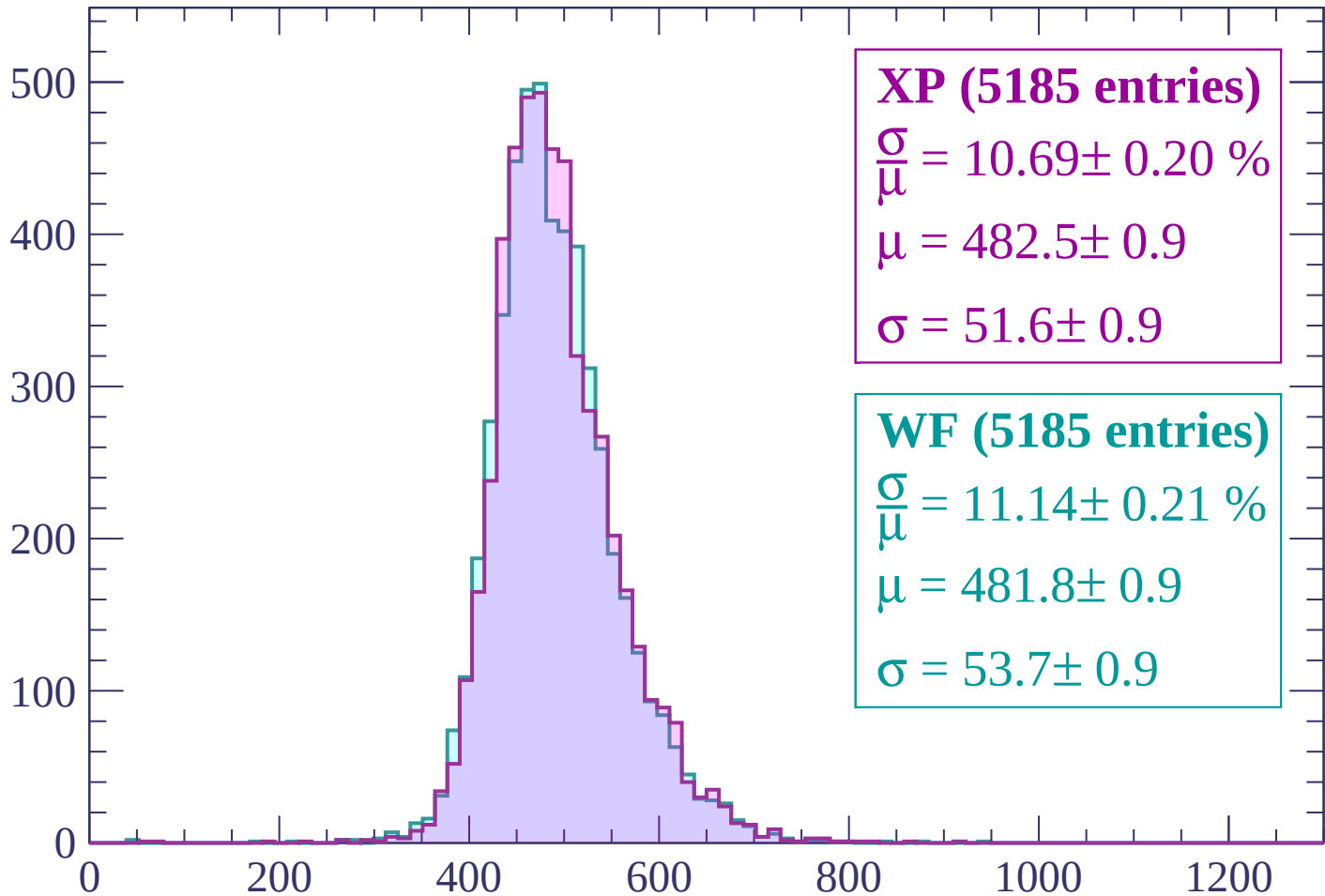
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (5185 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.20 \%$$

$$\mu = 482.5 \pm 0.9$$

$$\sigma = 51.6 \pm 0.9$$

WF (5185 entries)

$$\frac{\sigma}{\mu} = 11.14 \pm 0.21 \%$$

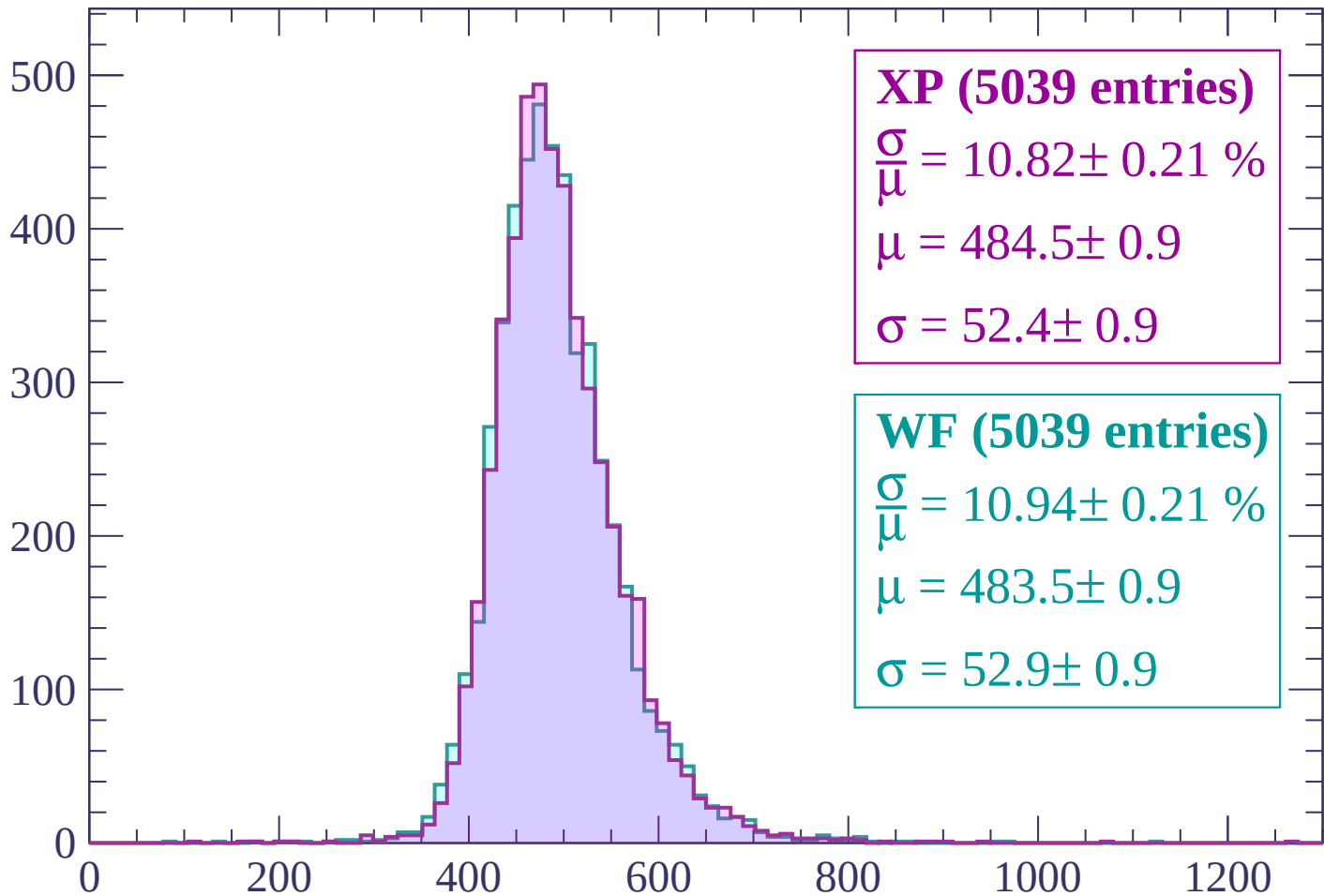
$$\mu = 481.8 \pm 0.9$$

$$\sigma = 53.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count



XP (5039 entries)

$\frac{\sigma}{\mu} = 10.82 \pm 0.21 \%$

$\mu = 484.5 \pm 0.9$

$\sigma = 52.4 \pm 0.9$

WF (5039 entries)

$\frac{\sigma}{\mu} = 10.94 \pm 0.21 \%$

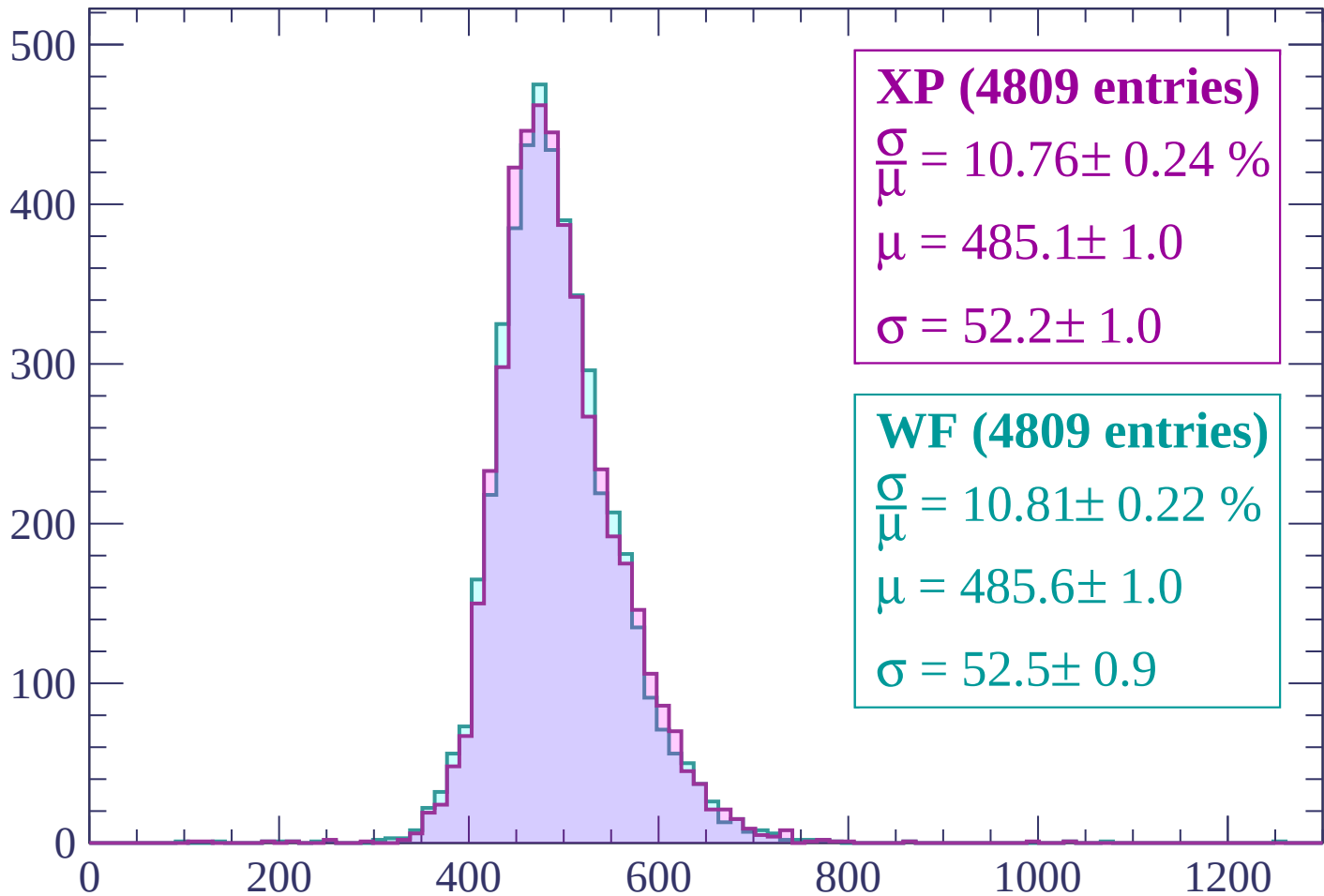
$\mu = 483.5 \pm 0.9$

$\sigma = 52.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (4809 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.24 \%$$

$$\mu = 485.1 \pm 1.0$$

$$\sigma = 52.2 \pm 1.0$$

WF (4809 entries)

$$\frac{\sigma}{\mu} = 10.81 \pm 0.22 \%$$

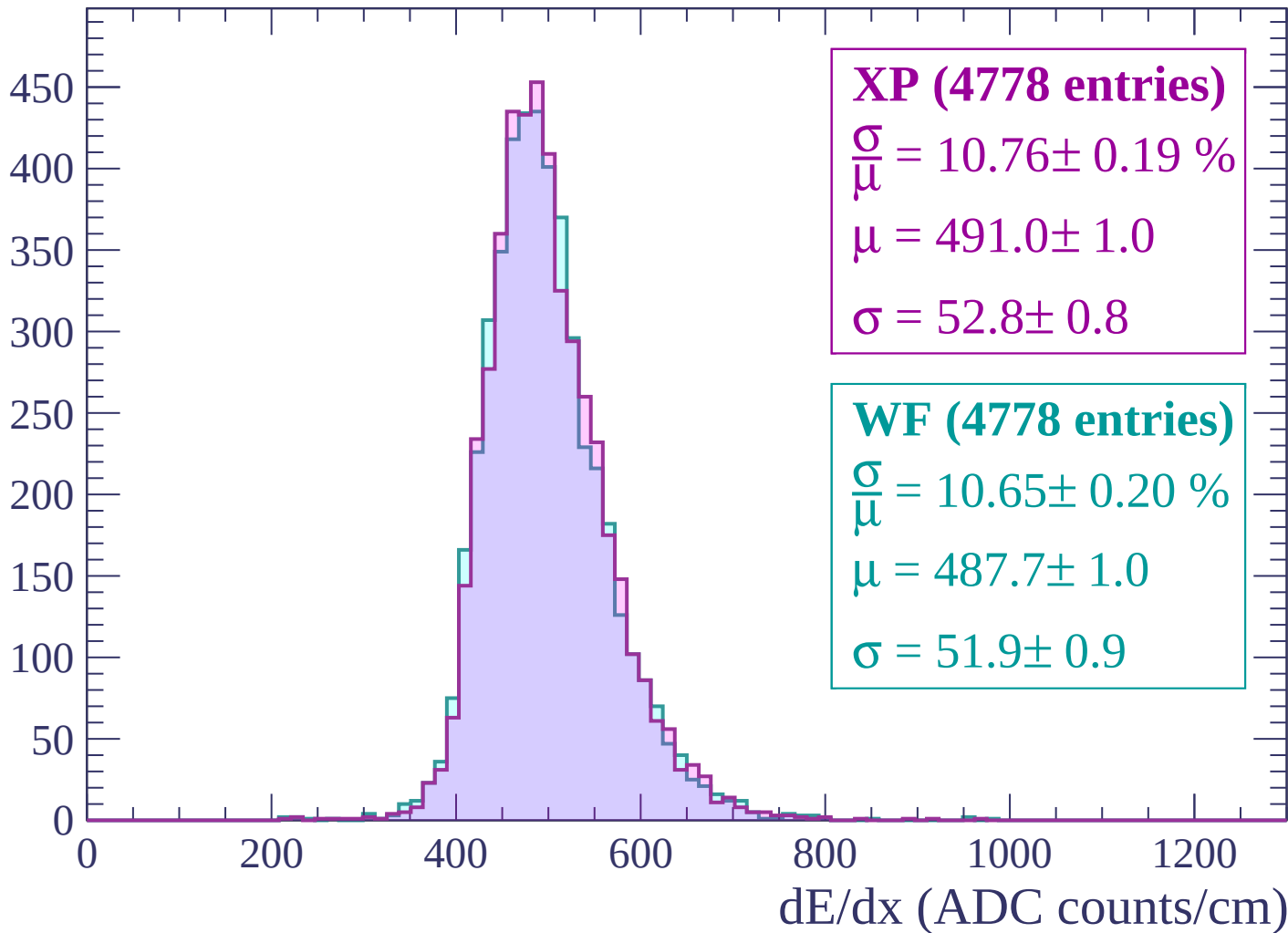
$$\mu = 485.6 \pm 1.0$$

$$\sigma = 52.5 \pm 0.9$$

dE/dx (ADC counts/cm)

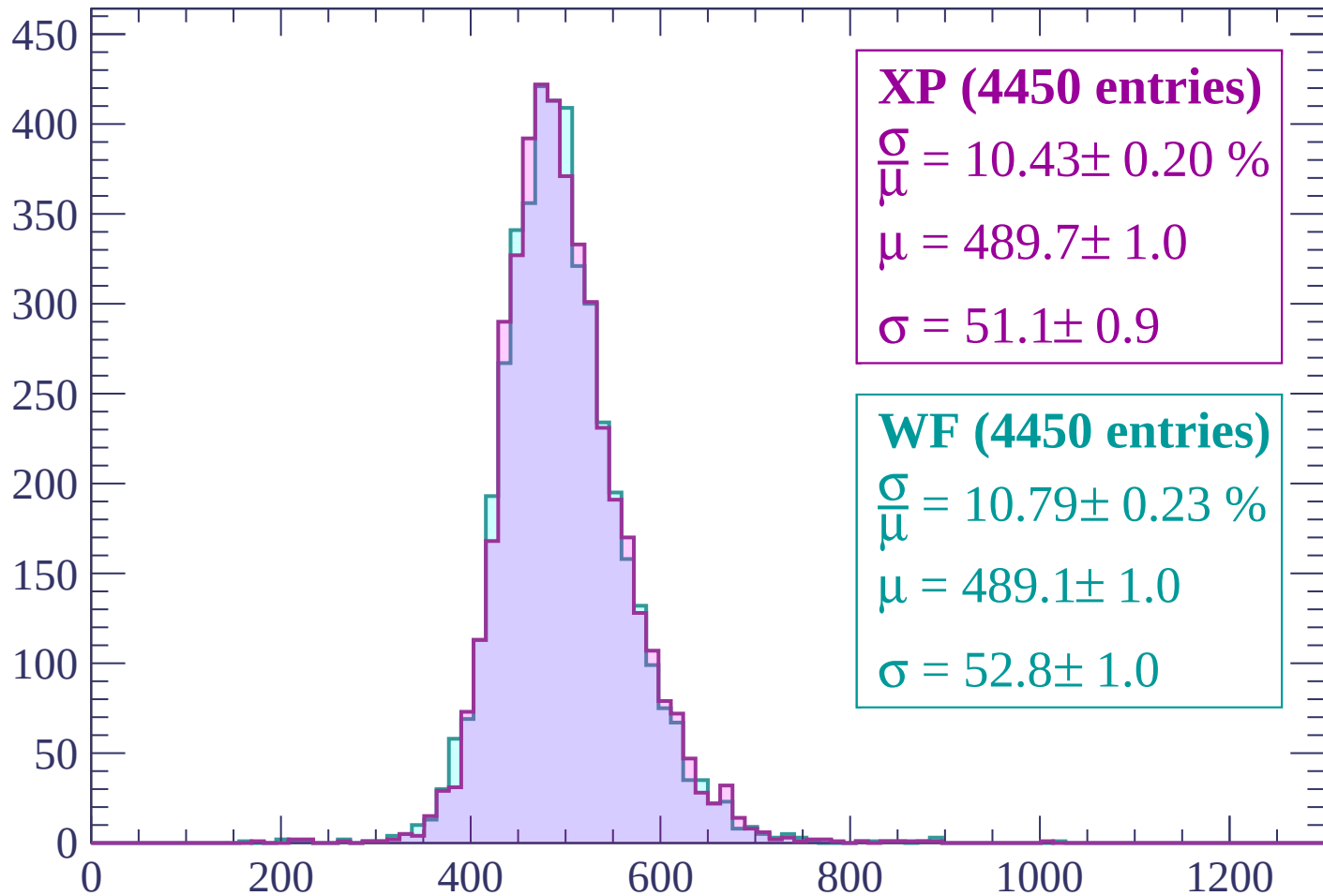
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (4450 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.20 \%$

$\mu = 489.7 \pm 1.0$

$\sigma = 51.1 \pm 0.9$

WF (4450 entries)

$\frac{\sigma}{\mu} = 10.79 \pm 0.23 \%$

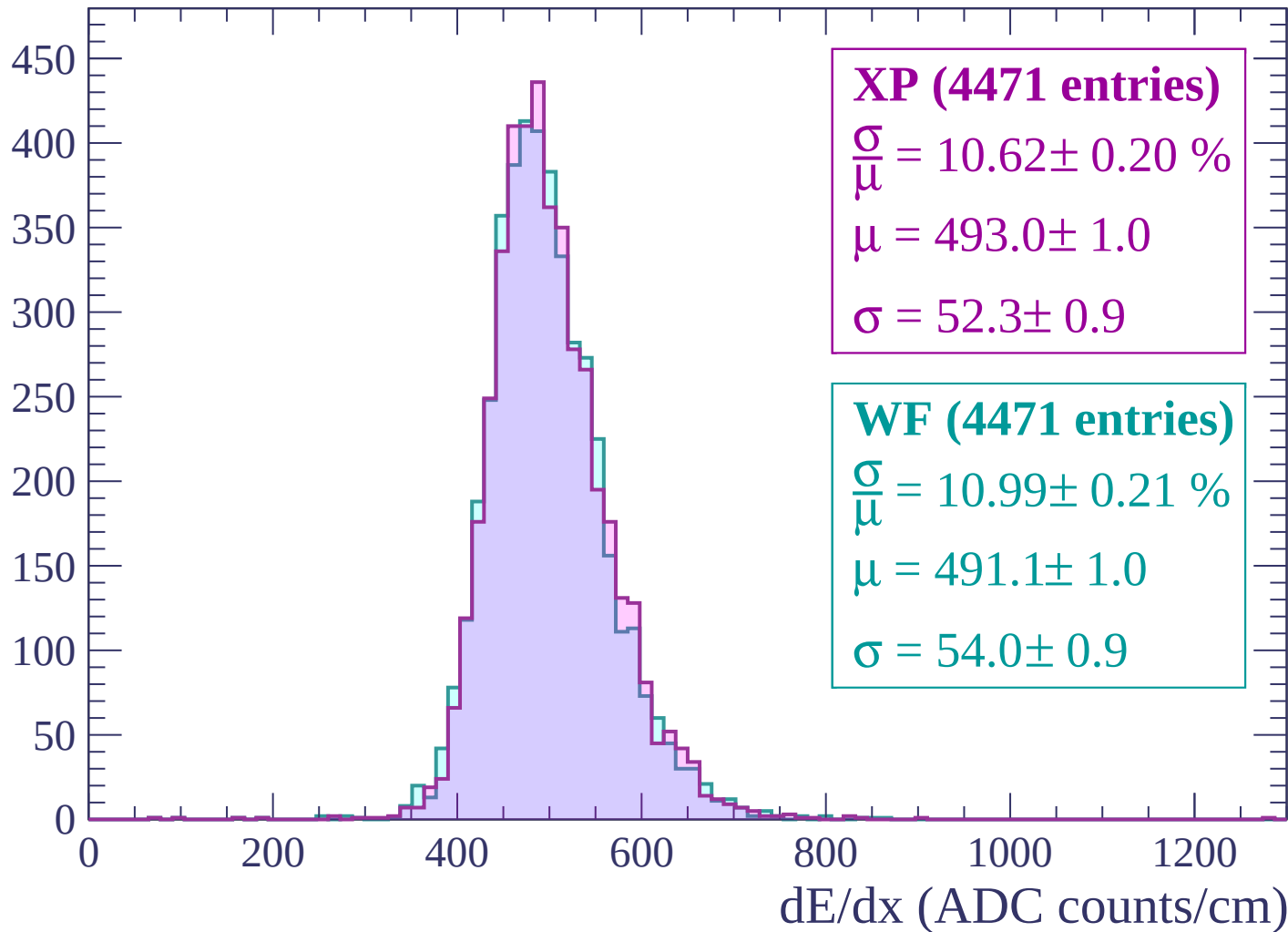
$\mu = 489.1 \pm 1.0$

$\sigma = 52.8 \pm 1.0$

dE/dx (ADC counts/cm)

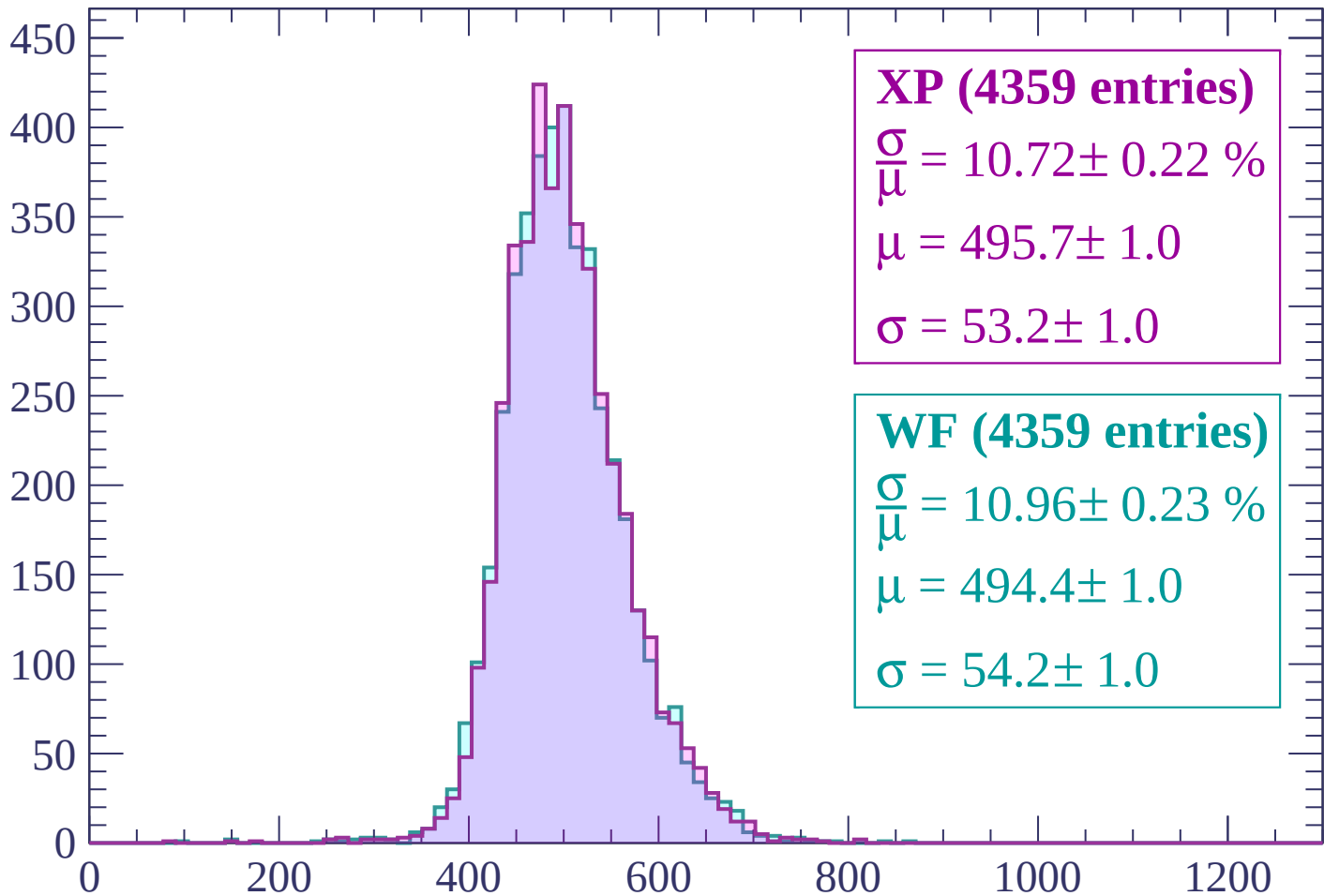
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

Count



XP (4359 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$

$\mu = 495.7 \pm 1.0$

$\sigma = 53.2 \pm 1.0$

WF (4359 entries)

$\frac{\sigma}{\mu} = 10.96 \pm 0.23 \%$

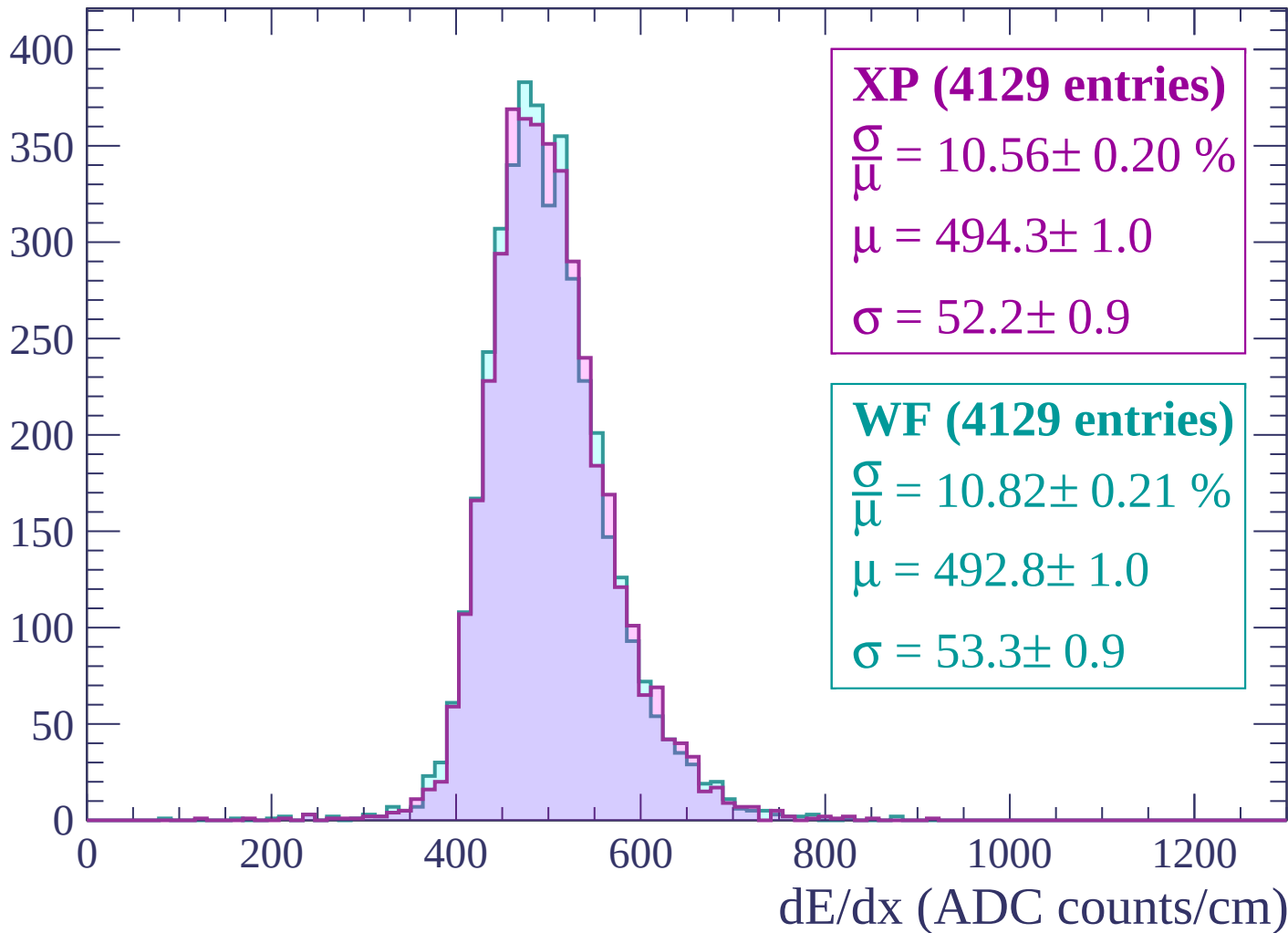
$\mu = 494.4 \pm 1.0$

$\sigma = 54.2 \pm 1.0$

dE/dx (ADC counts/cm)

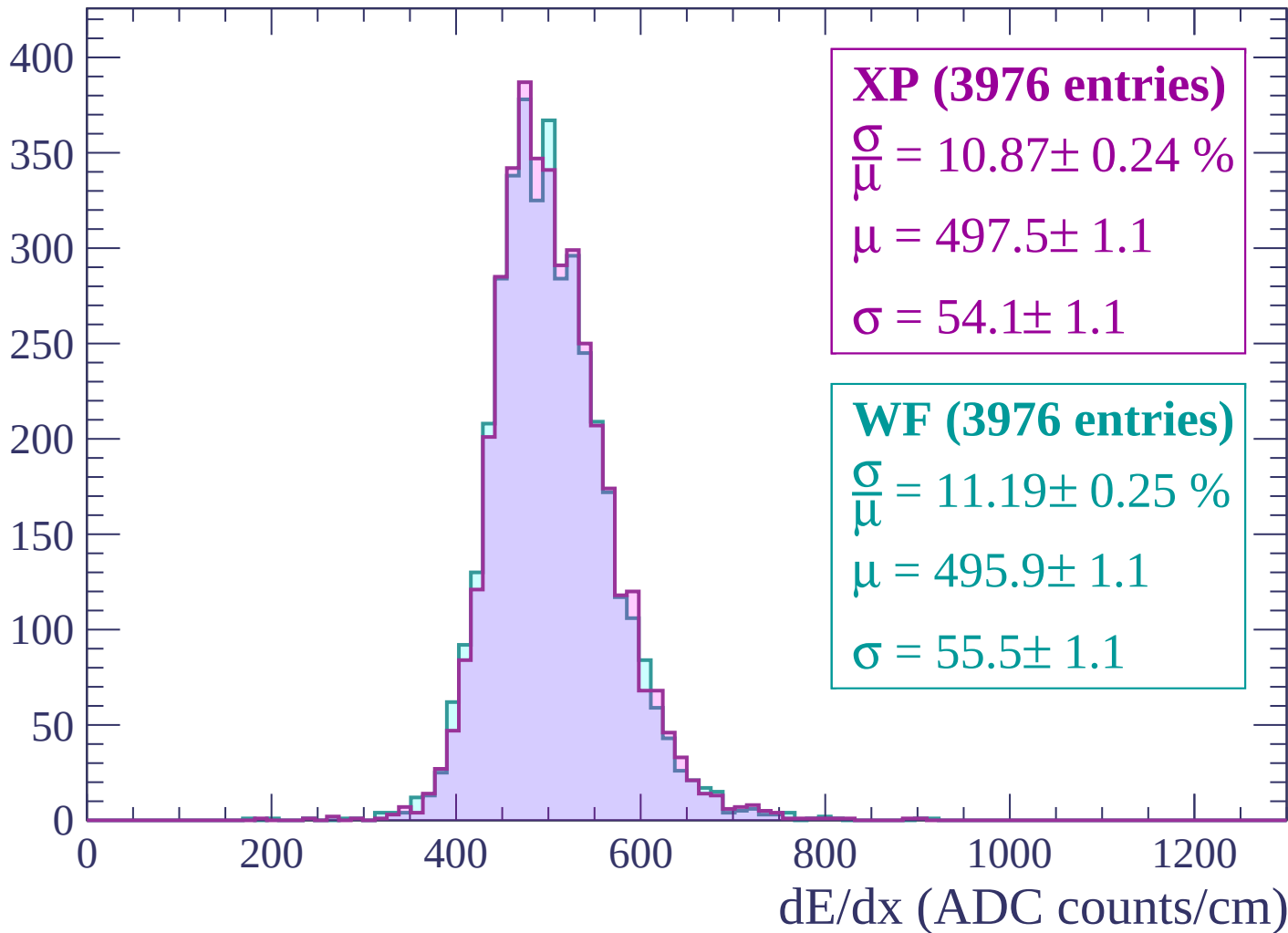
Energy loss | $2340 < p < 2400$

Count



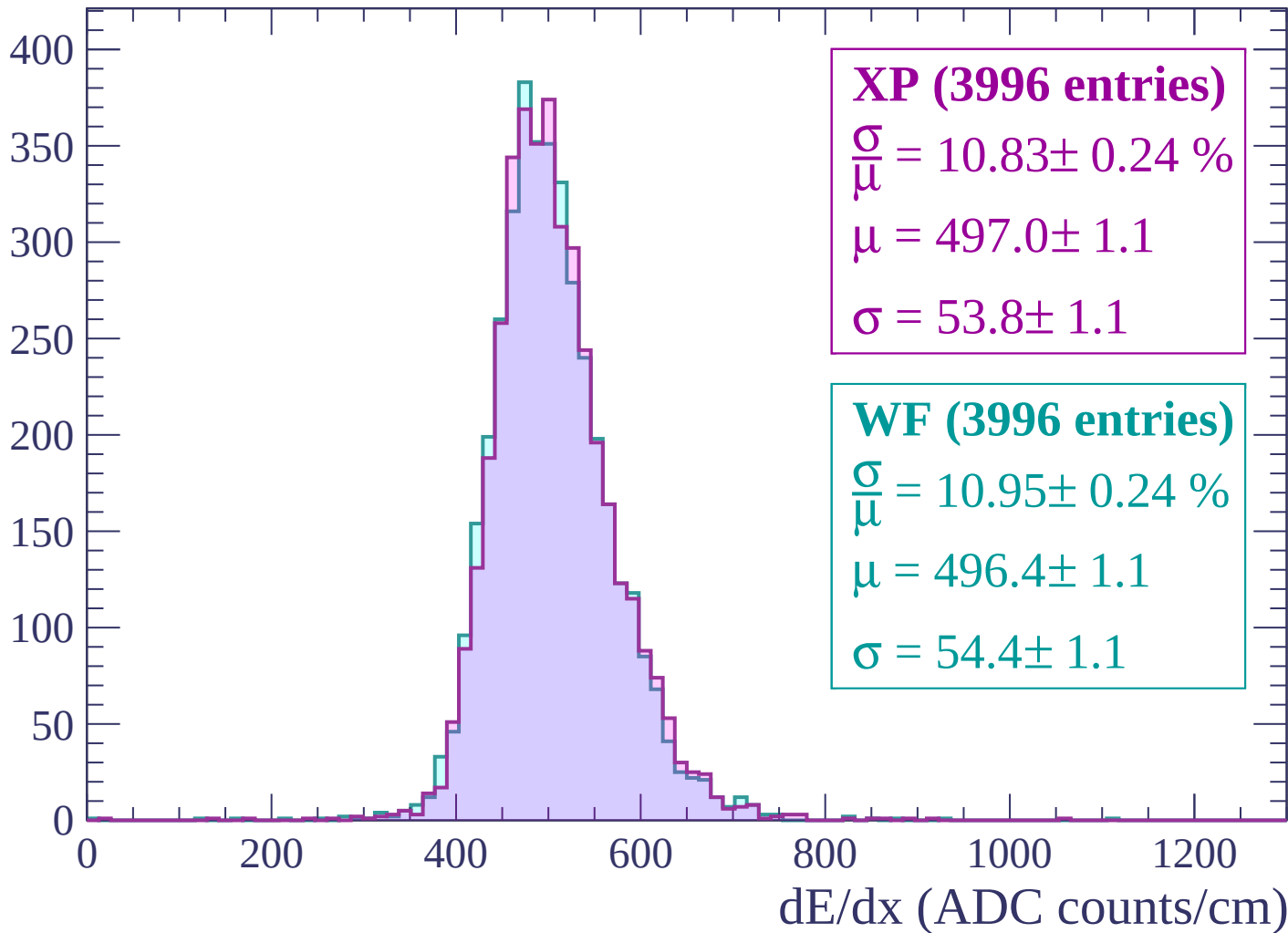
Energy loss | $2400 < p < 2460$

Count



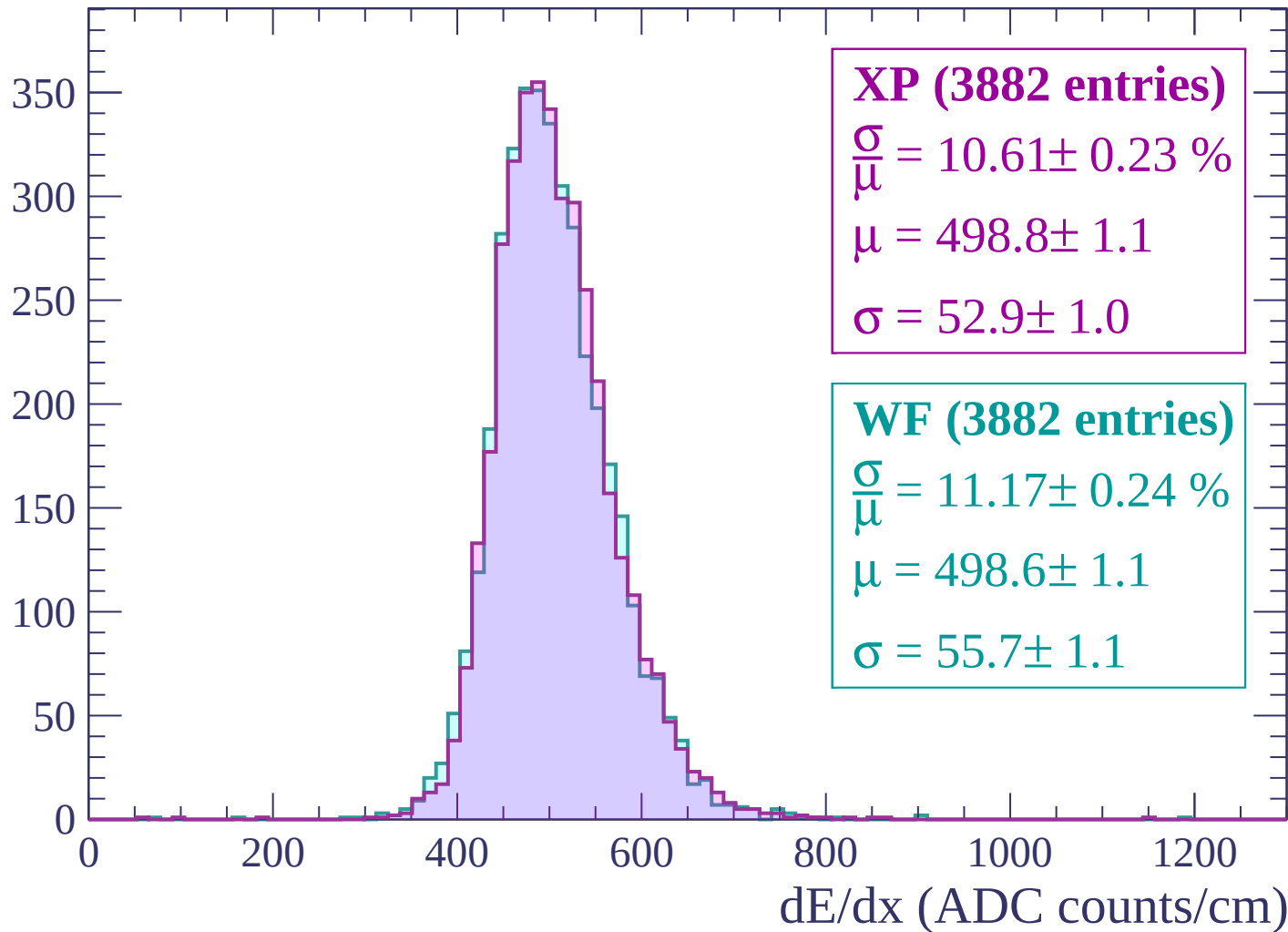
Energy loss | $2460 < p < 2520$

Count



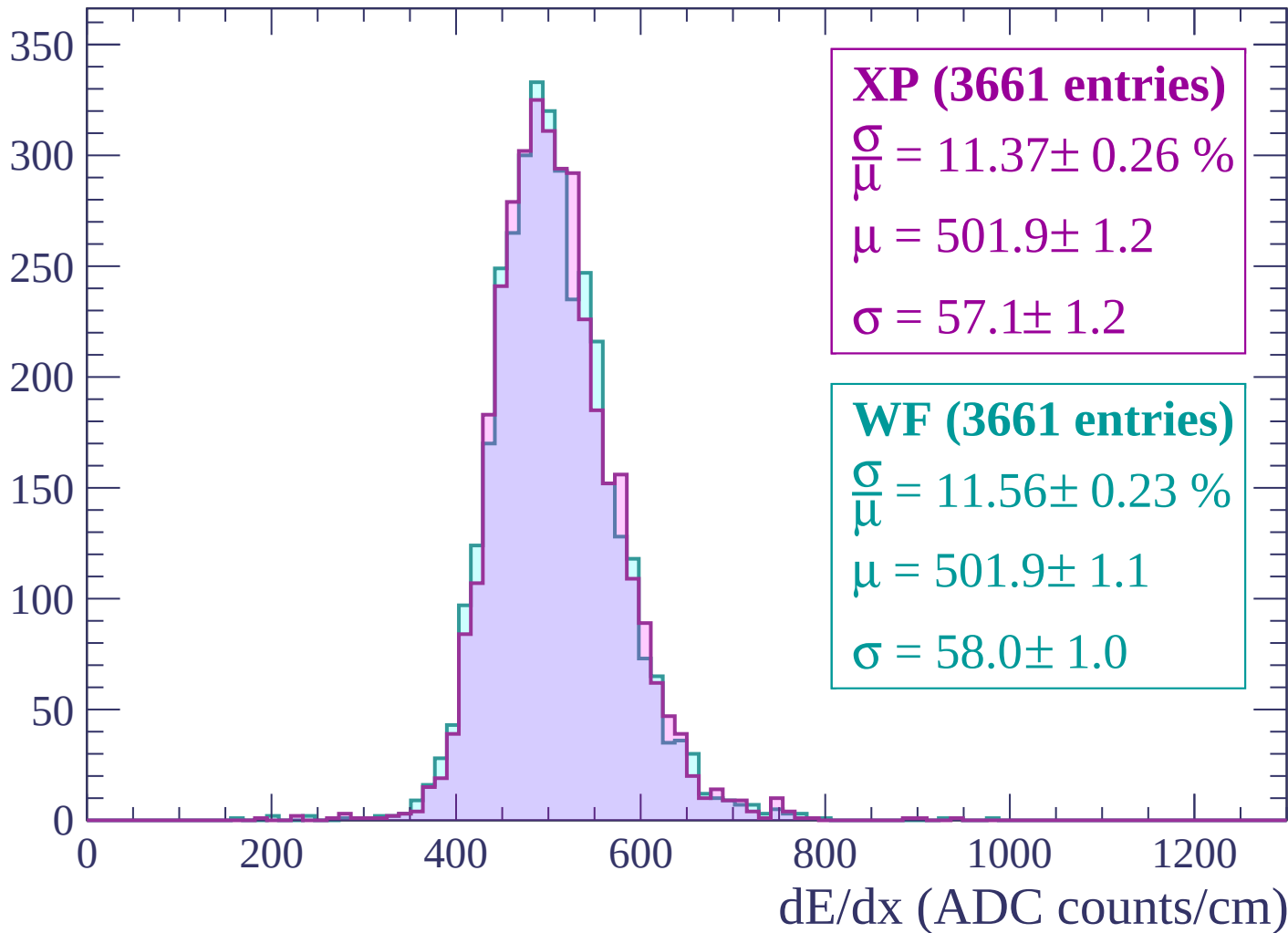
Energy loss | $2520 < p < 2580$

Count



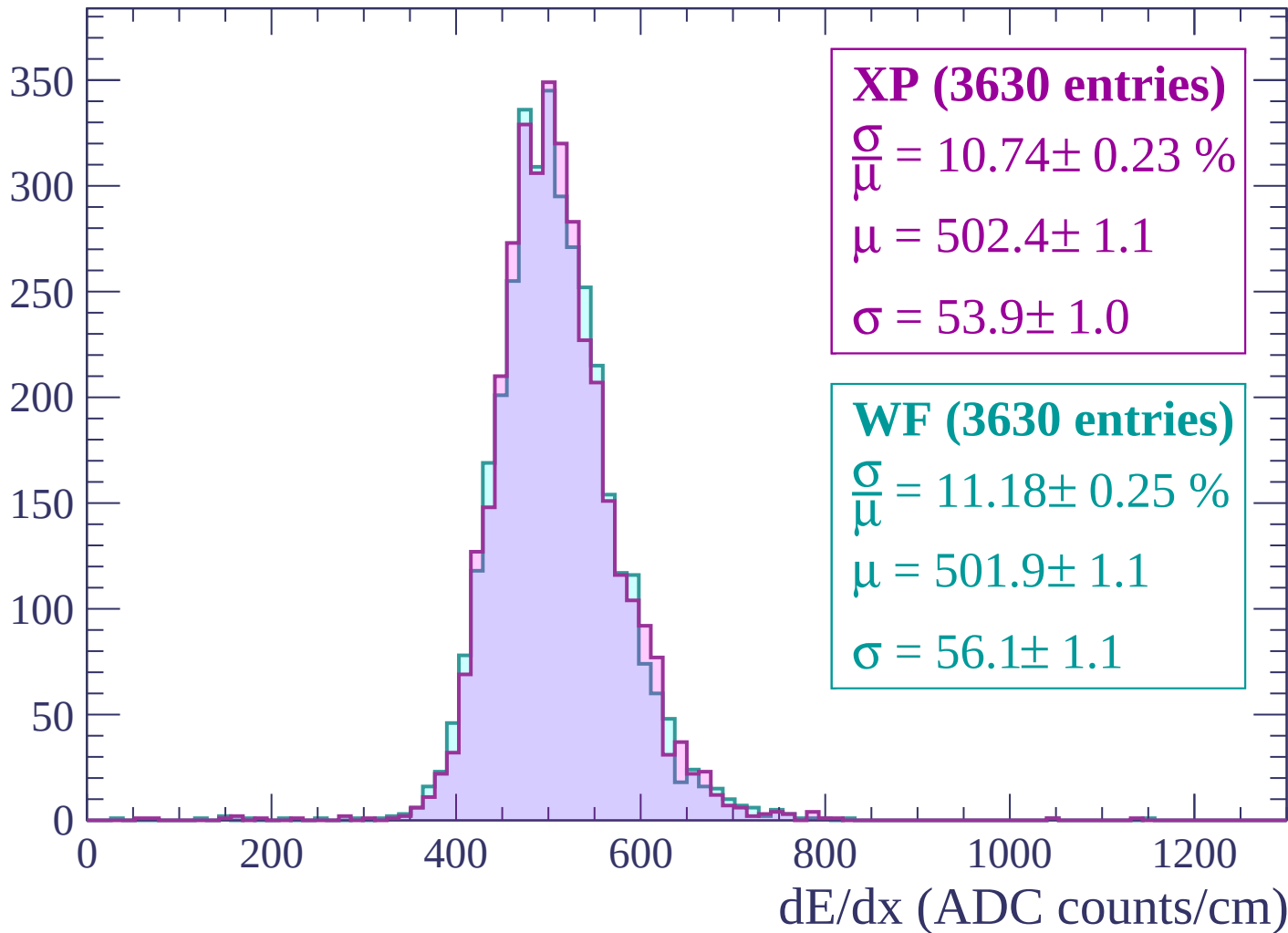
Energy loss | $2580 < p < 2640$

Count



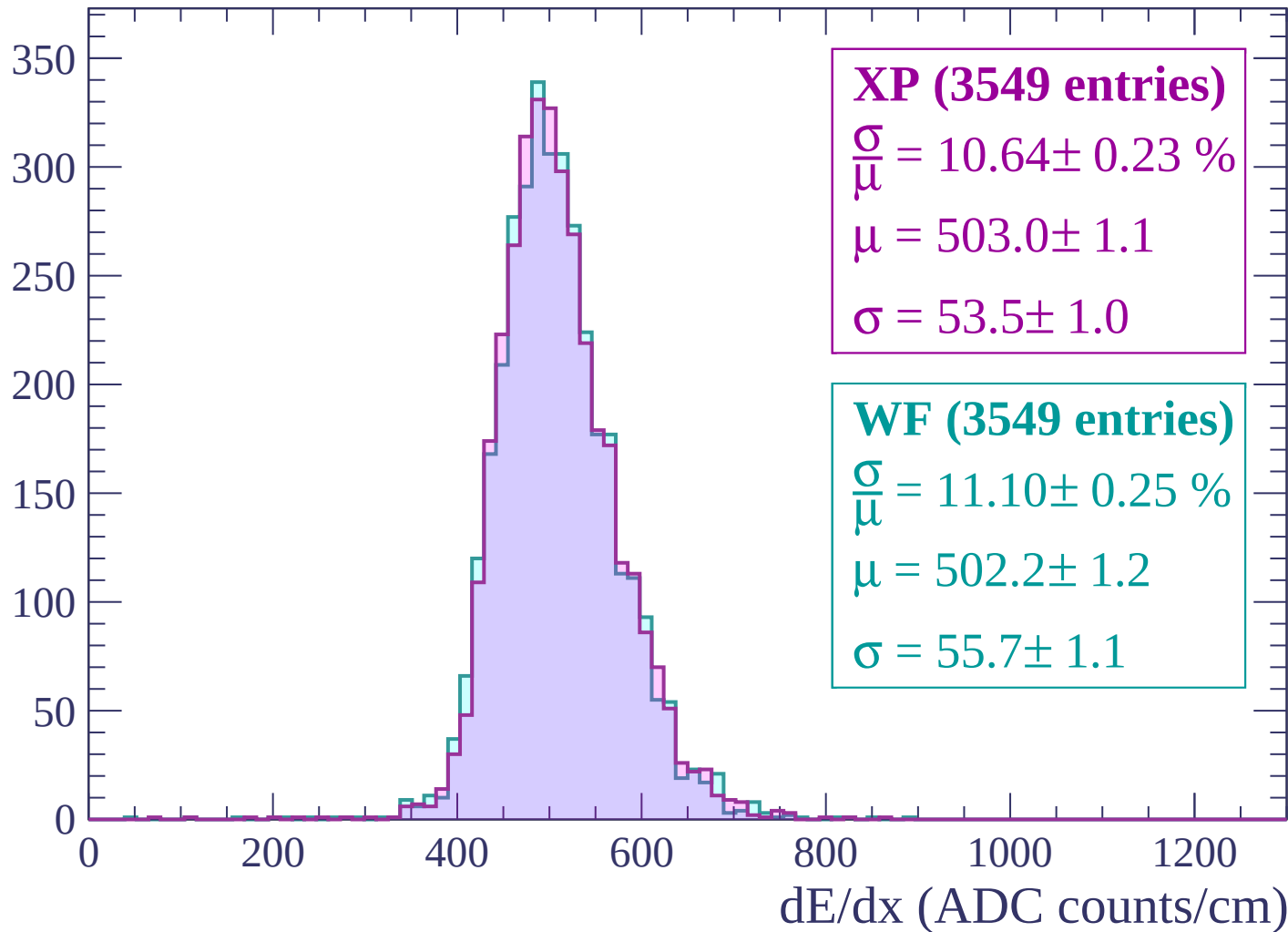
Energy loss | $2640 < p < 2700$

Count



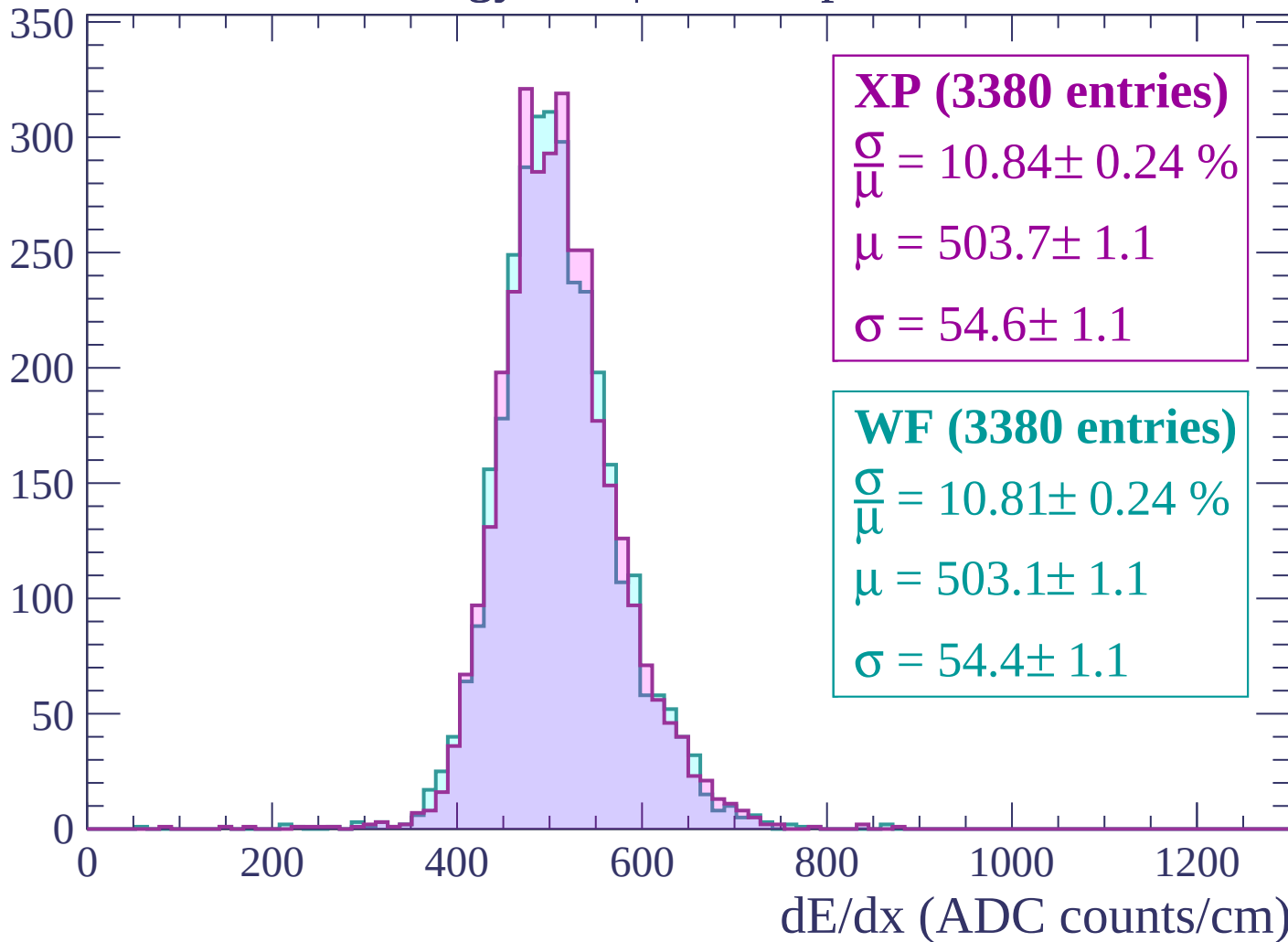
Energy loss | $2700 < p < 2760$

Count



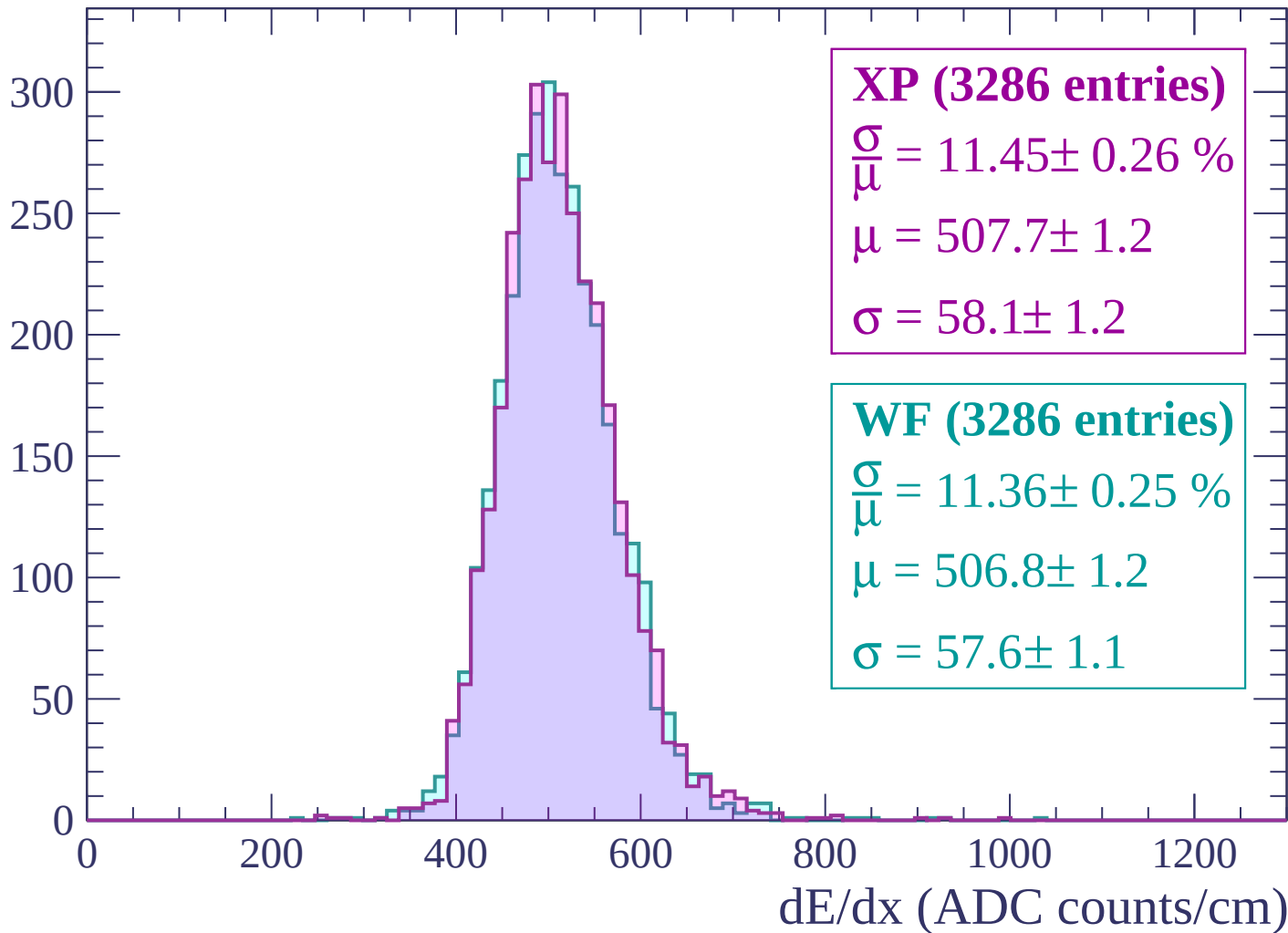
Energy loss | $2760 < p < 2820$

Count



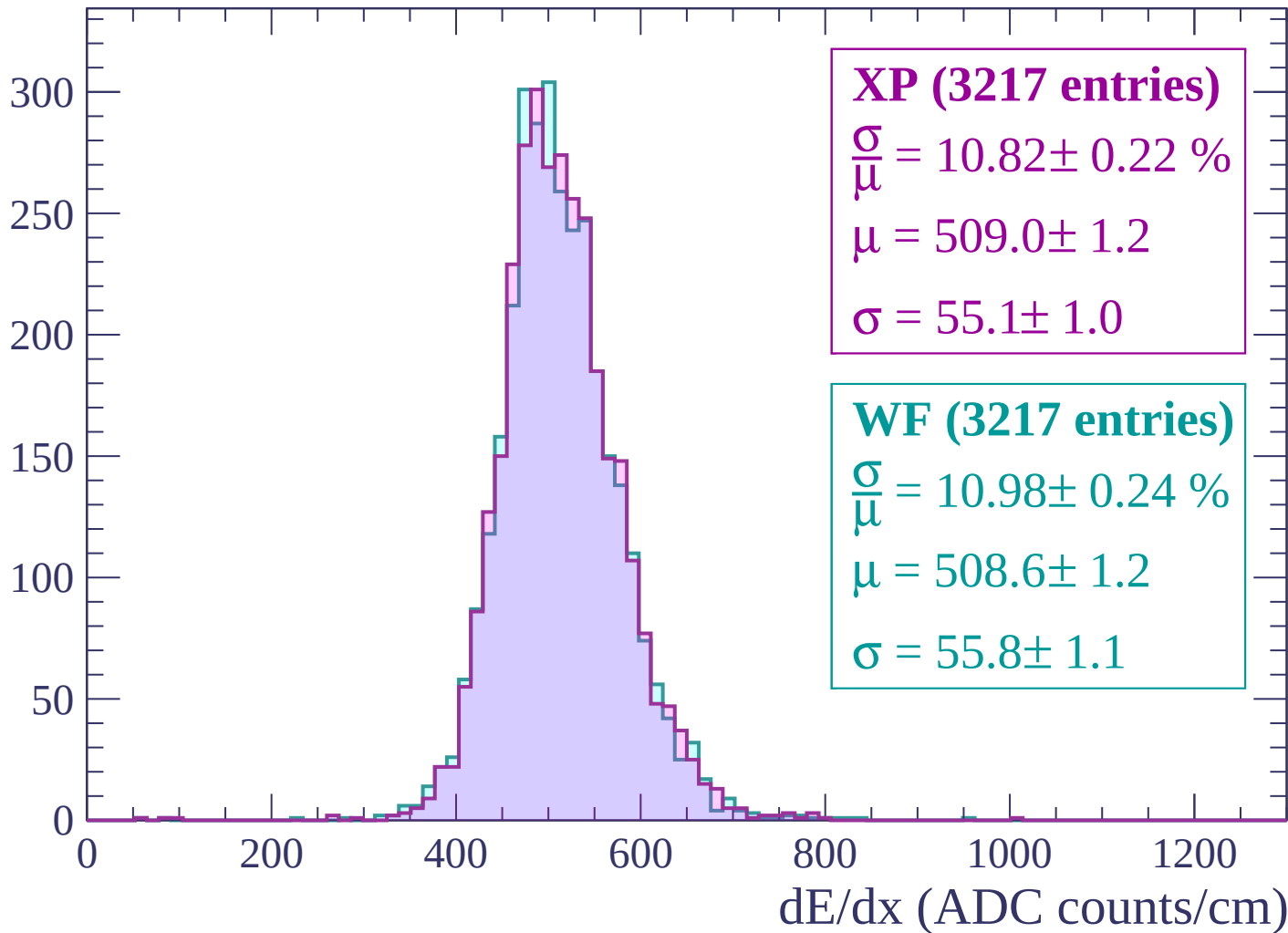
Energy loss | $2820 < p < 2880$

Count



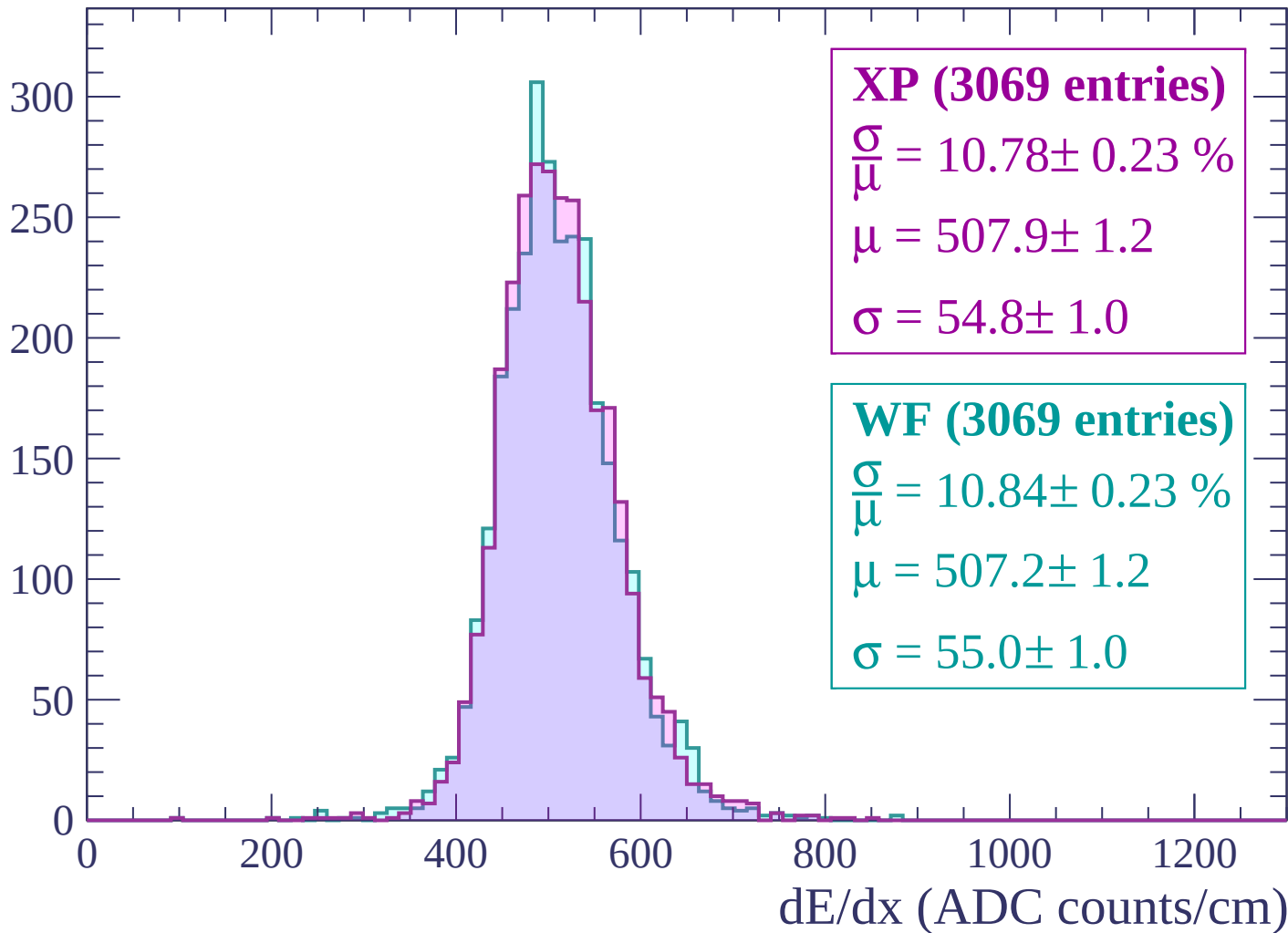
Energy loss | $2880 < p < 2940$

Count



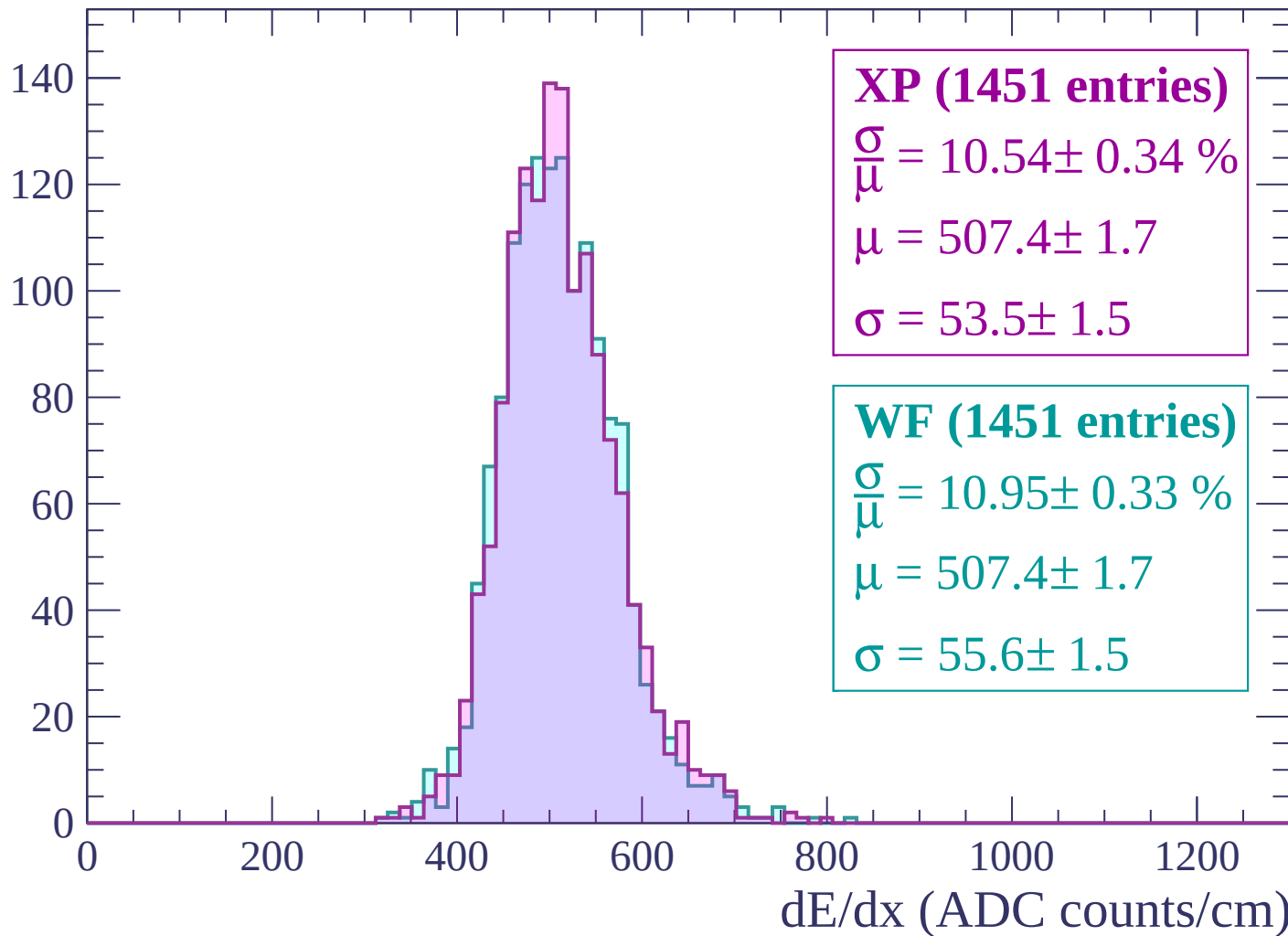
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$

$\mu = 6.2 \pm 2.4$

$\sigma = 0.3 \pm 1.3$

WF (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

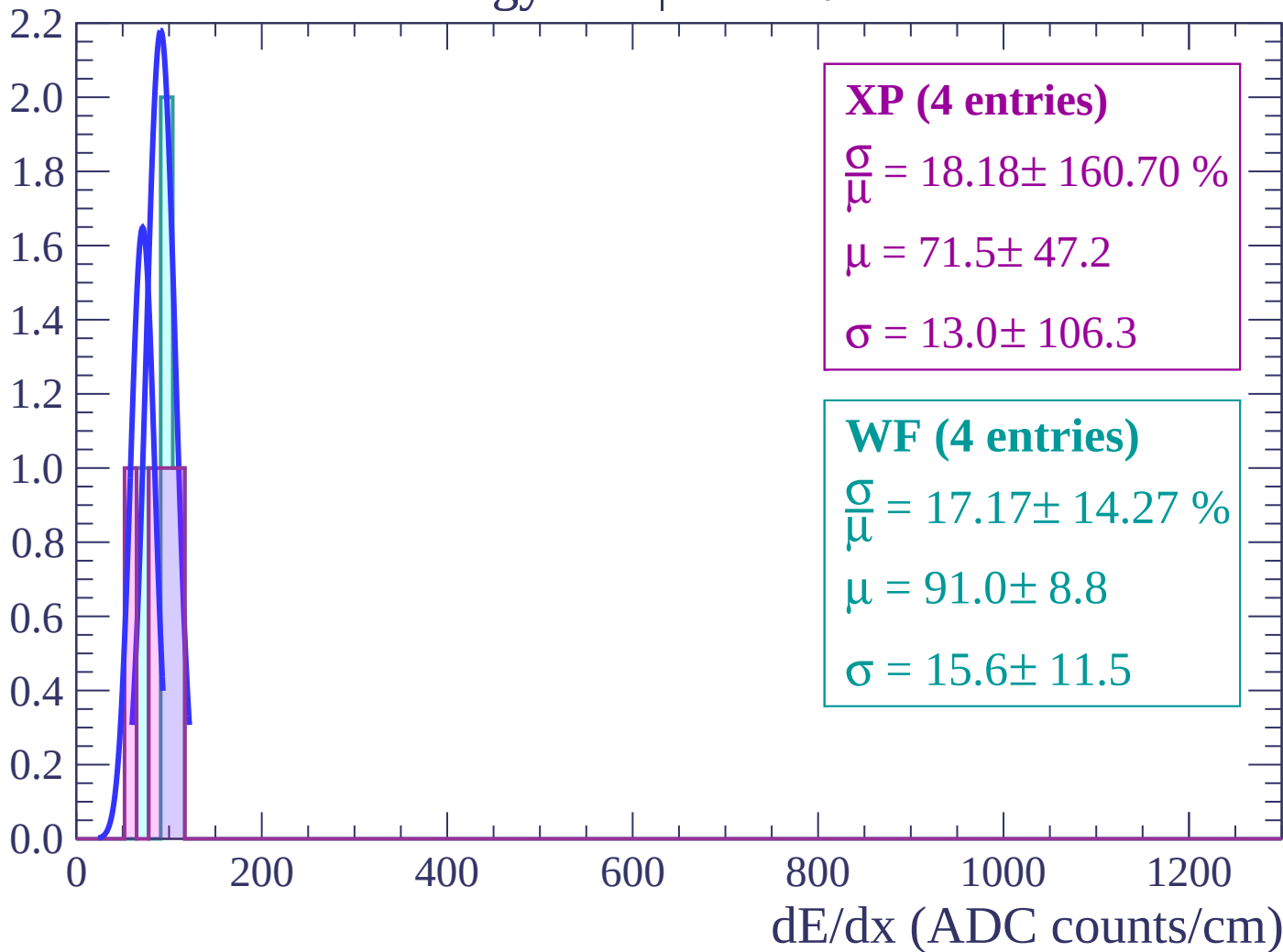
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

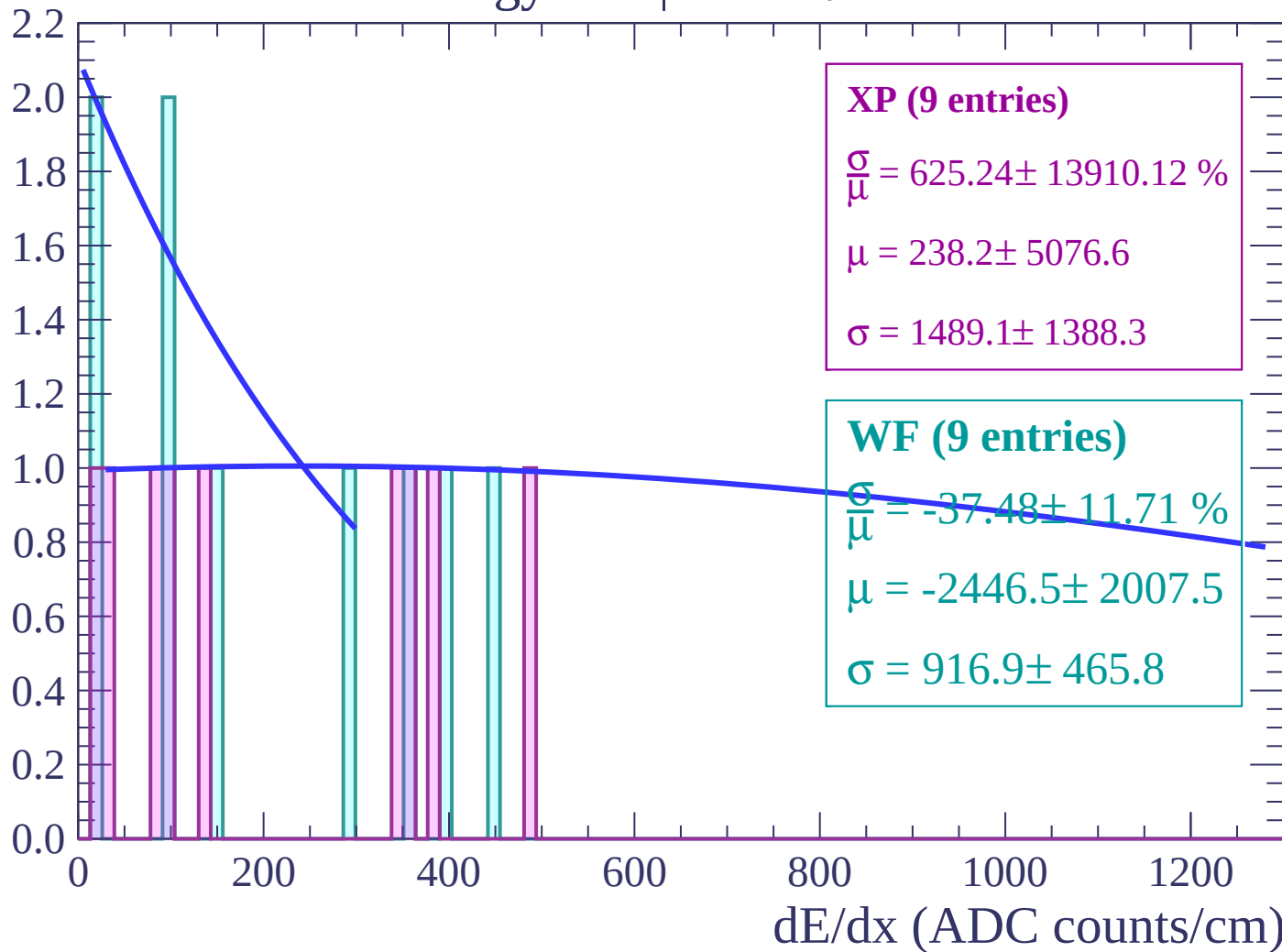
Energy loss | $-78 < \theta < -76$

Count



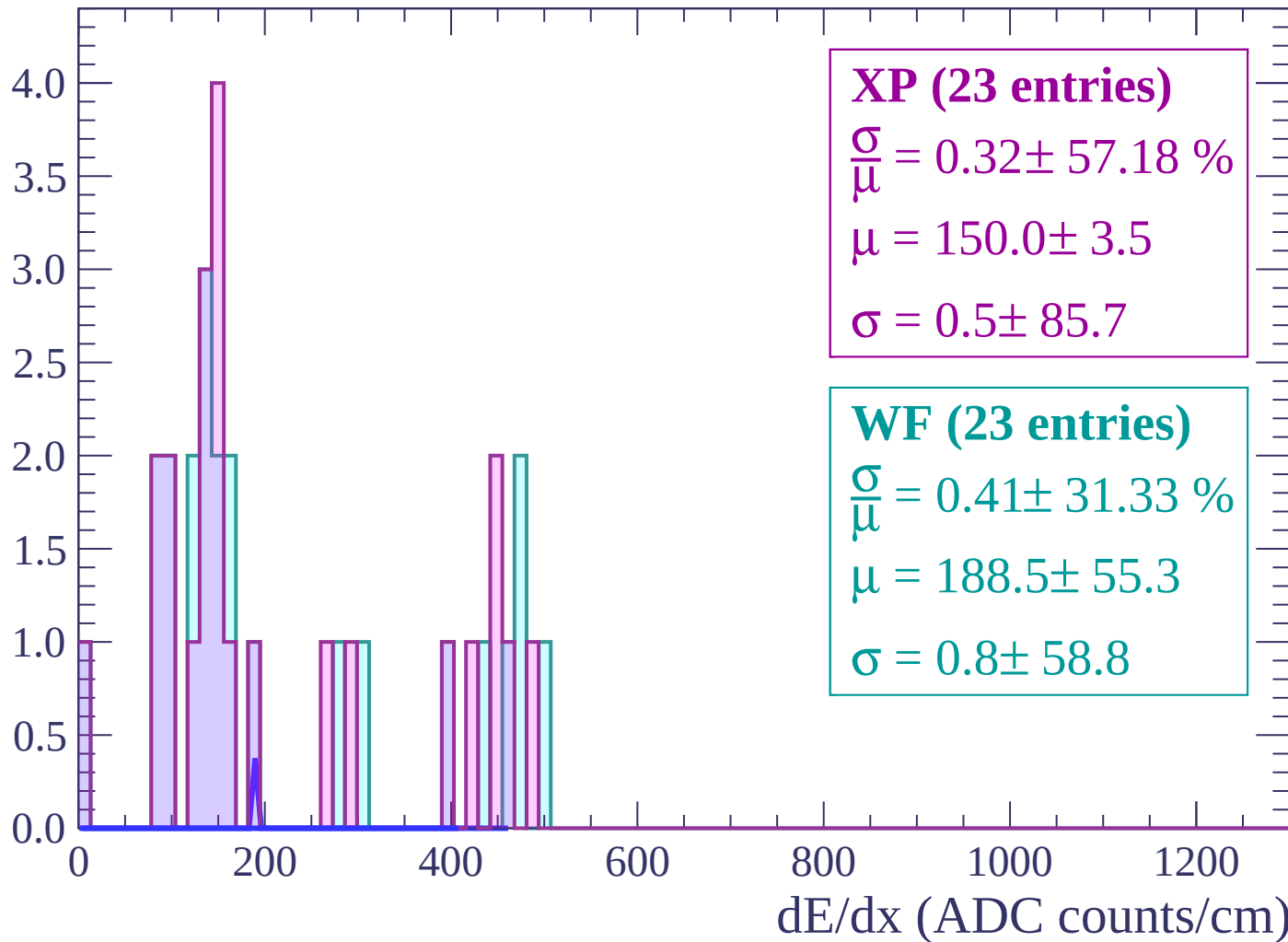
Energy loss | $-76 < \theta < -74$

Count



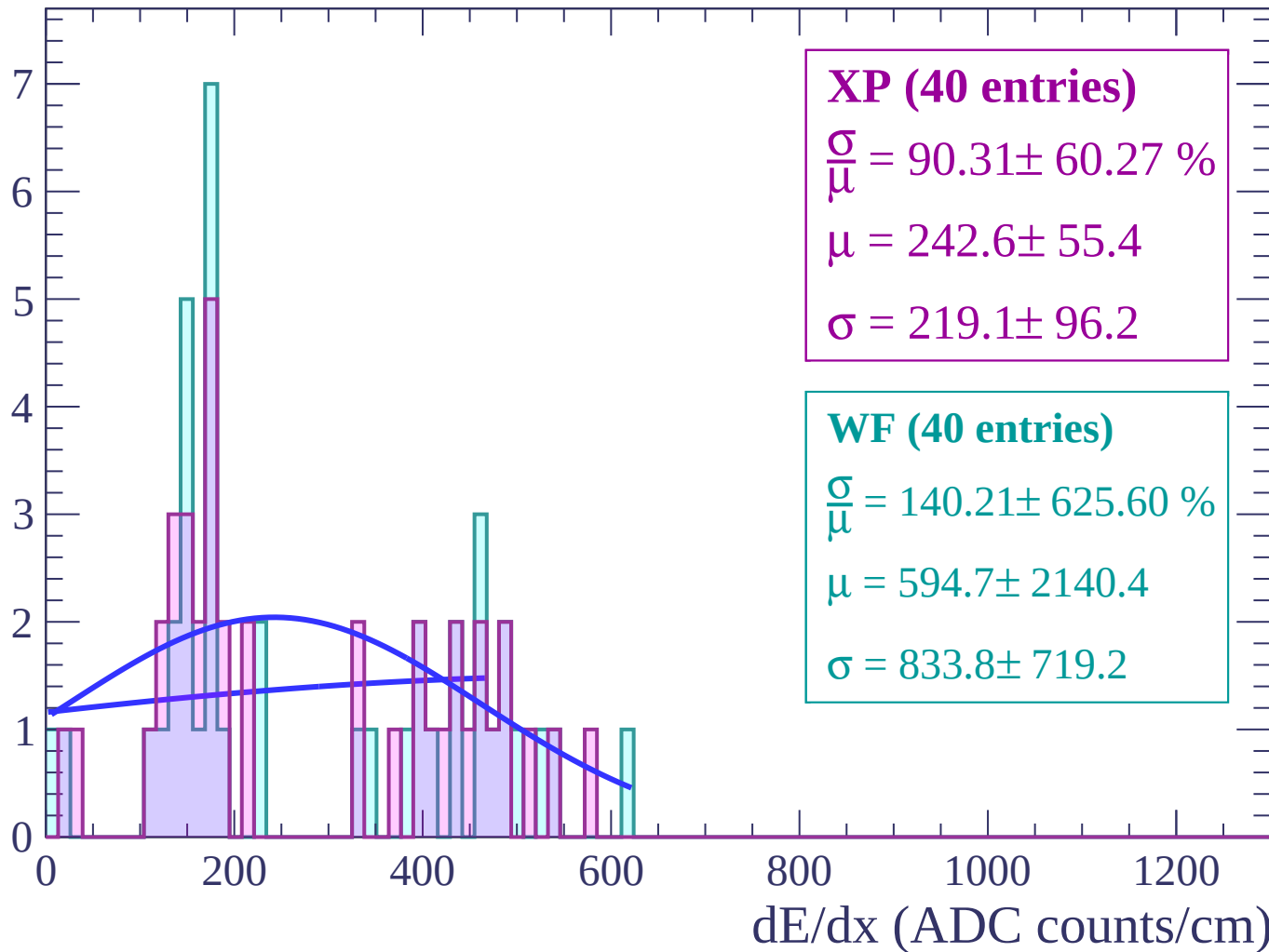
Energy loss | $-74 < \theta < -72$

Count



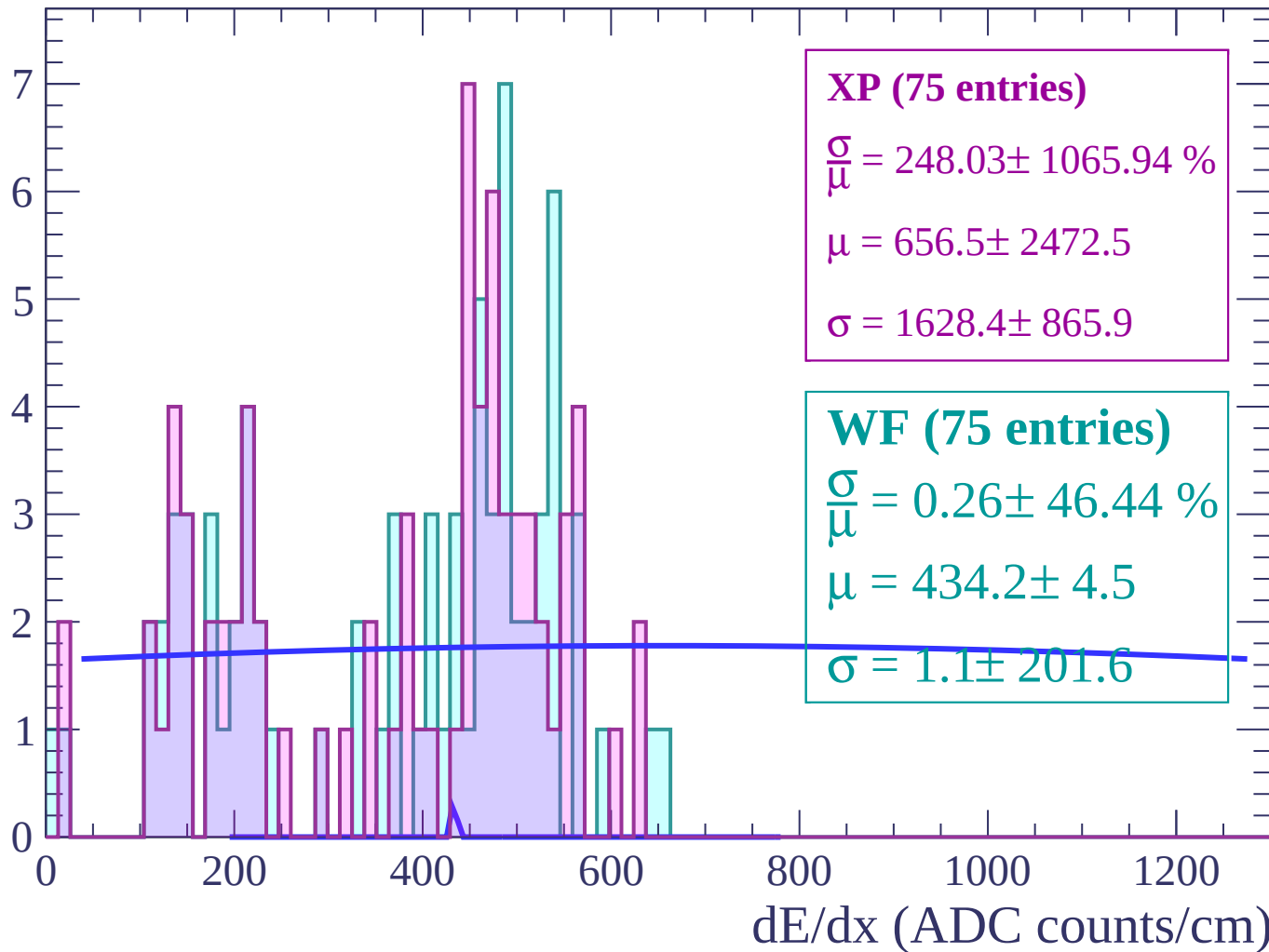
Energy loss | $-72 < \theta < -70$

Count



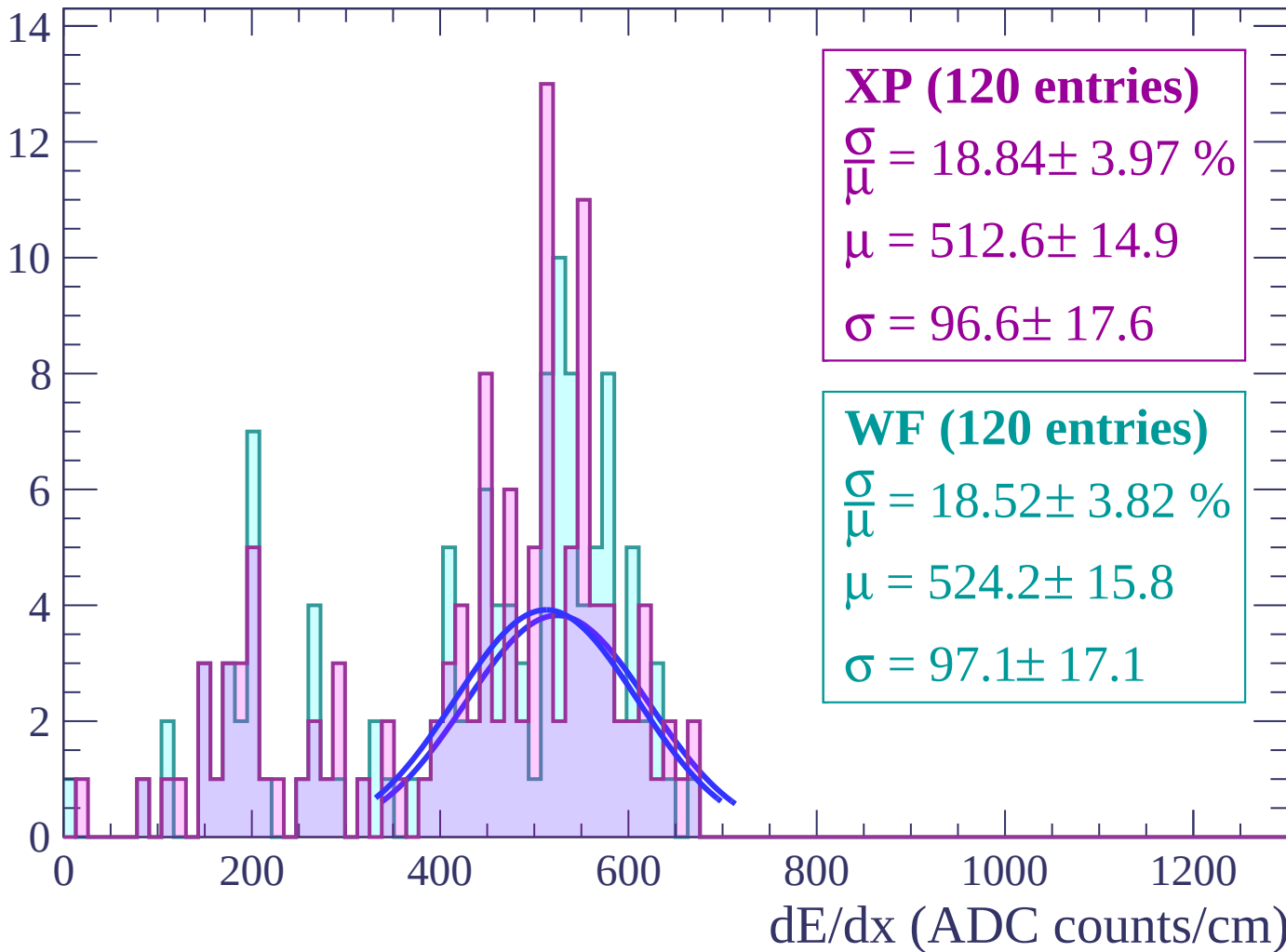
Energy loss | $-70 < \theta < -68$

Count



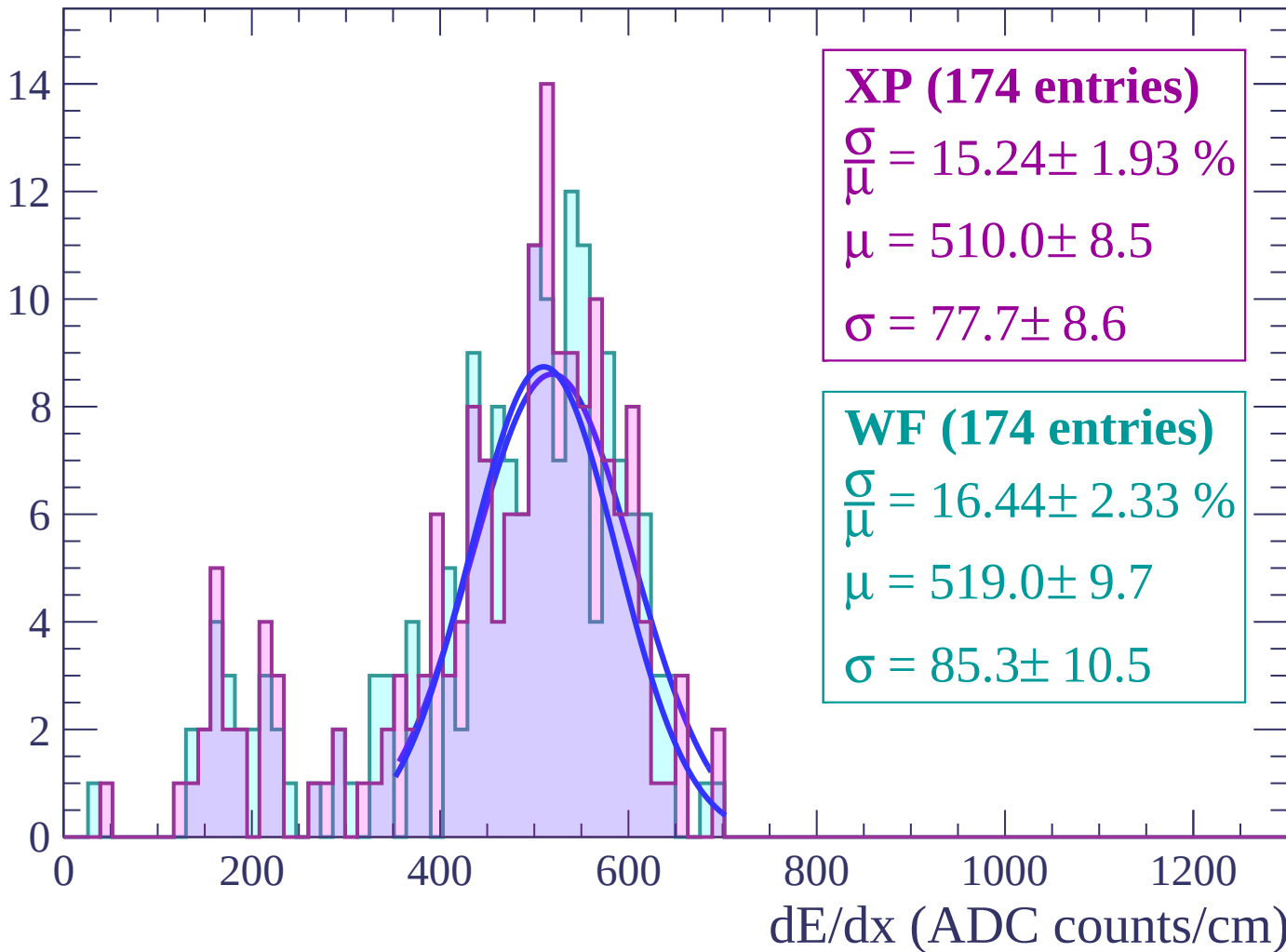
Energy loss | $-68 < \theta < -66$

Count



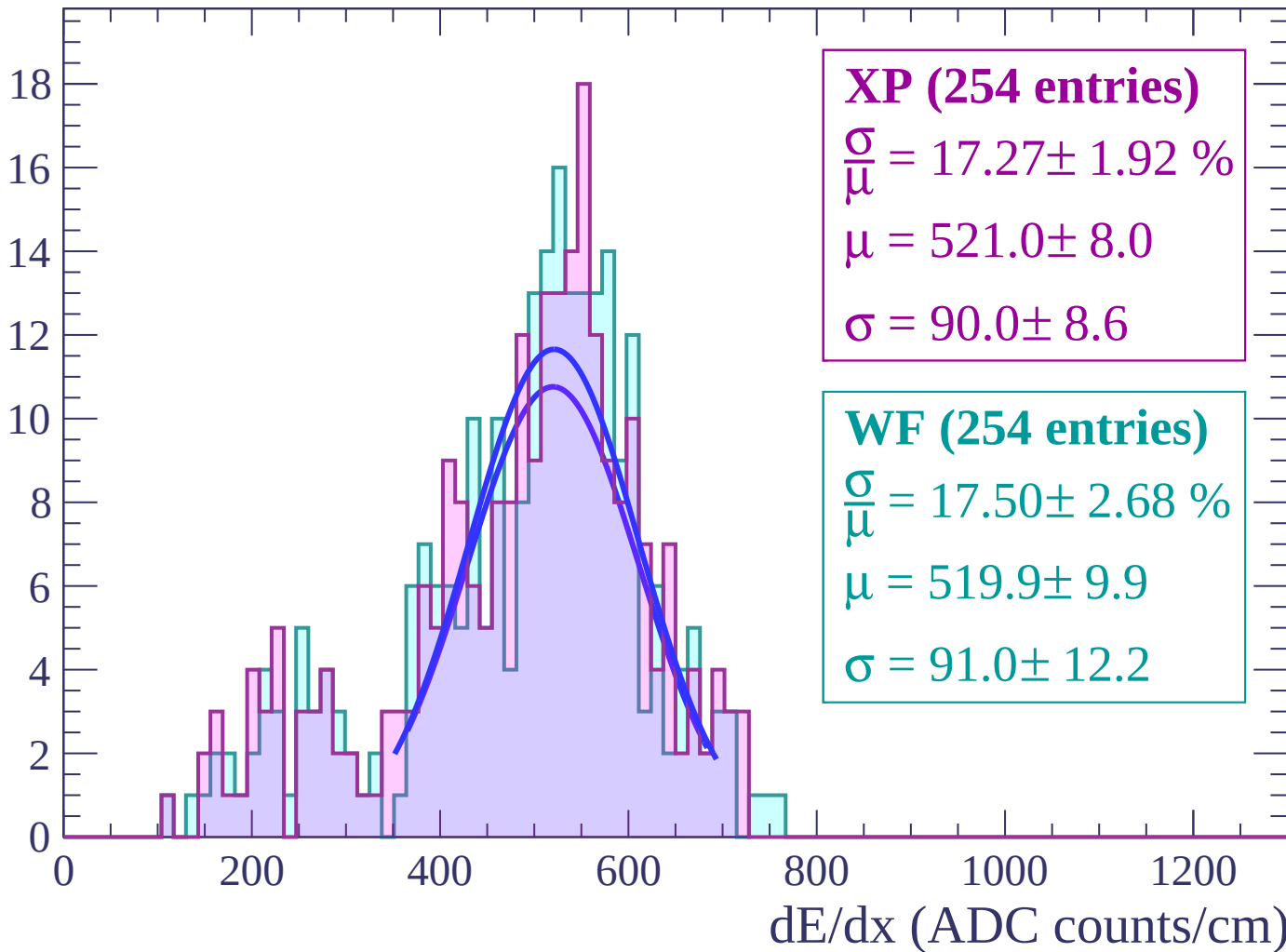
Energy loss | $-66 < \theta < -64$

Count



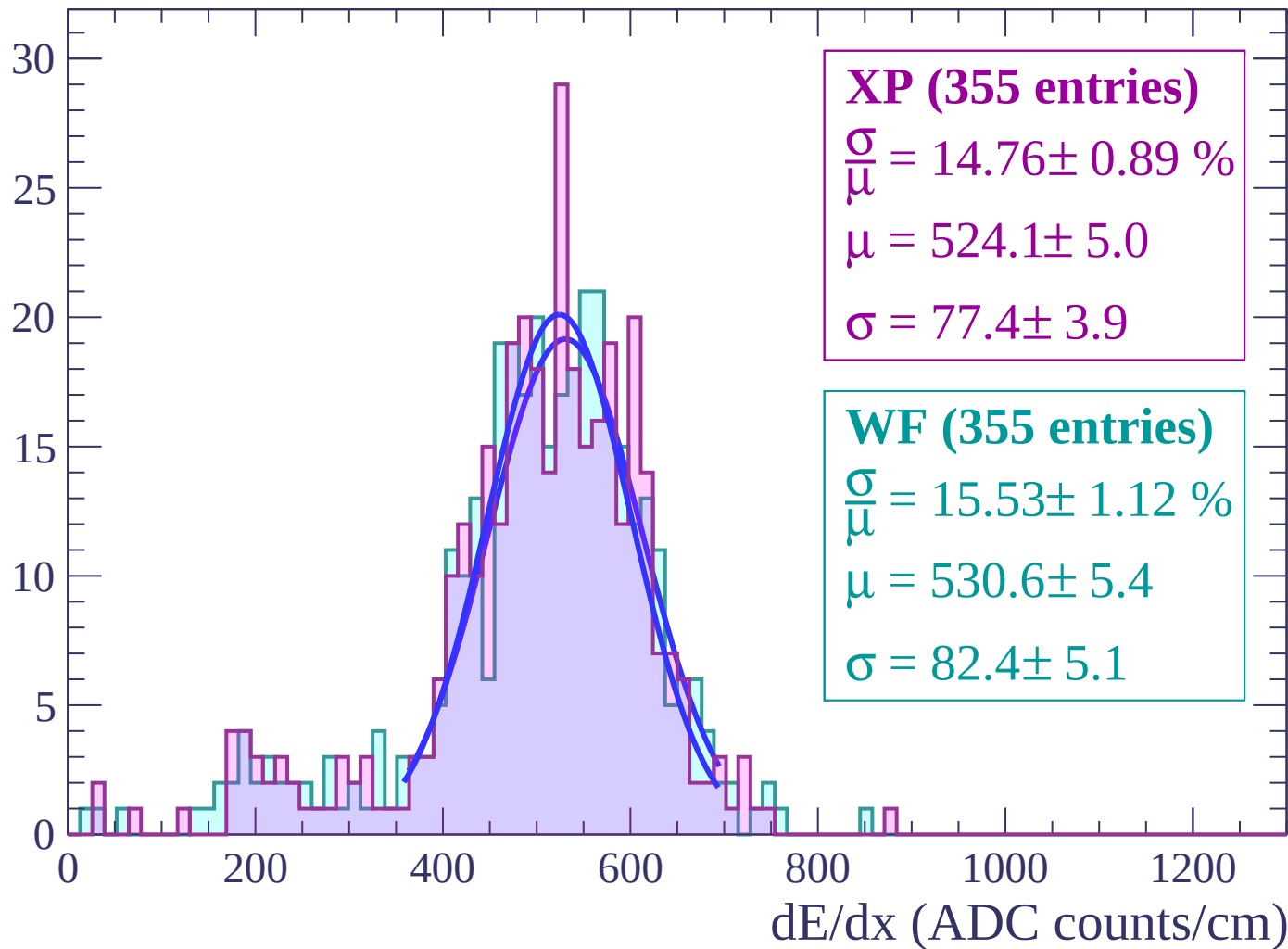
Energy loss | $-64 < \theta < -62$

Count



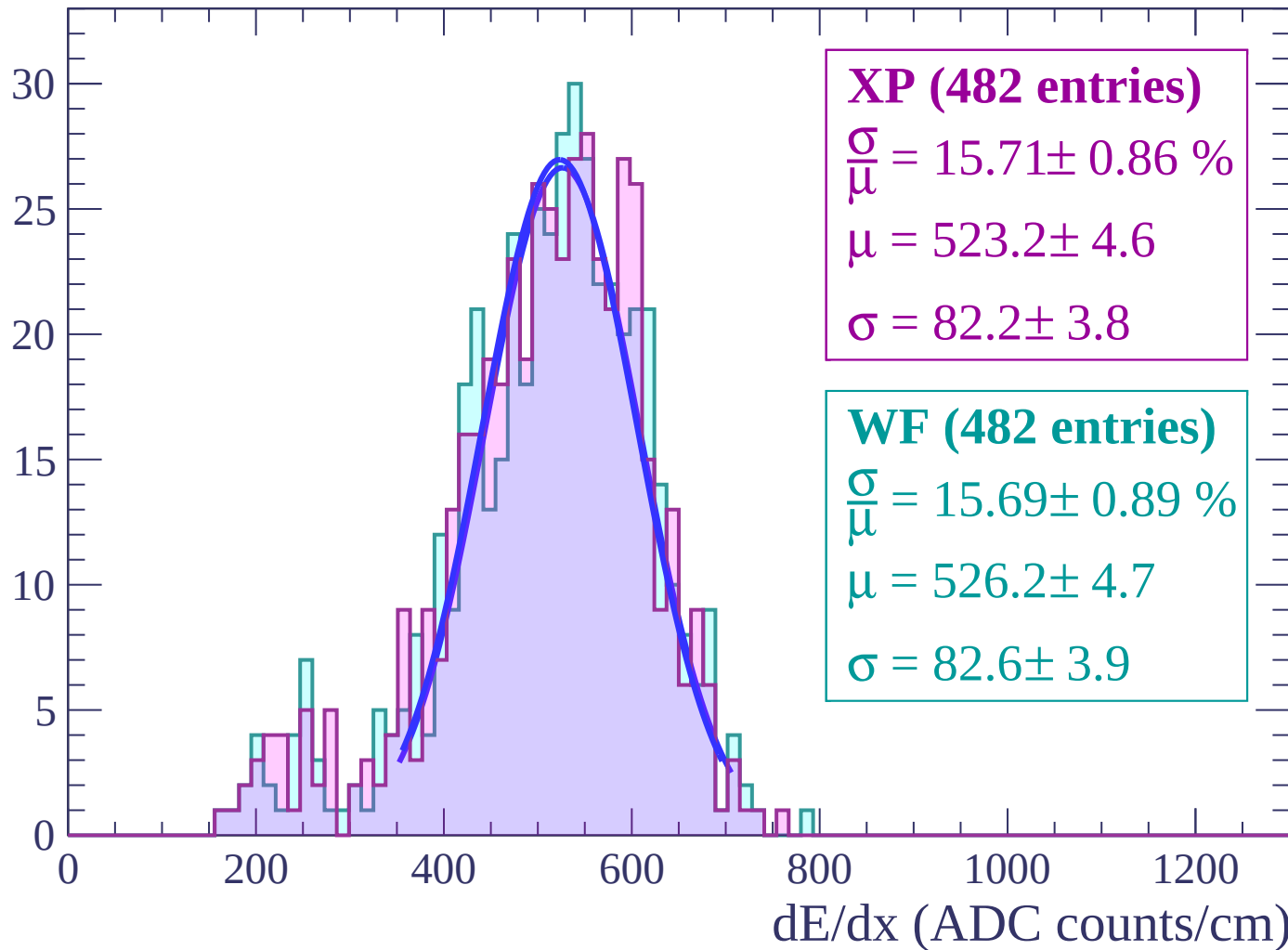
Energy loss | $-62 < \theta < -60$

Count



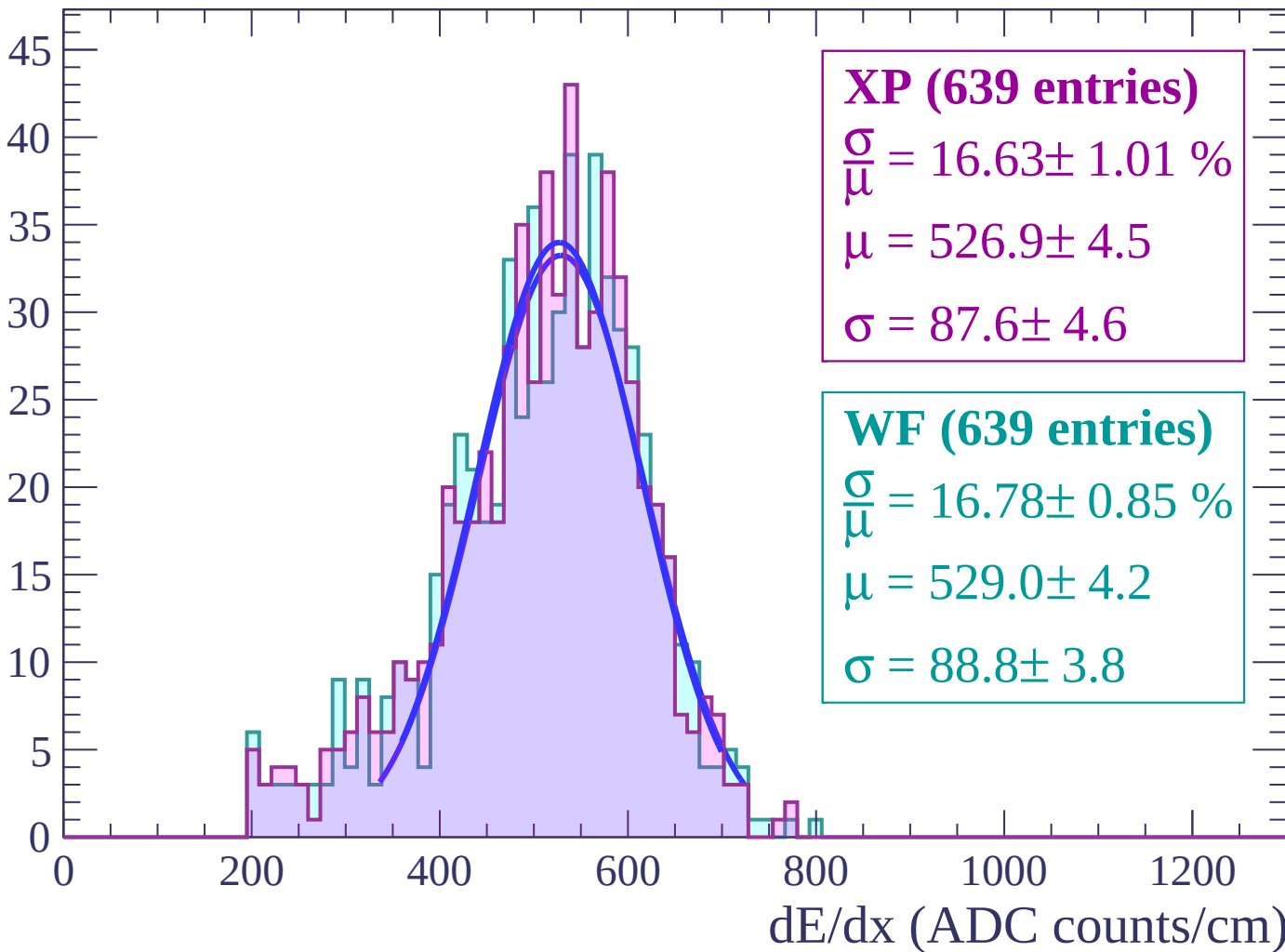
Energy loss | $-60 < \theta < -58$

Count



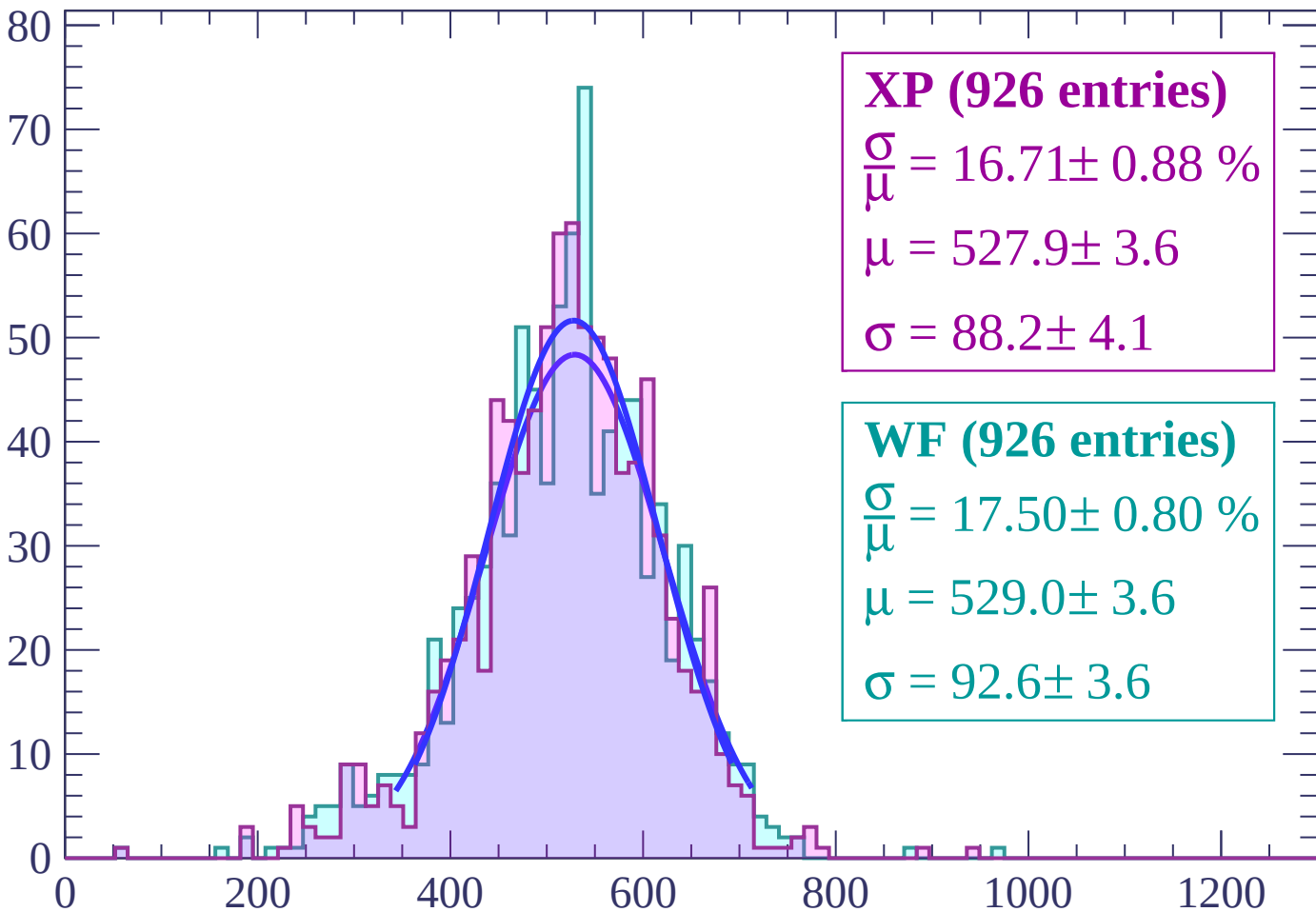
Energy loss | $-58 < \theta < -56$

Count



Energy loss | $-56 < \theta < -54$

Count



XP (926 entries)

$$\frac{\sigma}{\mu} = 16.71 \pm 0.88 \%$$

$$\mu = 527.9 \pm 3.6$$

$$\sigma = 88.2 \pm 4.1$$

WF (926 entries)

$$\frac{\sigma}{\mu} = 17.50 \pm 0.80 \%$$

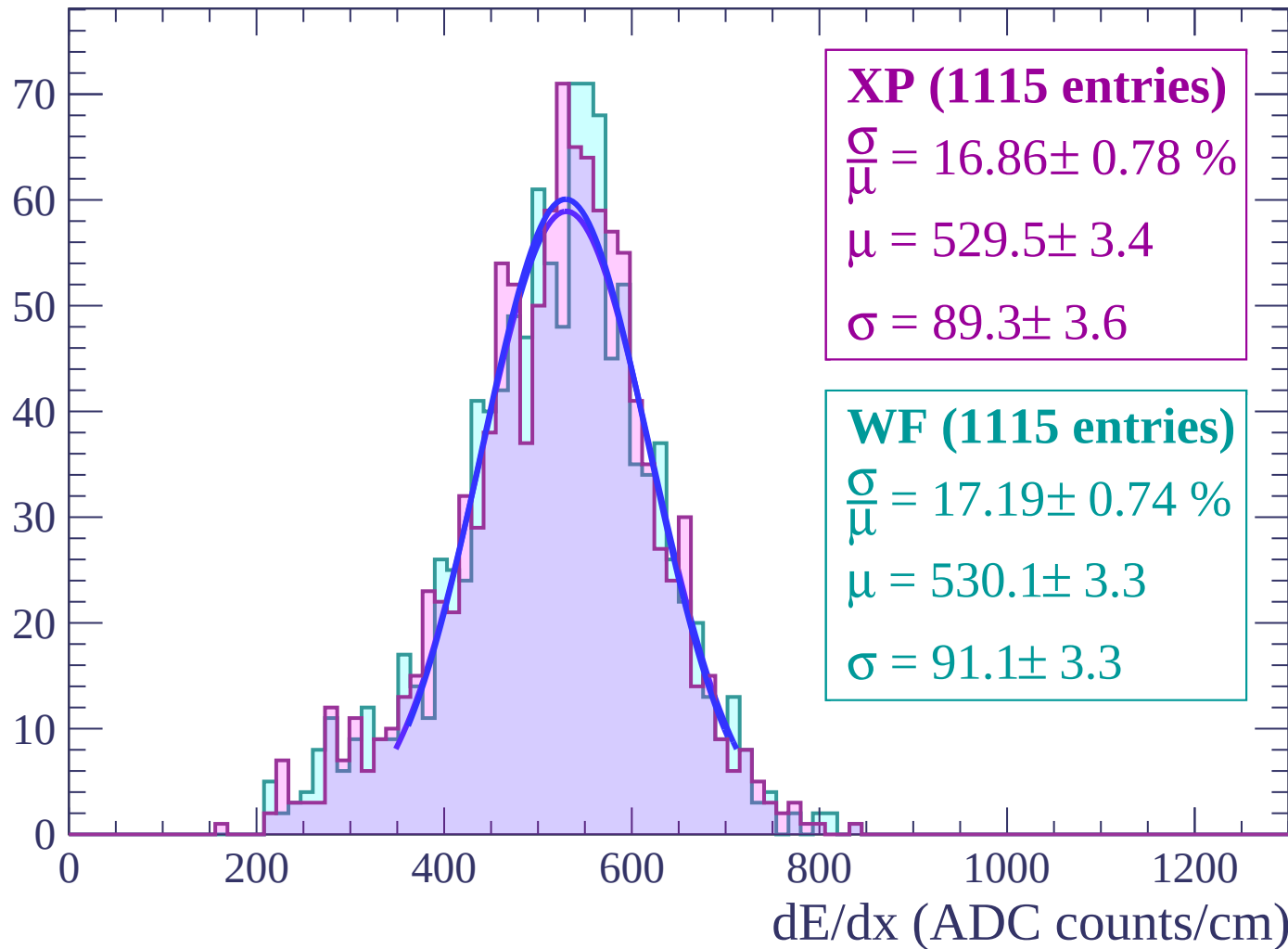
$$\mu = 529.0 \pm 3.6$$

$$\sigma = 92.6 \pm 3.6$$

dE/dx (ADC counts/cm)

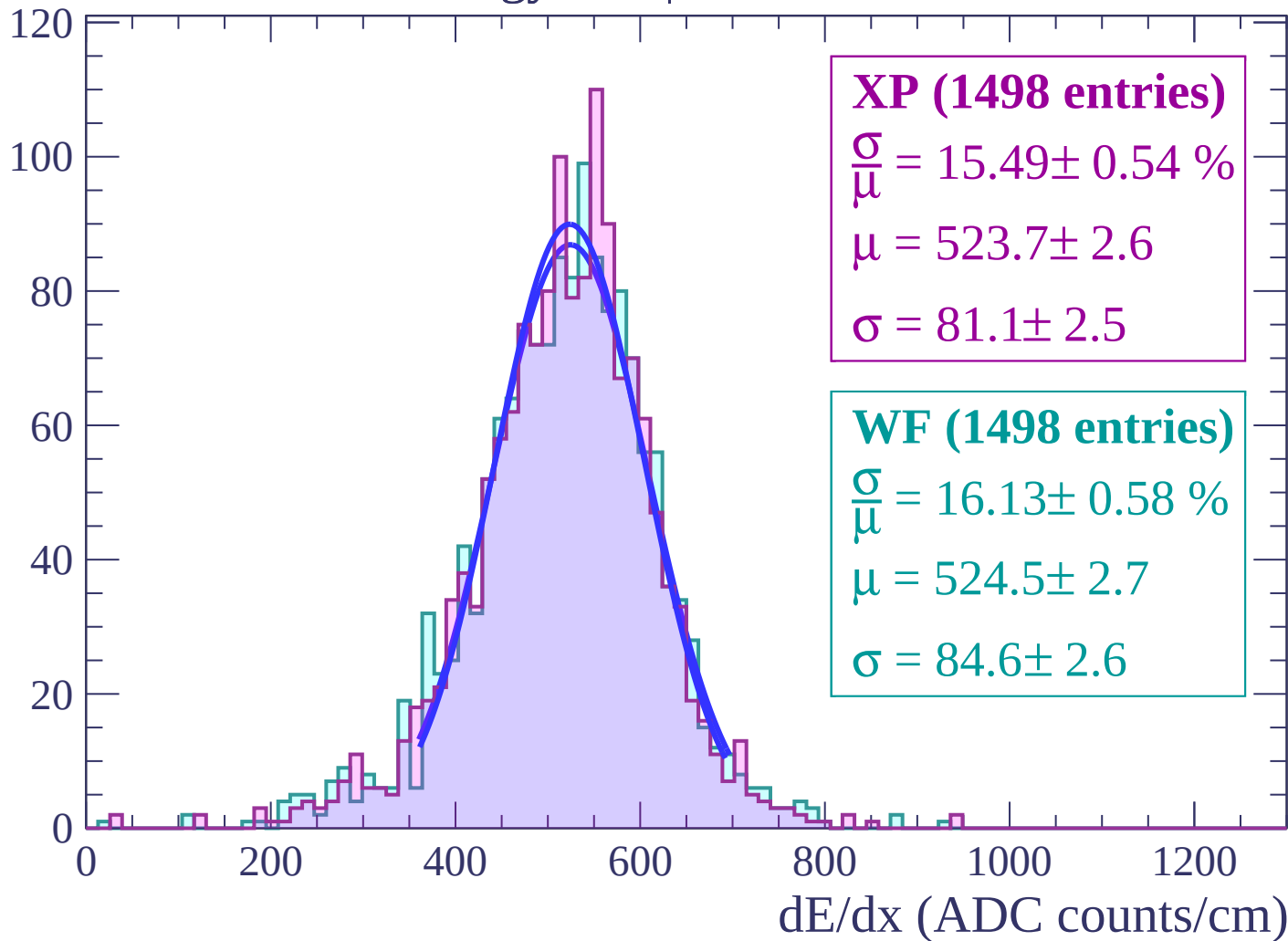
Energy loss | $-54 < \theta < -52$

Count



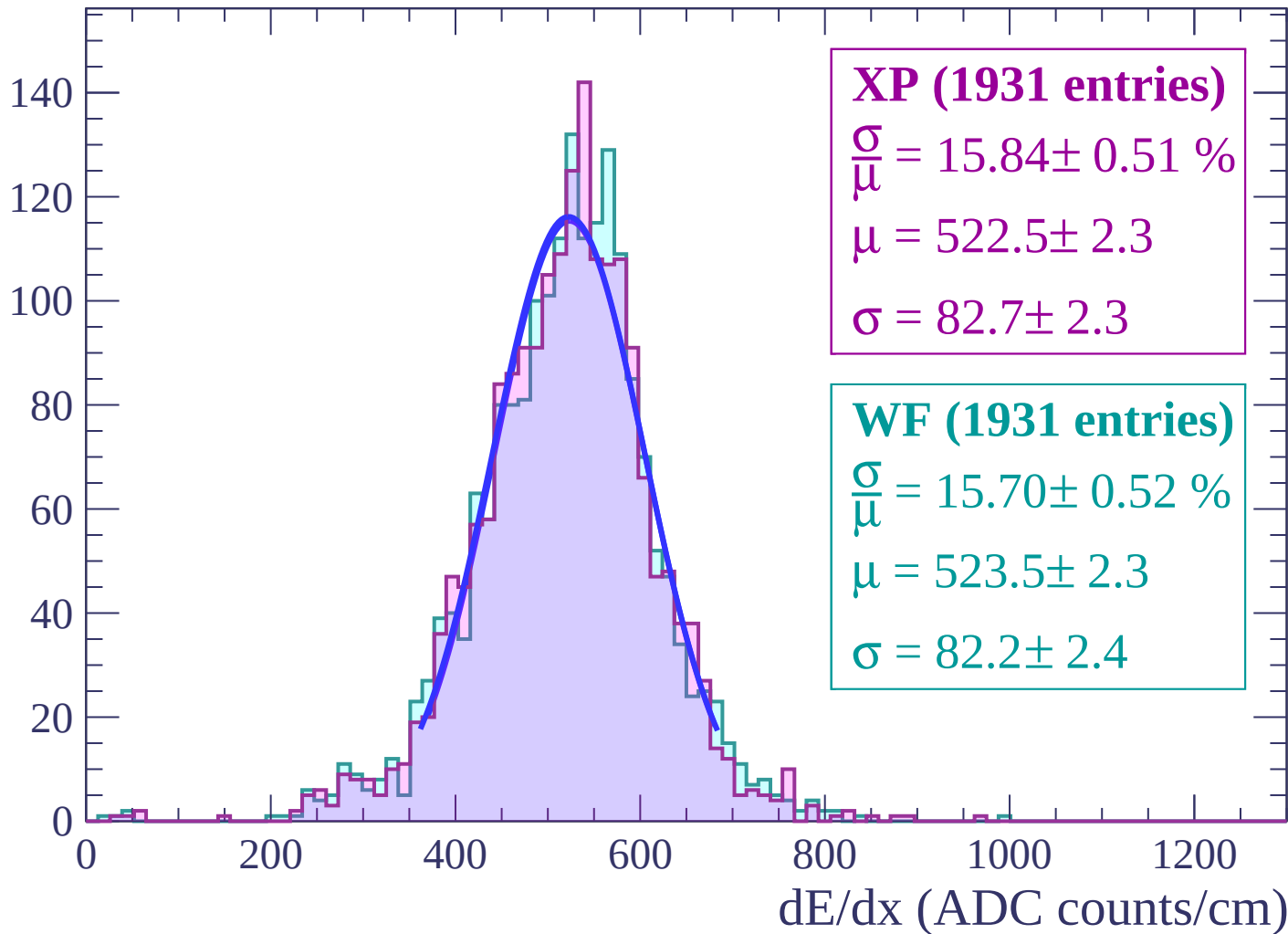
Energy loss | $-52 < \theta < -50$

Count



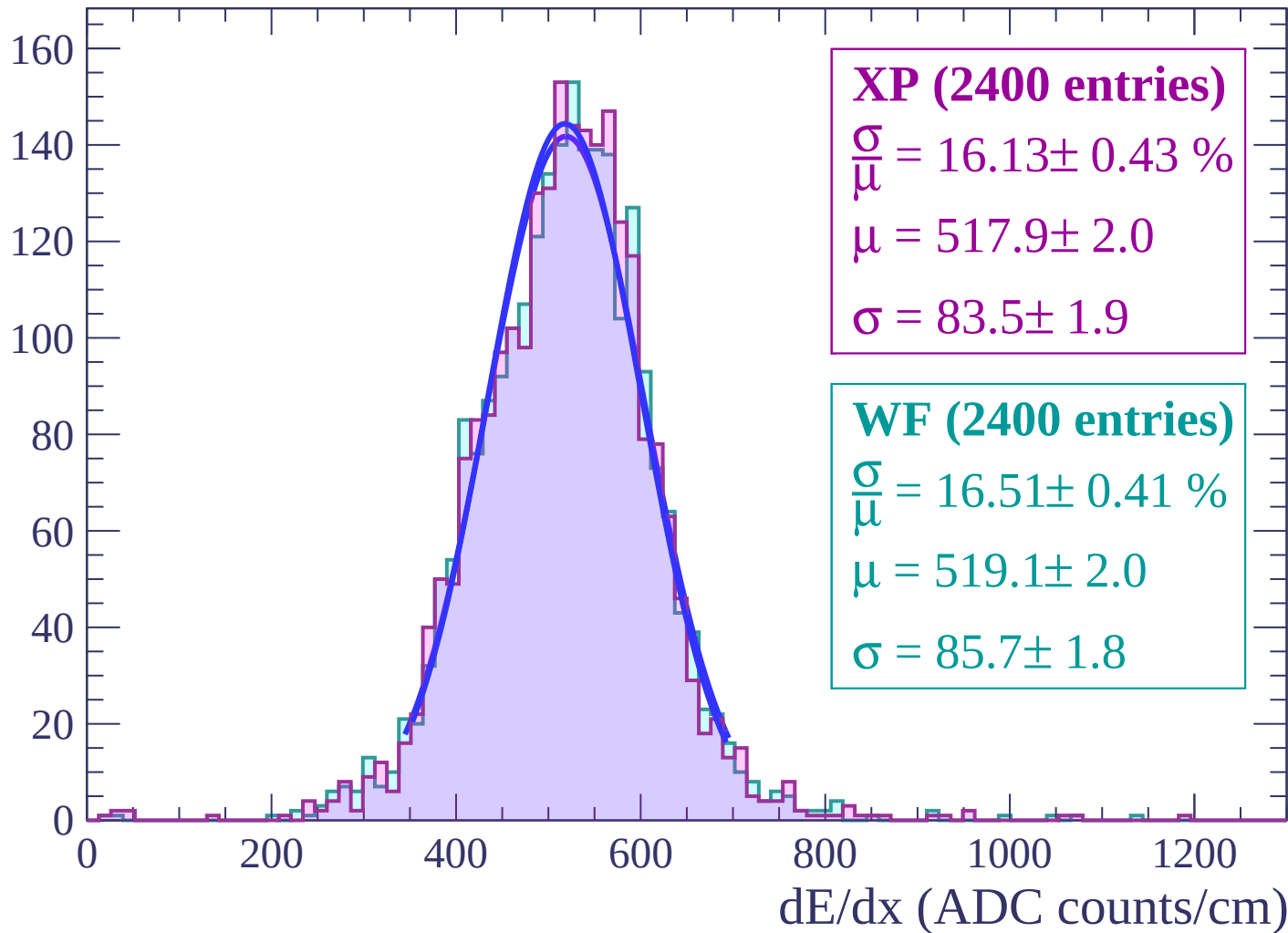
Energy loss | $-50 < \theta < -48$

Count

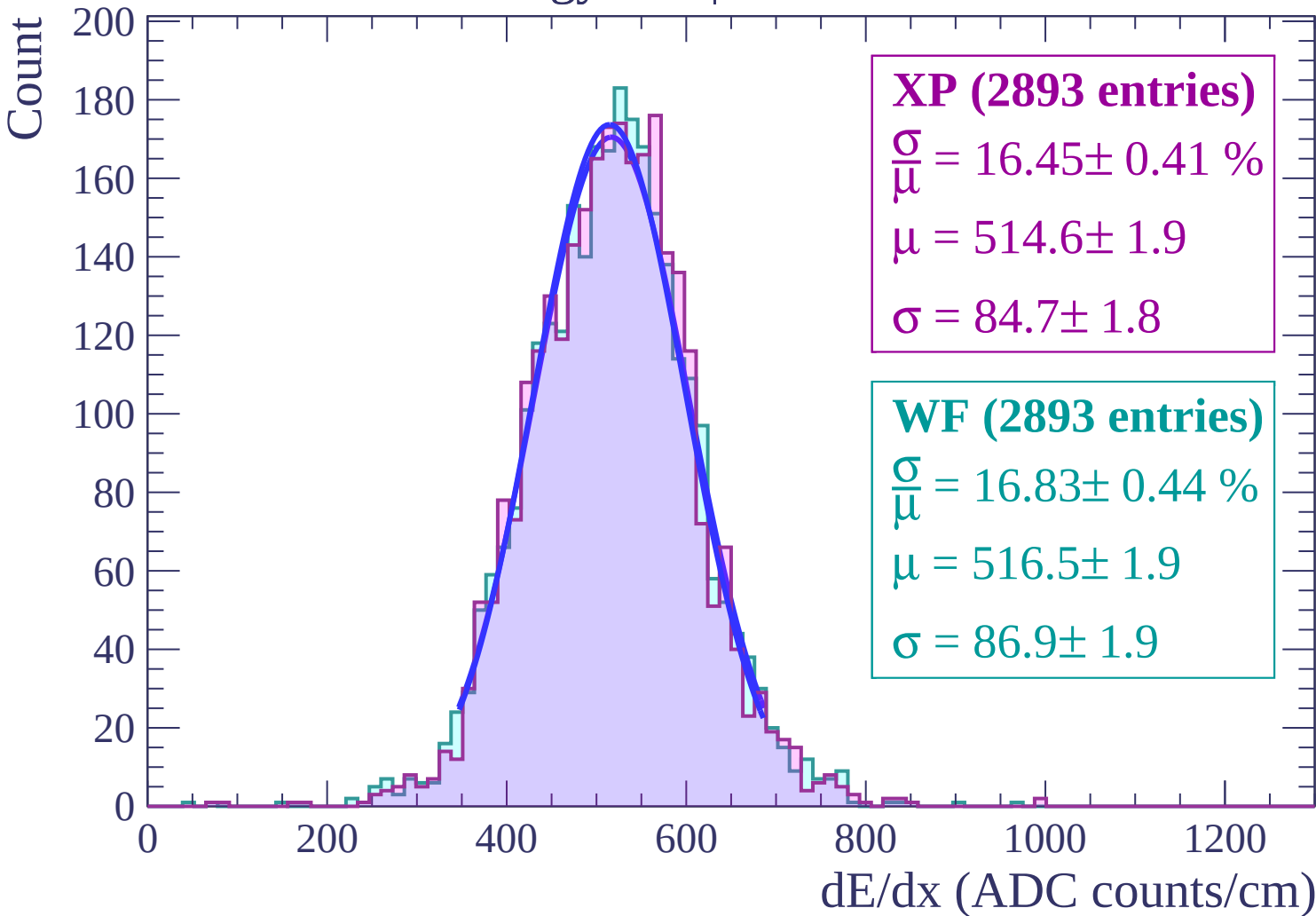


Energy loss | $-48 < \theta < -46$

Count

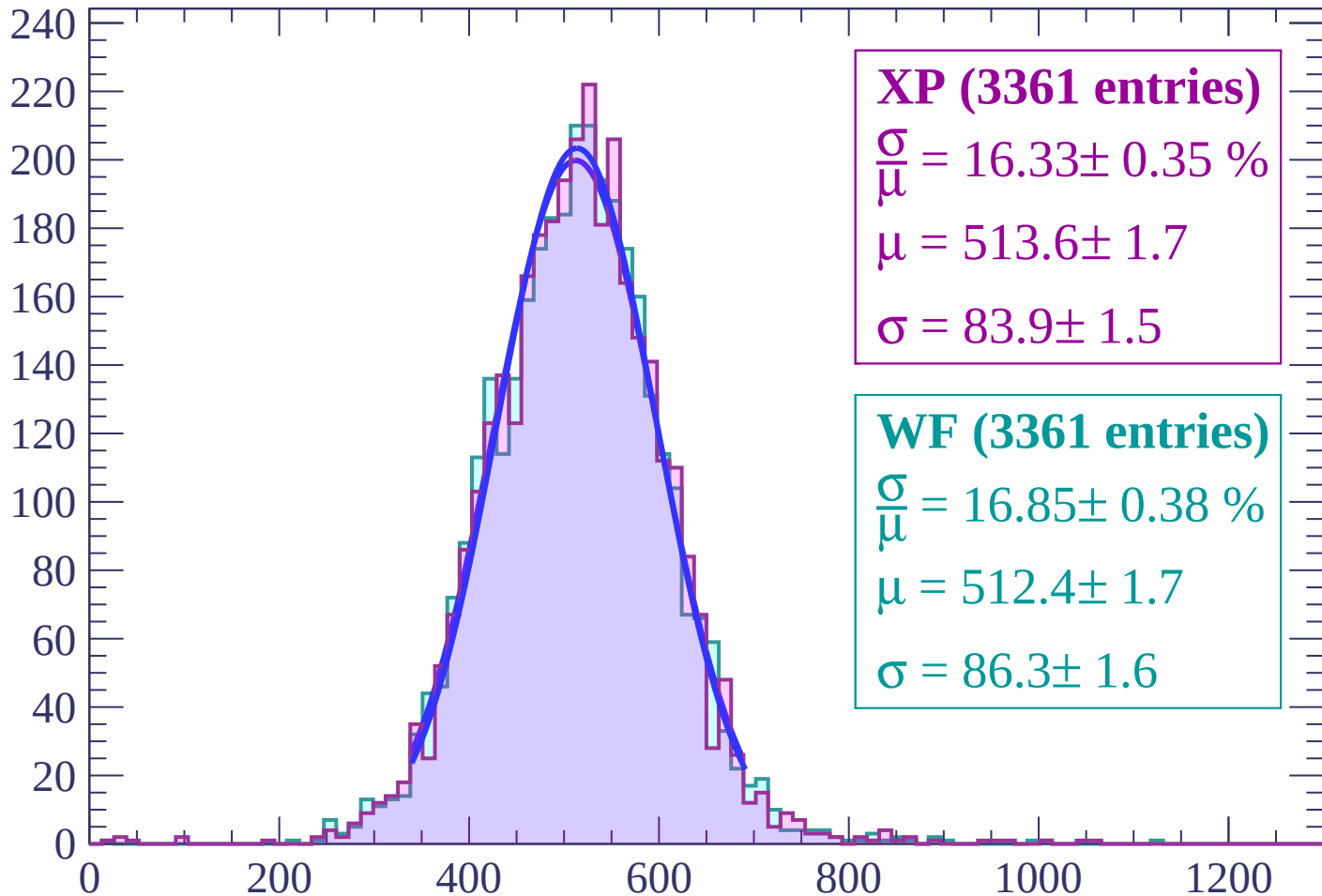


Energy loss | $-46 < \theta < -44$



Energy loss | $-44 < \theta < -42$

Count



XP (3361 entries)

$$\frac{\sigma}{\mu} = 16.33 \pm 0.35 \%$$

$$\mu = 513.6 \pm 1.7$$

$$\sigma = 83.9 \pm 1.5$$

WF (3361 entries)

$$\frac{\sigma}{\mu} = 16.85 \pm 0.38 \%$$

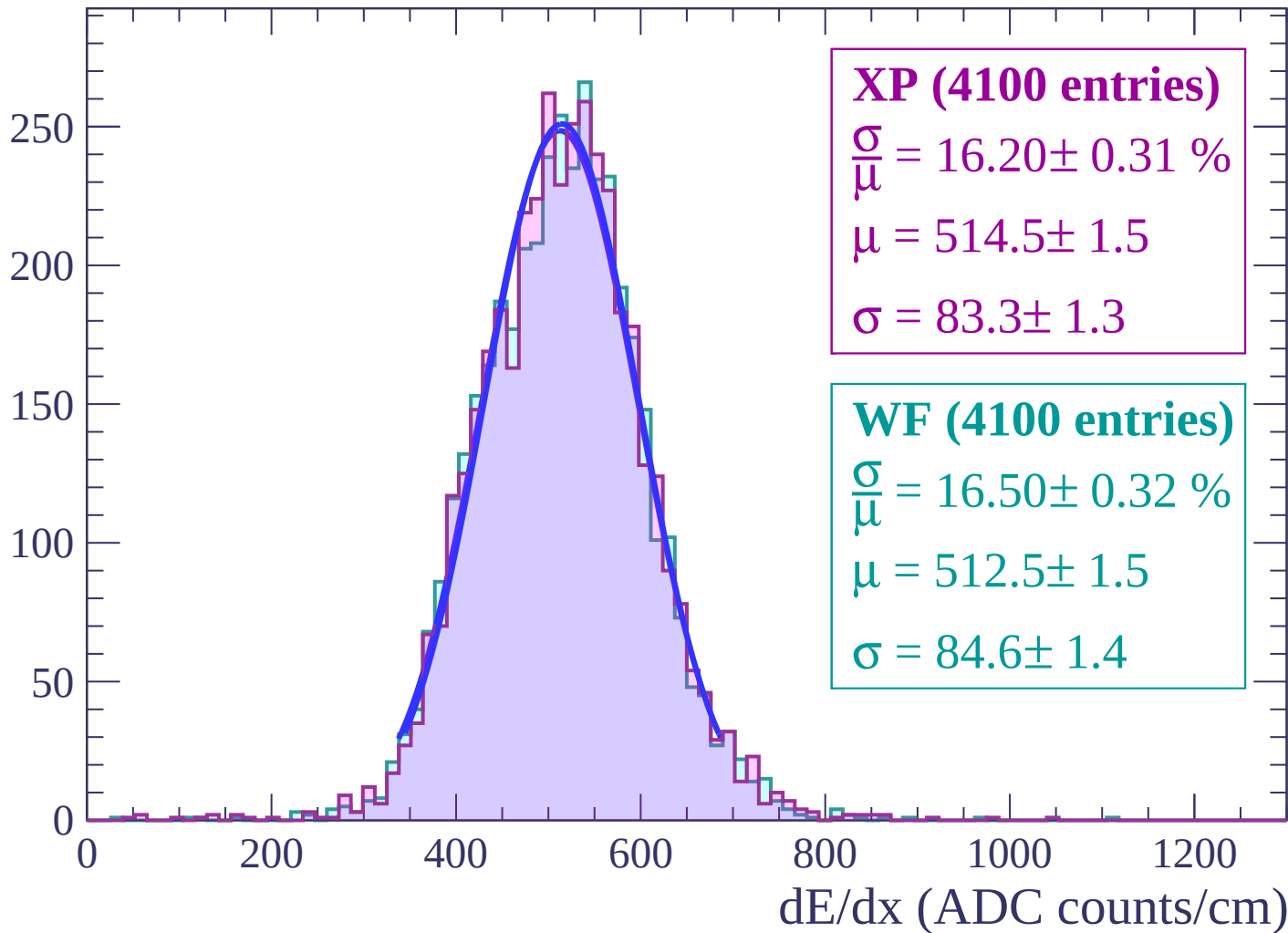
$$\mu = 512.4 \pm 1.7$$

$$\sigma = 86.3 \pm 1.6$$

dE/dx (ADC counts/cm)

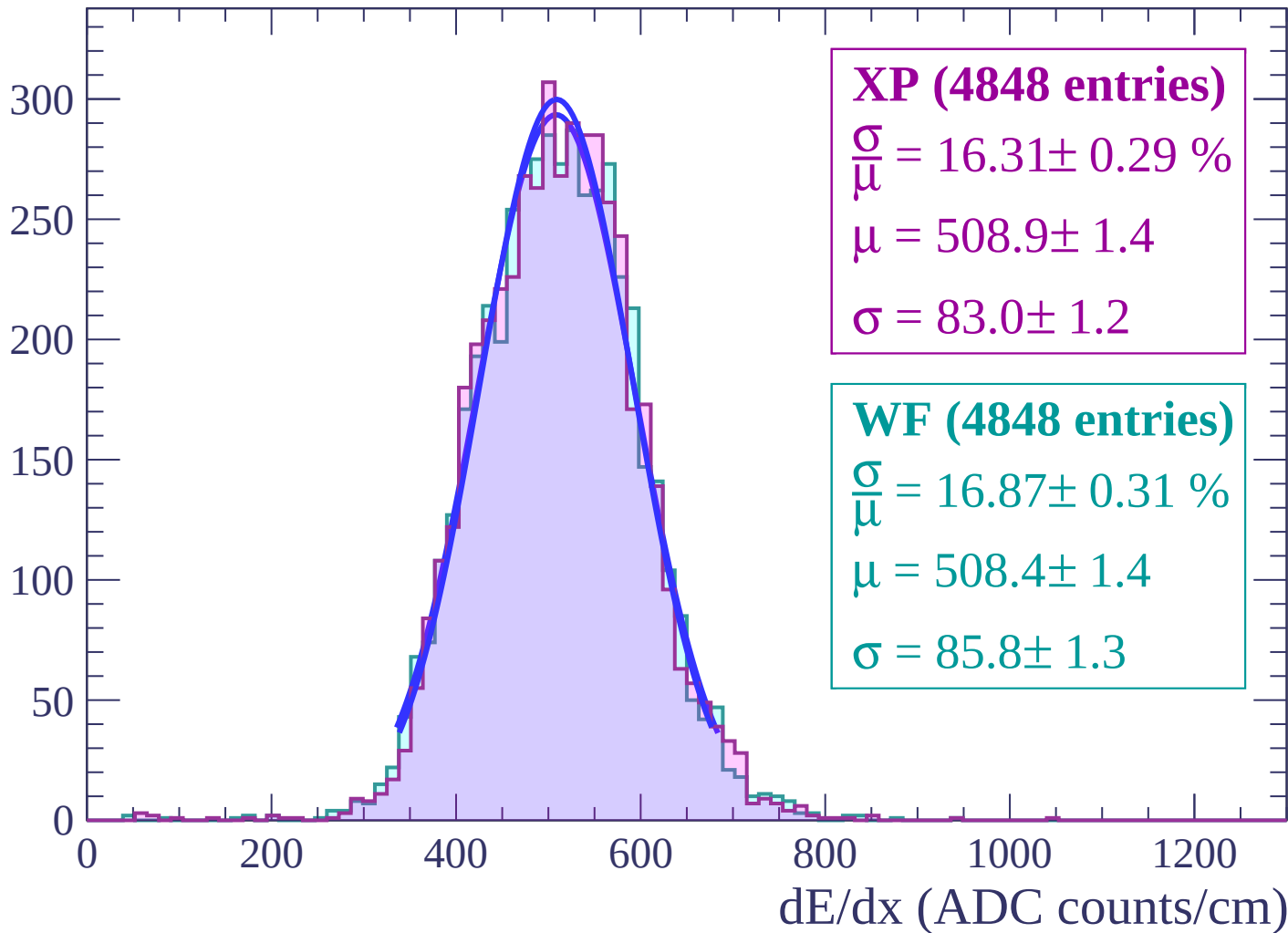
Energy loss | $-42 < \theta < -40$

Count



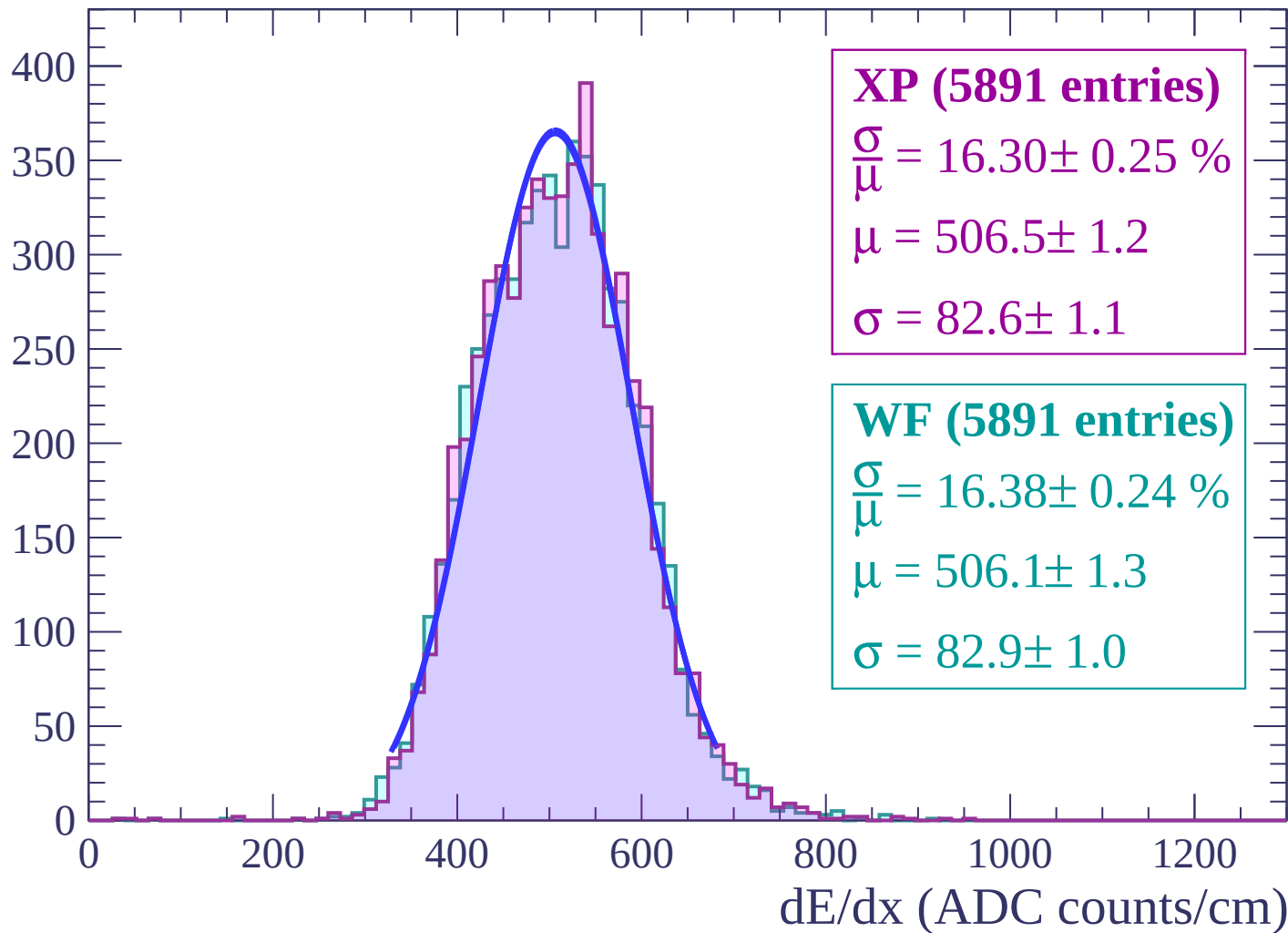
Energy loss | $-40 < \theta < -38$

Count



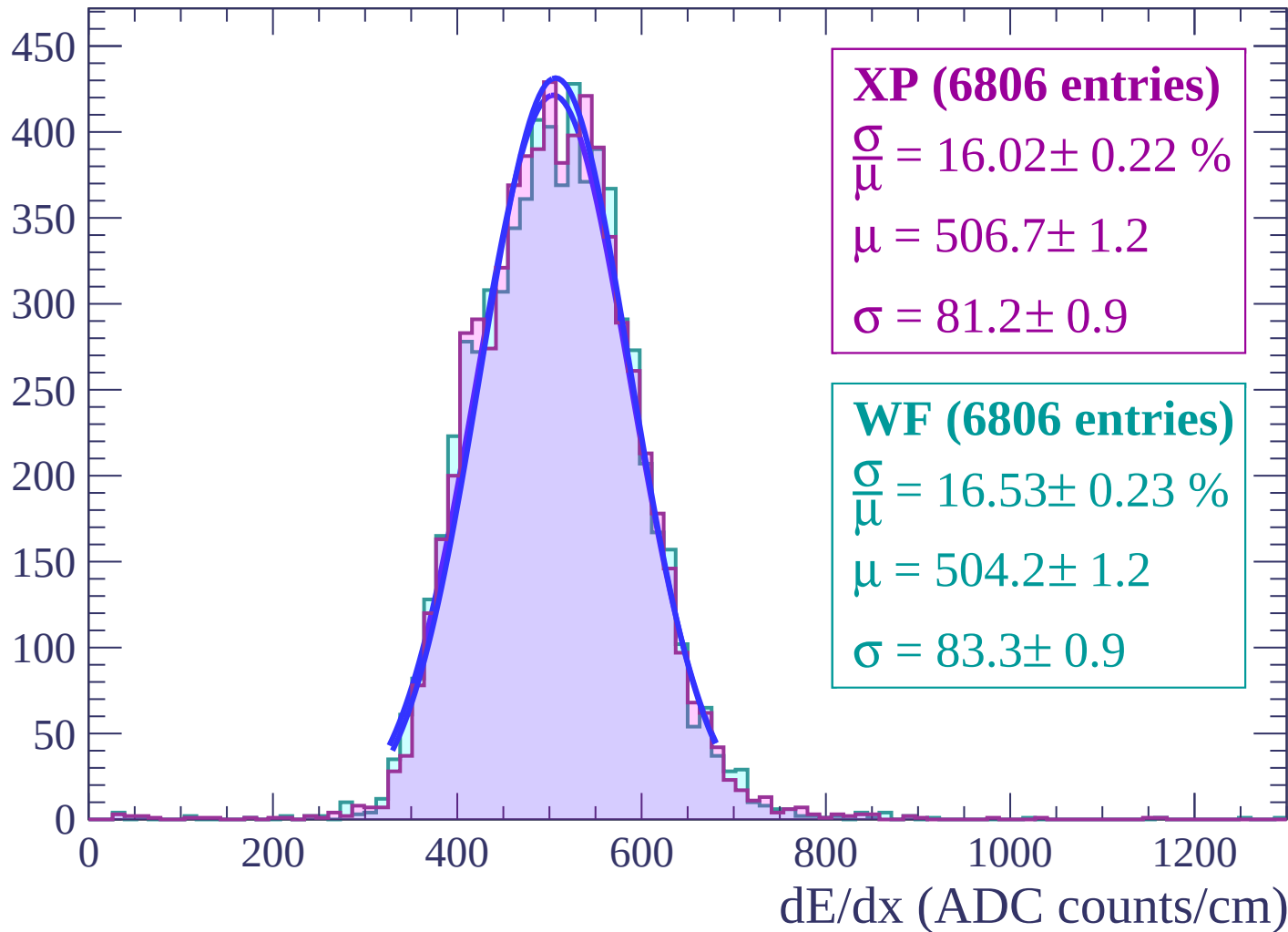
Energy loss | $-38 < \theta < -36$

Count



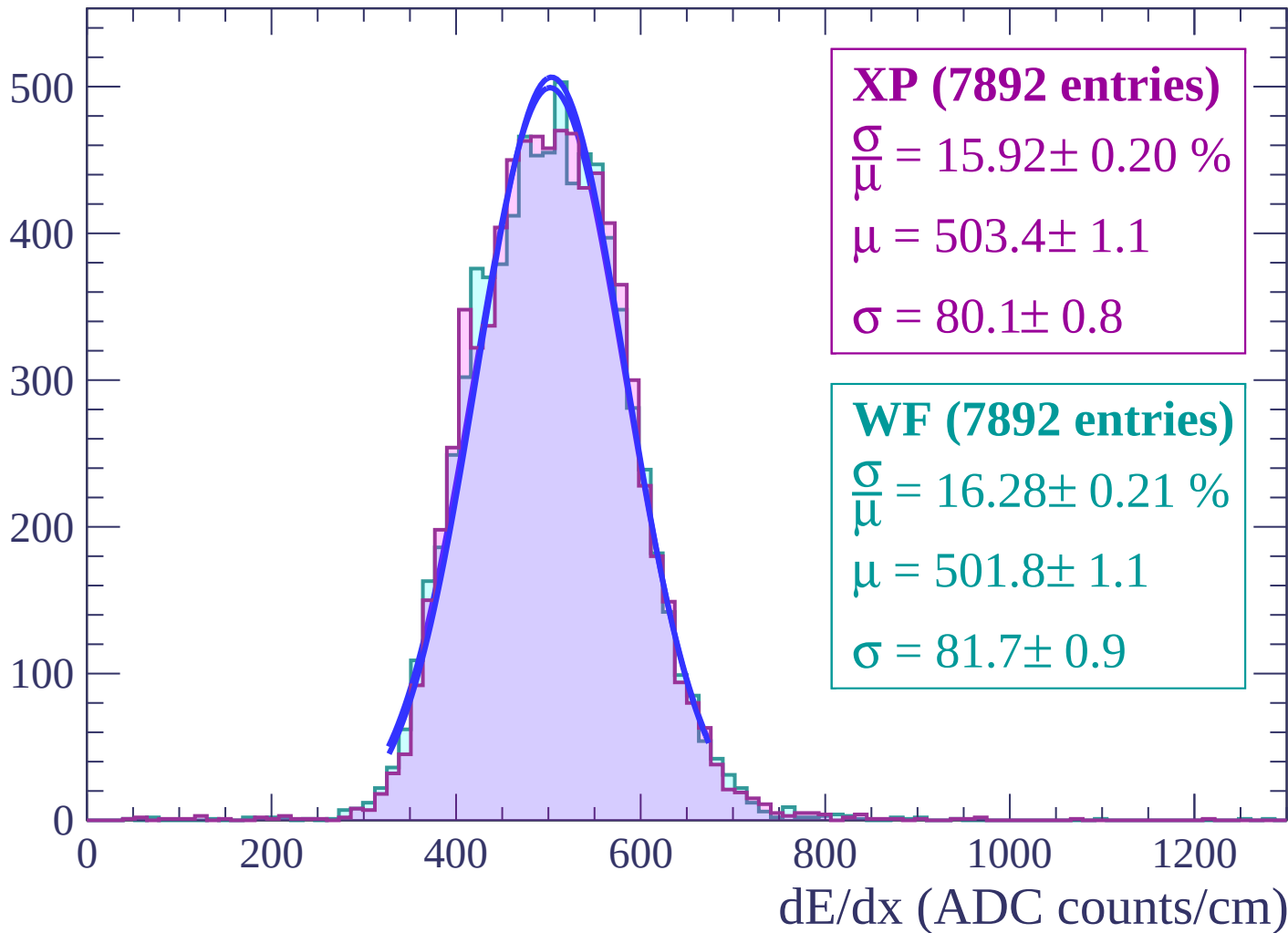
Energy loss | $-36 < \theta < -34$

Count



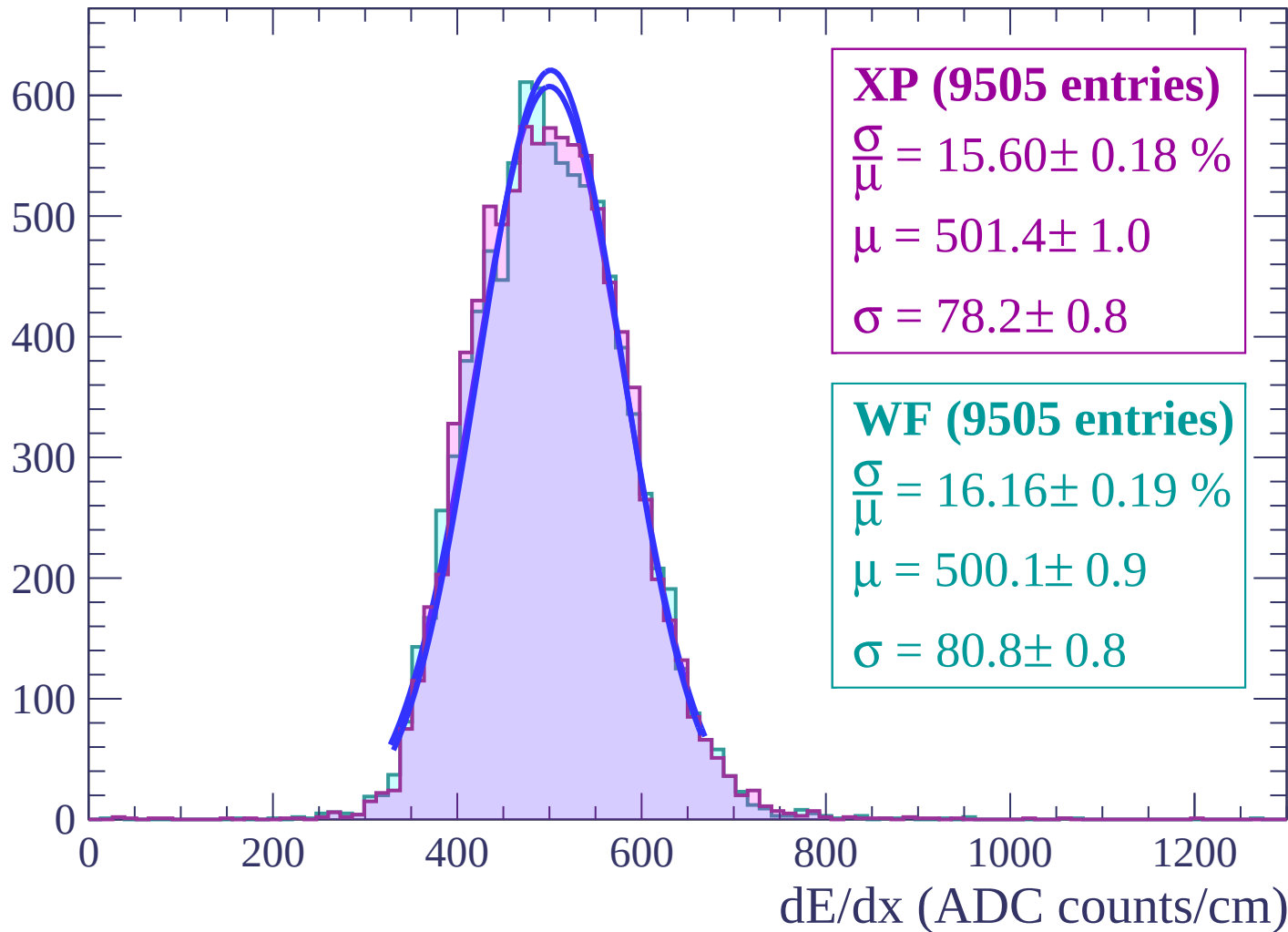
Energy loss | $-34 < \theta < -32$

Count



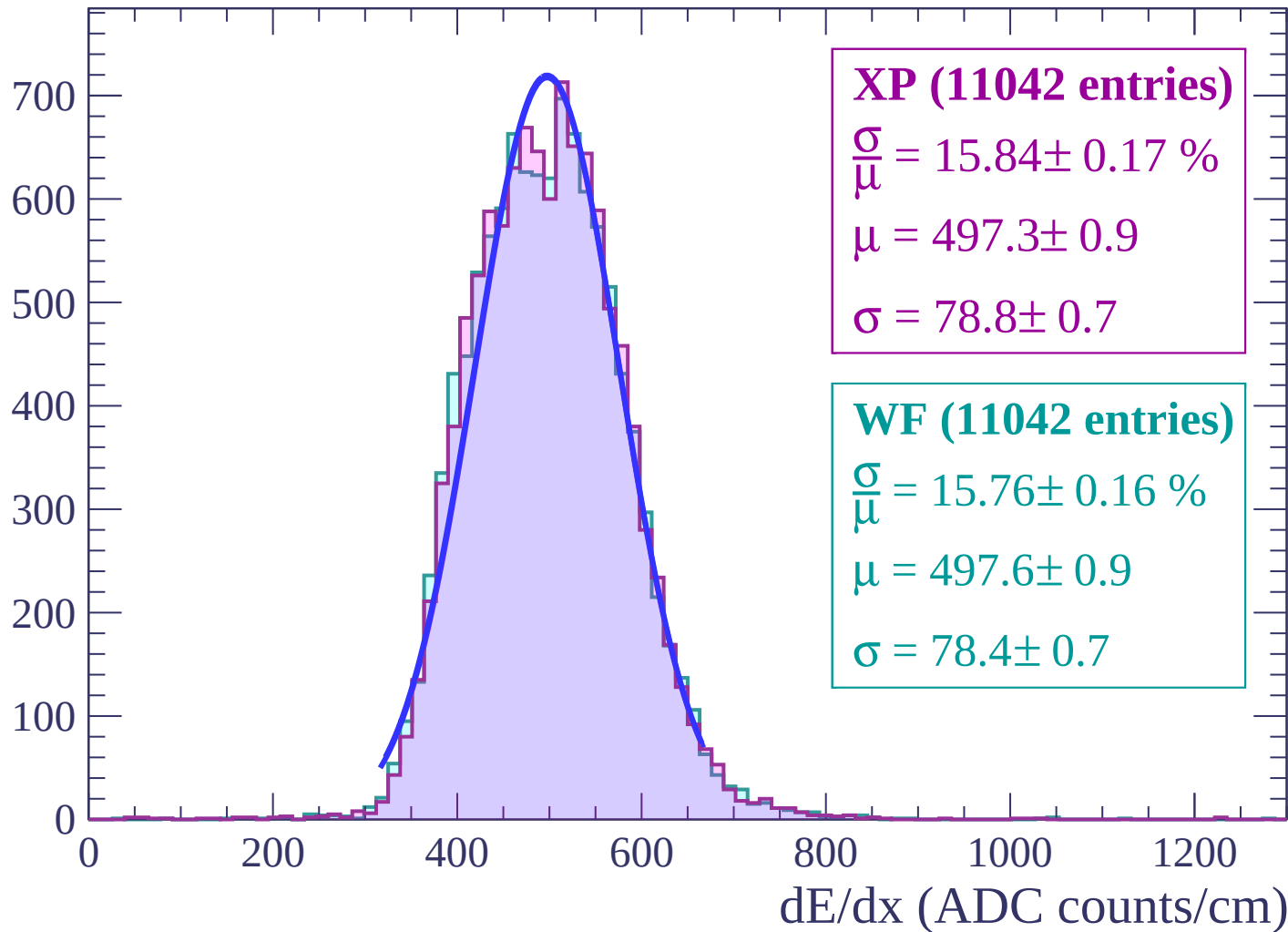
Energy loss | $-32 < \theta < -30$

Count



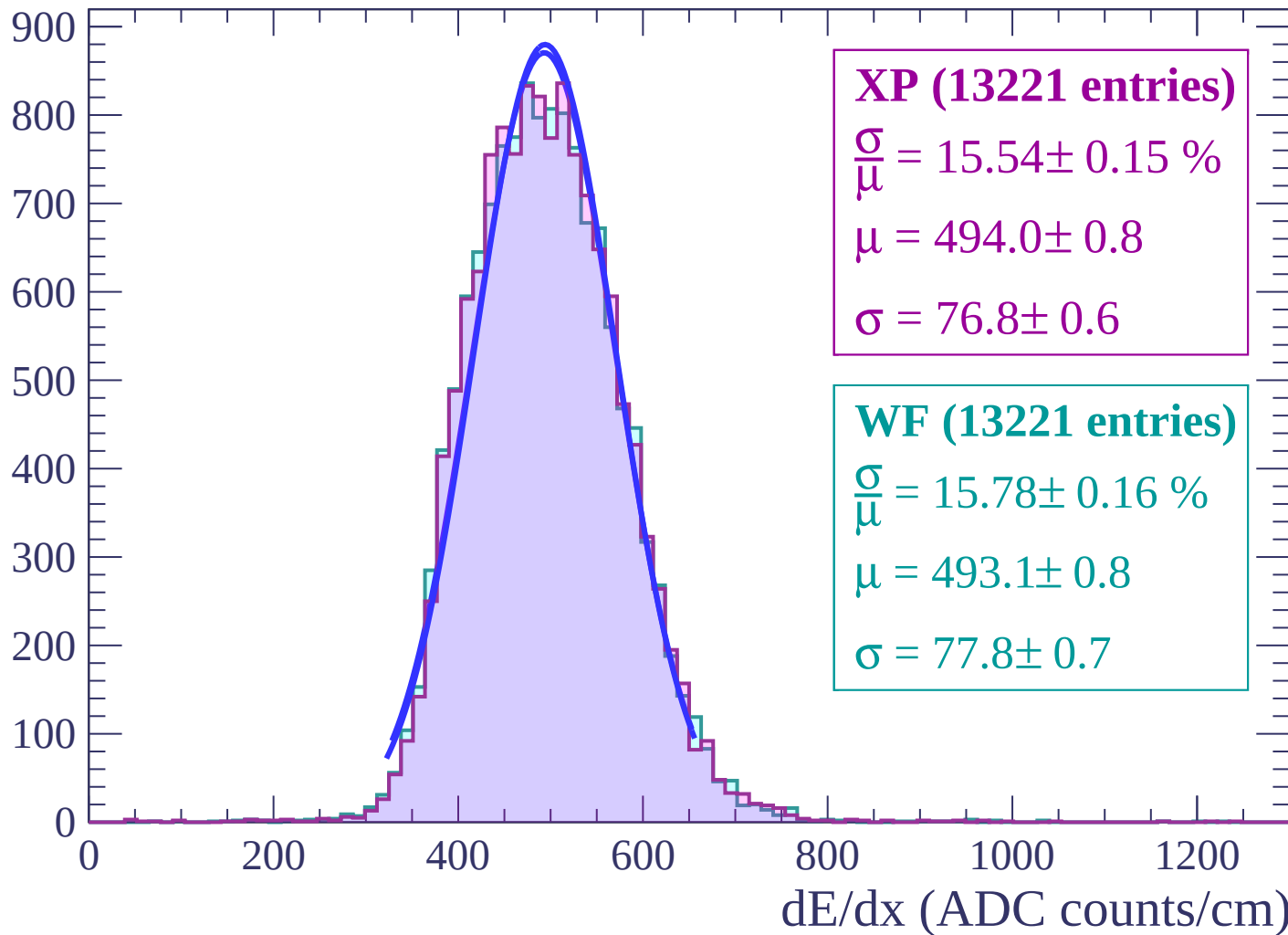
Energy loss | $-30 < \theta < -28$

Count



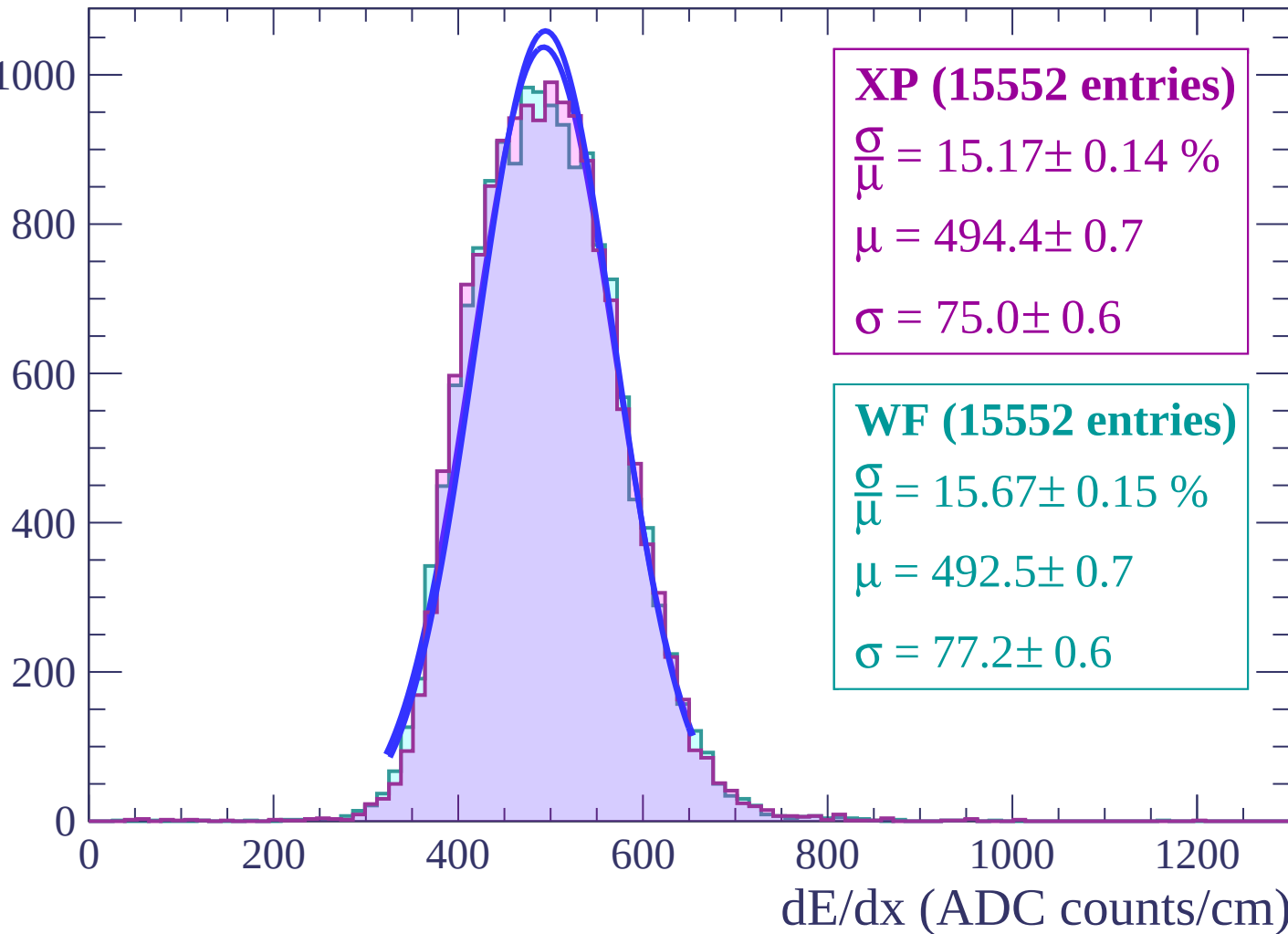
Energy loss | $-28 < \theta < -26$

Count



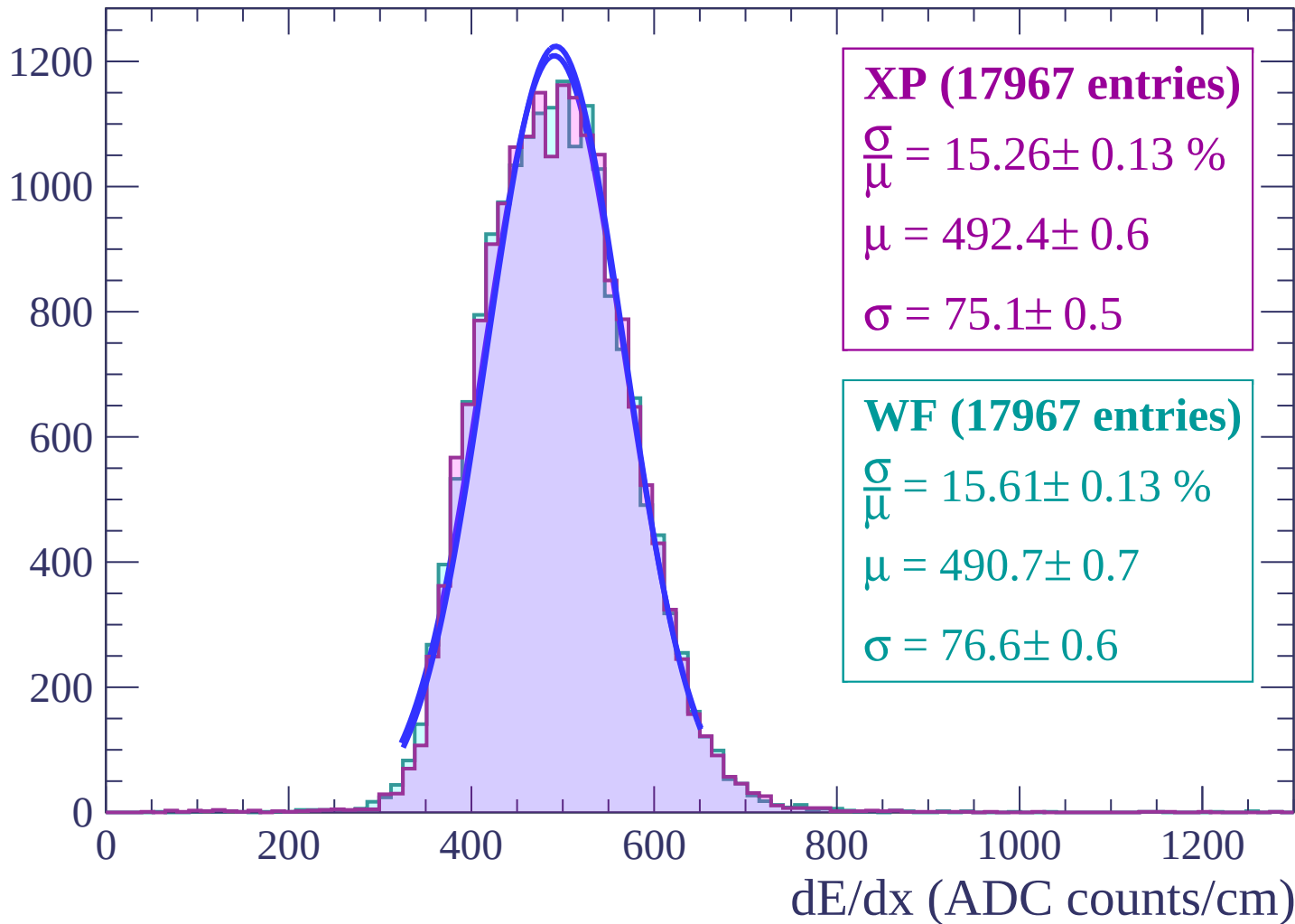
Energy loss | $-26 < \theta < -24$

Count

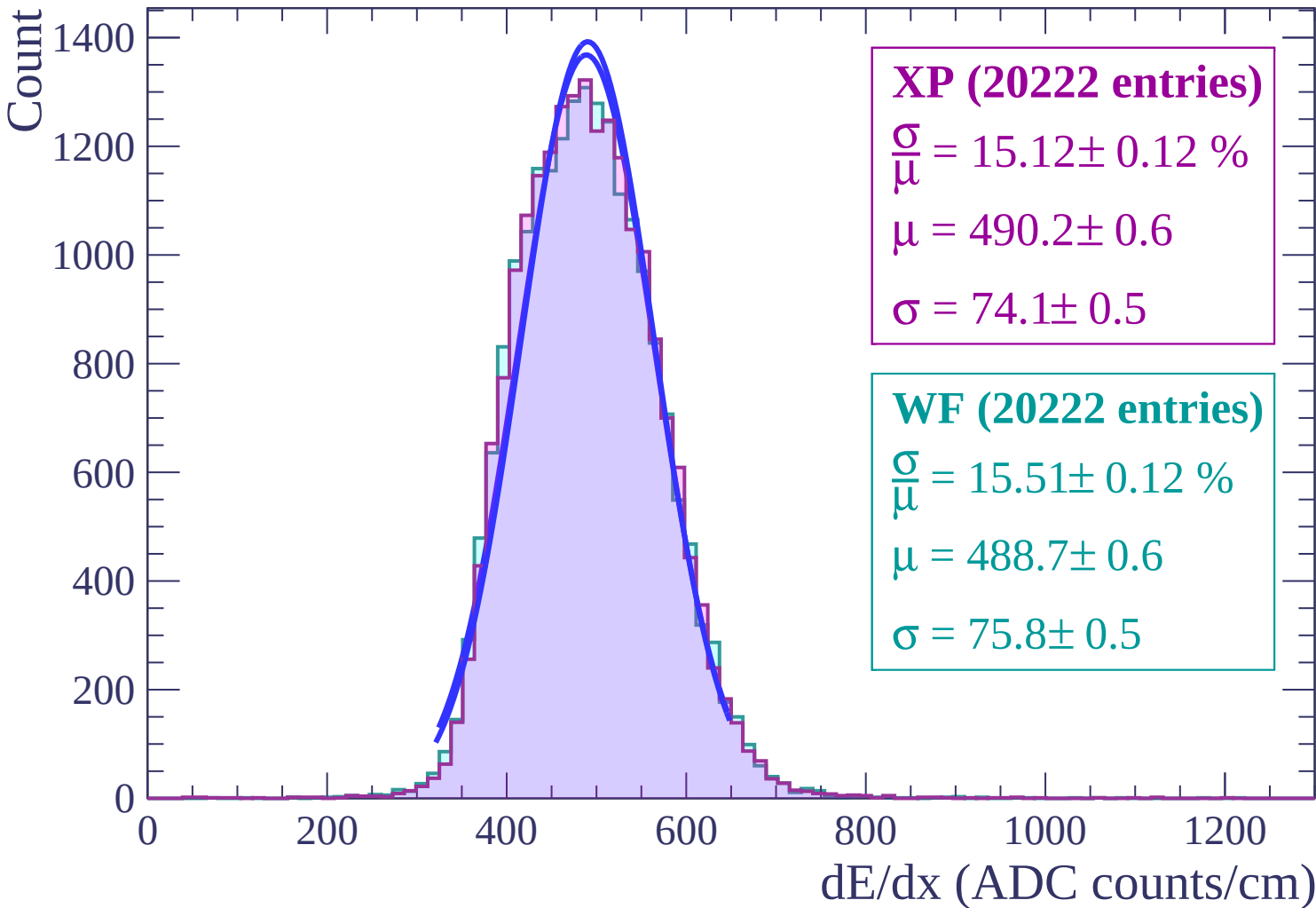


Energy loss | $-24 < \theta < -22$

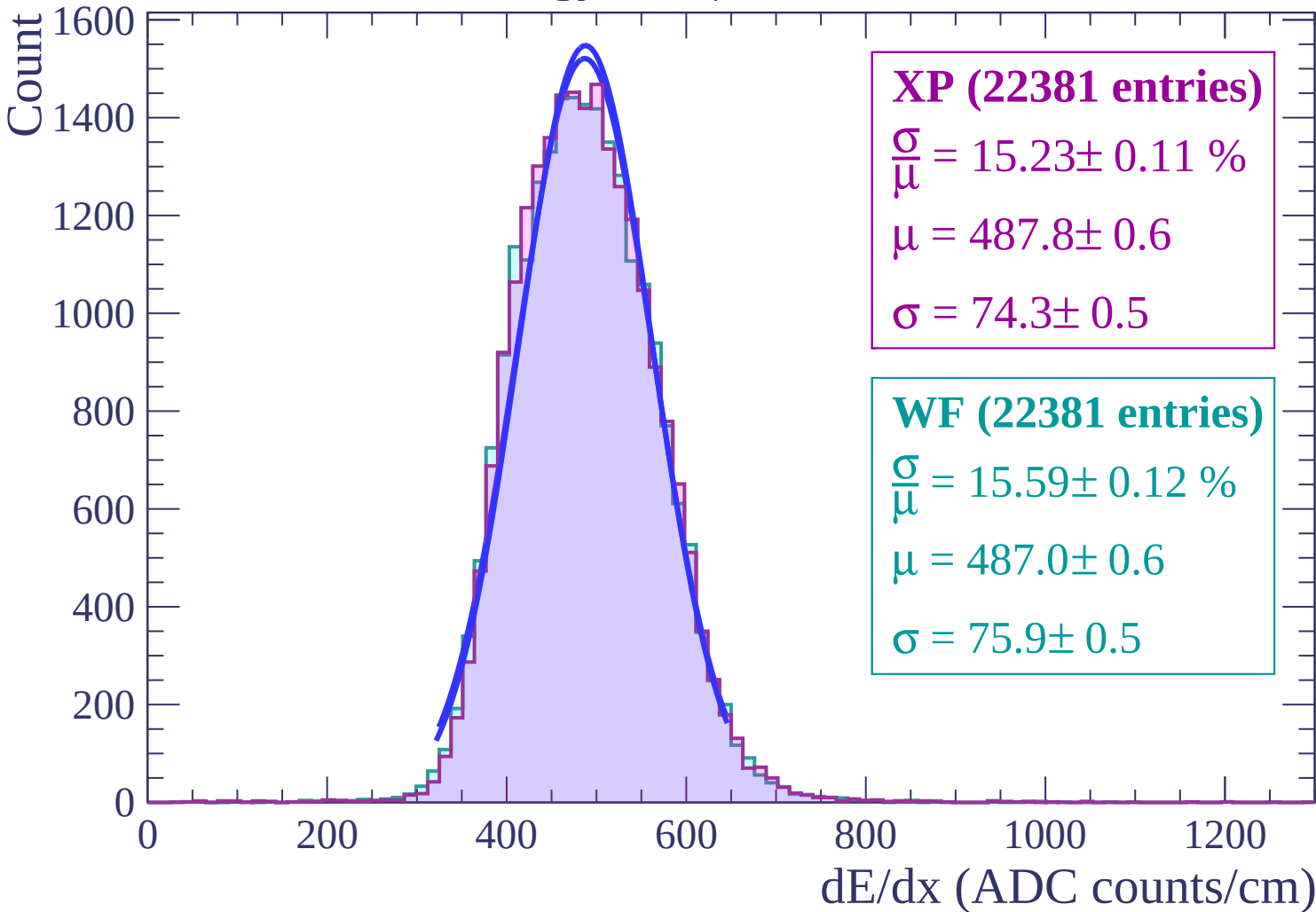
Count



Energy loss | $-22 < \theta < -20$



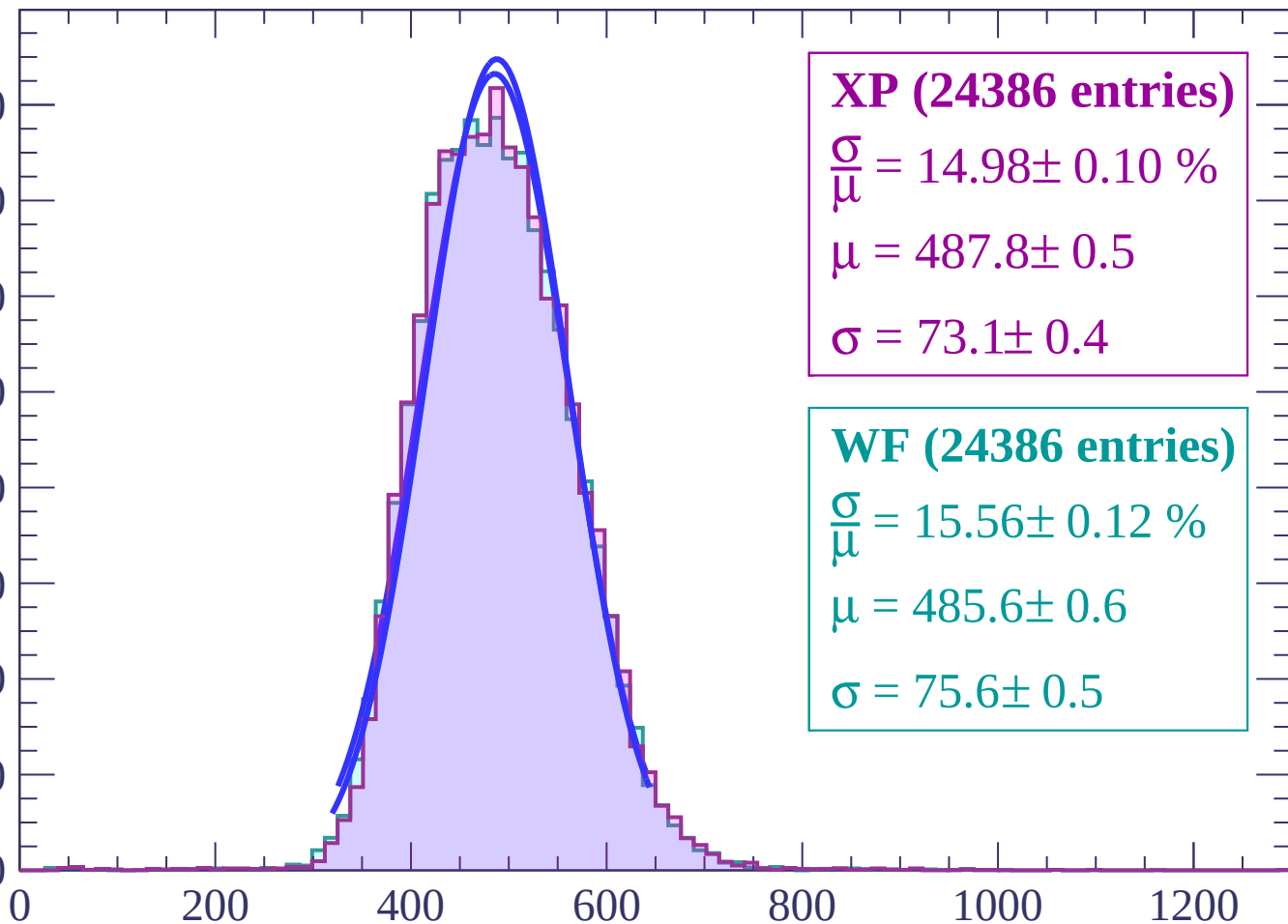
Energy loss | $-20 < \theta < -18$



Energy loss | $-18 < \theta < -16$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (24386 entries)

$\frac{\sigma}{\mu} = 14.98 \pm 0.10 \%$

$\mu = 487.8 \pm 0.5$

$\sigma = 73.1 \pm 0.4$

WF (24386 entries)

$\frac{\sigma}{\mu} = 15.56 \pm 0.12 \%$

$\mu = 485.6 \pm 0.6$

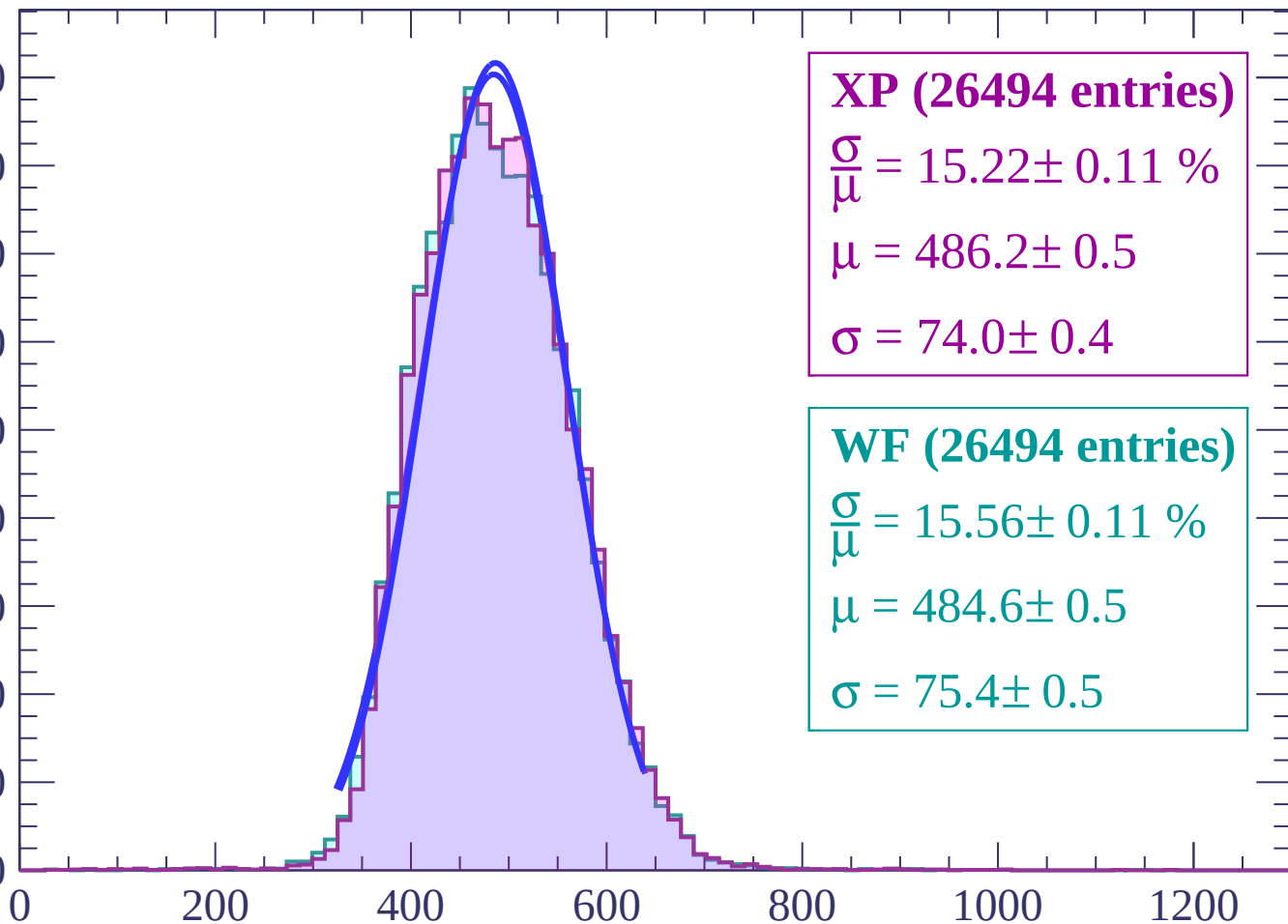
$\sigma = 75.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-16 < \theta < -14$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26494 entries)

$\frac{\sigma}{\mu} = 15.22 \pm 0.11 \%$

$\mu = 486.2 \pm 0.5$

$\sigma = 74.0 \pm 0.4$

WF (26494 entries)

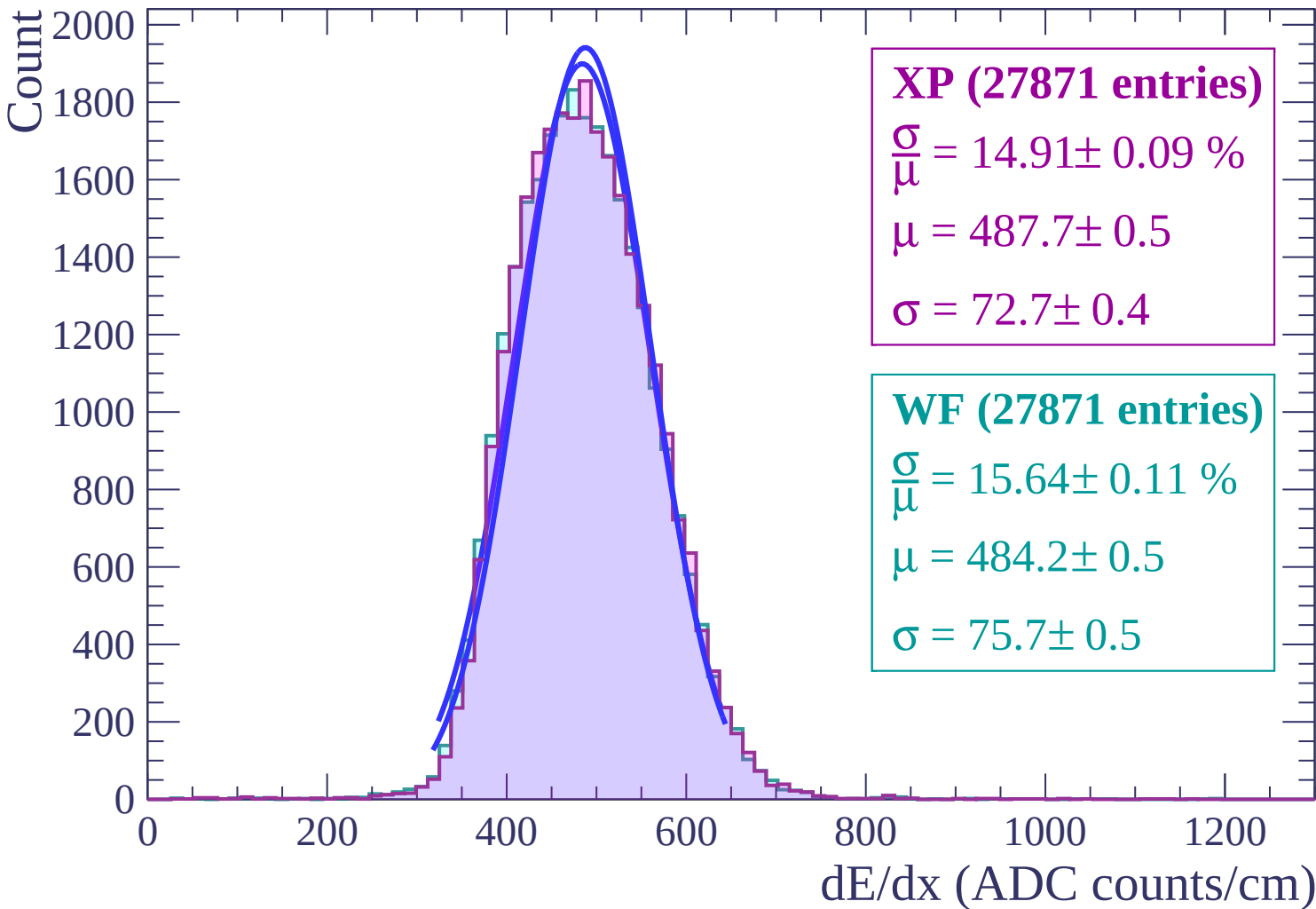
$\frac{\sigma}{\mu} = 15.56 \pm 0.11 \%$

$\mu = 484.6 \pm 0.5$

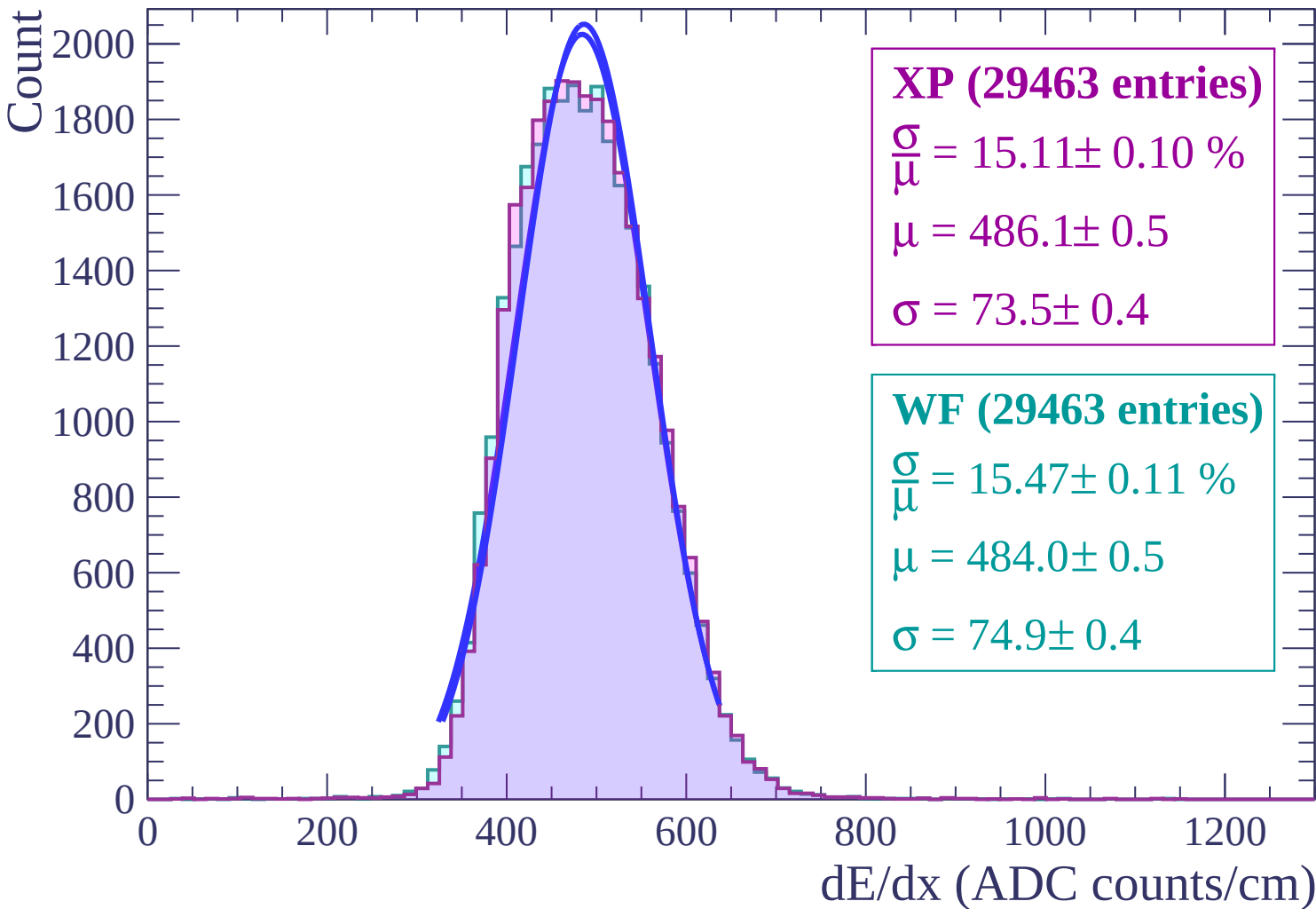
$\sigma = 75.4 \pm 0.5$

dE/dx (ADC counts/cm)

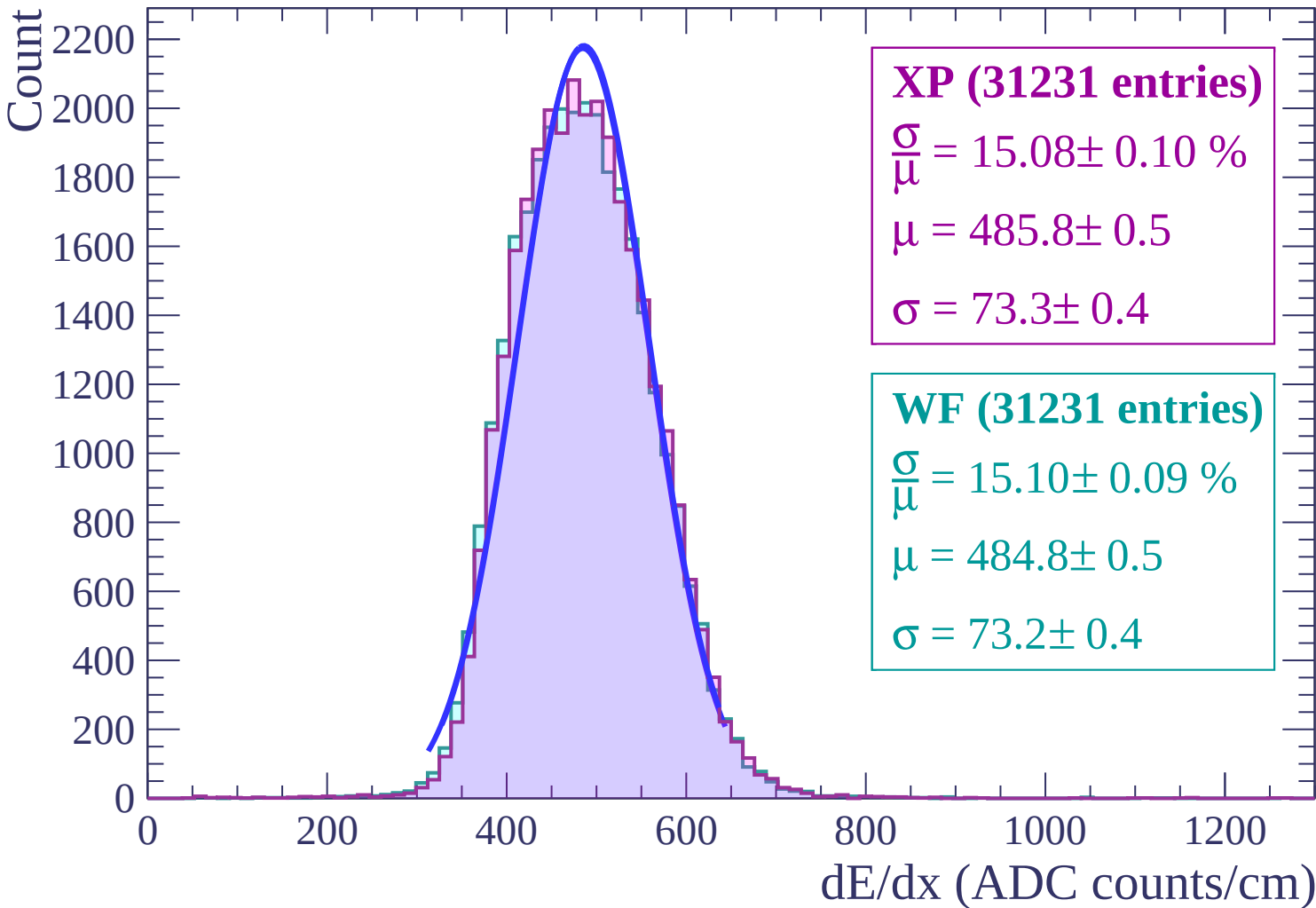
Energy loss | $-14 < \theta < -12$



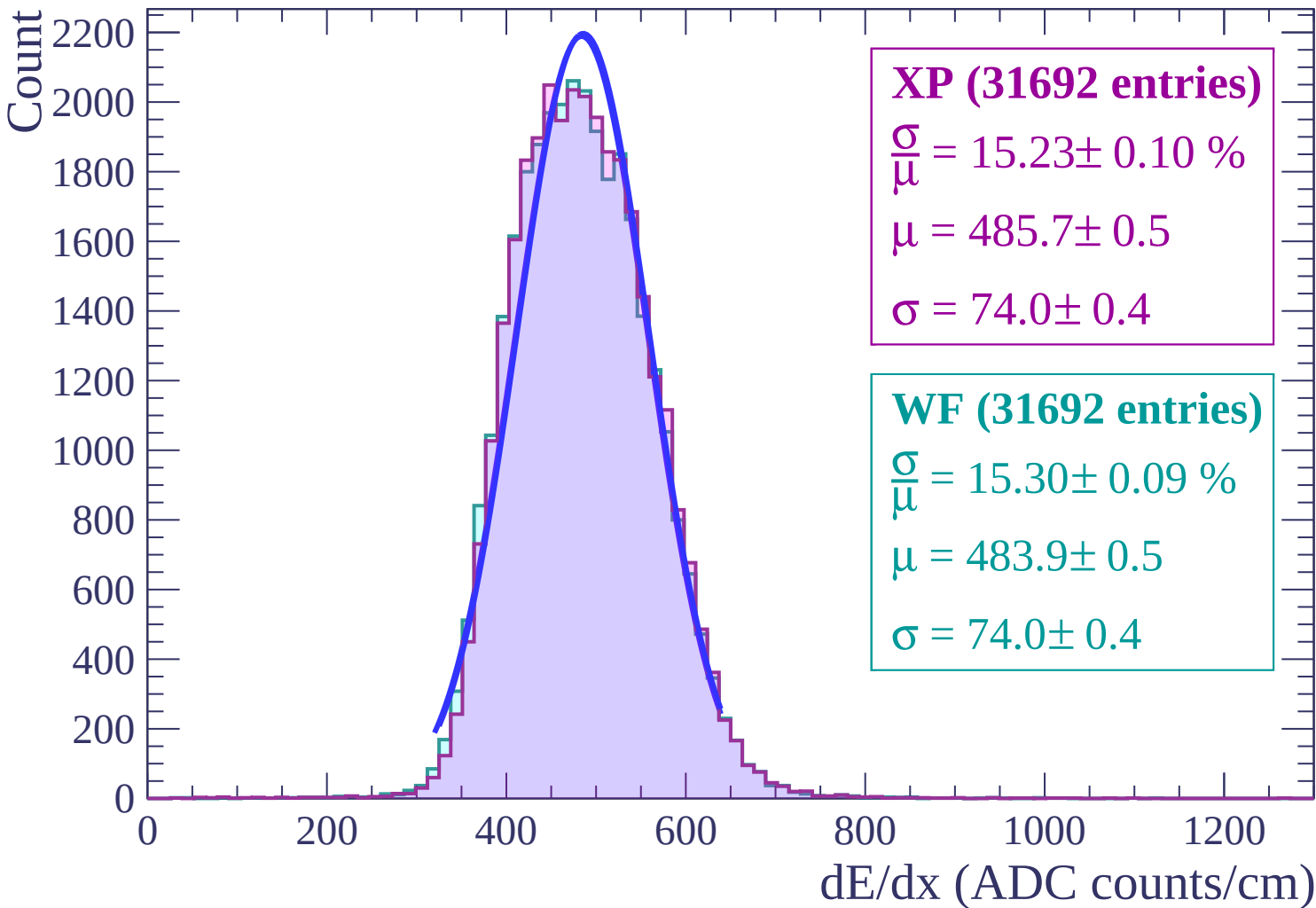
Energy loss | $-12 < \theta < -10$



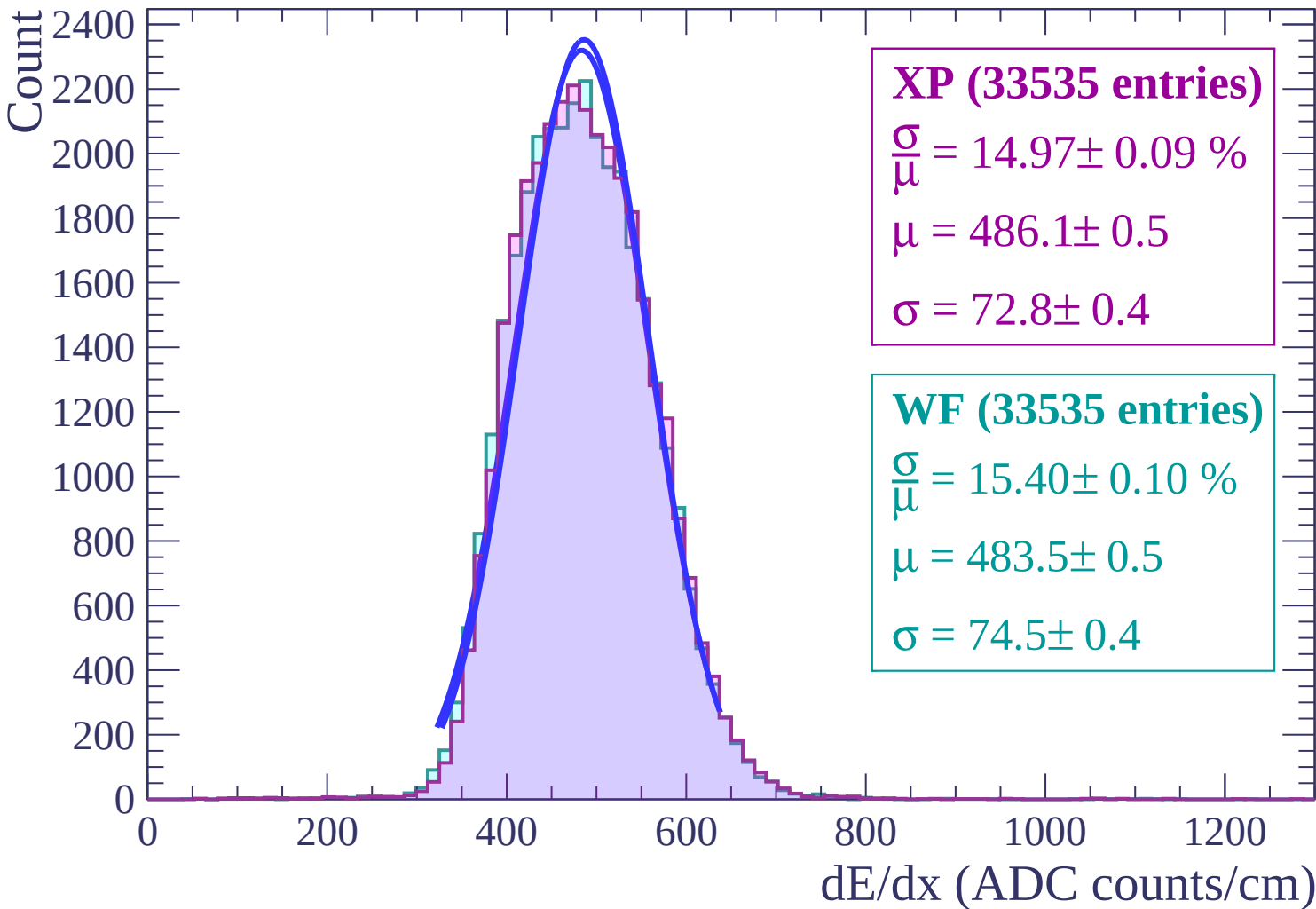
Energy loss | $-10 < \theta < -8$



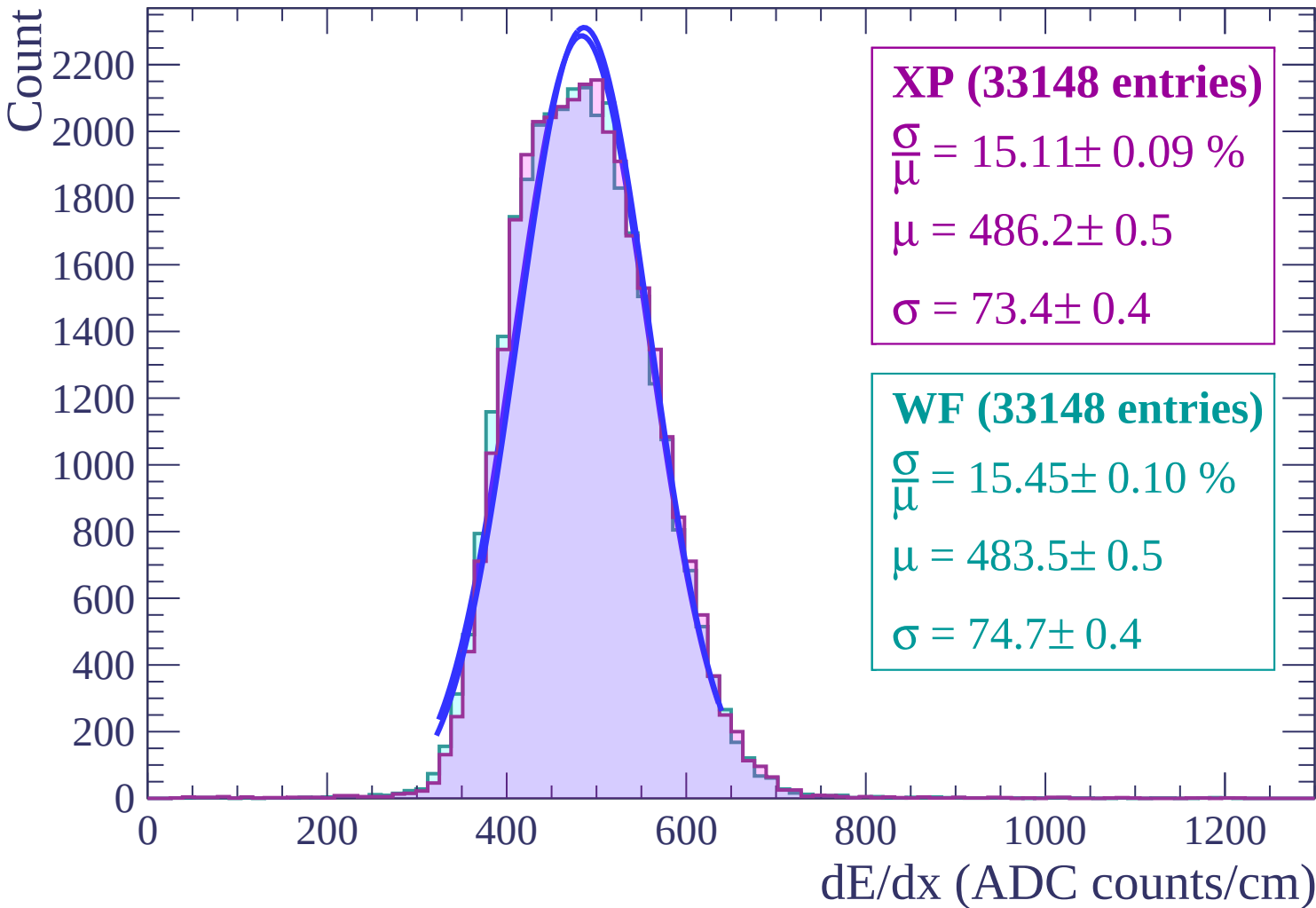
Energy loss | $-8 < \theta < -6$



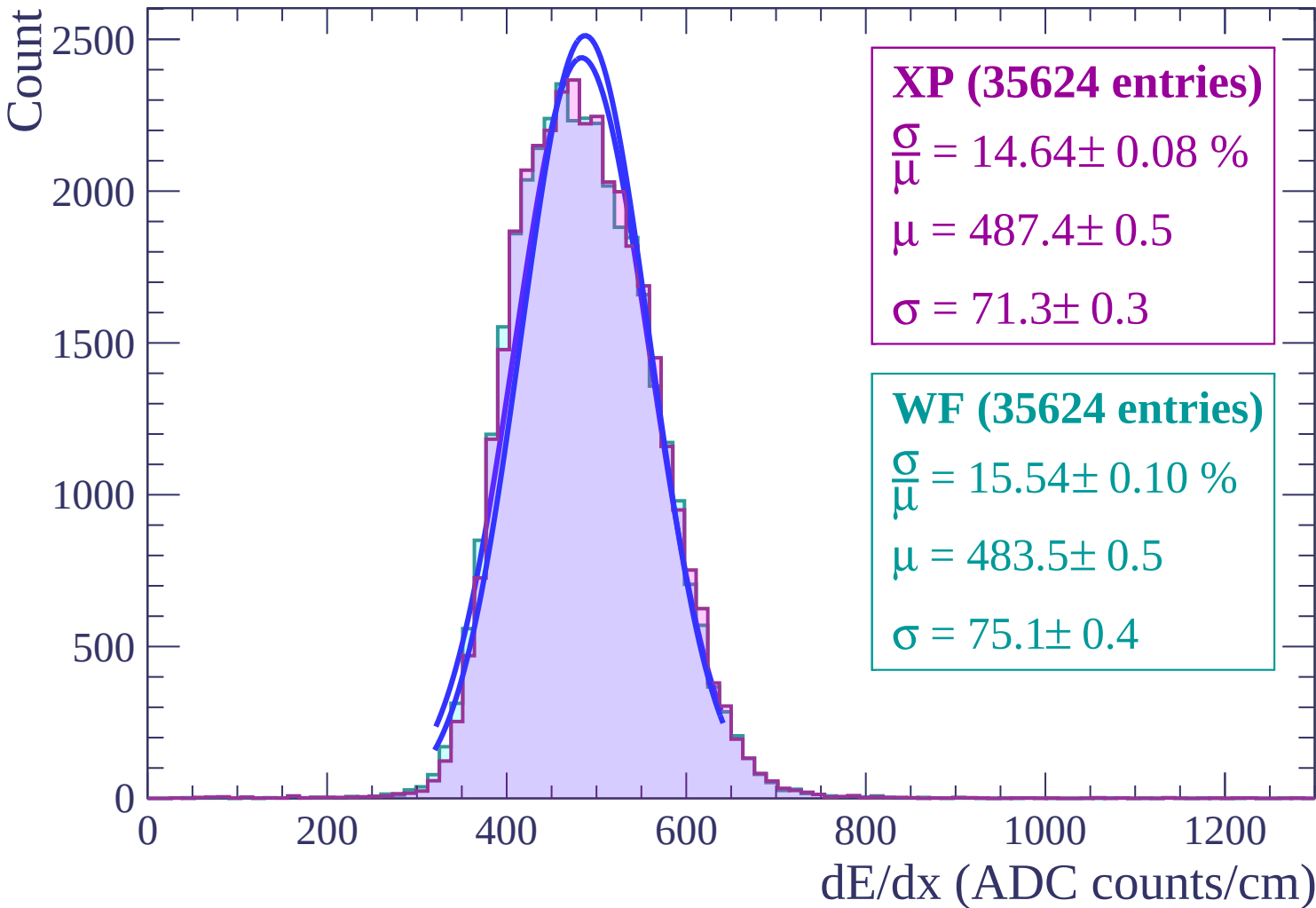
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$

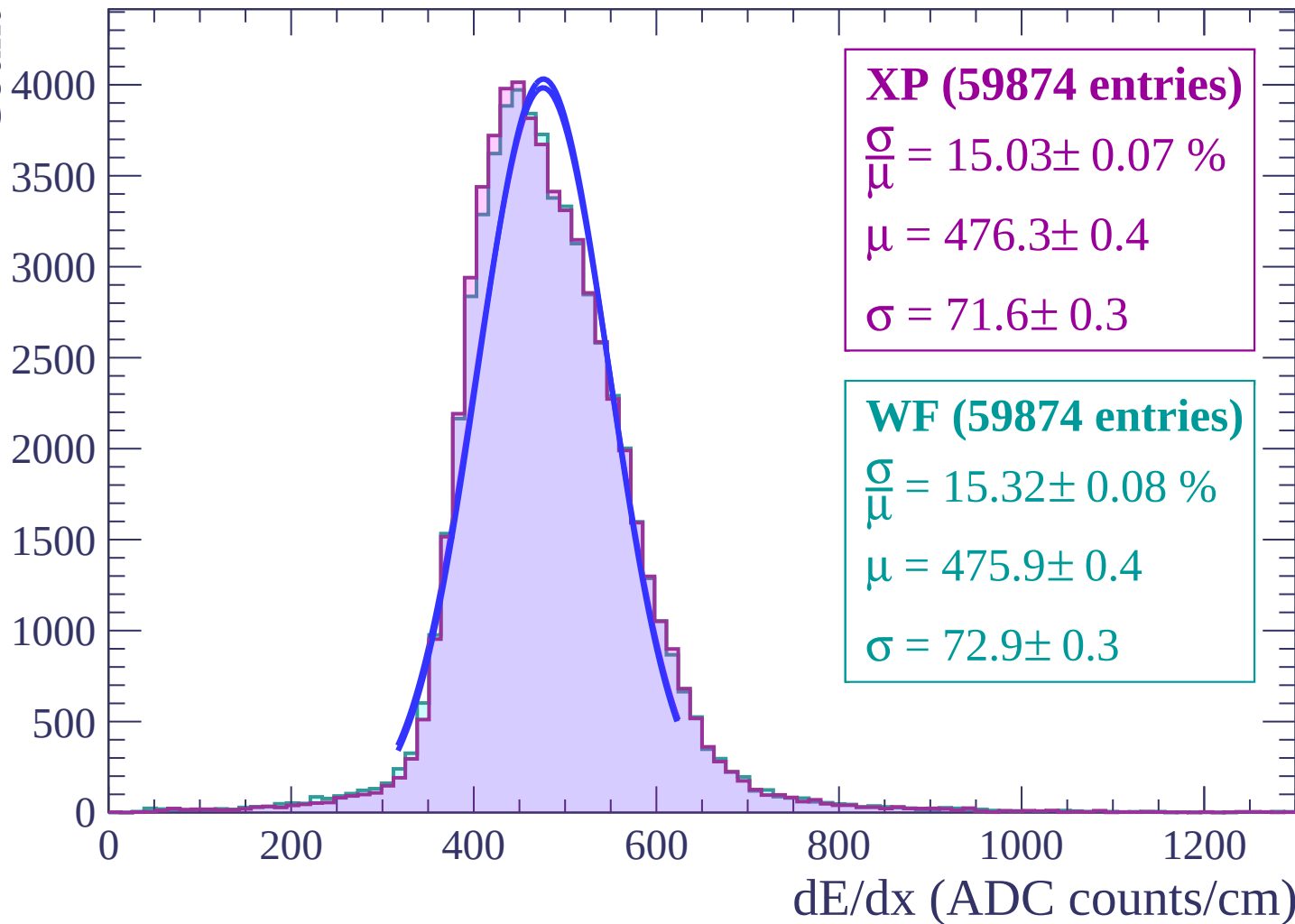


Energy loss $|-2 < \theta < 0$

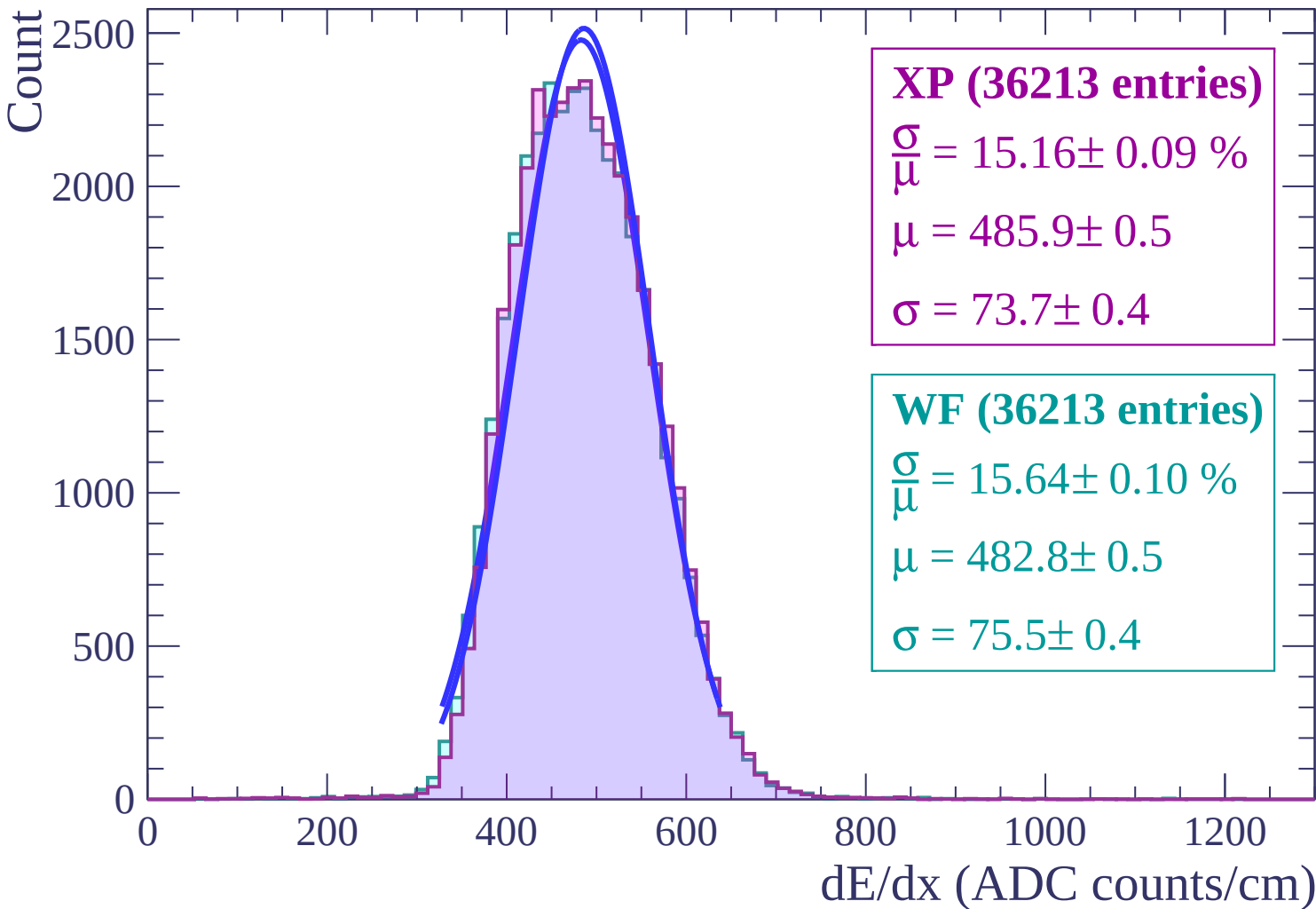


Energy loss | $0 < \theta < 2$

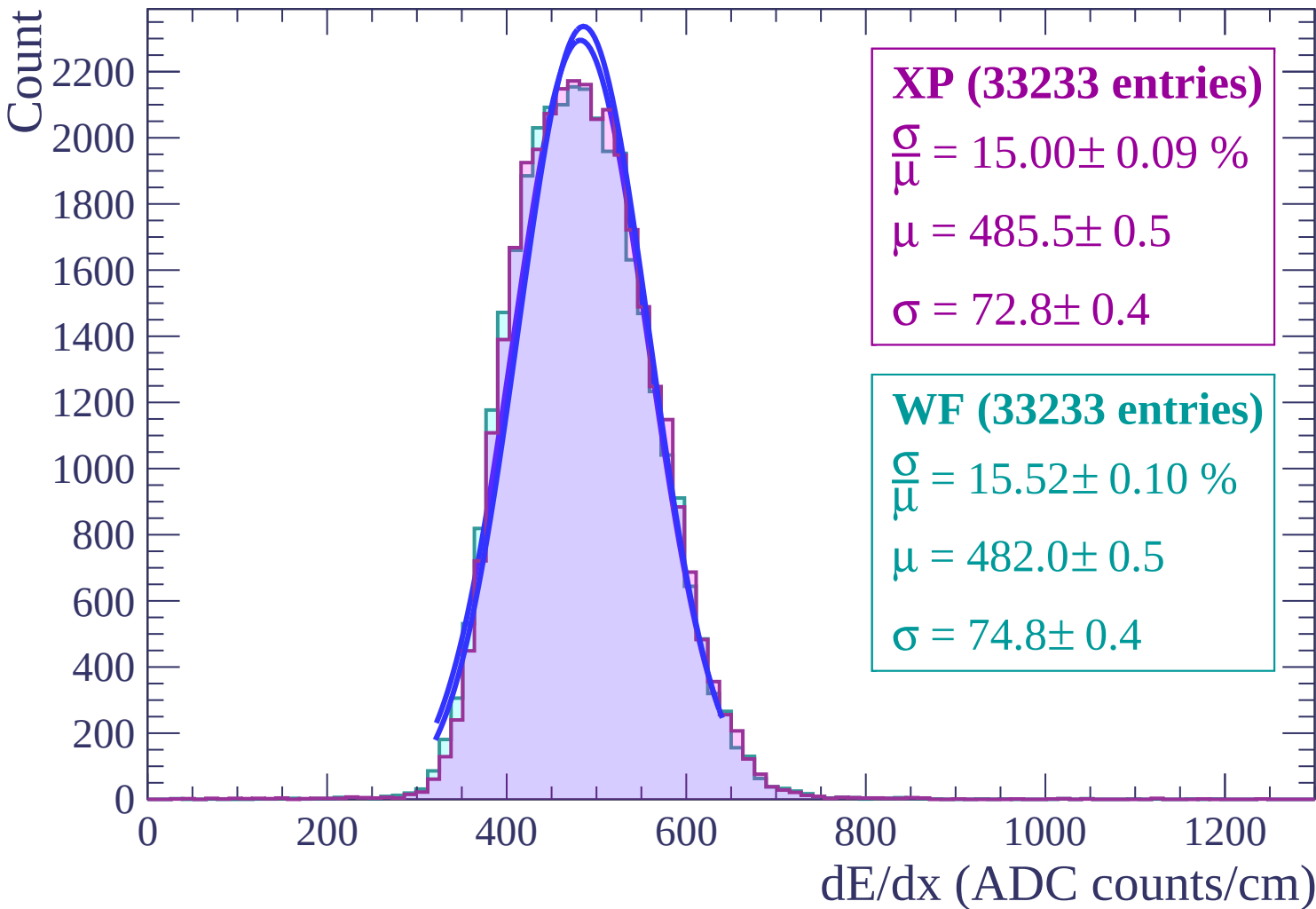
Count



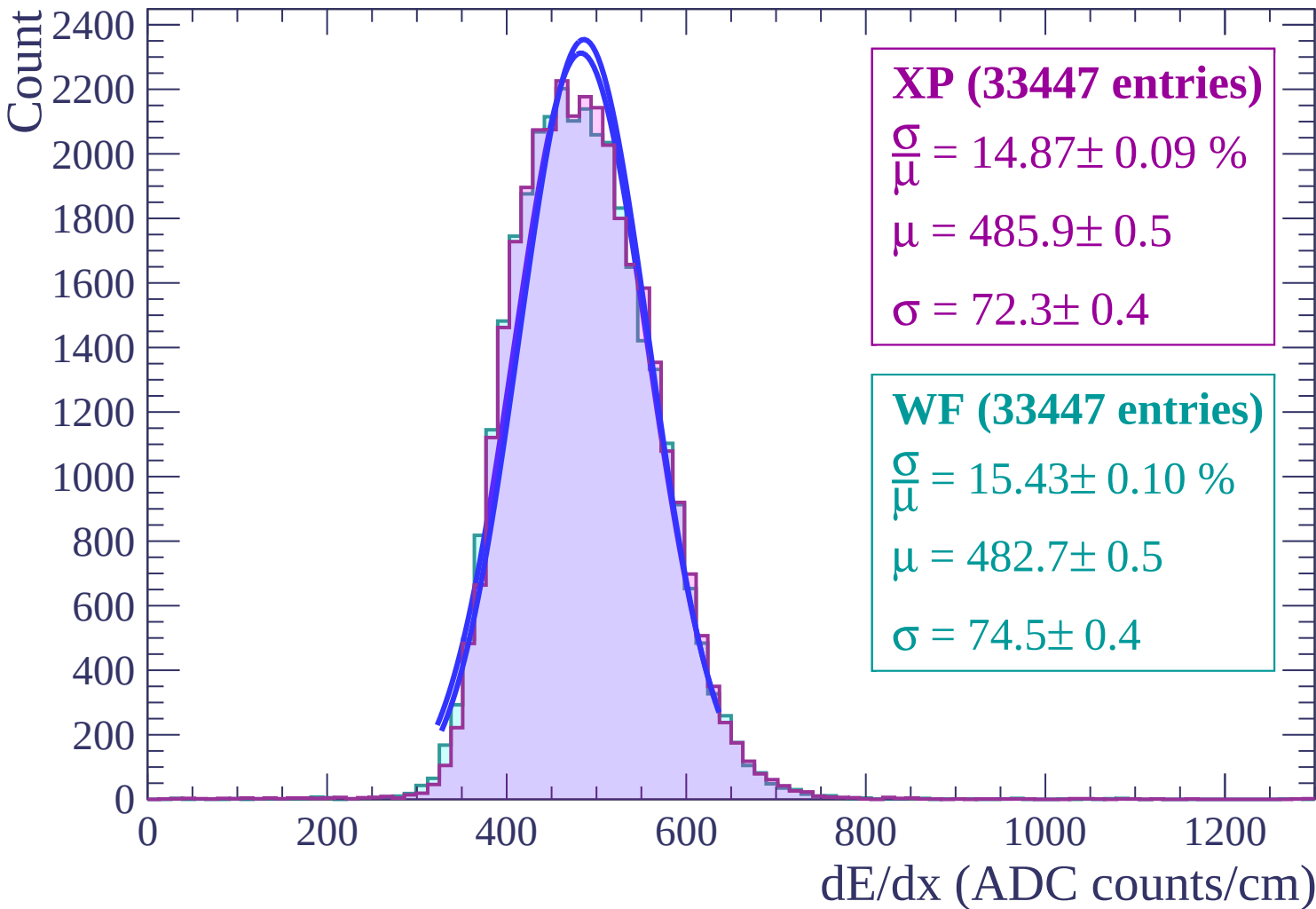
Energy loss | $2 < \theta < 4$



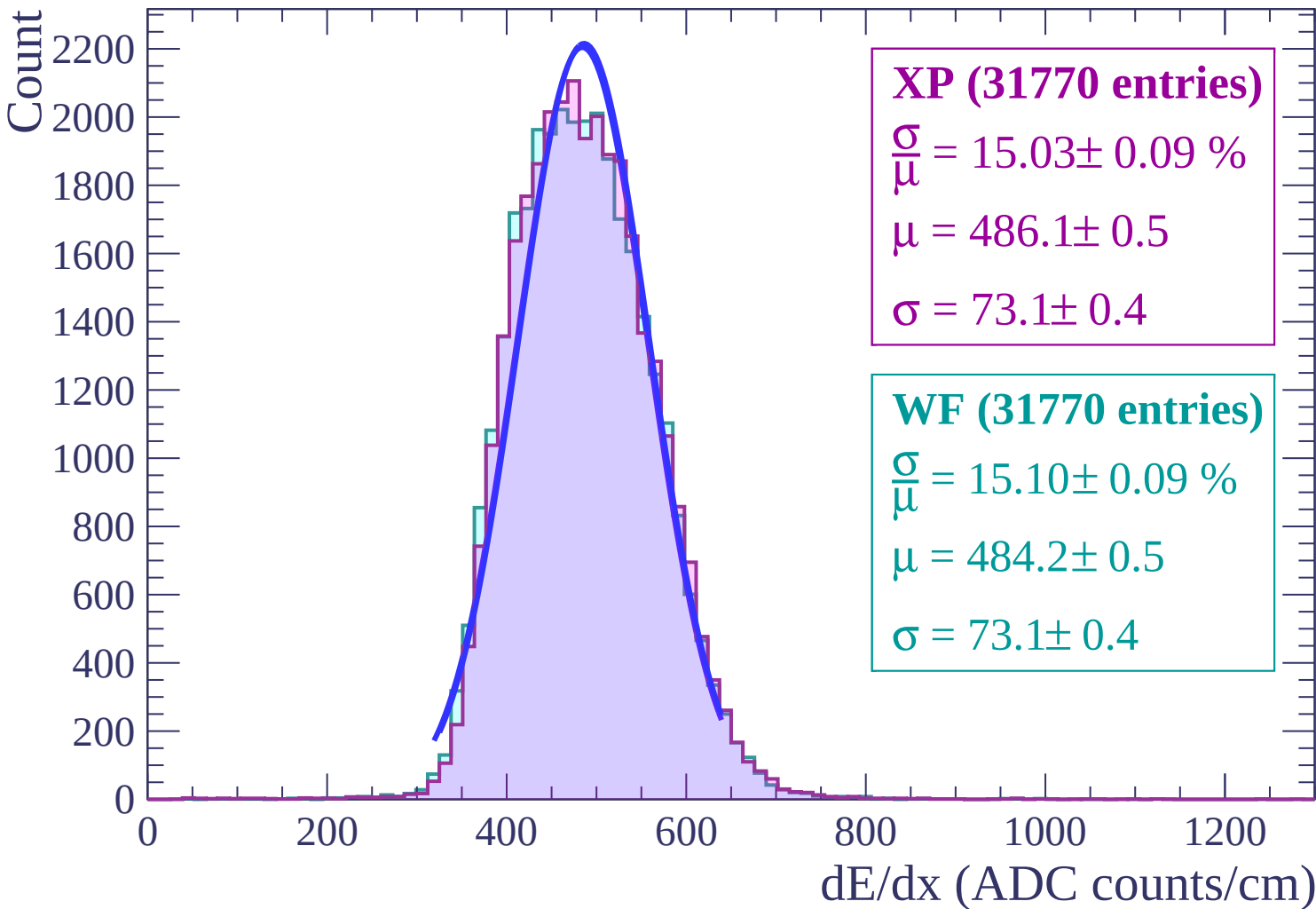
Energy loss | $4 < \theta < 6$



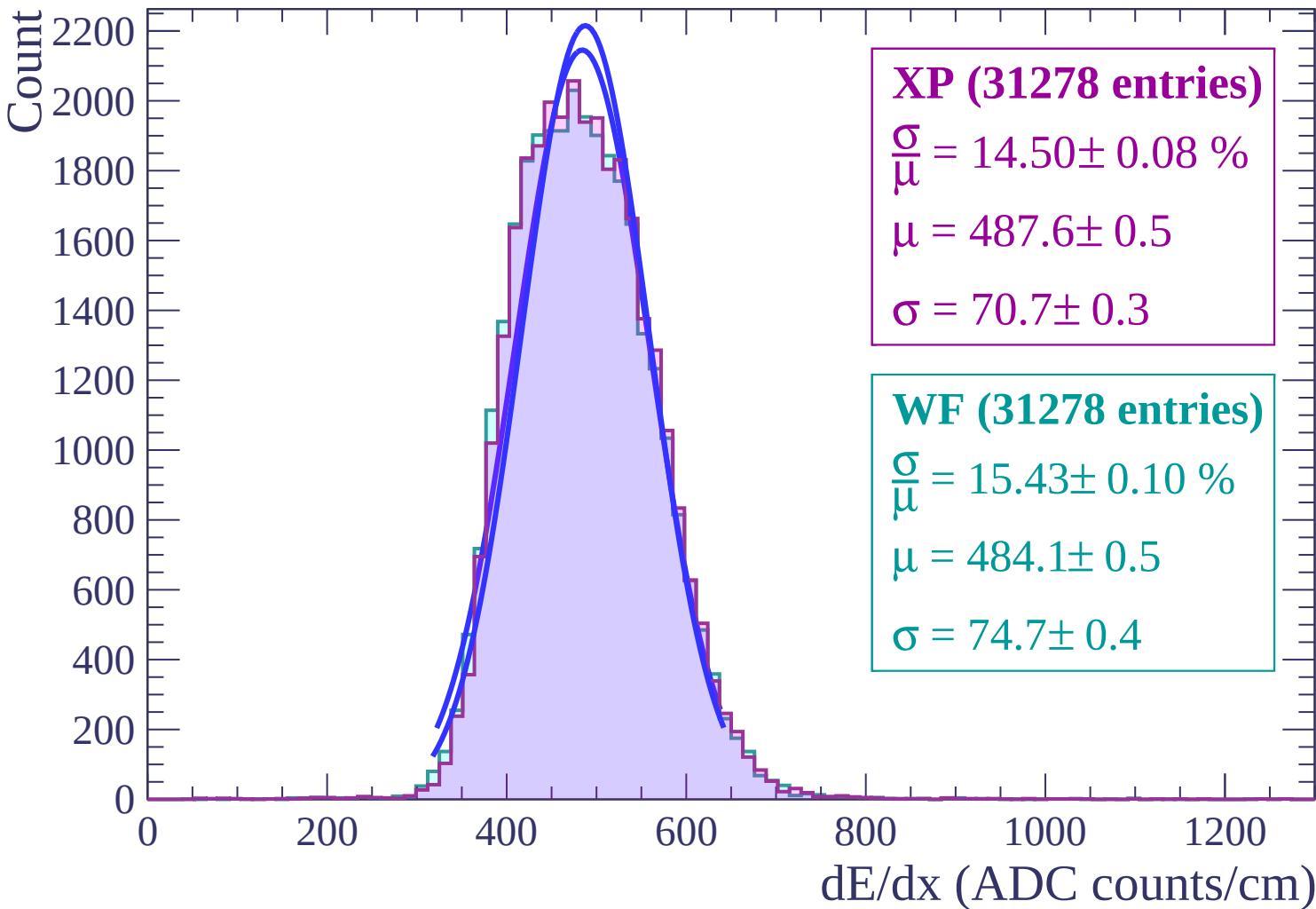
Energy loss | $6 < \theta < 8$



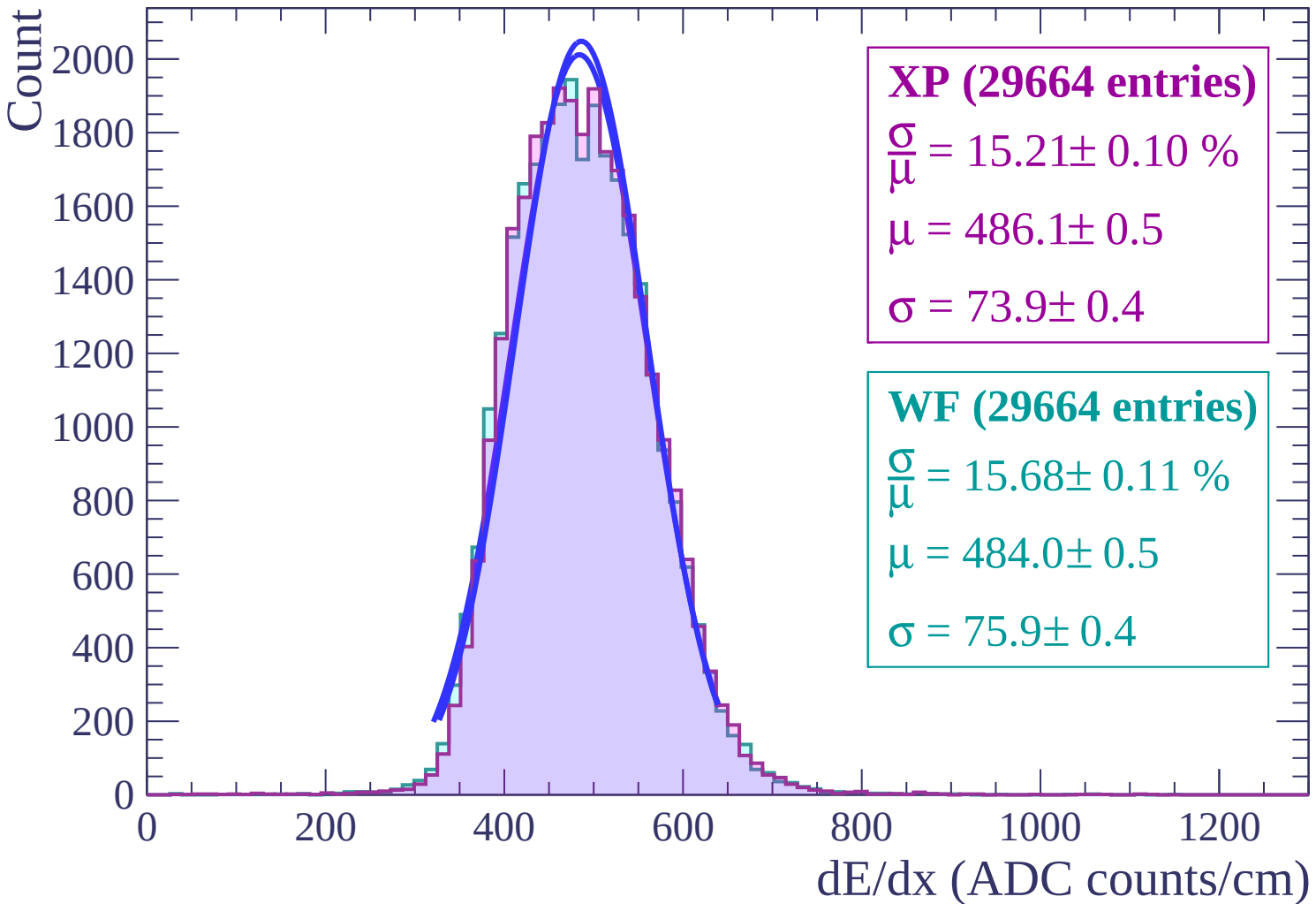
Energy loss | $8 < \theta < 10$



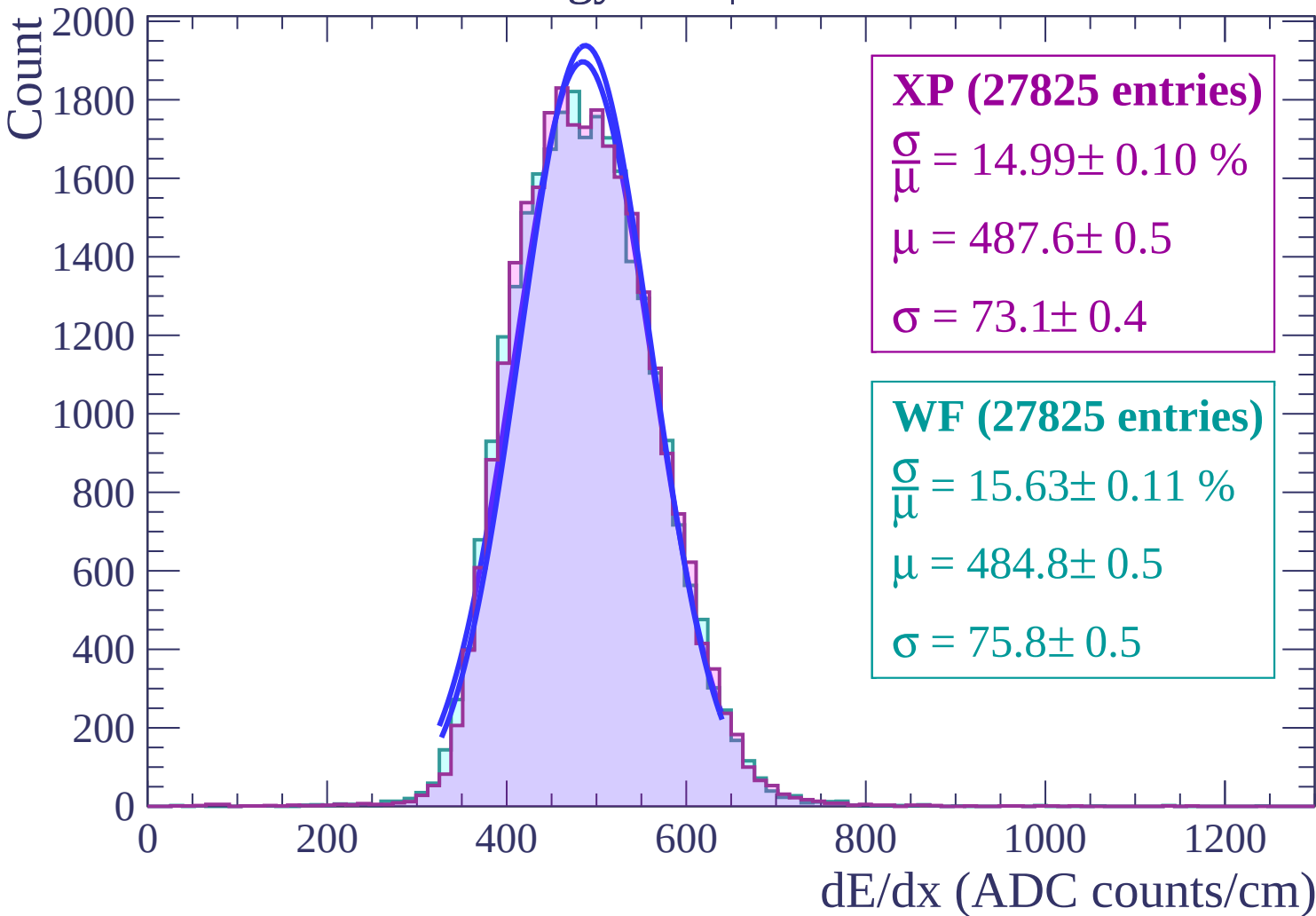
Energy loss | $10 < \theta < 12$



Energy loss | $12 < \theta < 14$



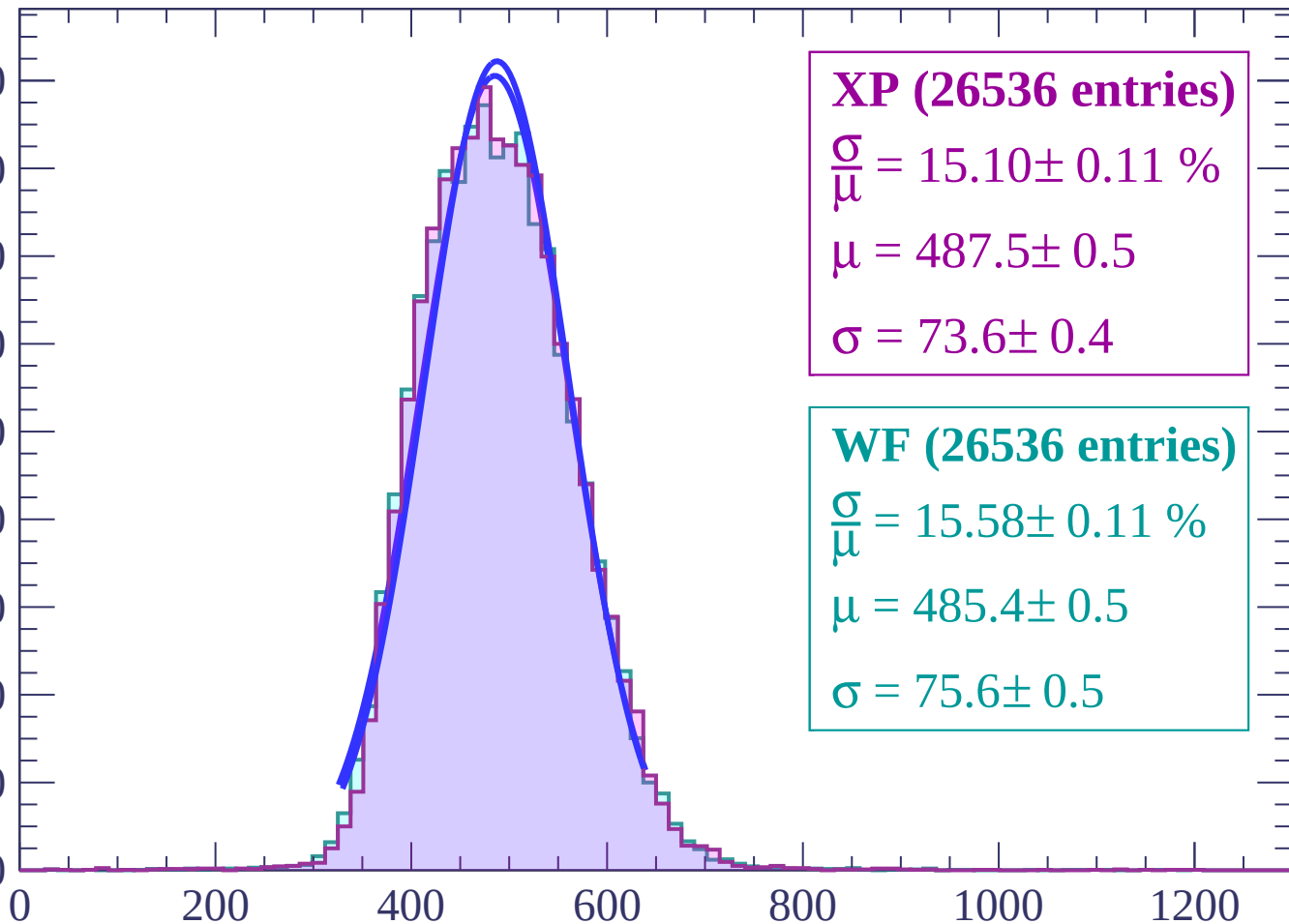
Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26536 entries)

$$\frac{\sigma}{\mu} = 15.10 \pm 0.11 \%$$

$$\mu = 487.5 \pm 0.5$$

$$\sigma = 73.6 \pm 0.4$$

WF (26536 entries)

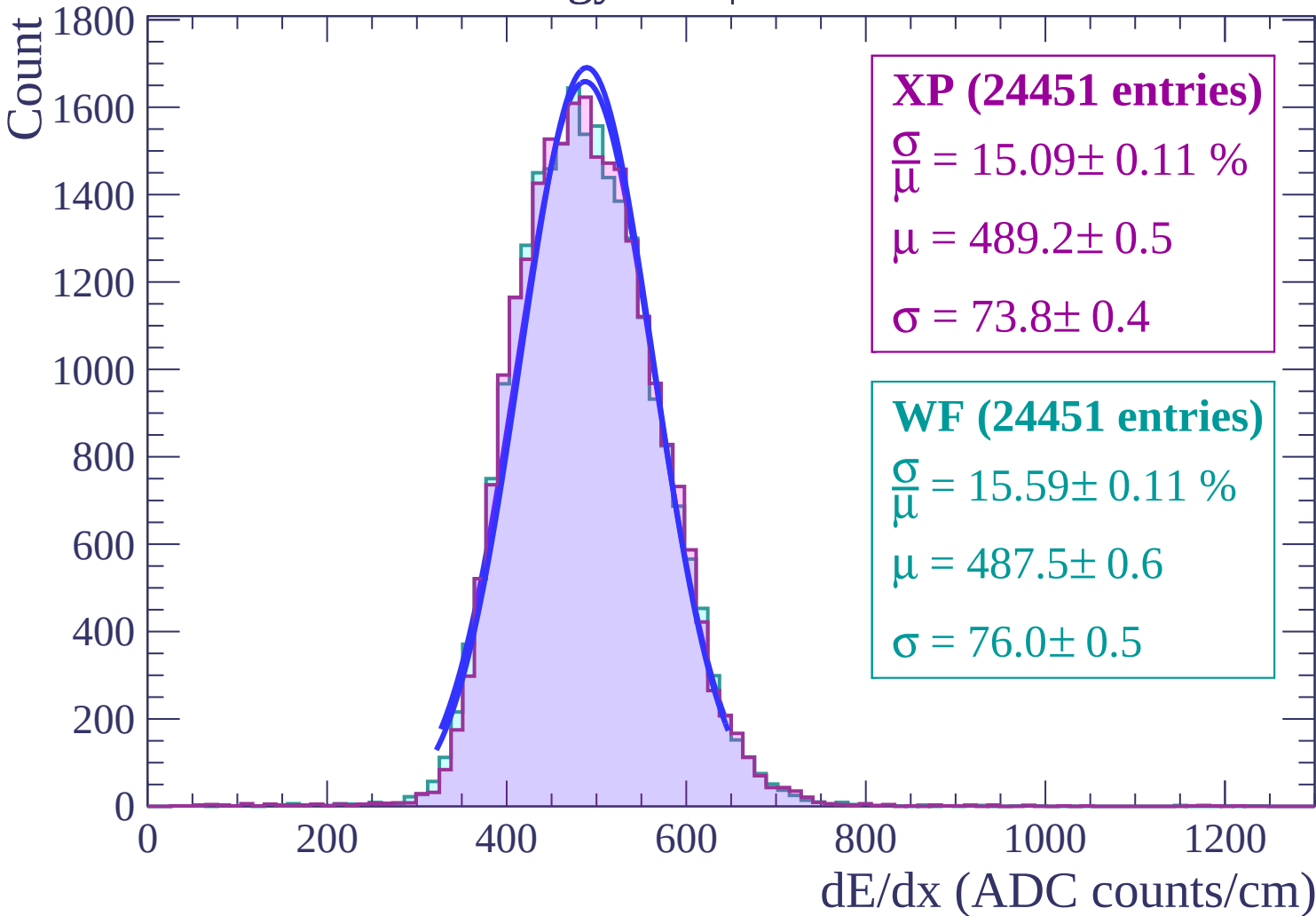
$$\frac{\sigma}{\mu} = 15.58 \pm 0.11 \%$$

$$\mu = 485.4 \pm 0.5$$

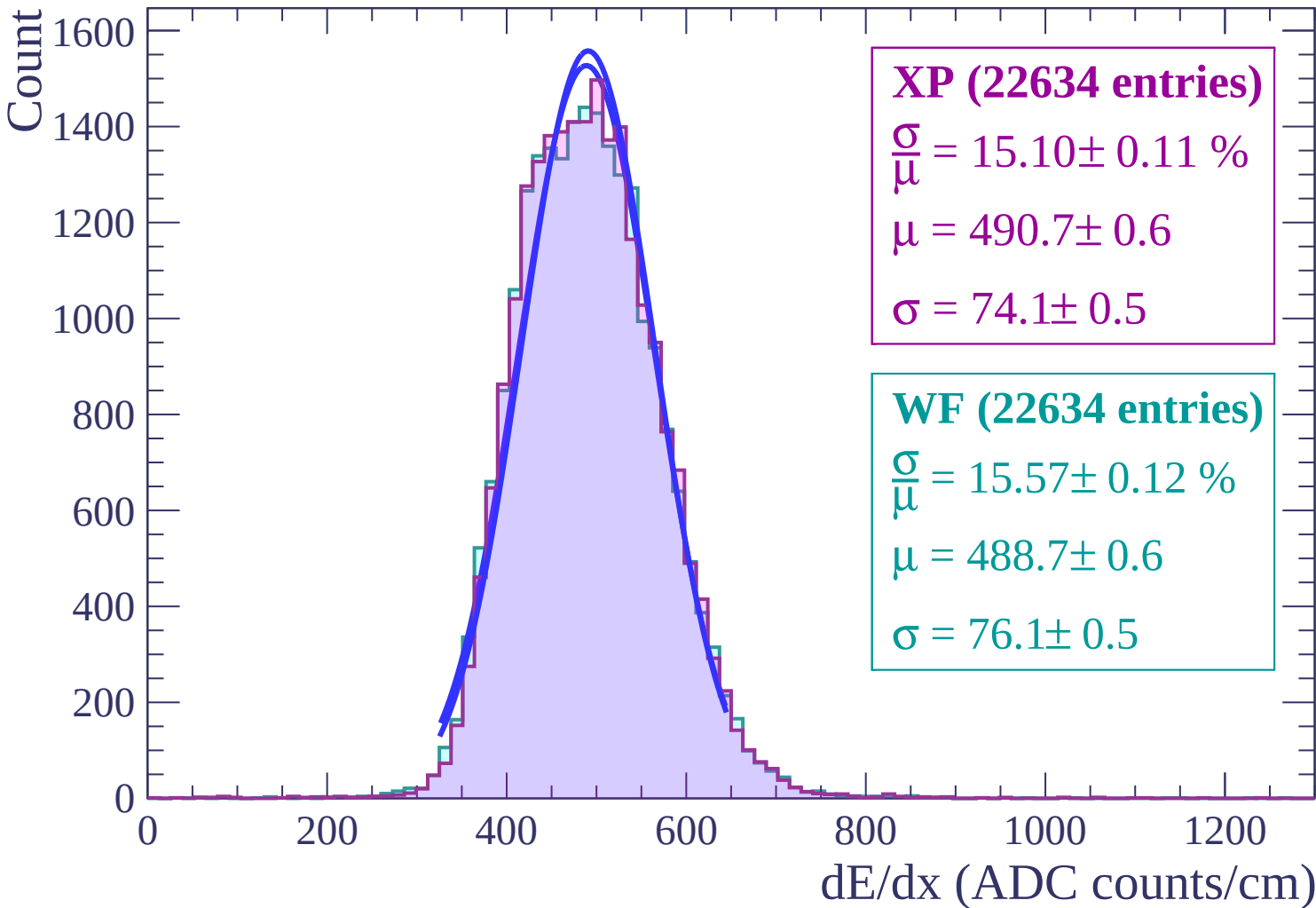
$$\sigma = 75.6 \pm 0.5$$

dE/dx (ADC counts/cm)

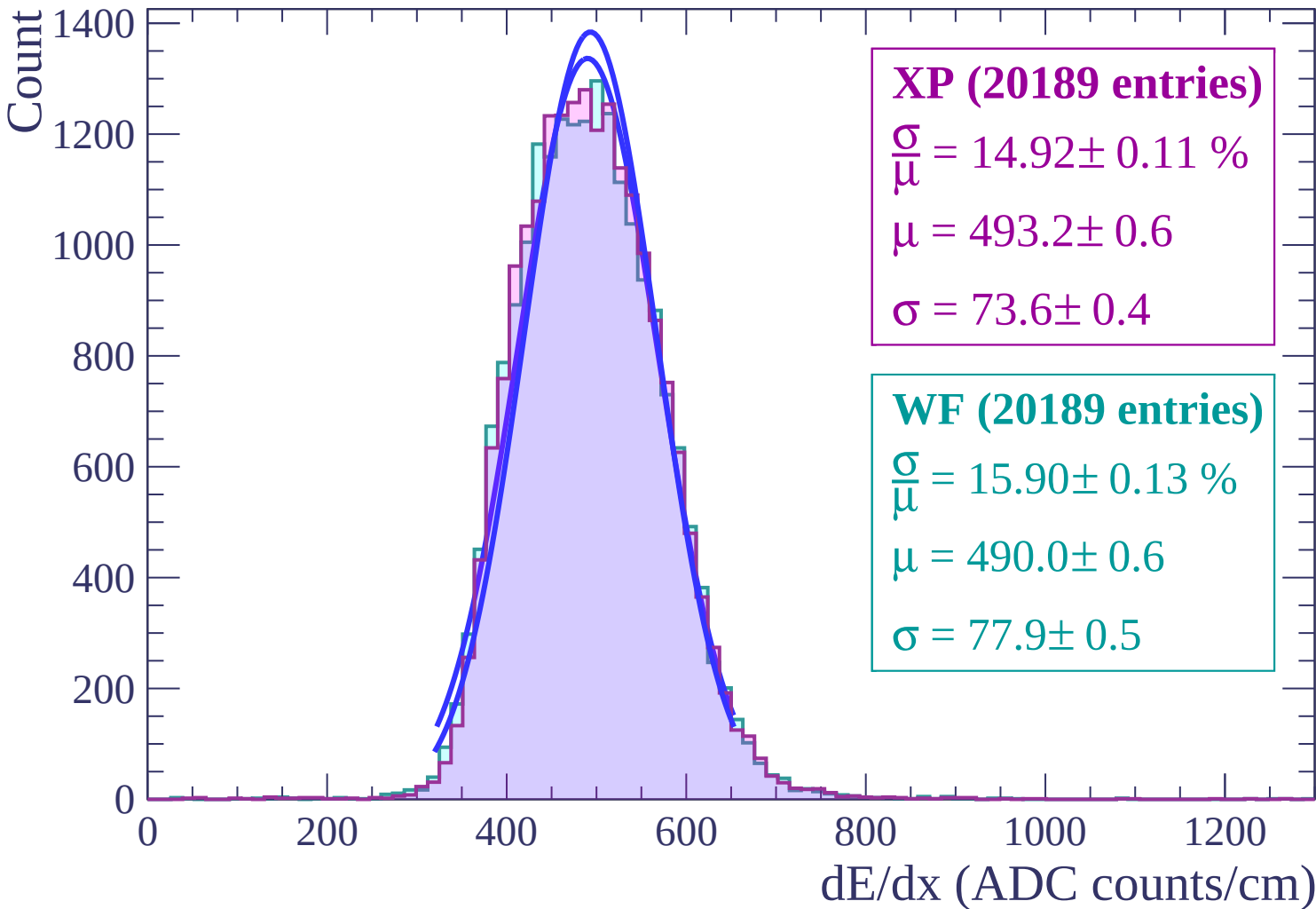
Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

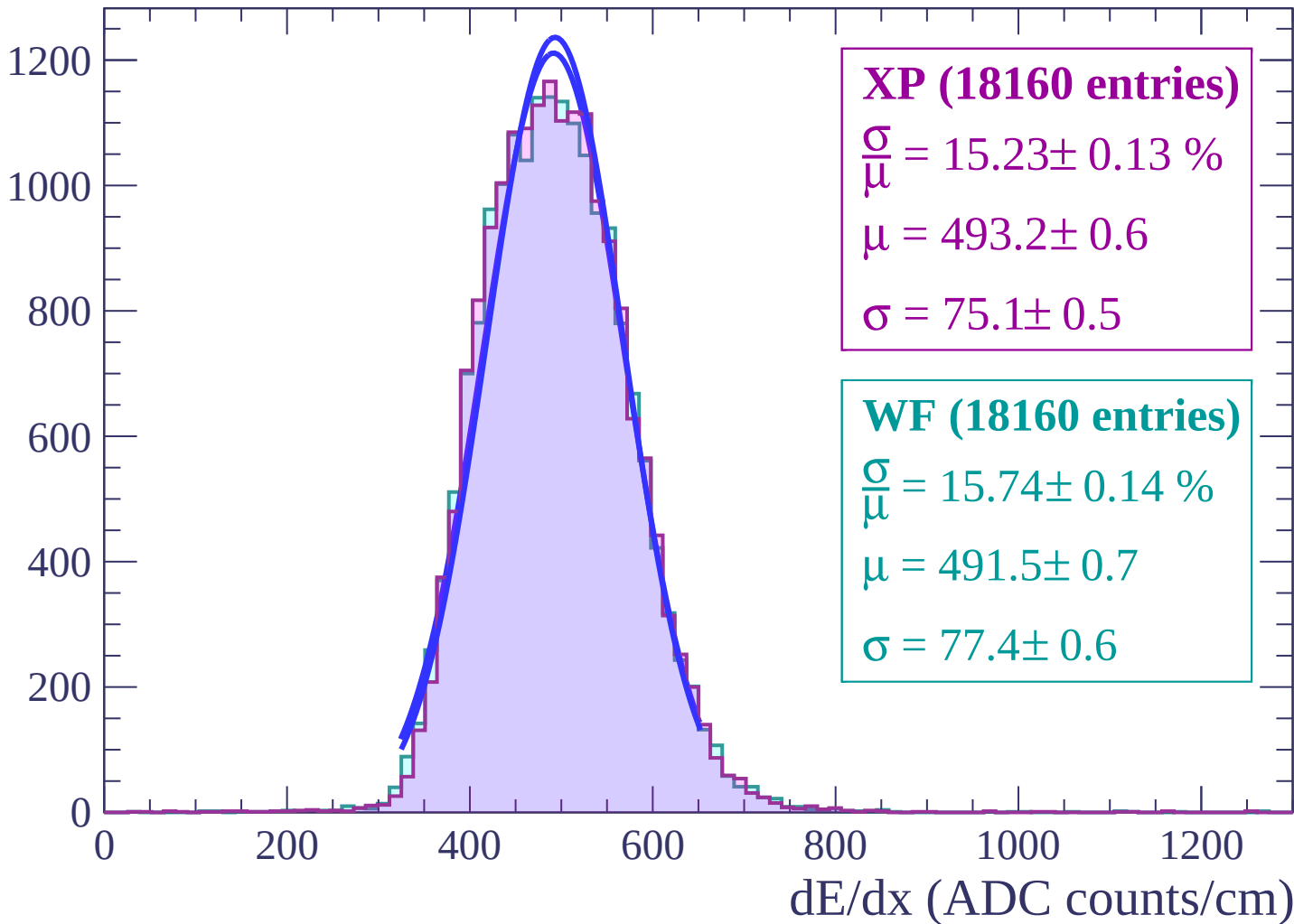


Energy loss | $22 < \theta < 24$



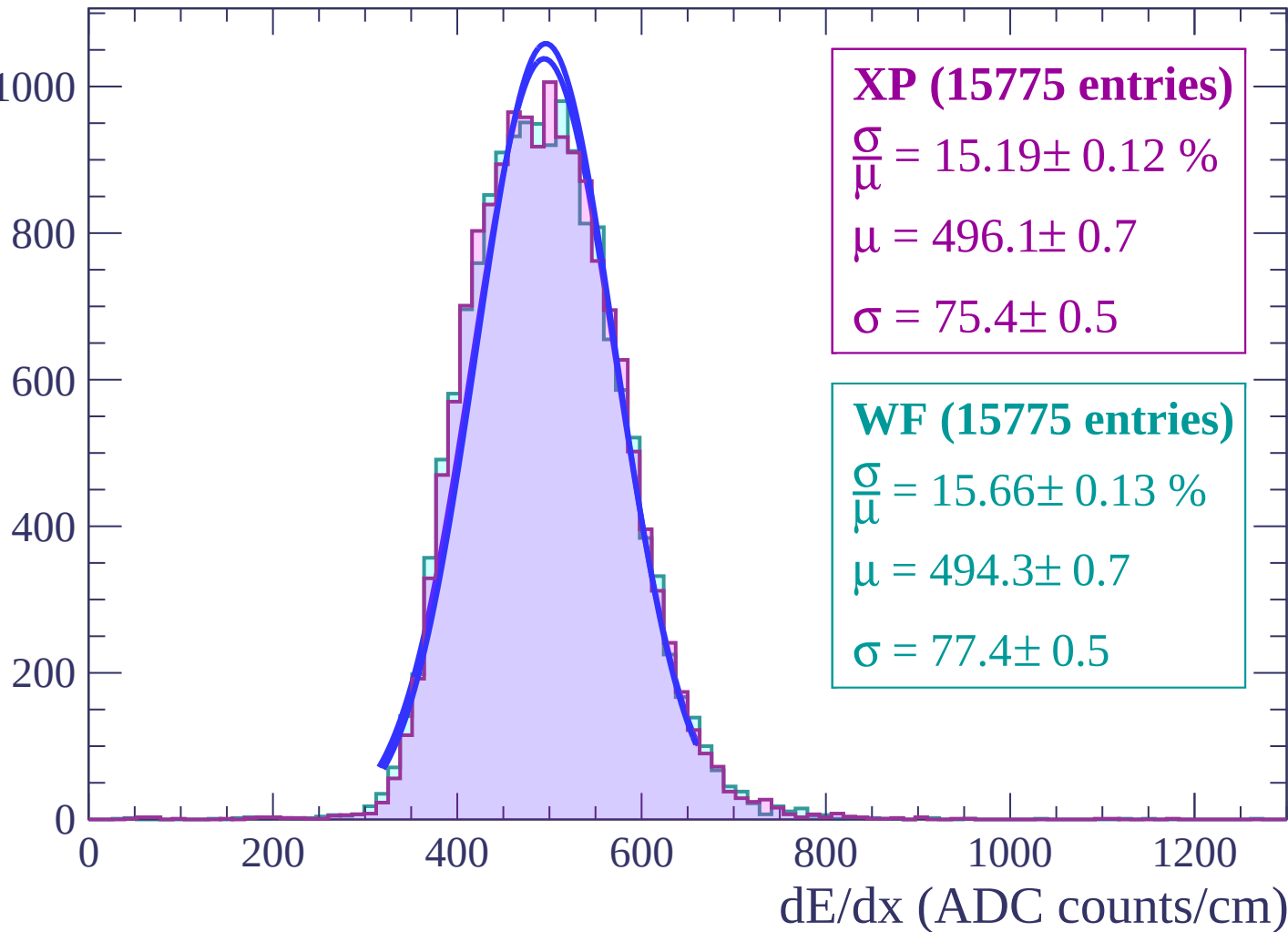
Energy loss | $24 < \theta < 26$

Count



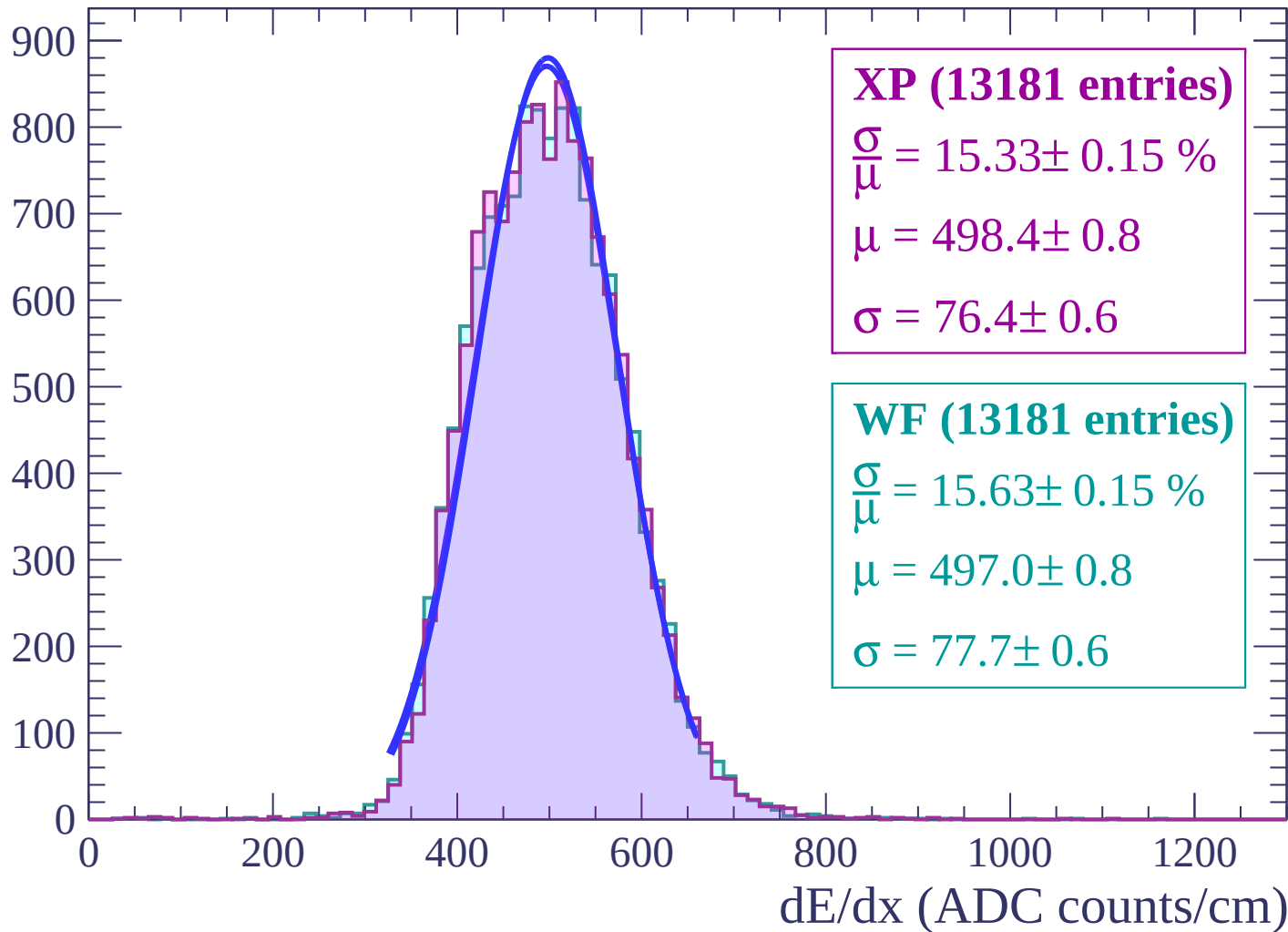
Energy loss | $26 < \theta < 28$

Count



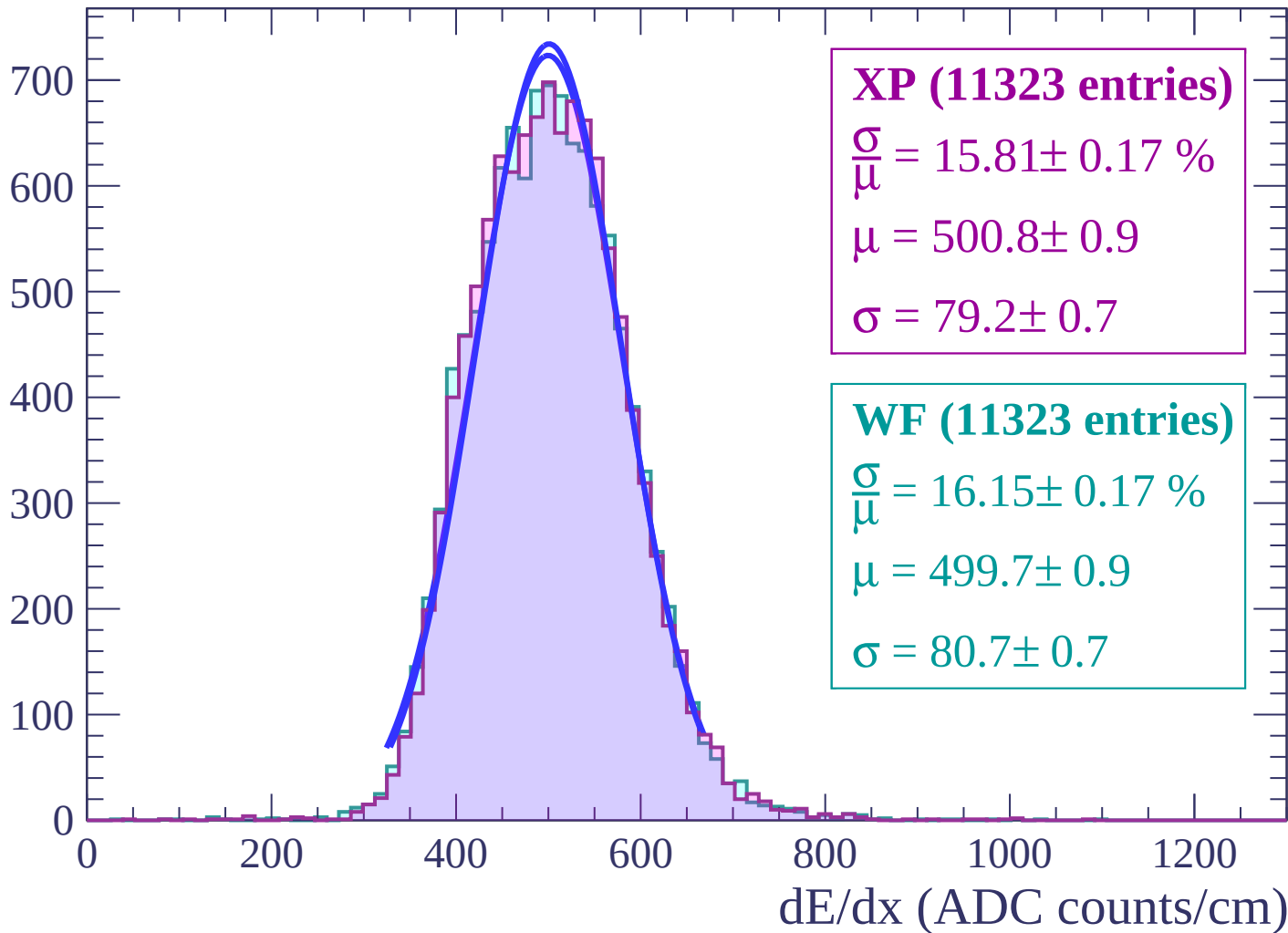
Energy loss | $28 < \theta < 30$

Count



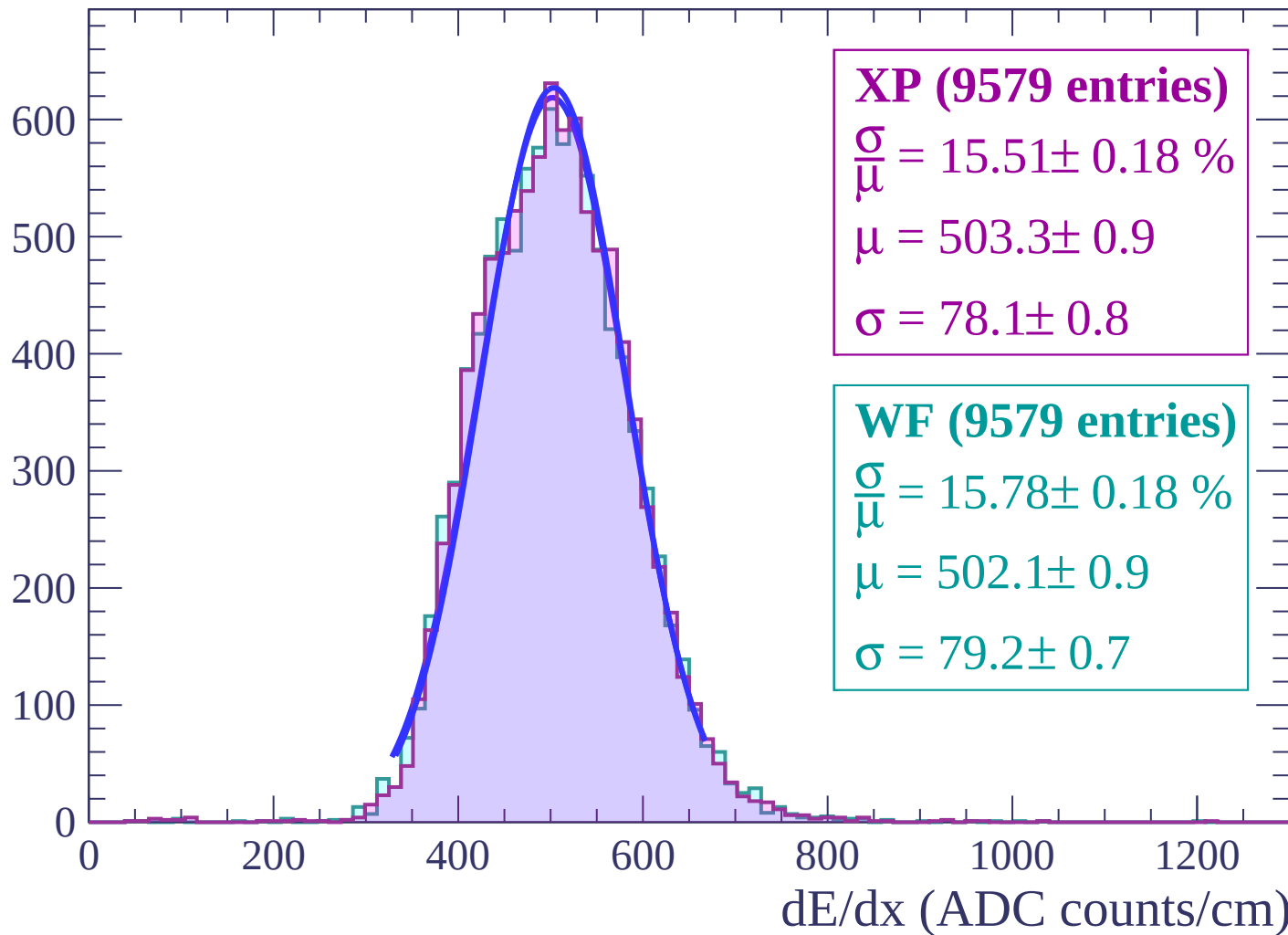
Energy loss | $30 < \theta < 32$

Count



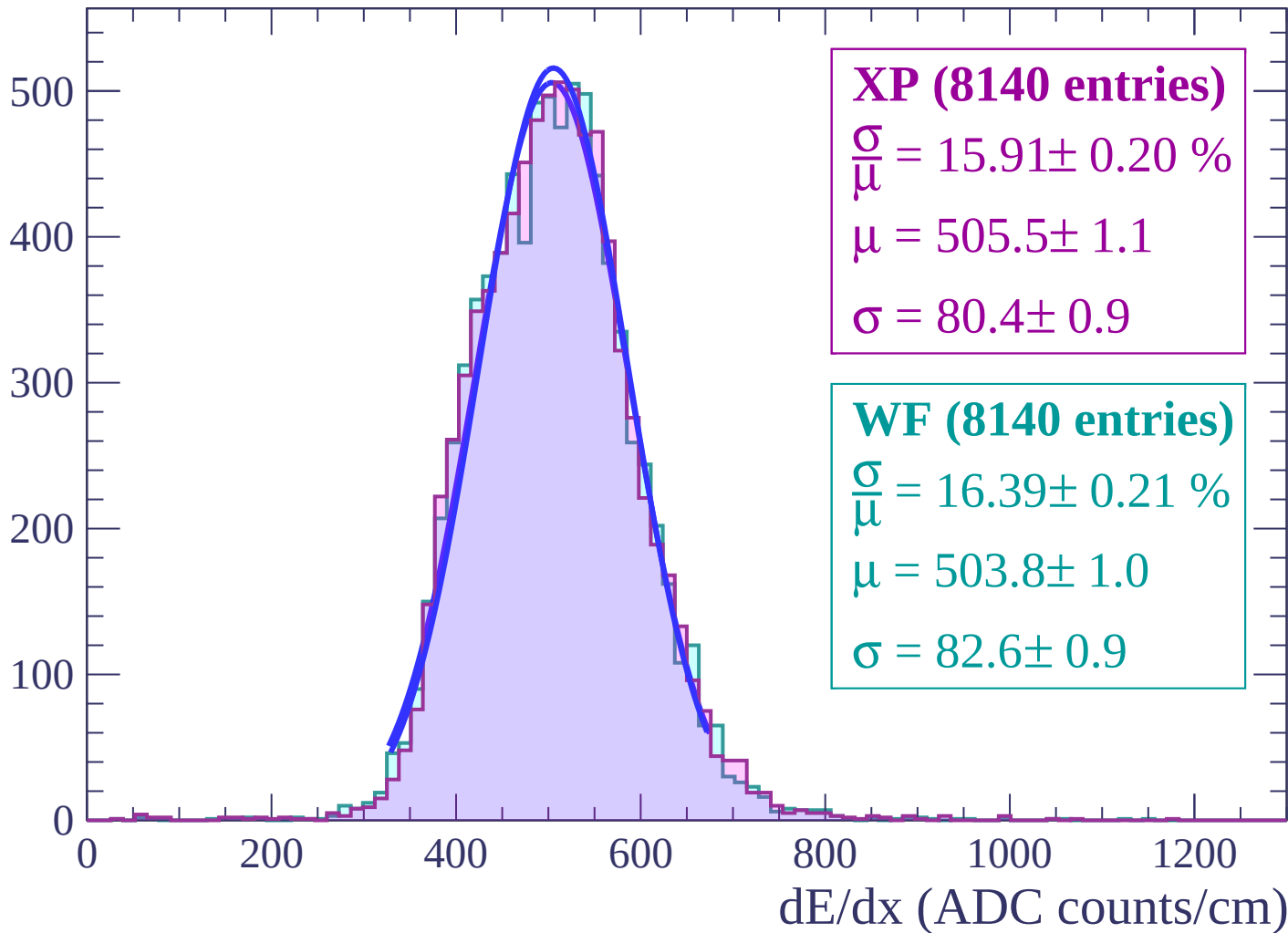
Energy loss | $32 < \theta < 34$

Count



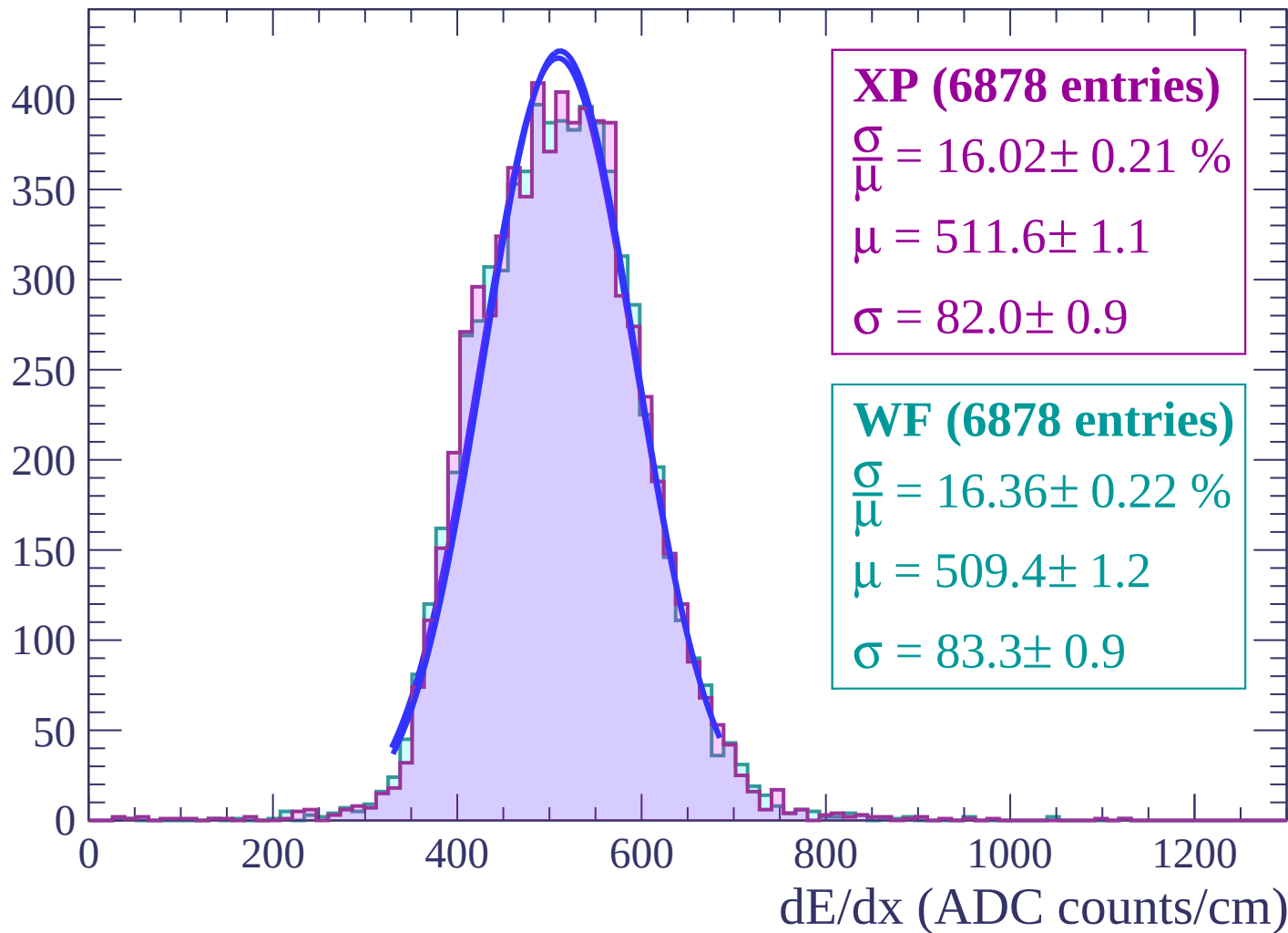
Energy loss | $34 < \theta < 36$

Count



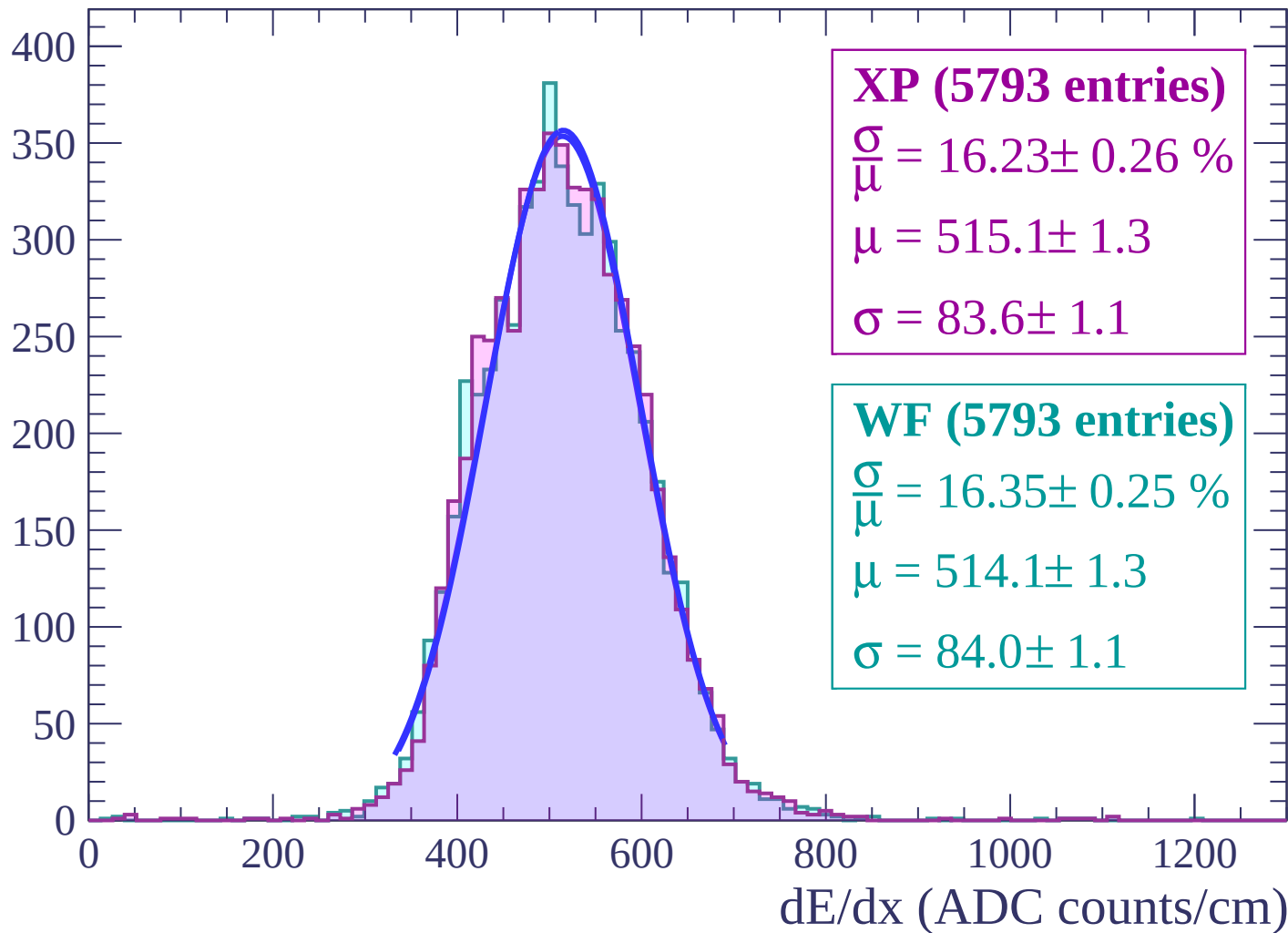
Energy loss | $36 < \theta < 38$

Count



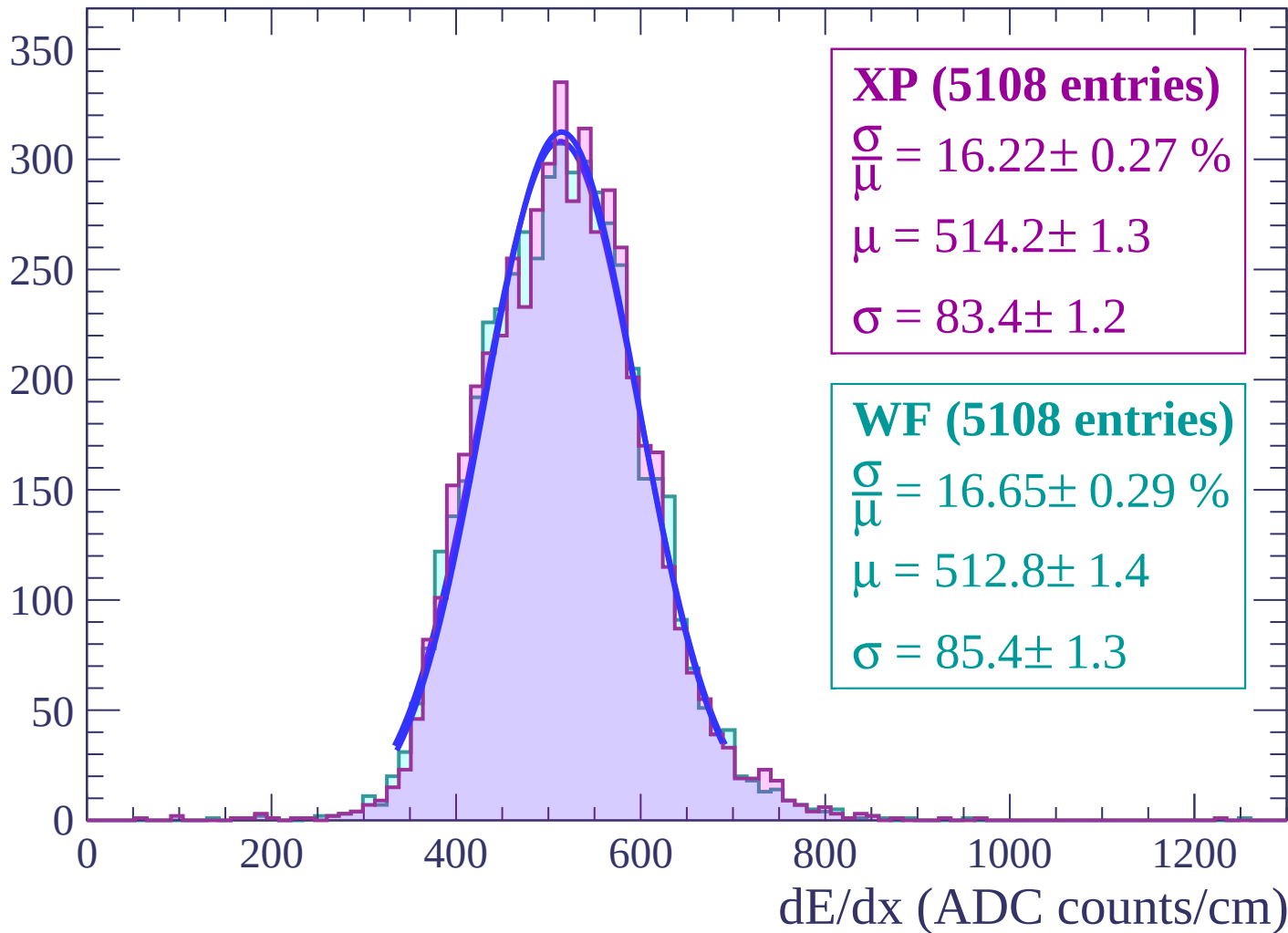
Energy loss | $38 < \theta < 40$

Count



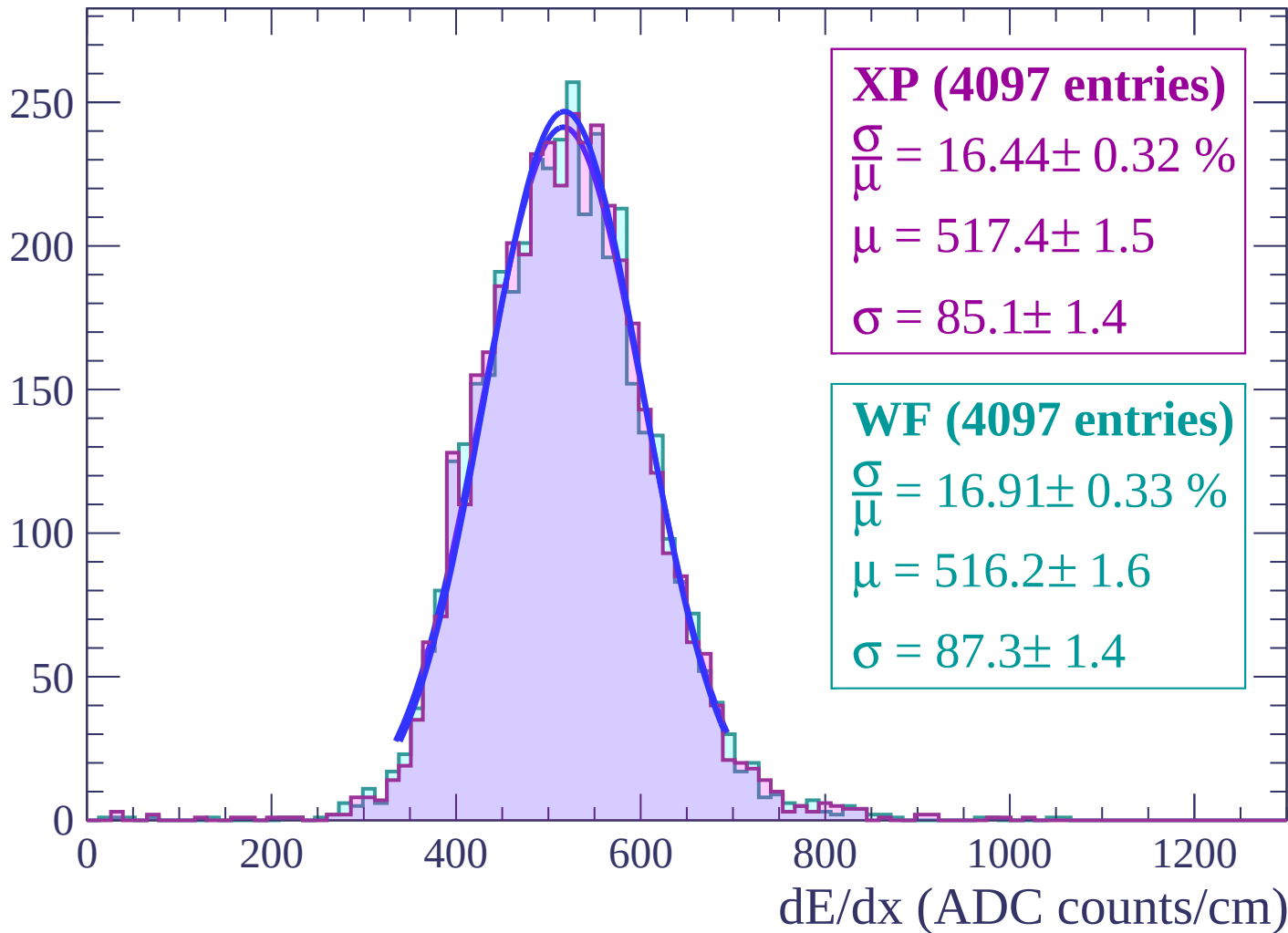
Energy loss | $40 < \theta < 42$

Count



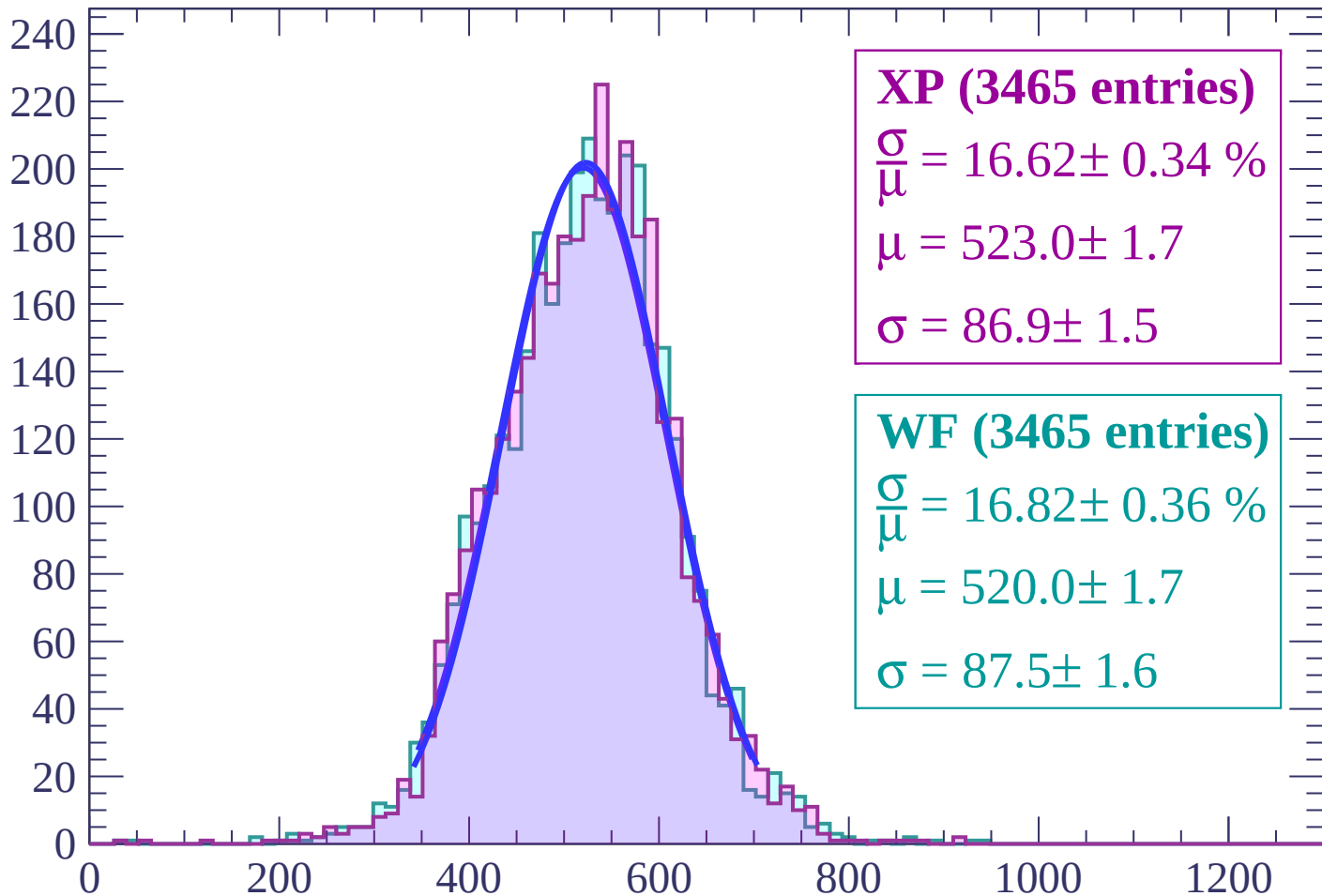
Energy loss | $42 < \theta < 44$

Count



Energy loss | $44 < \theta < 46$

Count



XP (3465 entries)

$\frac{\sigma}{\mu} = 16.62 \pm 0.34 \%$

$\mu = 523.0 \pm 1.7$

$\sigma = 86.9 \pm 1.5$

WF (3465 entries)

$\frac{\sigma}{\mu} = 16.82 \pm 0.36 \%$

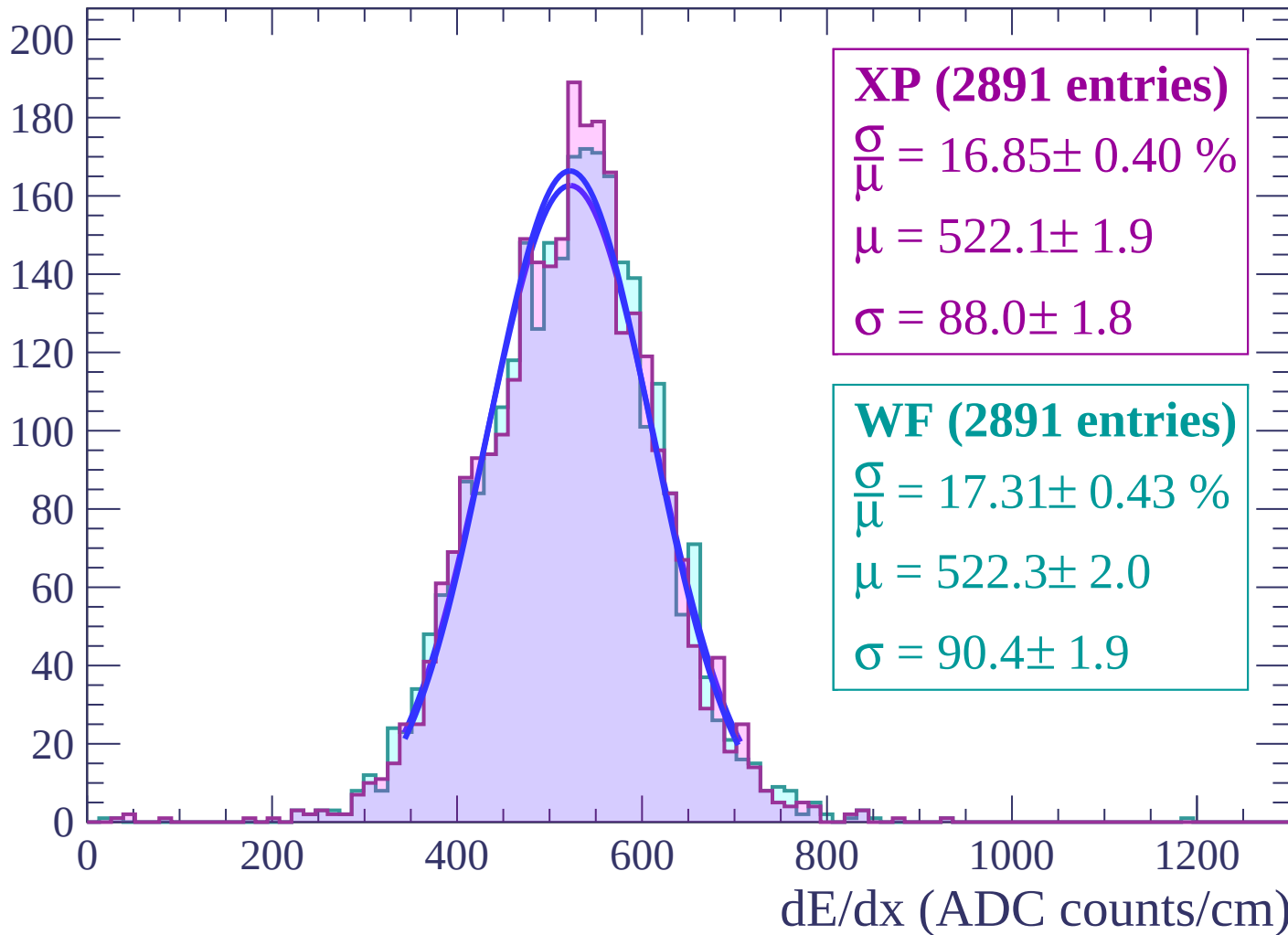
$\mu = 520.0 \pm 1.7$

$\sigma = 87.5 \pm 1.6$

dE/dx (ADC counts/cm)

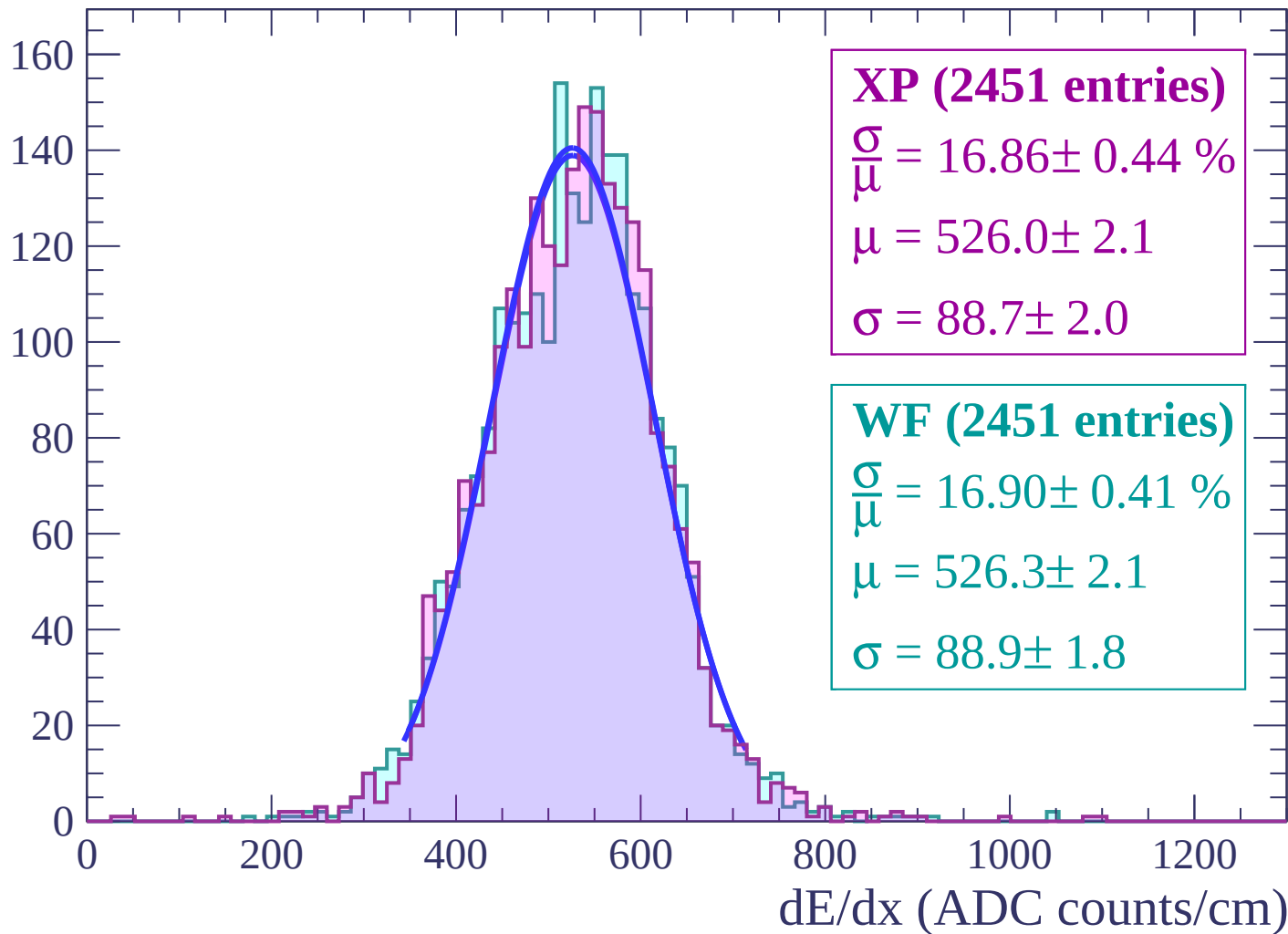
Energy loss | $46 < \theta < 48$

Count



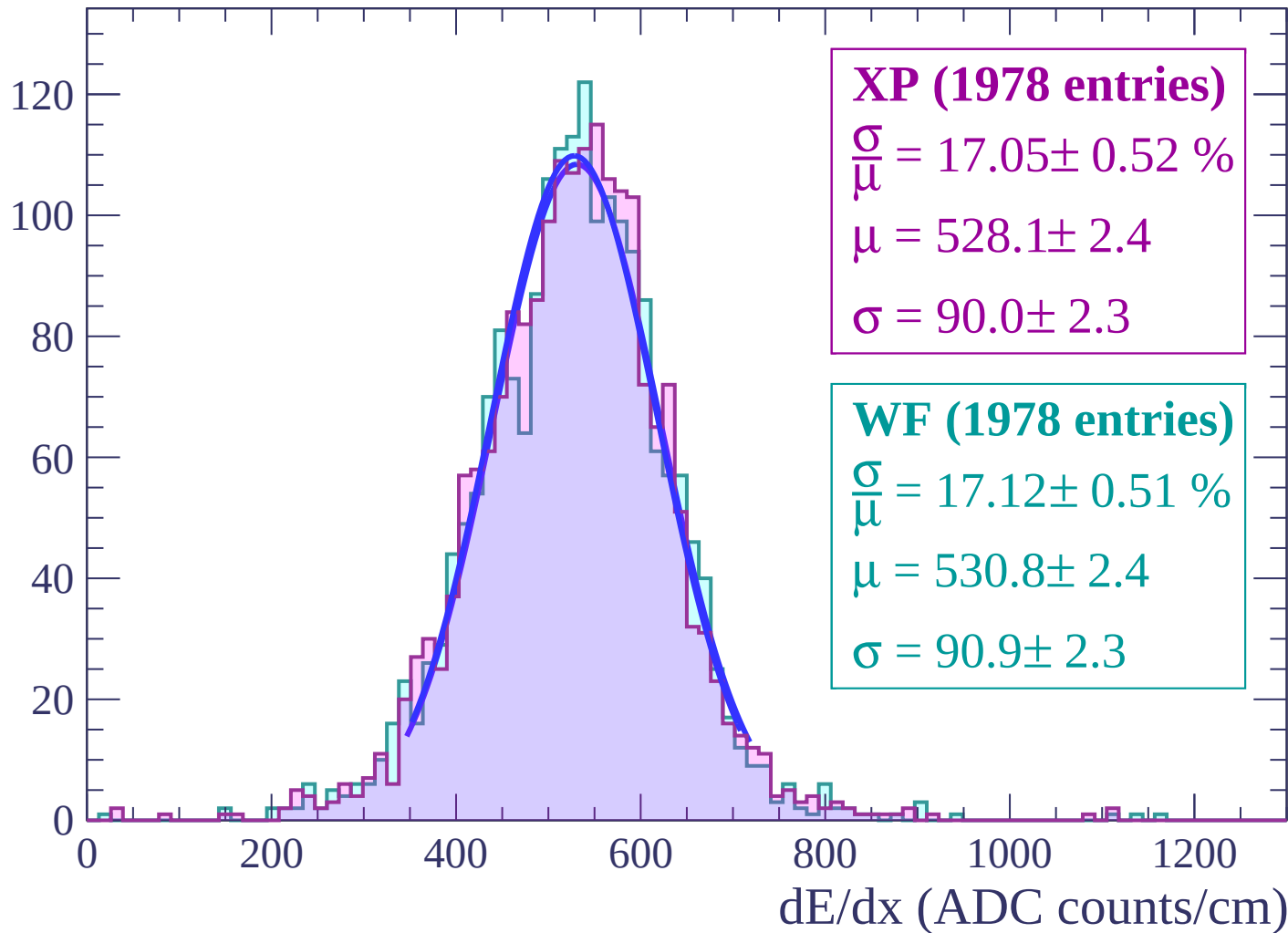
Energy loss | $48 < \theta < 50$

Count



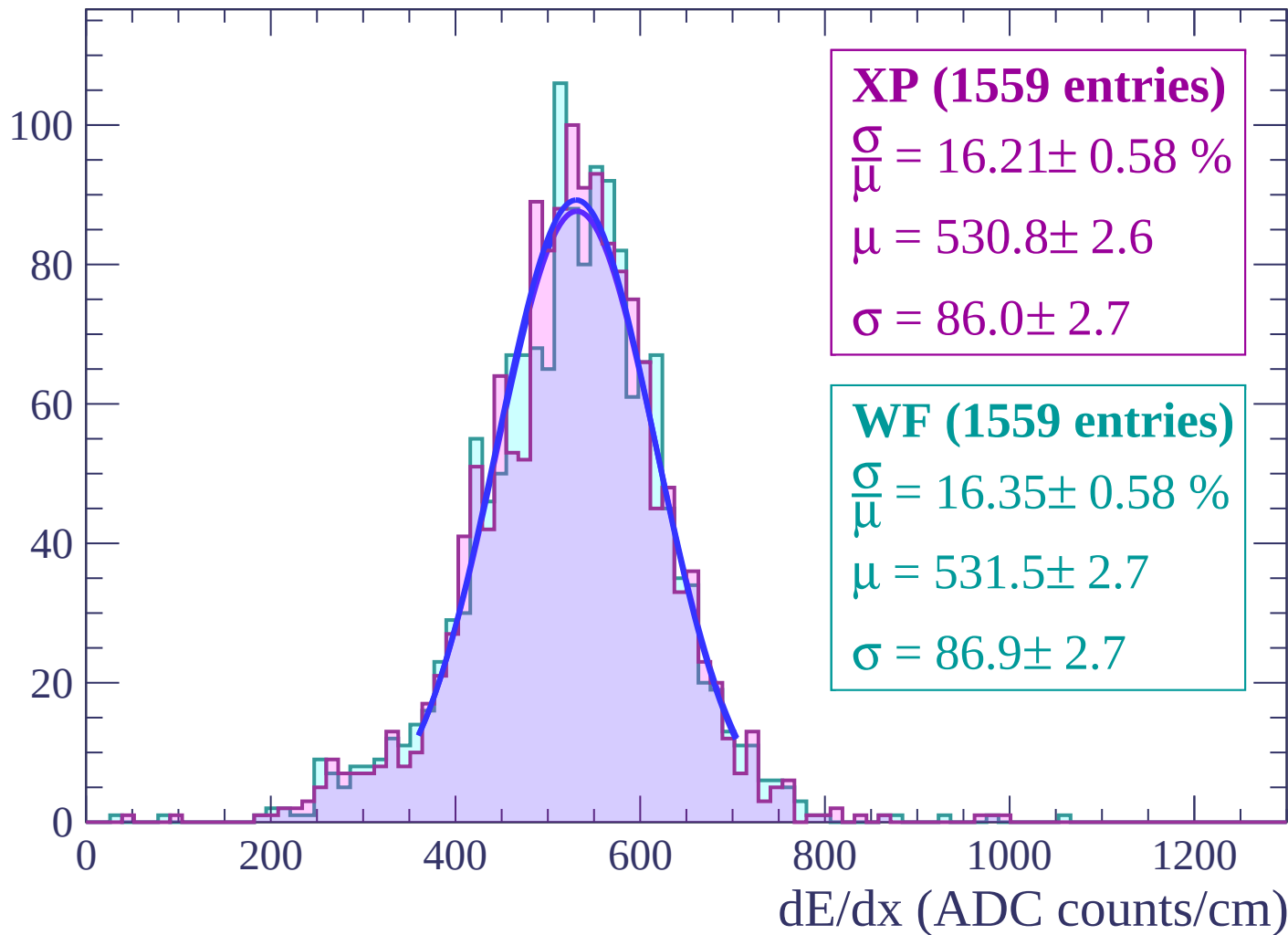
Energy loss | $50 < \theta < 52$

Count



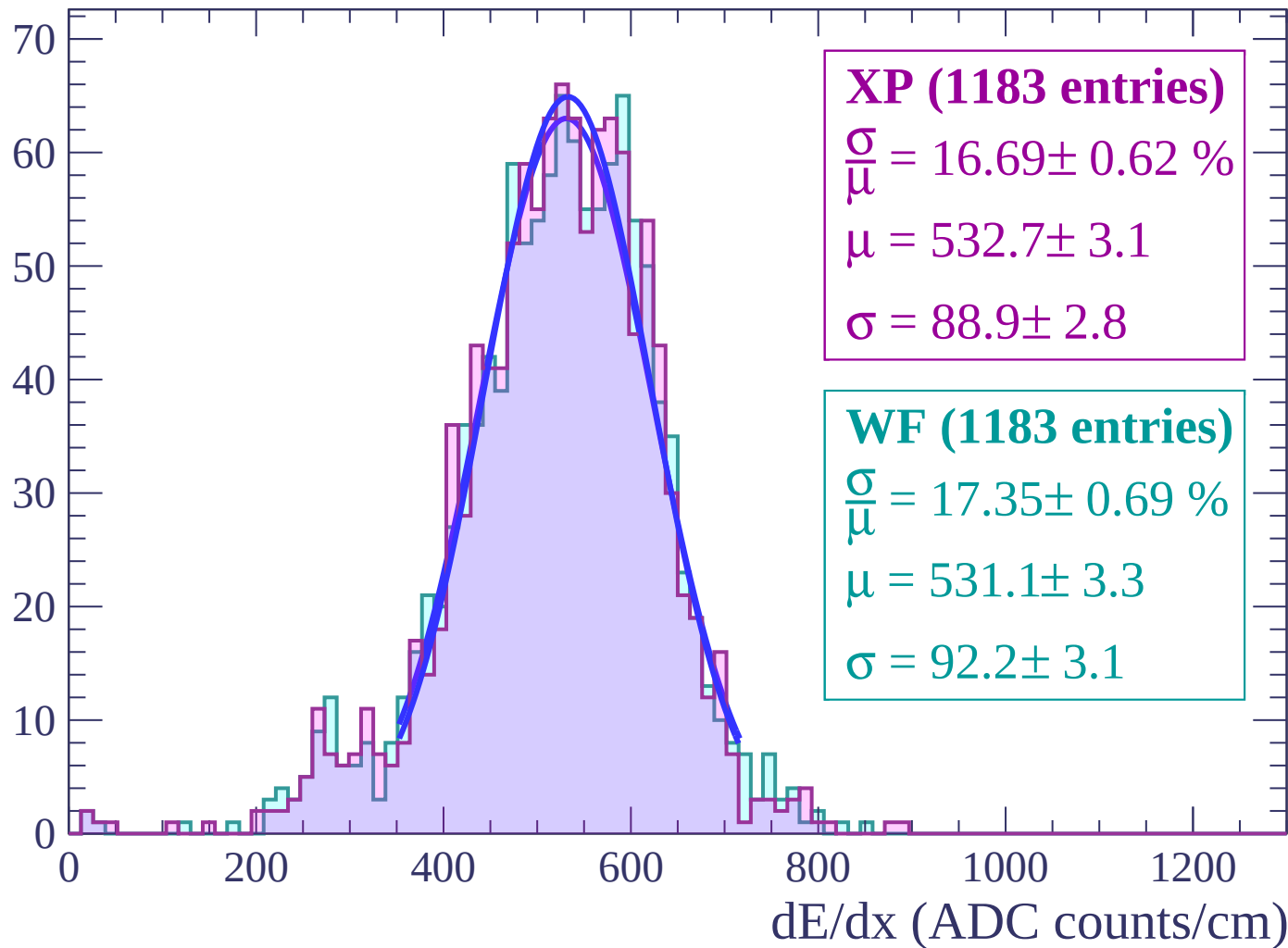
Energy loss | $52 < \theta < 54$

Count



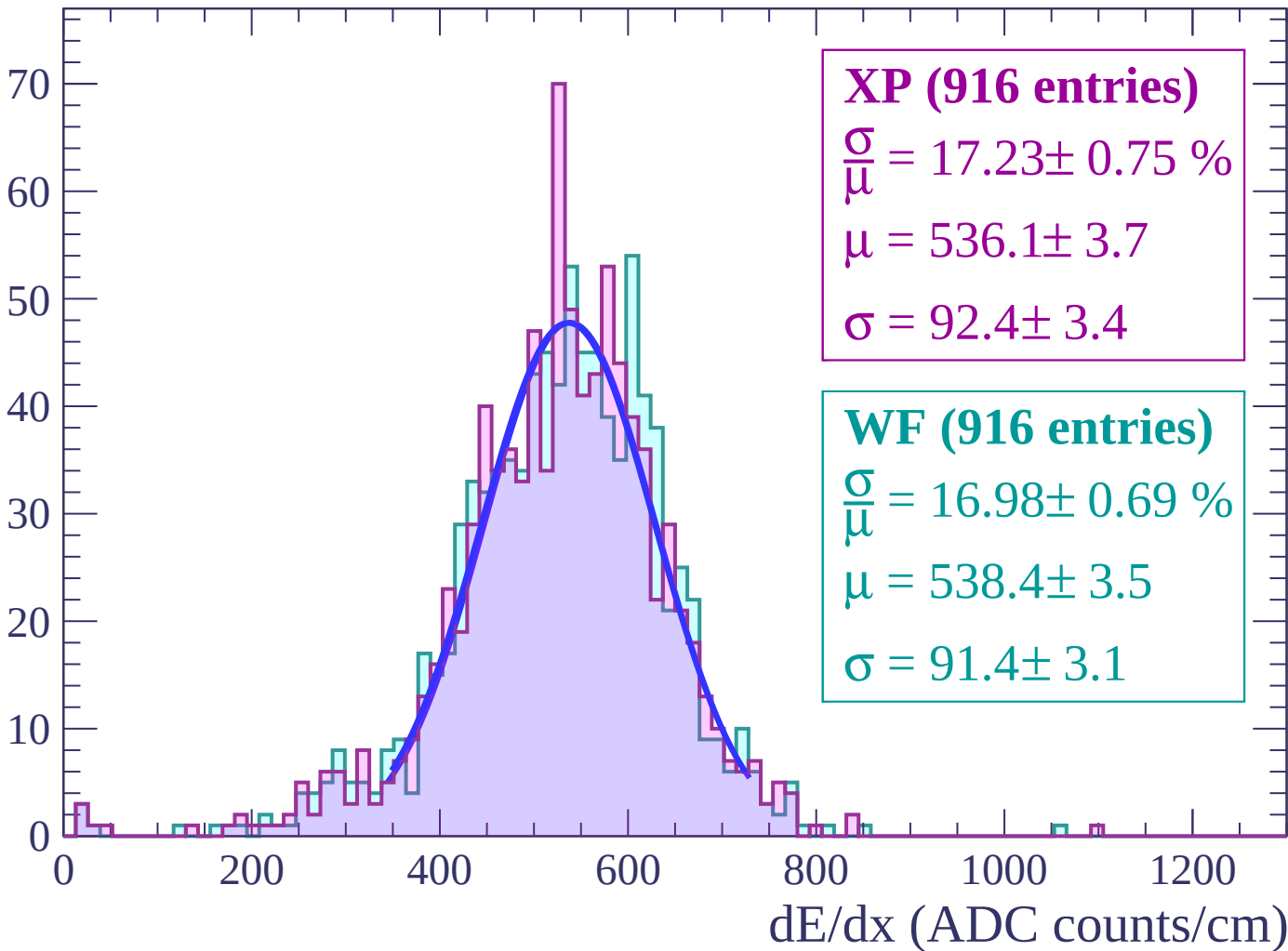
Energy loss | $54 < \theta < 56$

Count



Energy loss | $56 < \theta < 58$

Count



Energy loss | $58 < \theta < 60$

Count

XP (653 entries)

$$\frac{\sigma}{\mu} = 15.59 \pm 0.92 \%$$

$$\mu = 527.8 \pm 4.0$$

$$\sigma = 82.3 \pm 4.2$$

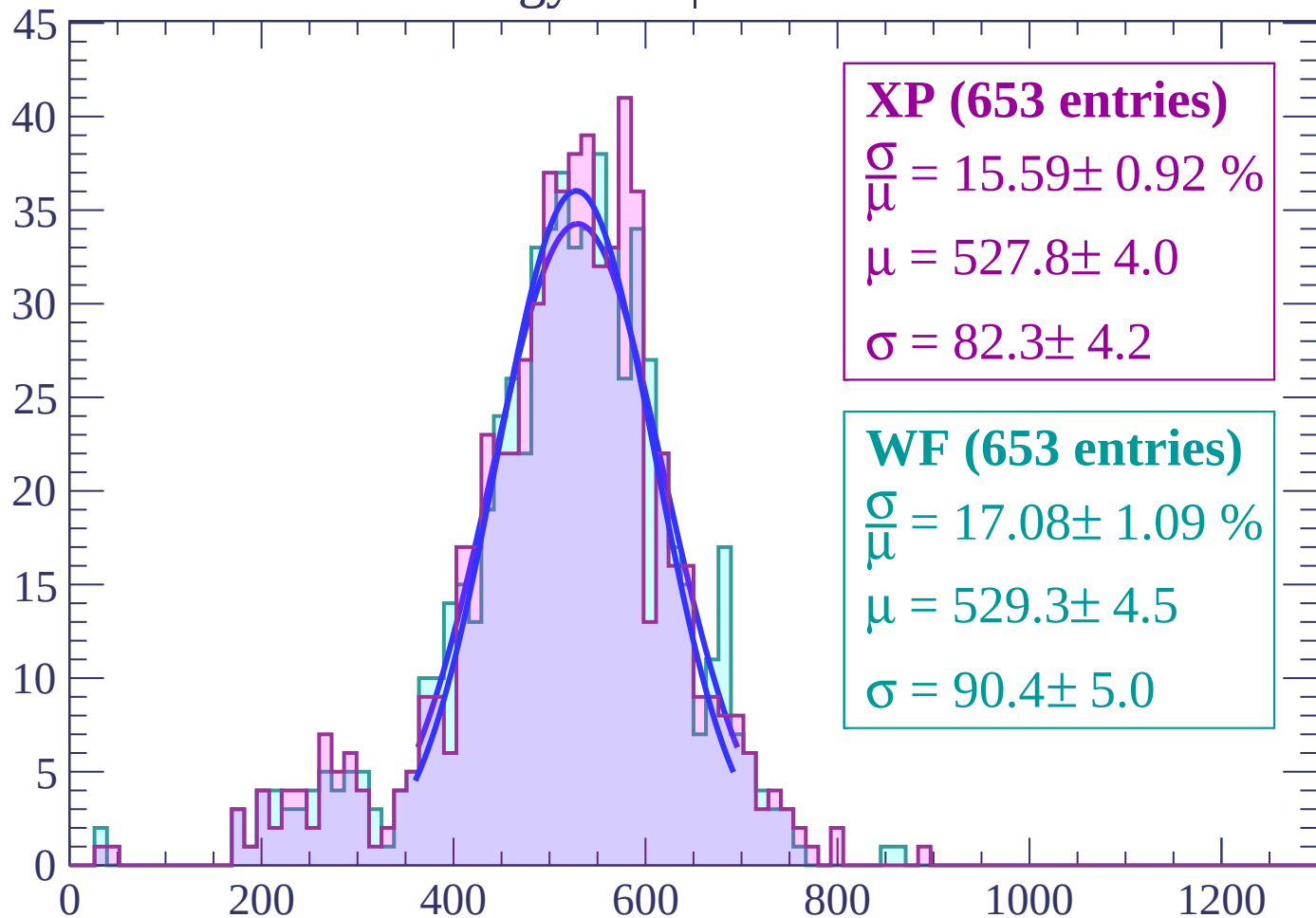
WF (653 entries)

$$\frac{\sigma}{\mu} = 17.08 \pm 1.09 \%$$

$$\mu = 529.3 \pm 4.5$$

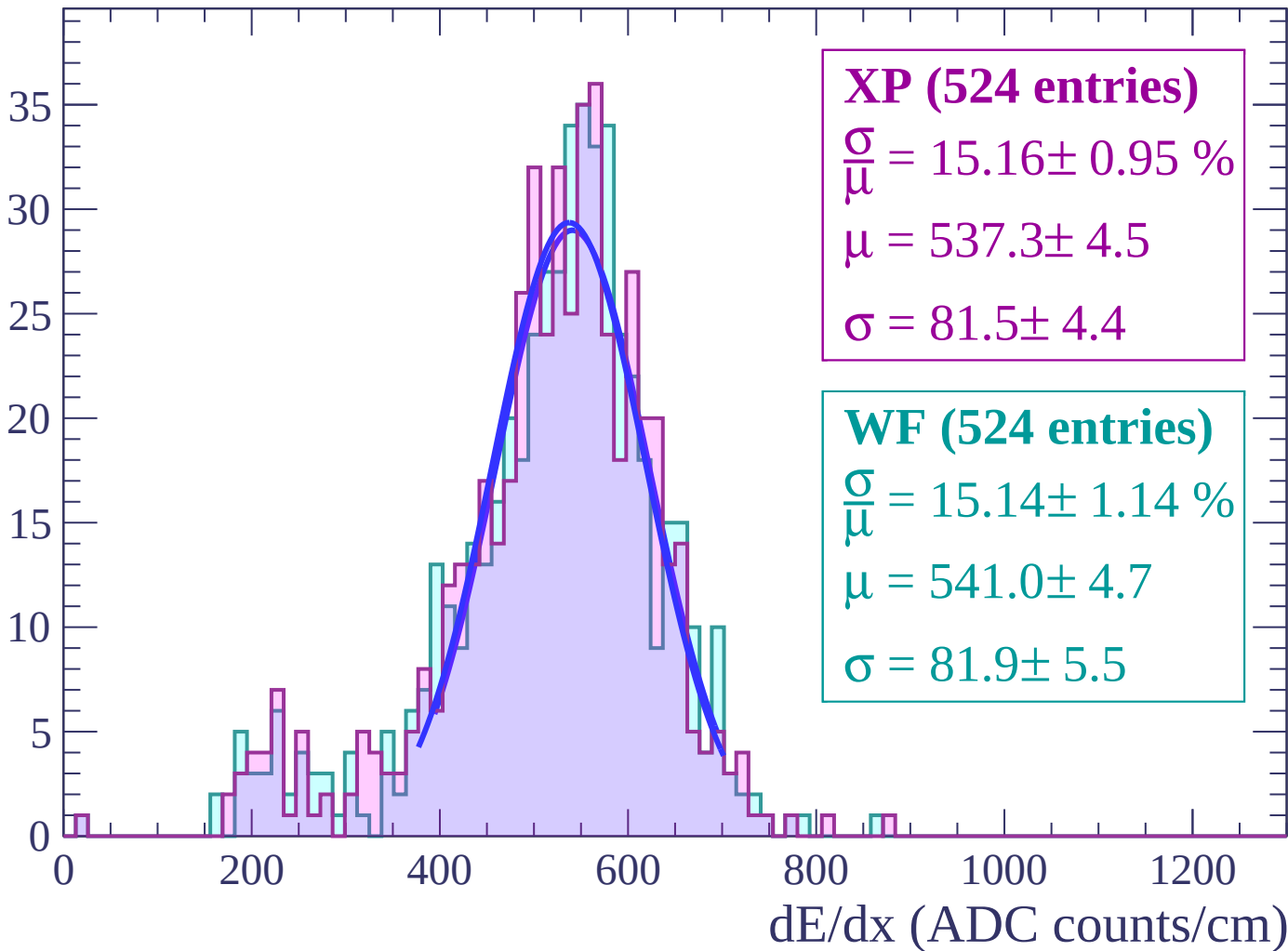
$$\sigma = 90.4 \pm 5.0$$

dE/dx (ADC counts/cm)



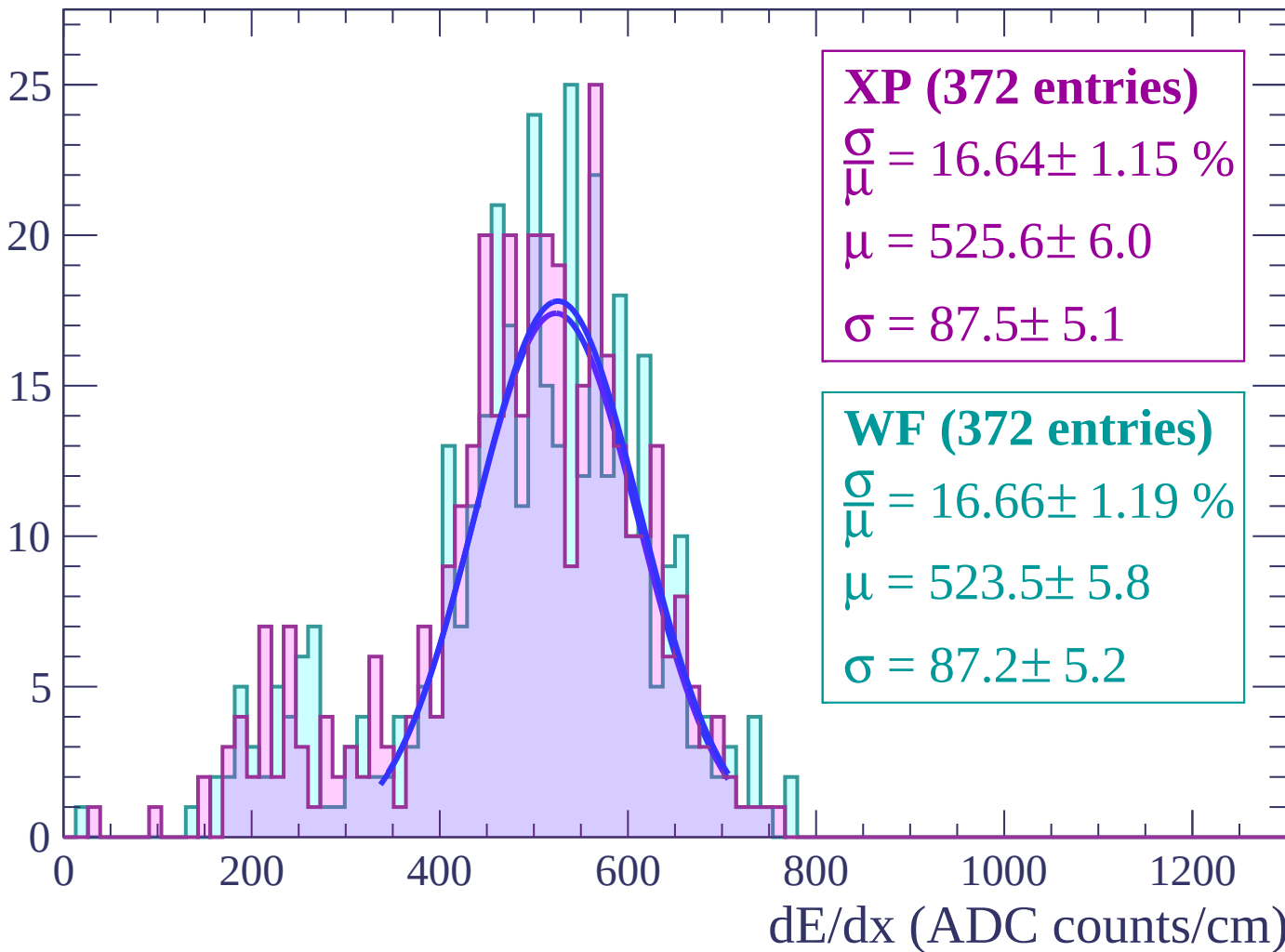
Energy loss | $60 < \theta < 62$

Count



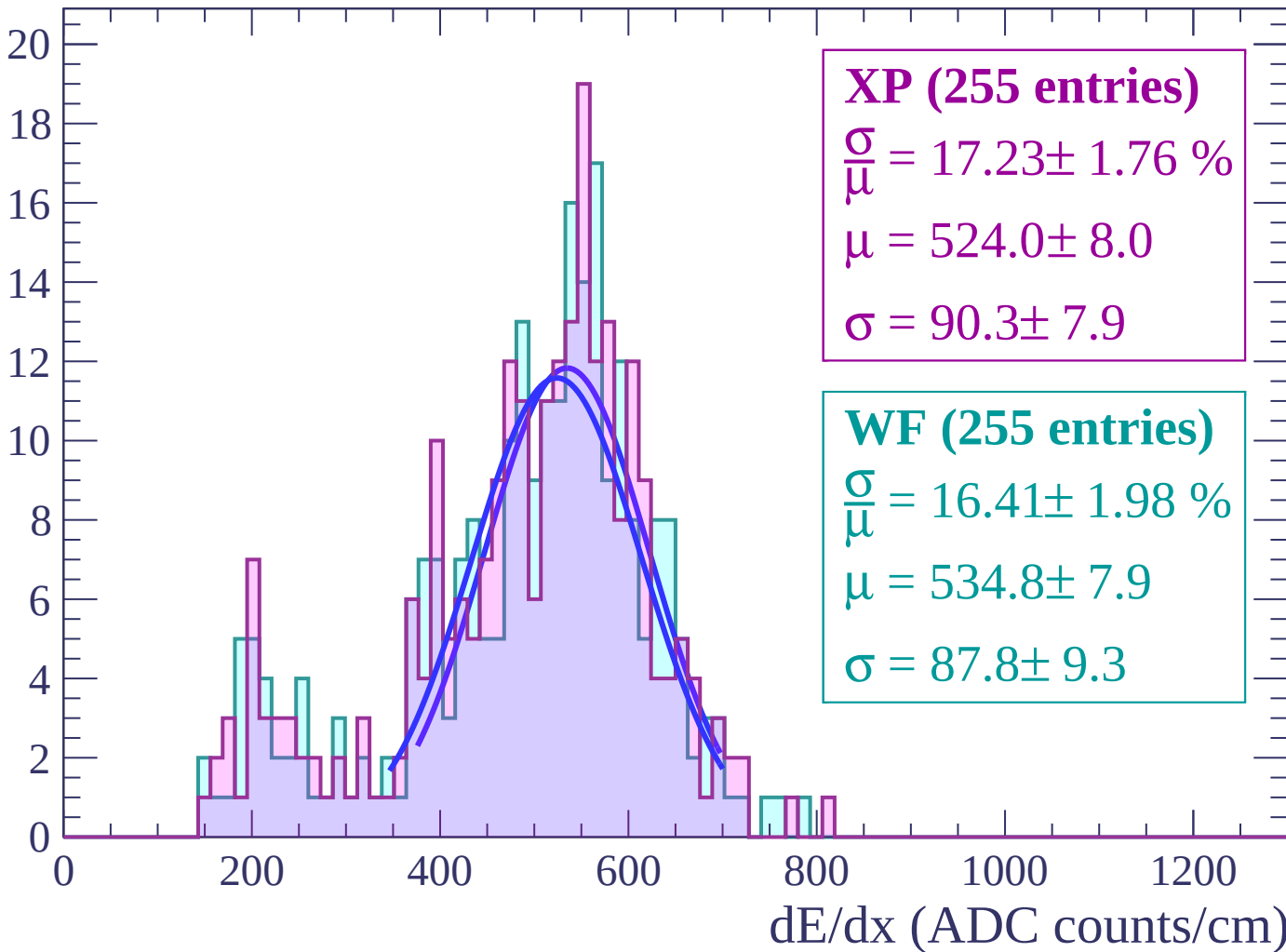
Energy loss | $62 < \theta < 64$

Count



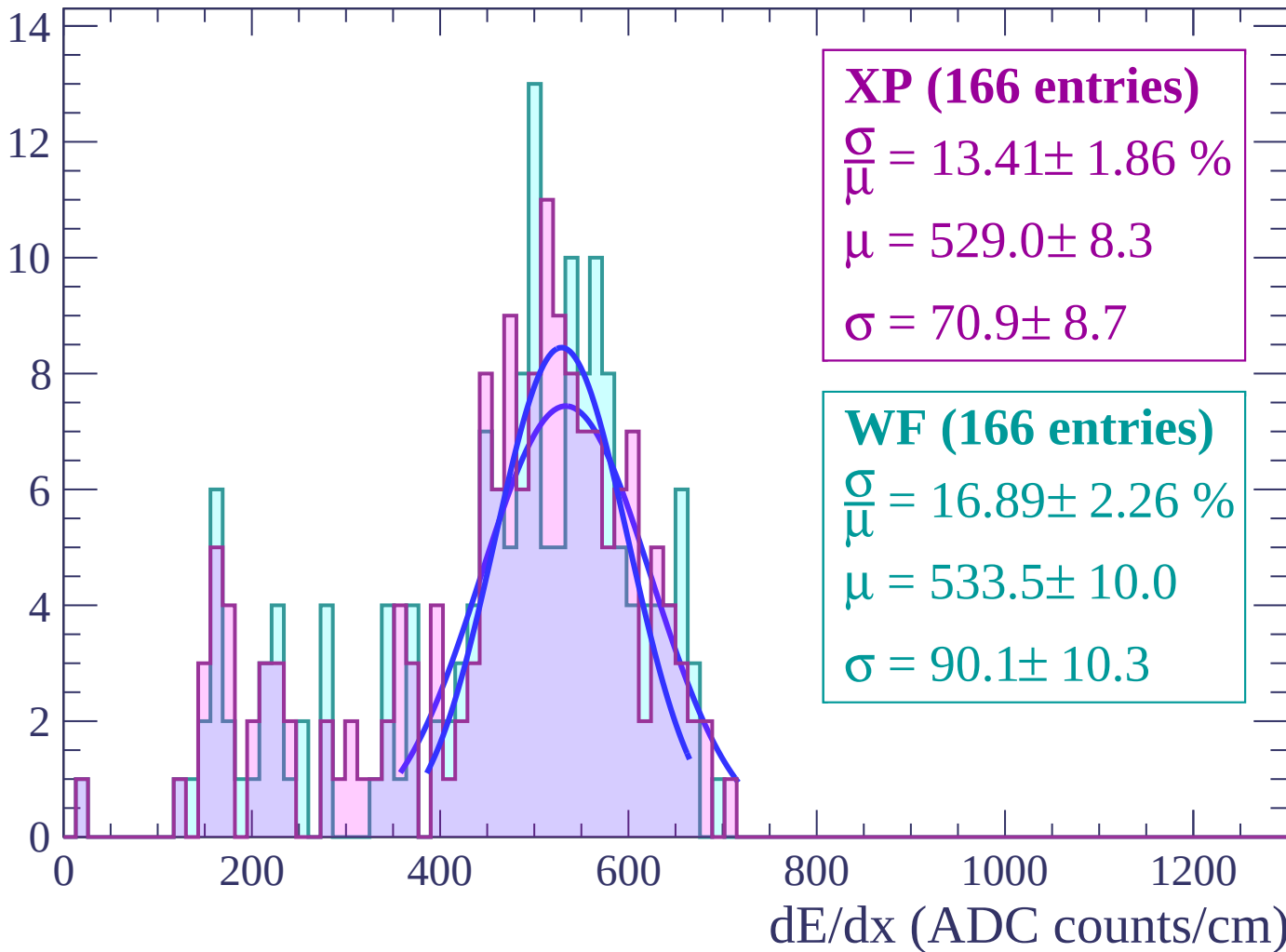
Energy loss | $64 < \theta < 66$

Count



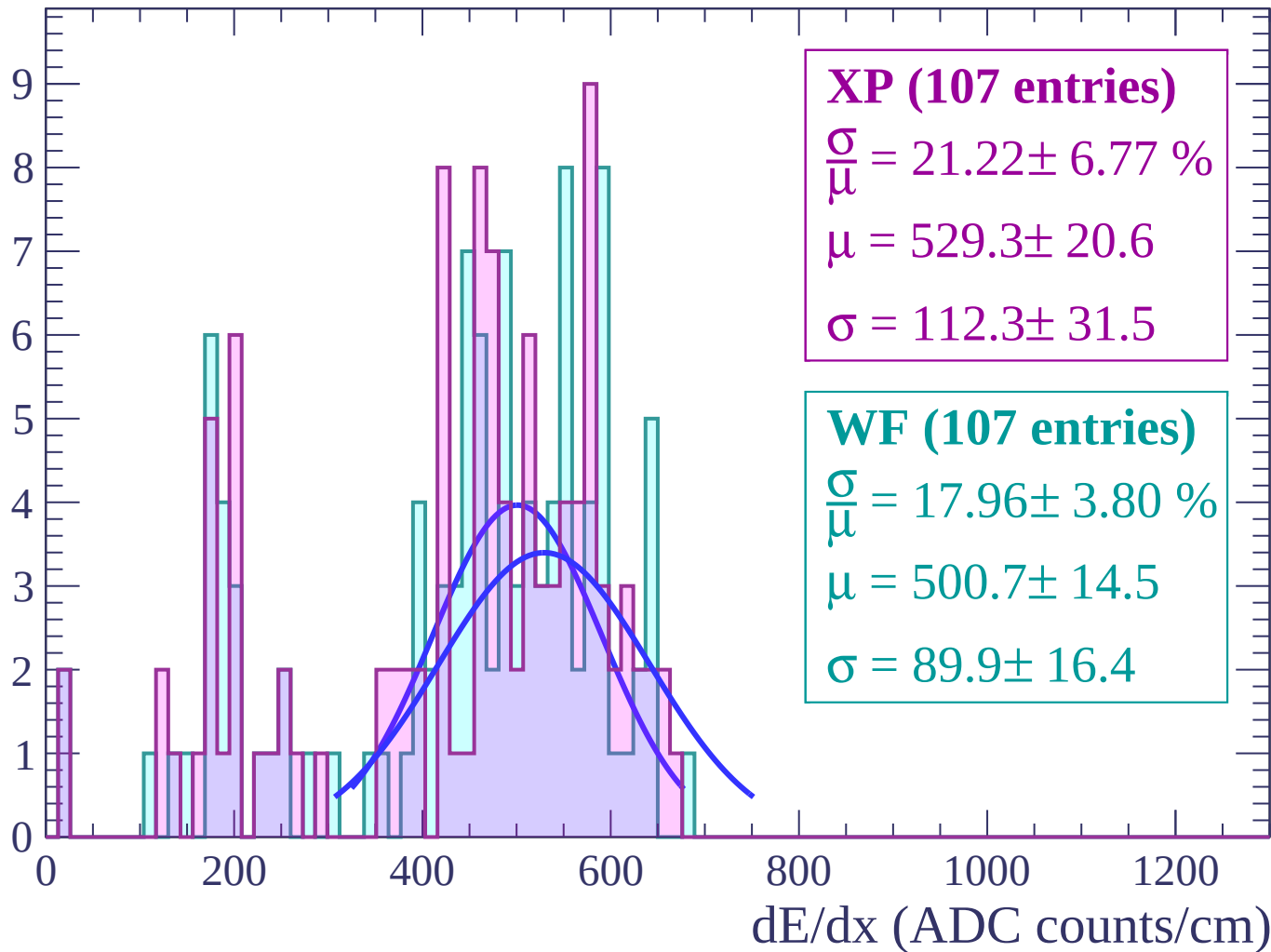
Energy loss | $66 < \theta < 68$

Count



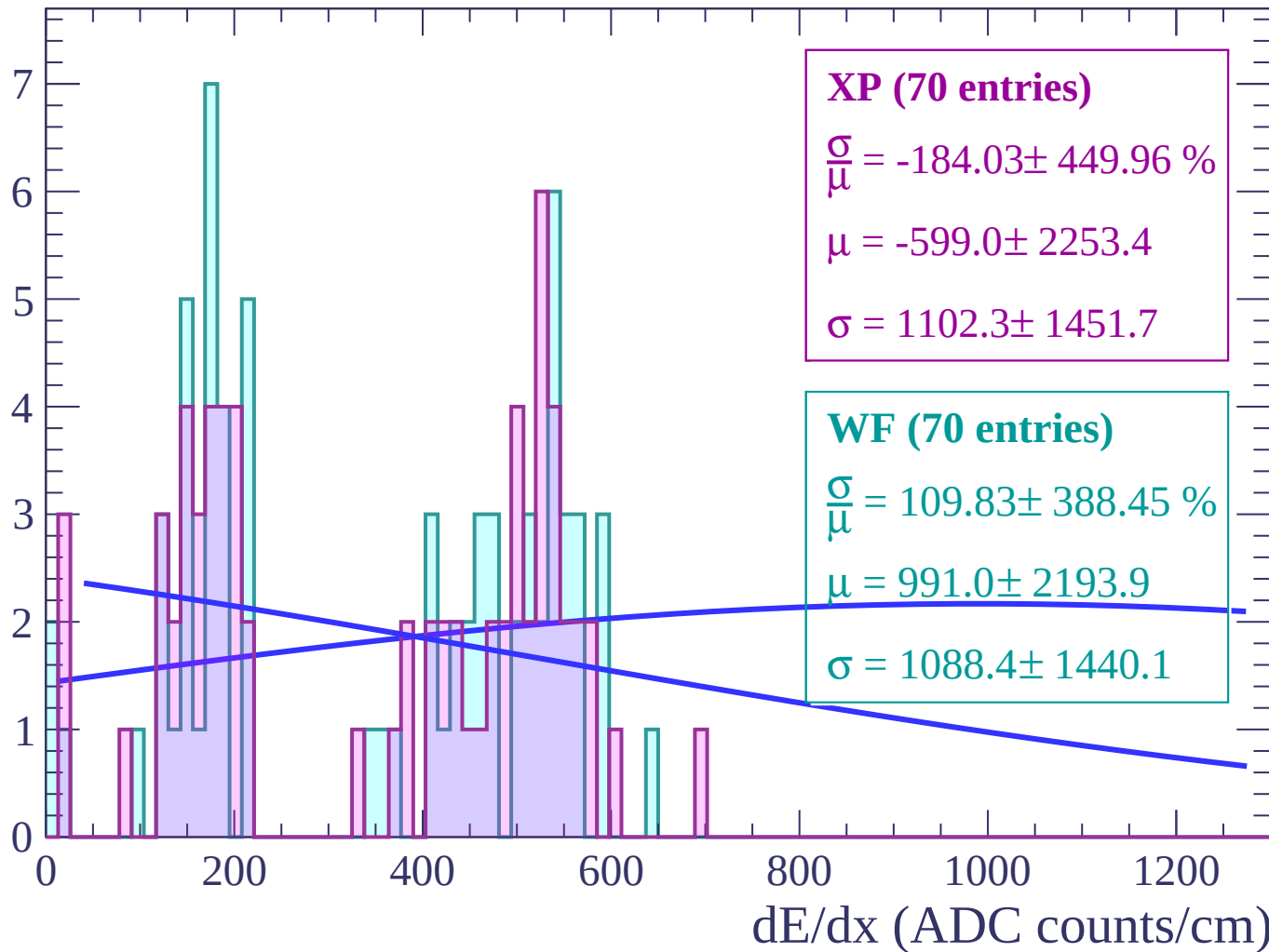
Energy loss | $68 < \theta < 70$

Count



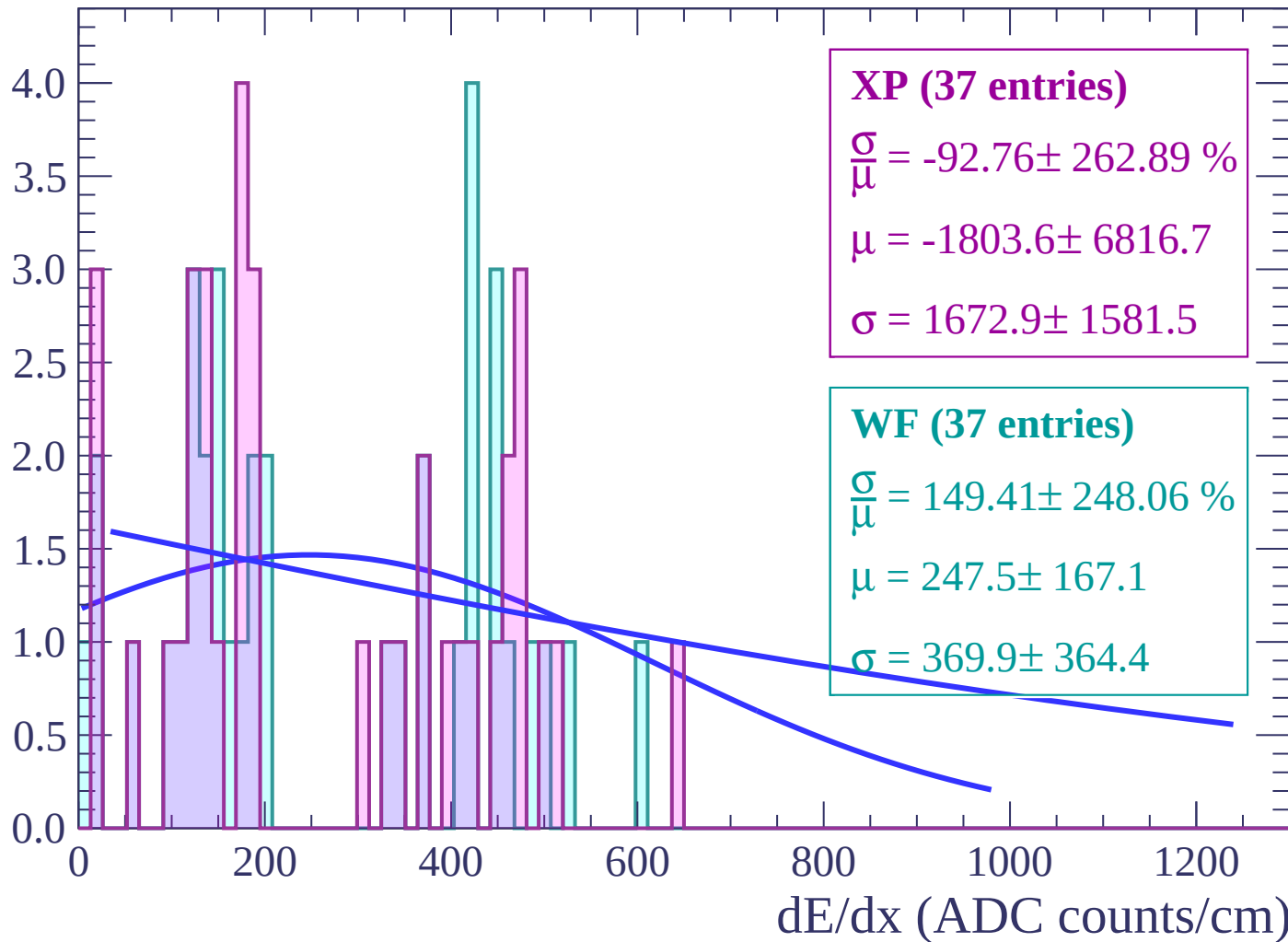
Energy loss | $70 < \theta < 72$

Count



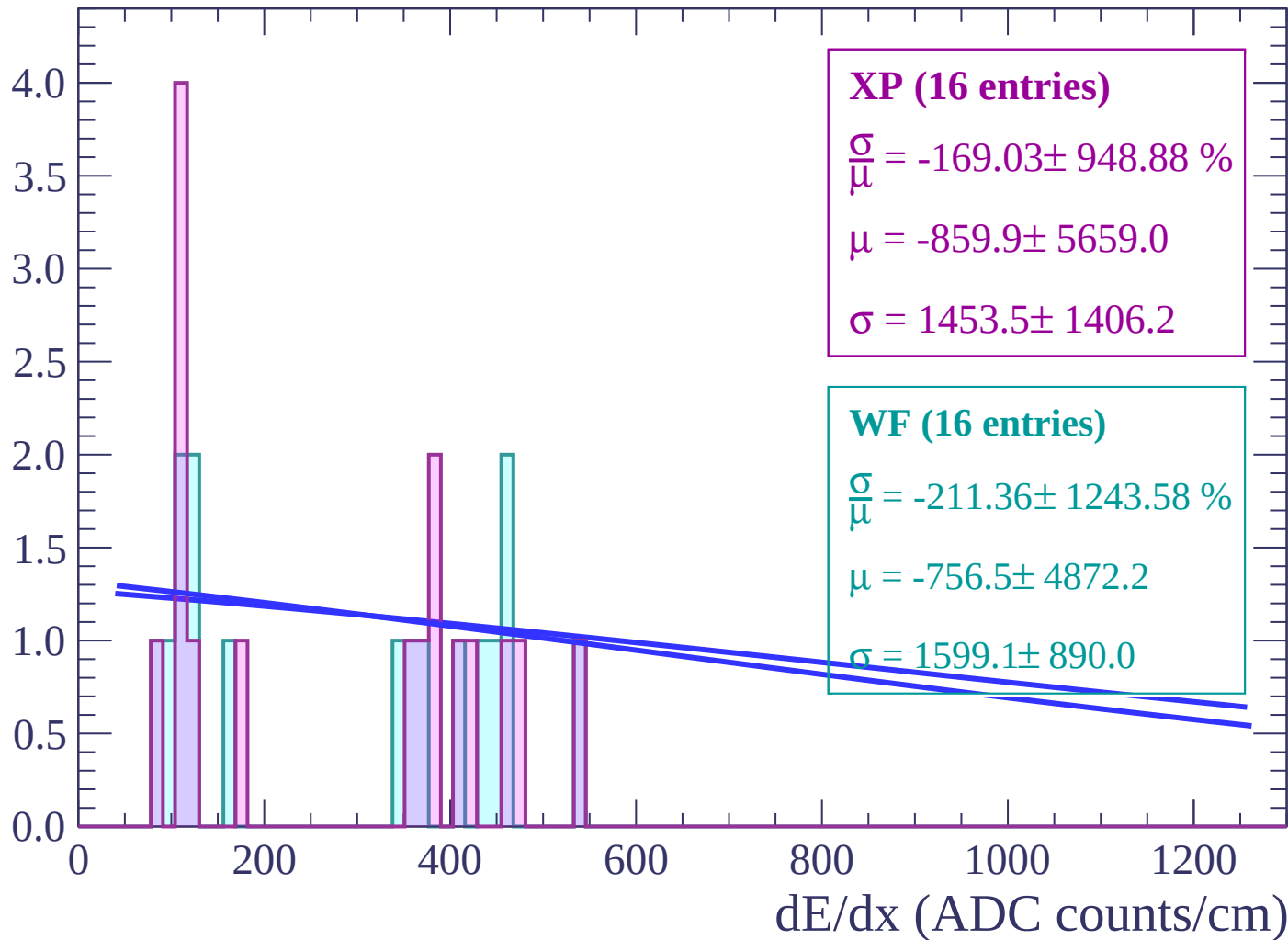
Energy loss | $72 < \theta < 74$

Count



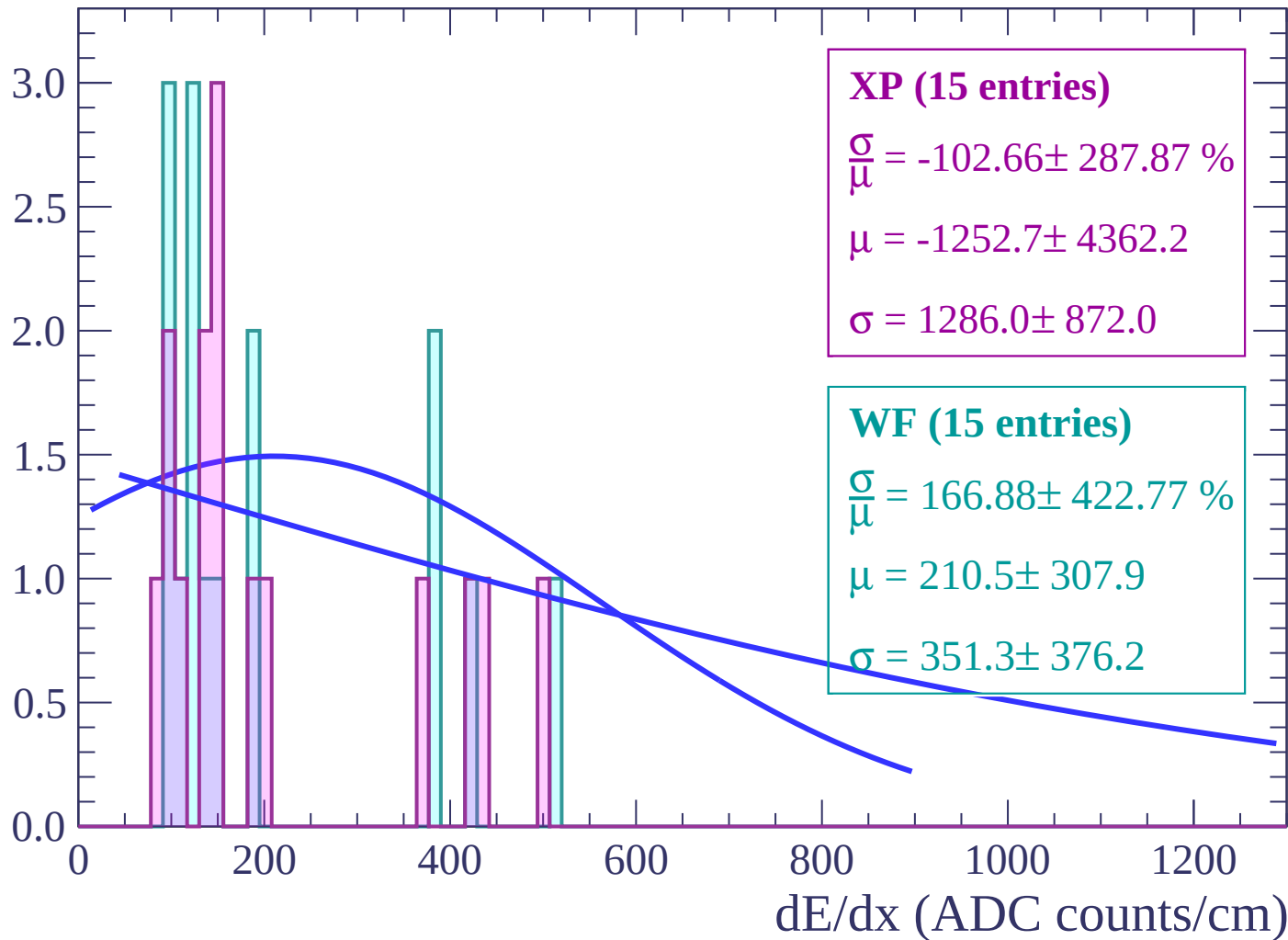
Energy loss | $74 < \theta < 76$

Count



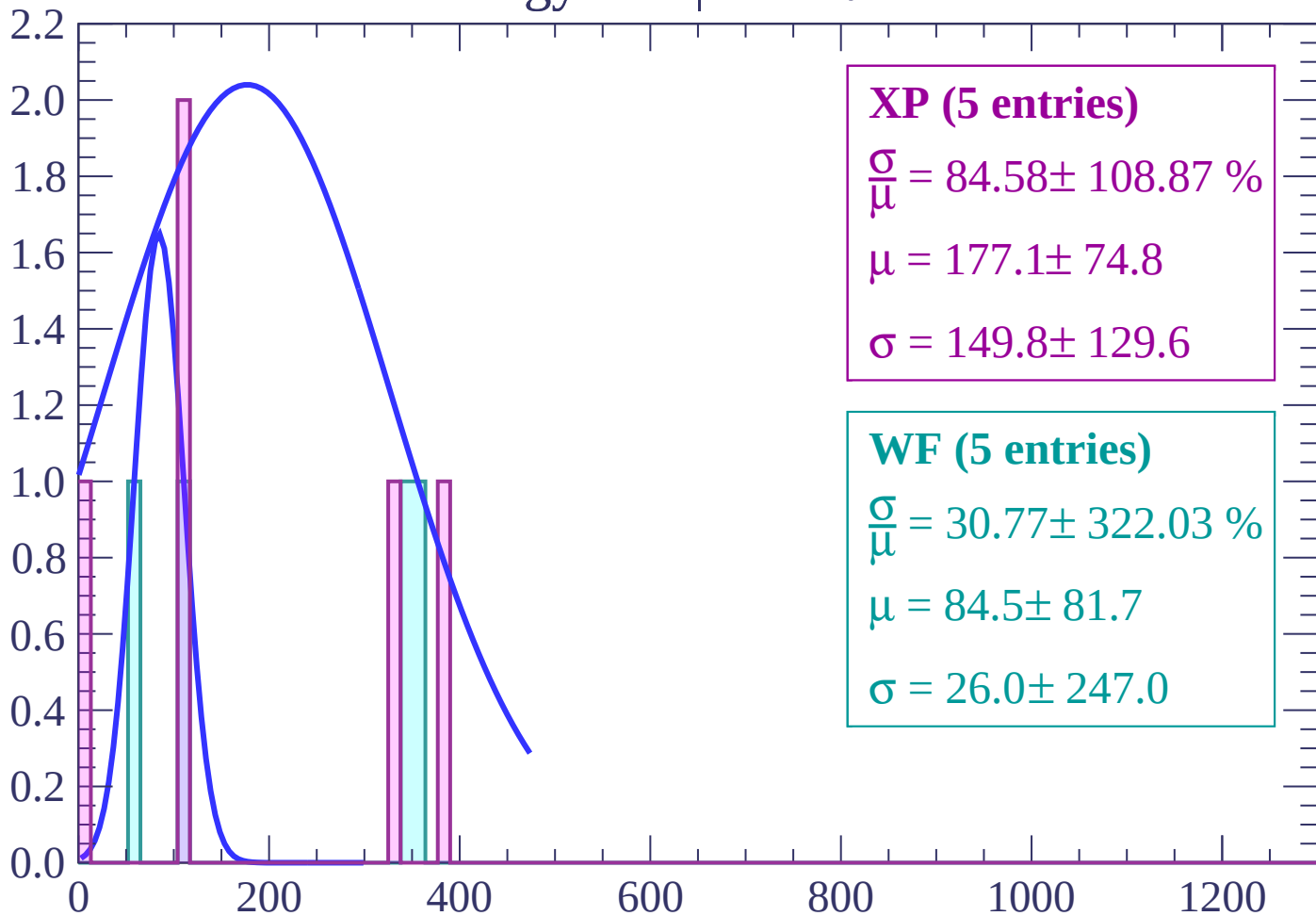
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count



XP (5 entries)

$\frac{\sigma}{\mu} = 84.58 \pm 108.87 \%$

$\mu = 177.1 \pm 74.8$

$\sigma = 149.8 \pm 129.6$

WF (5 entries)

$\frac{\sigma}{\mu} = 30.77 \pm 322.03 \%$

$\mu = 84.5 \pm 81.7$

$\sigma = 26.0 \pm 247.0$

dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $82 < \theta < 84$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

